

(12) **United States Patent**
Kozuma et al.

(10) **Patent No.:** **US 12,400,569 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **DISPLAY DEVICE AND ELECTRONIC DEVICE**

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Takanori Matsuzaki, Kanagawa (JP)

(73) Assignee: **Semiconductor Energy Laboratory Co., Ltd.**, Kanagawa-ken (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/273,095**

(22) PCT Filed: **Jan. 17, 2022**

(86) PCT No.: **PCT/IB2022/050333**
§ 371 (c)(1),
(2) Date: **Jul. 19, 2023**

(87) PCT Pub. No.: **WO2022/162490**
PCT Pub. Date: **Aug. 4, 2022**

(65) **Prior Publication Data**
US 2024/0090284 A1 Mar. 14, 2024

(30) **Foreign Application Priority Data**
Jan. 28, 2021 (JP) 2021-011834
Feb. 19, 2021 (JP) 2021-024802
Feb. 25, 2021 (JP) 2021-028973

(51) **Int. Cl.**
G09G 3/20 (2006.01)
G09G 3/3258 (2016.01)
(Continued)

(52) **U.S. Cl.**
CPC **G09G 3/20** (2013.01); **G09G 3/3258** (2013.01); **H10K 59/1213** (2023.02);
(Continued)

(58) **Field of Classification Search**
CPC G09G 3/32-3291; G09G 2300/02; G09G 2300/04-0408; G09G 2300/0421;
(Continued)

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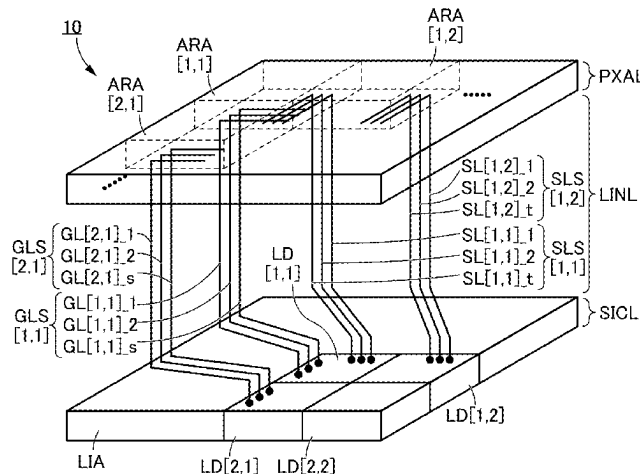
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Primary Examiner — Nathan Danielsen
(74) *Attorney, Agent, or Firm* — NIXON PEABODY LLP; Jeffrey L. Costellia

(57) **ABSTRACT**
A high-resolution display device in which delay of input signals to pixels is reduced is provided. In the display device, a first layer, a second layer, and a third layer are formed in this order from the bottom. The first layer includes a driver circuit and a plurality of first wirings, the second layer includes a plurality of first contact portions, and the third layer includes a pixel array and a plurality of second wirings. The pixel array includes a plurality of pixel circuits. The plurality of second wirings are parallel to each other and extended in the column direction of the pixel array, and the plurality of pixel circuits are electrically connected to the
(Continued)



plurality of second wirings. The driver circuit includes a plurality of output terminals positioned along a first direction. The plurality of first wirings are extended perpendicular to the first direction, and the plurality of output terminals are electrically connected to the plurality of first wirings. The plurality of first wirings are electrically connected to the plurality of second wirings through the plurality of first contact portions.

4 Claims, 40 Drawing Sheets

(51) Int. Cl.

H10K 59/121 (2023.01)
H10K 59/131 (2023.01)
H10K 59/80 (2023.01)

(52) U.S. Cl.

CPC *G09G 2300/0426* (2013.01); *G09G 2310/0297* (2013.01); *H10K 59/131* (2023.02); *H10K 59/871* (2023.02); *H10K 59/872* (2023.02); *H10K 59/873* (2023.02)

(58) Field of Classification Search

CPC *G09G 2300/0426*; *G09G 2310/0264-0278*; *G09G 2310/0297*; *G09G 2310/04*; *G09G 2310/06*; *G09G 2310/08*; *G09G 2330/021*; *H10K 59/00-1216*; *H10K 59/131*

See application file for complete search history.

(56)

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FIG. 1

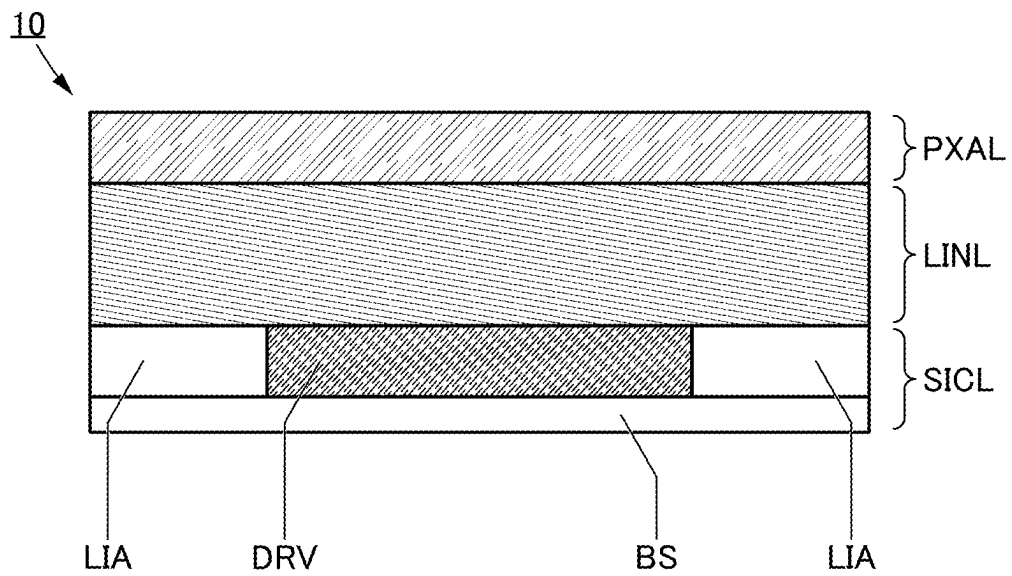


FIG. 2A

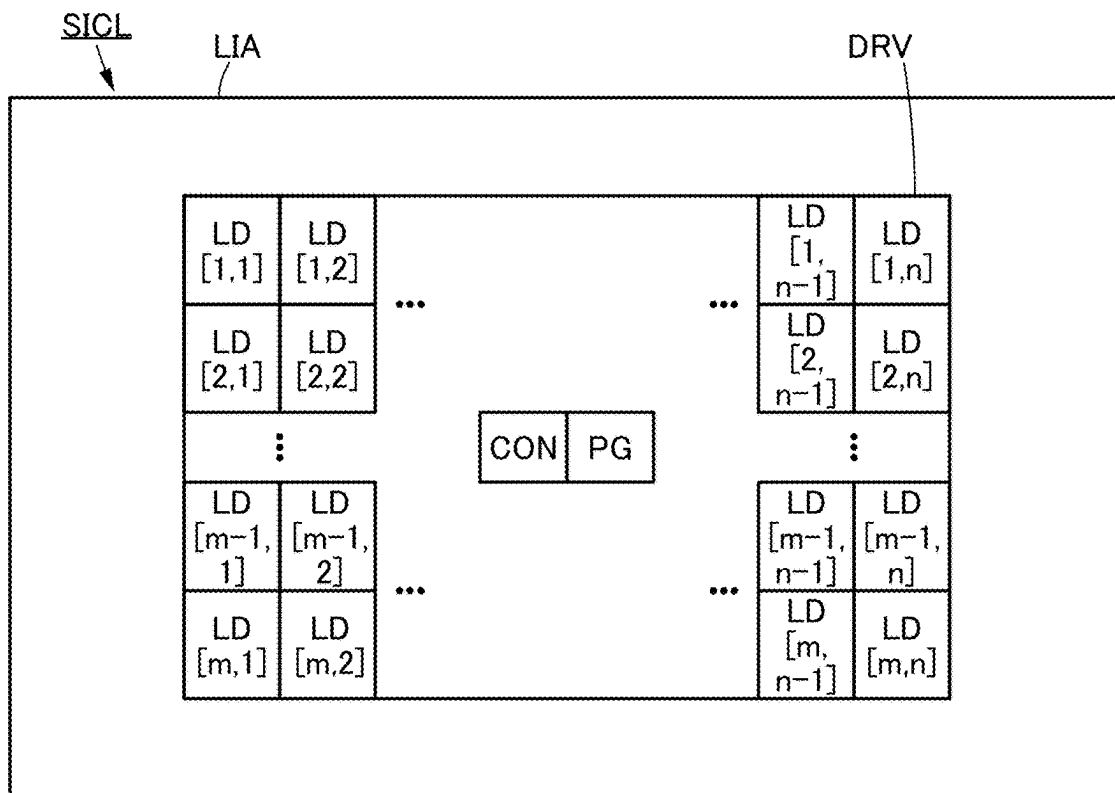


FIG. 2B

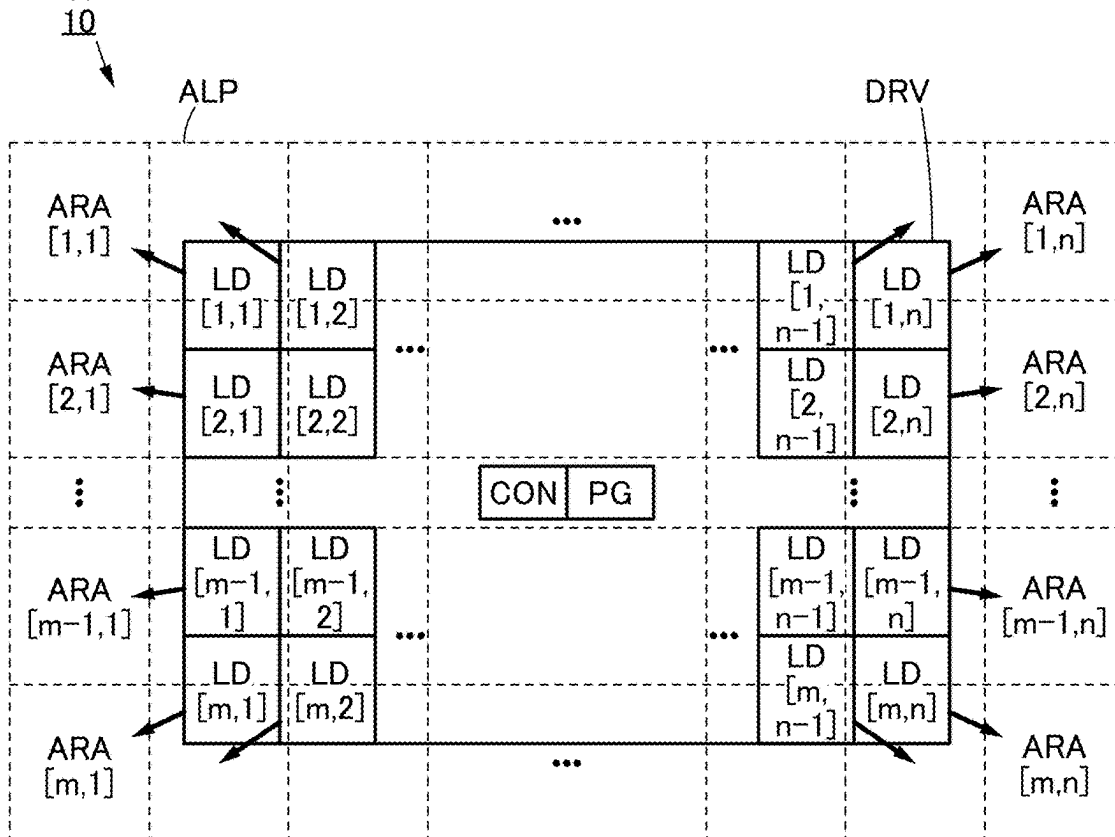


FIG. 3A

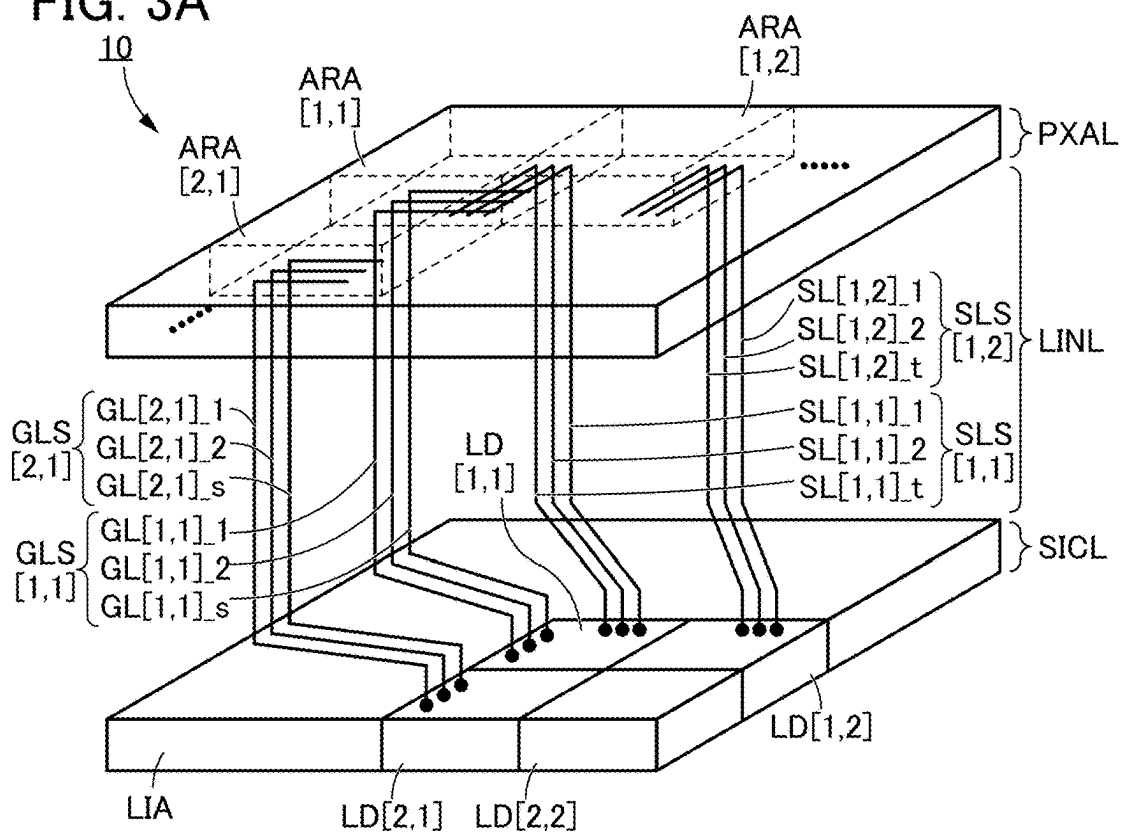


FIG. 3B

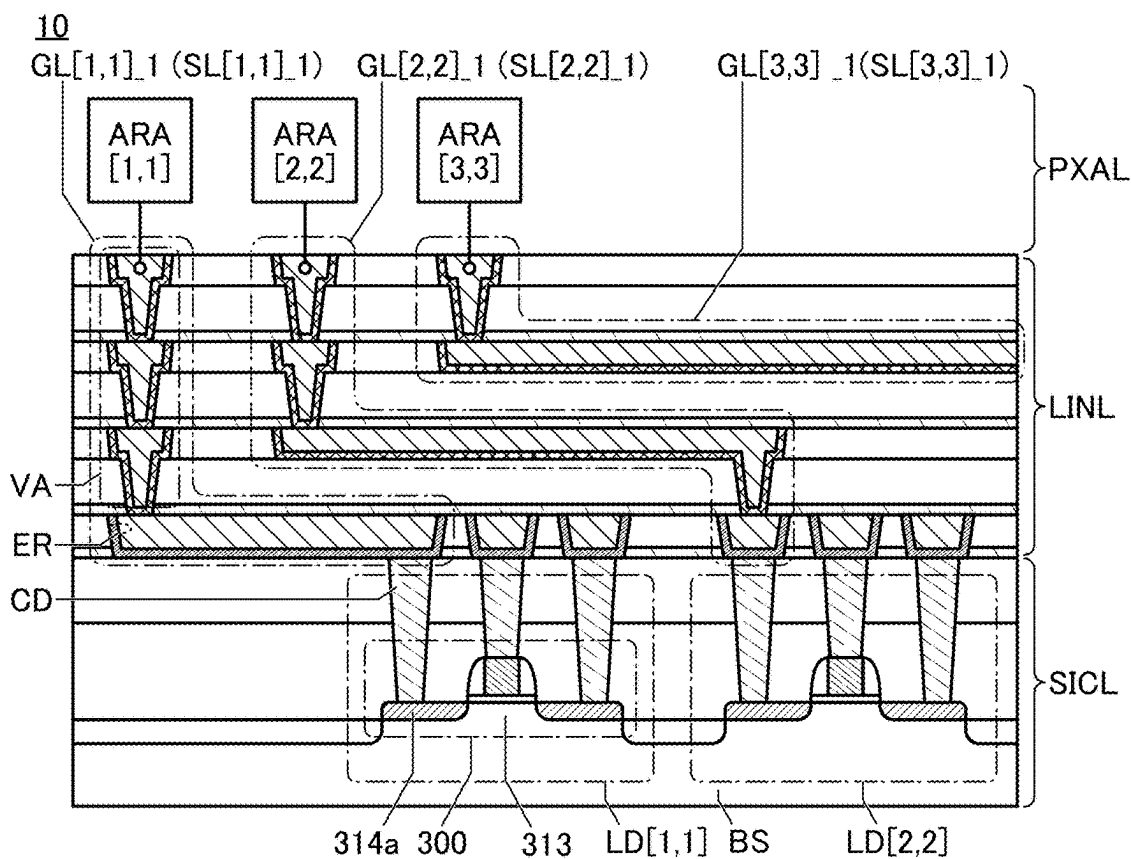


FIG. 4A

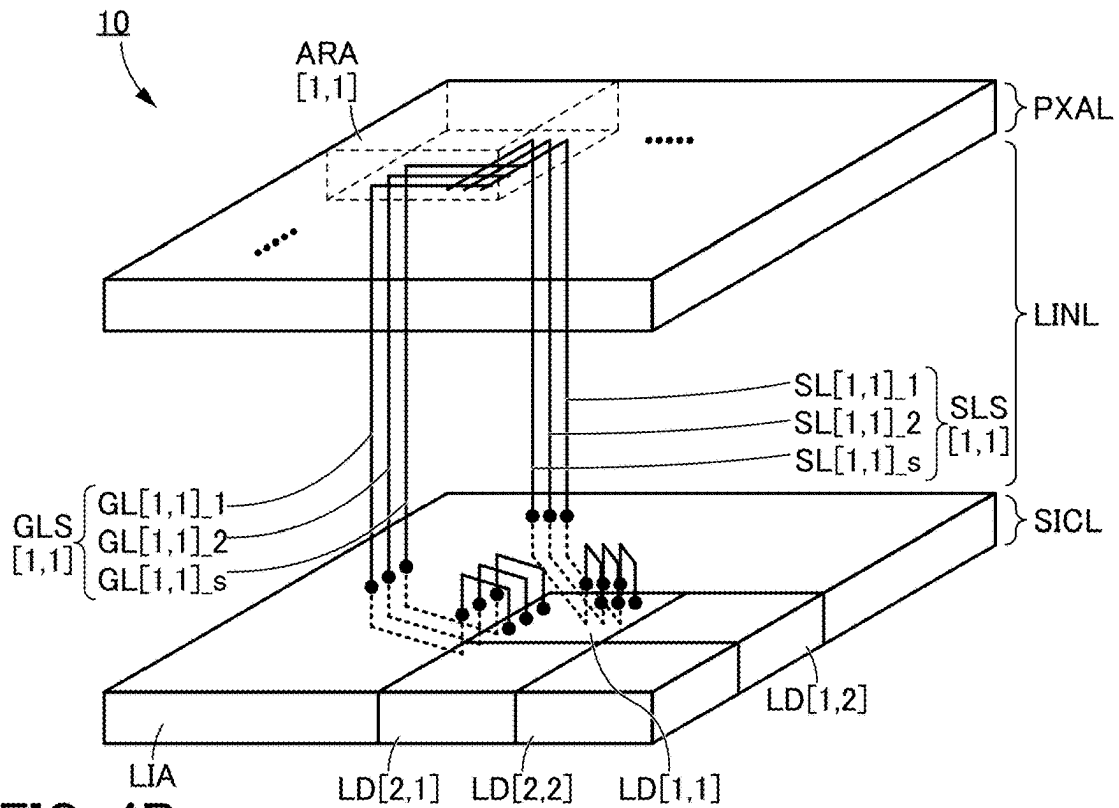


FIG. 4B

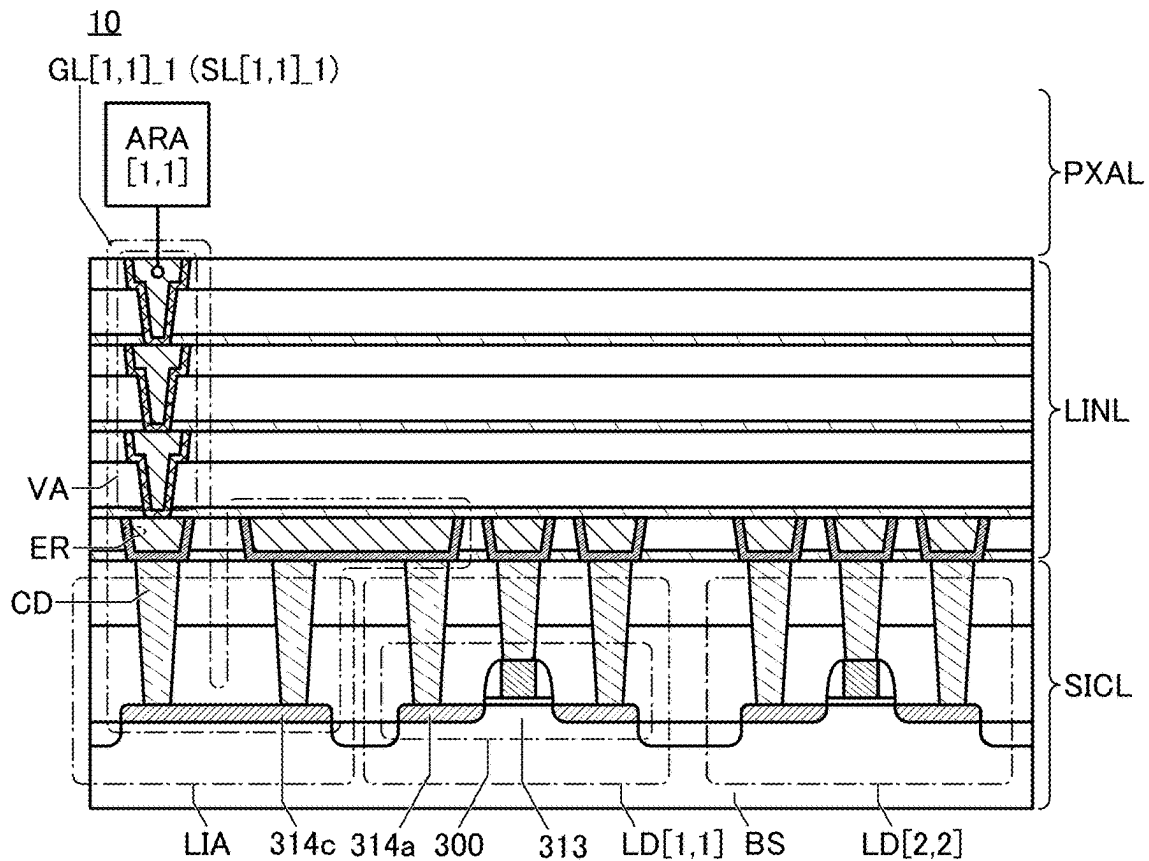


FIG. 5A

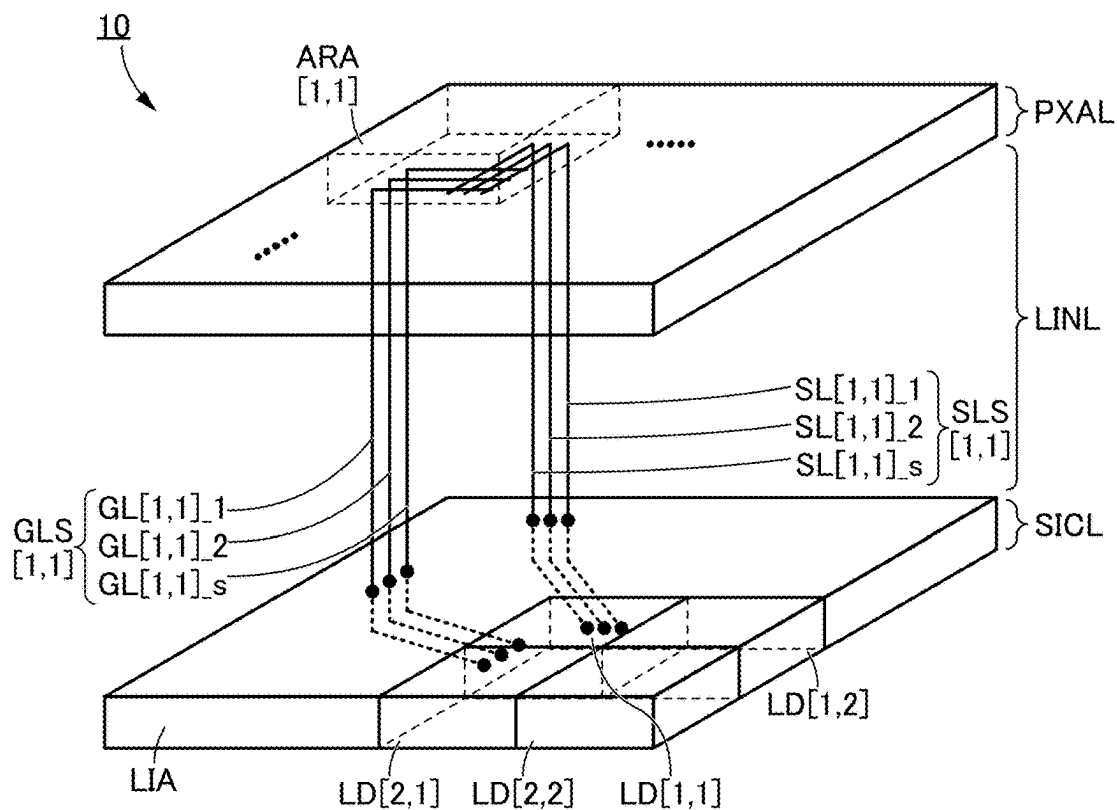


FIG. 5B

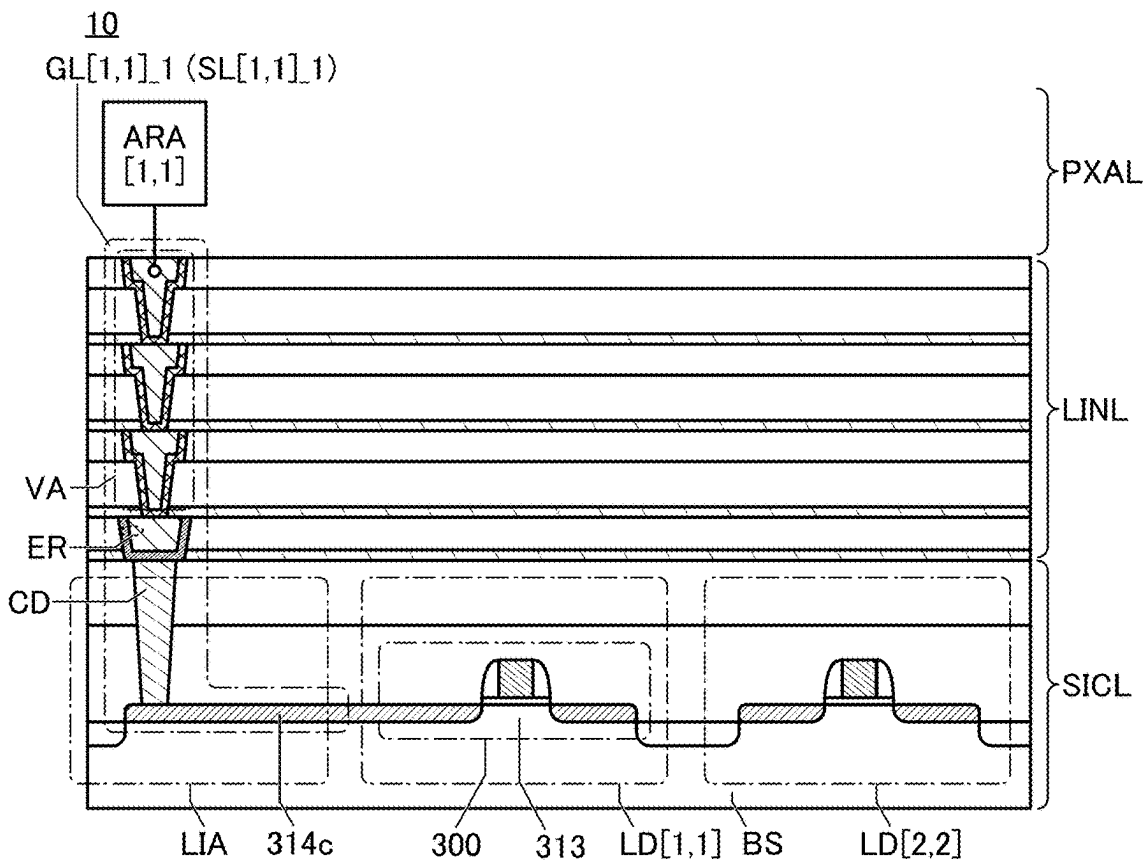


FIG. 6A

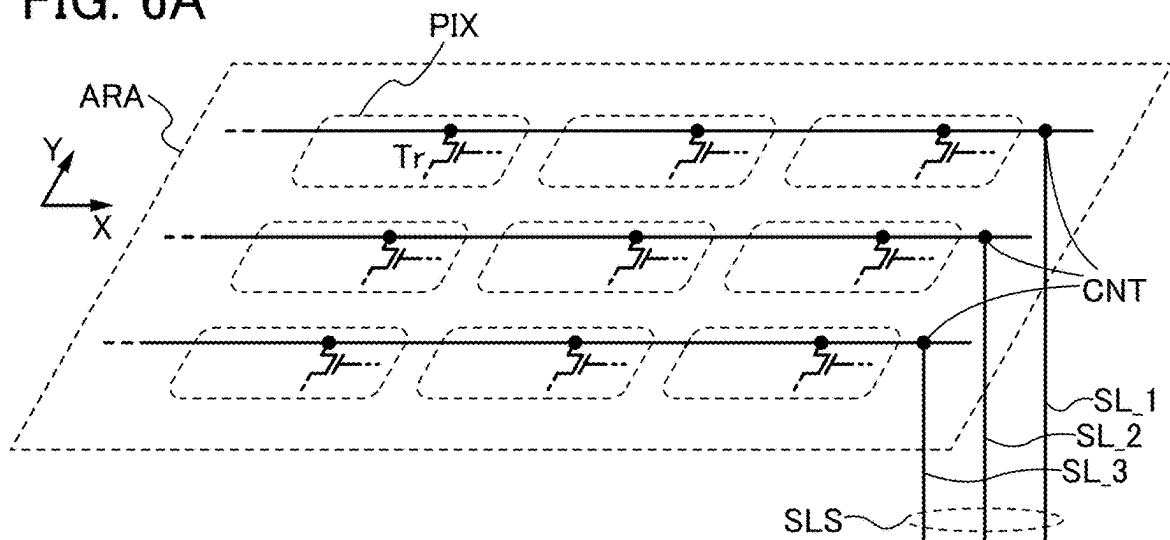


FIG. 6B

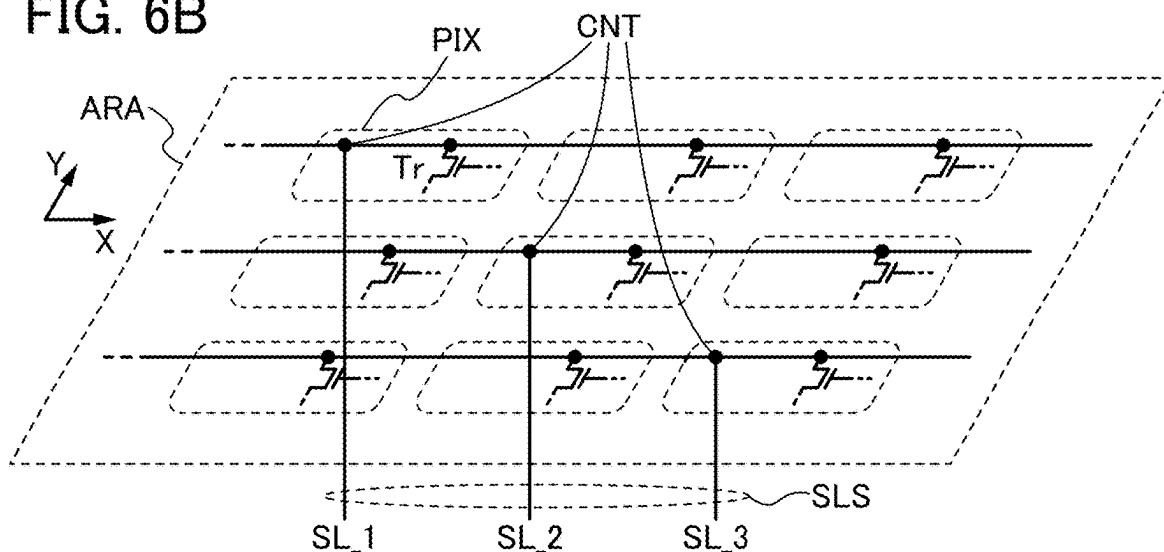


FIG. 6C

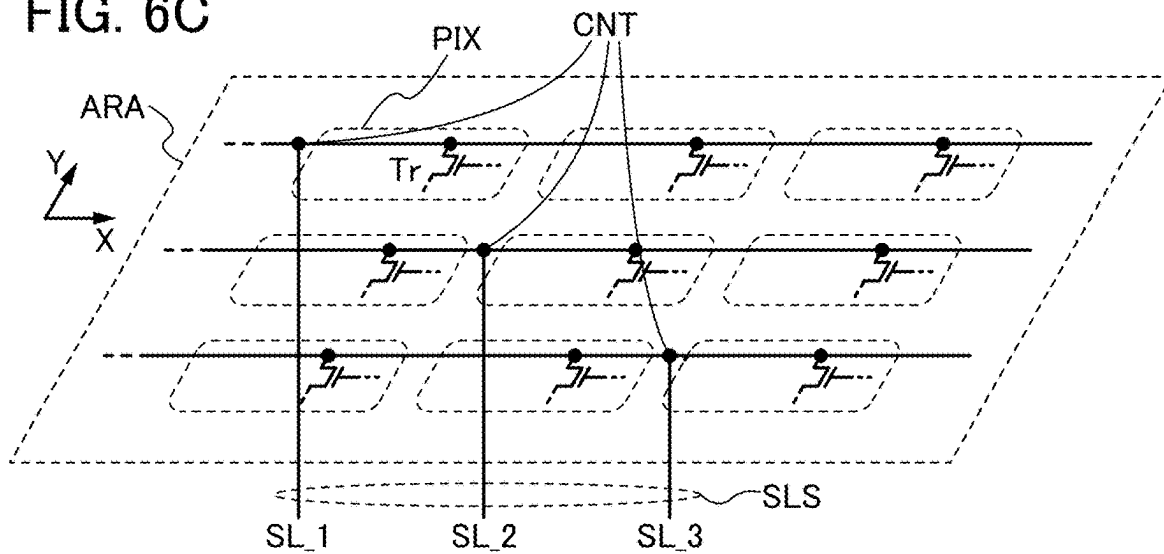


FIG. 7

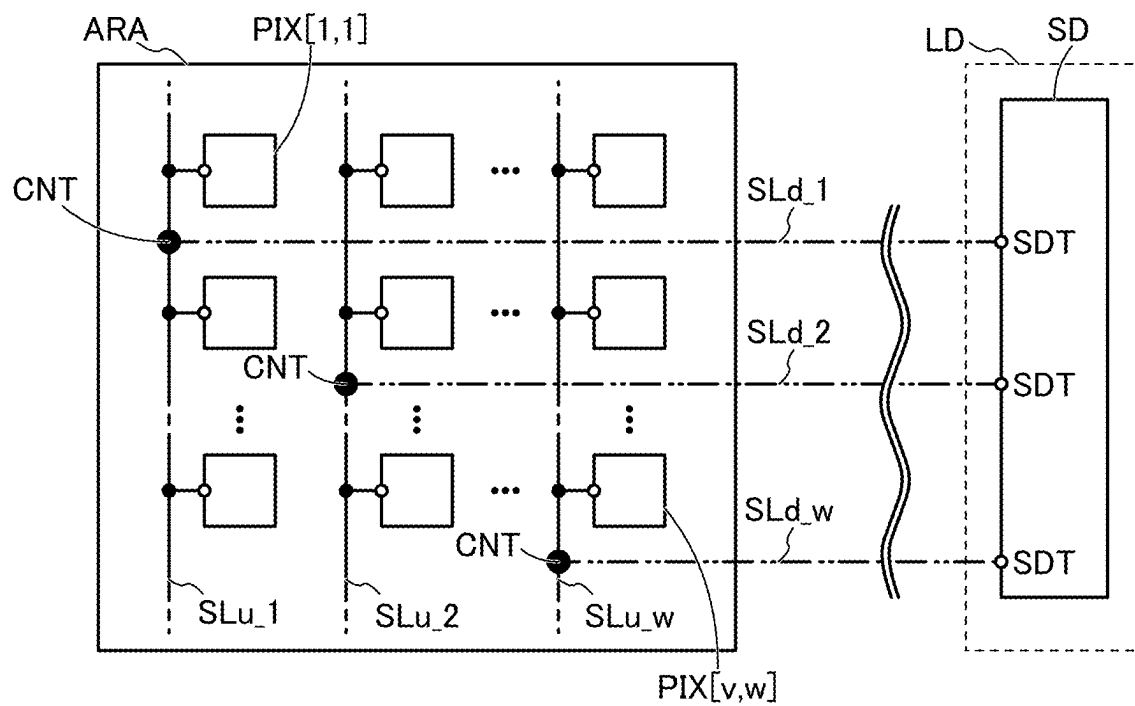


FIG. 8A

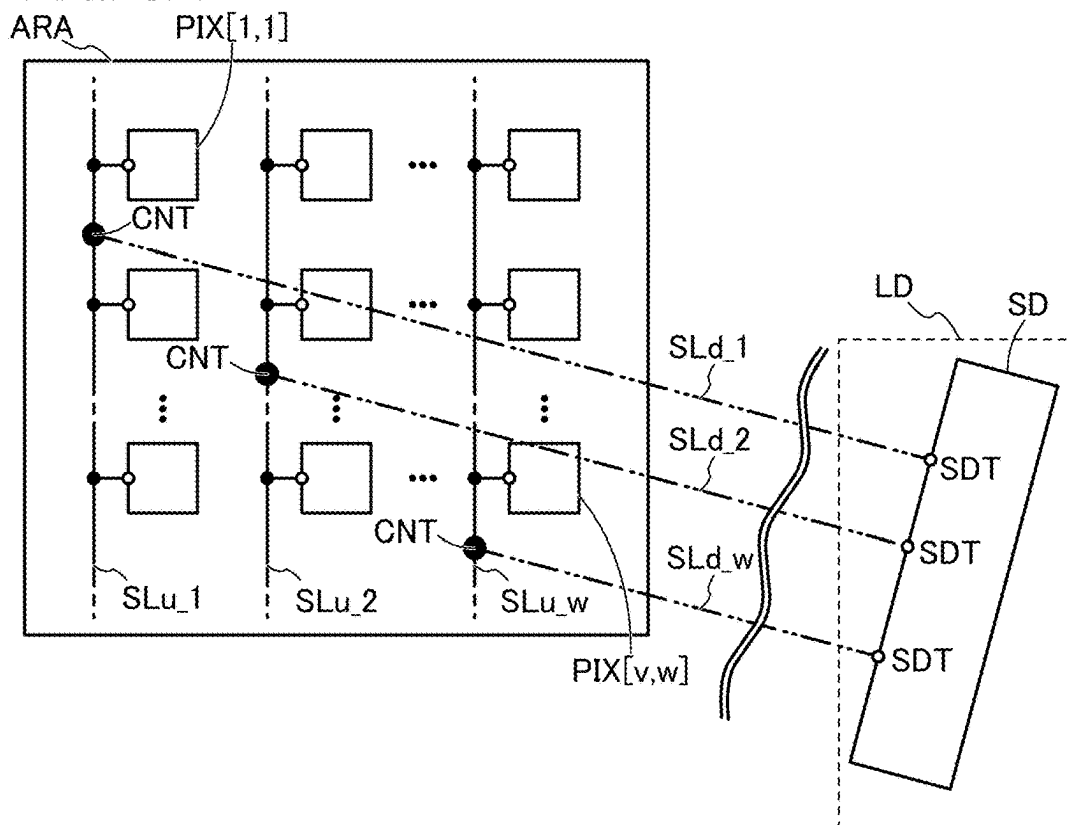


FIG. 8B

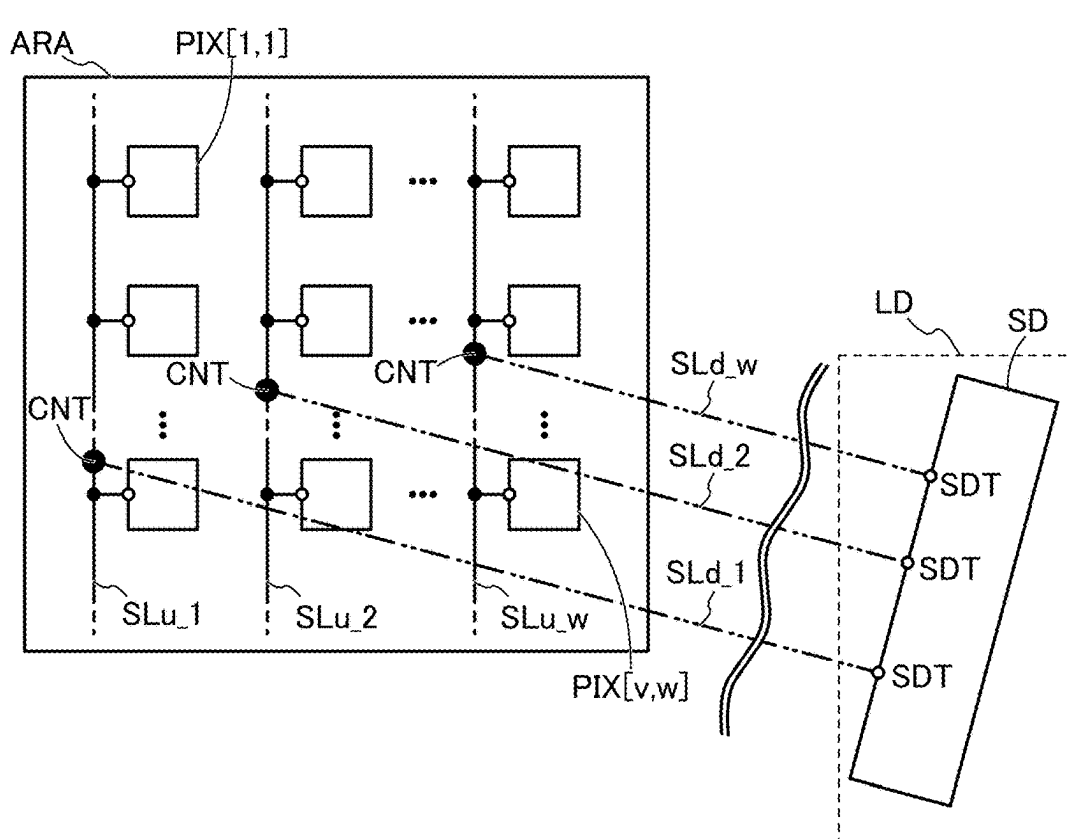


FIG. 9A

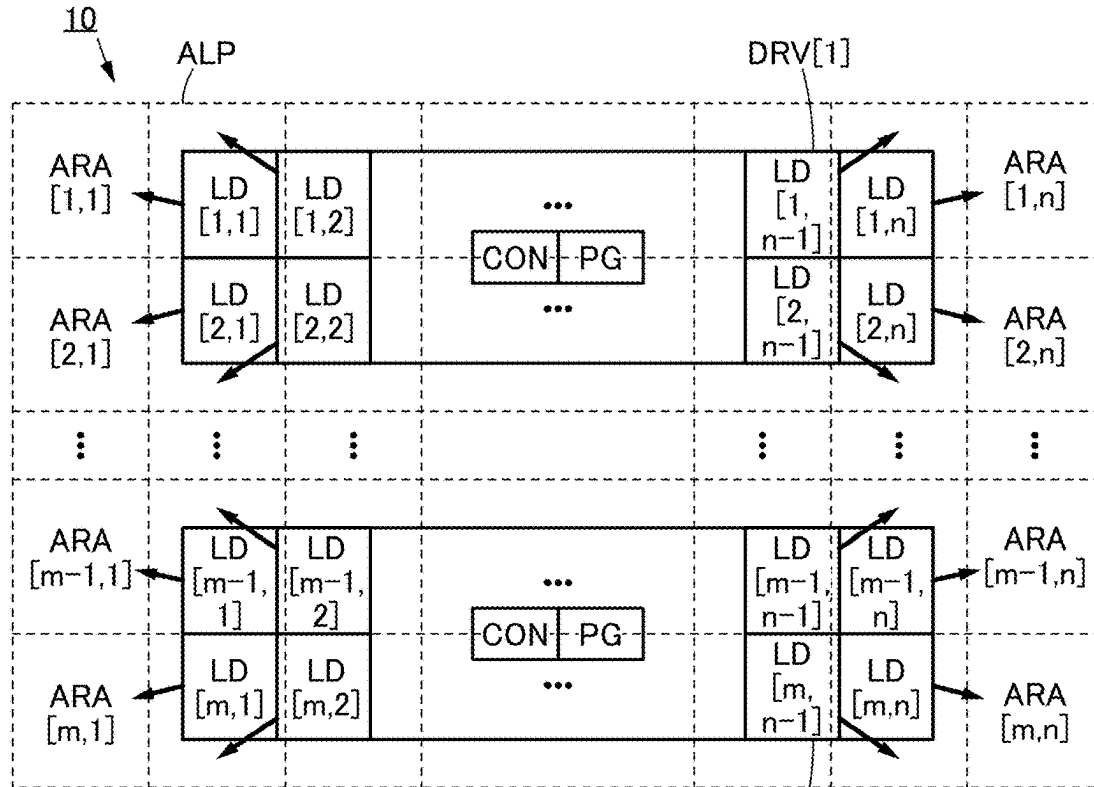


FIG. 9B

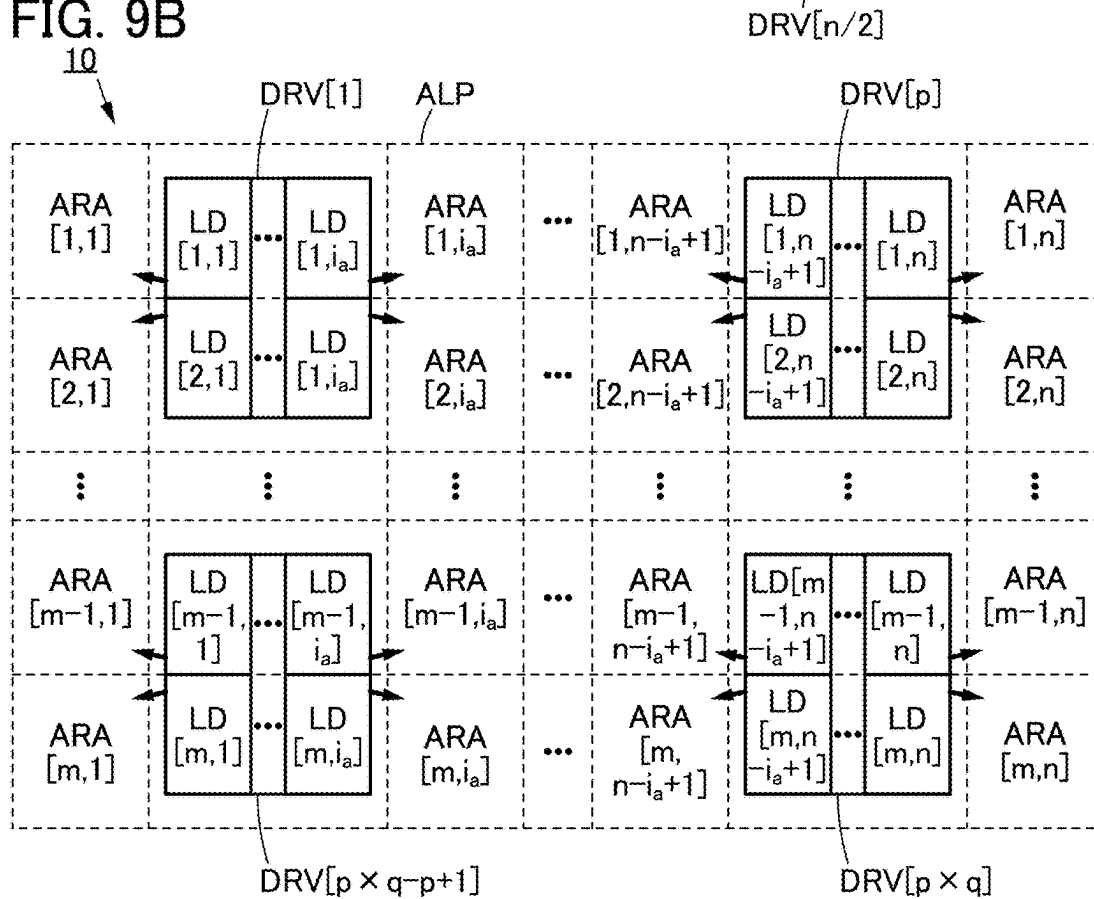


FIG. 10A

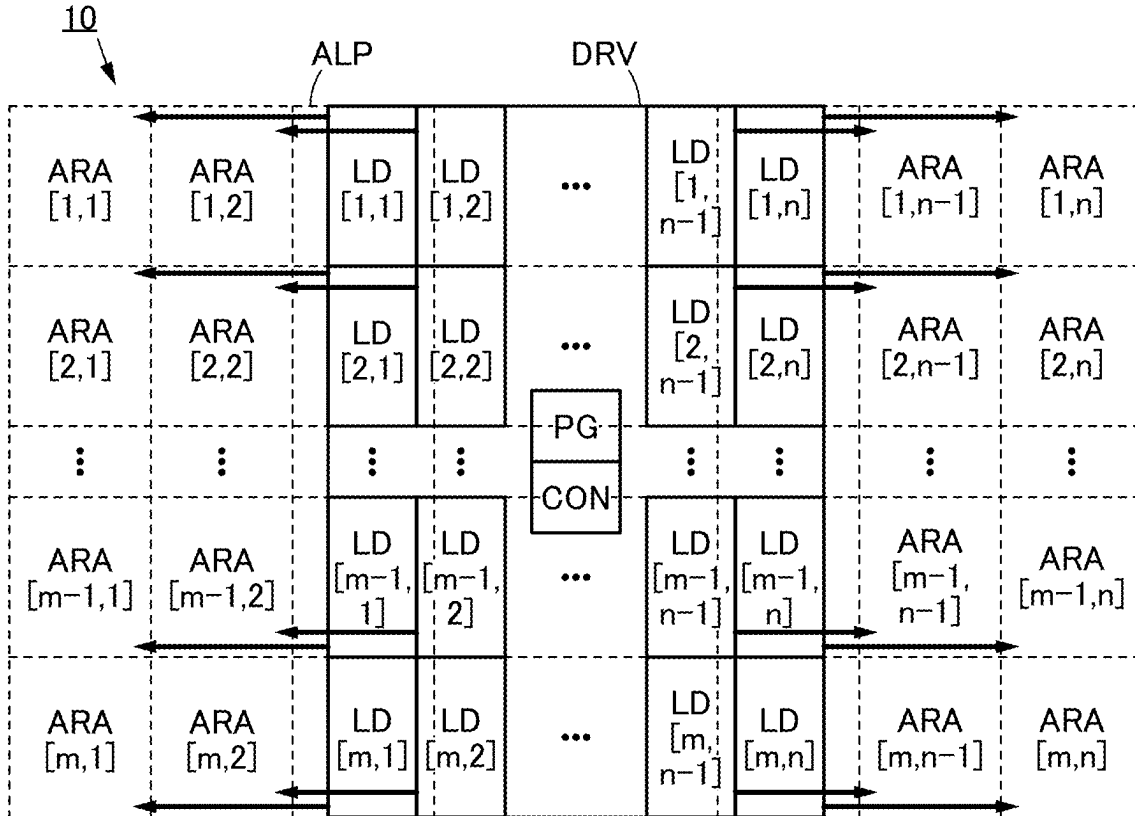


FIG. 10B

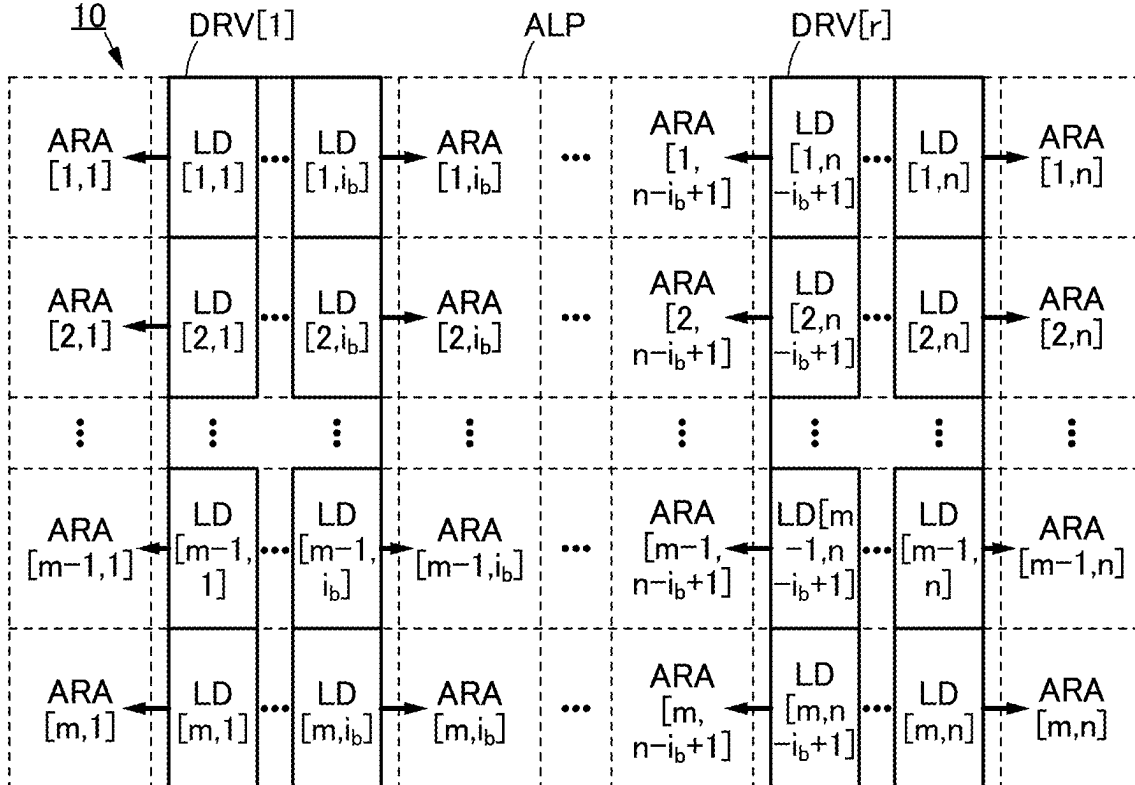


FIG. 11

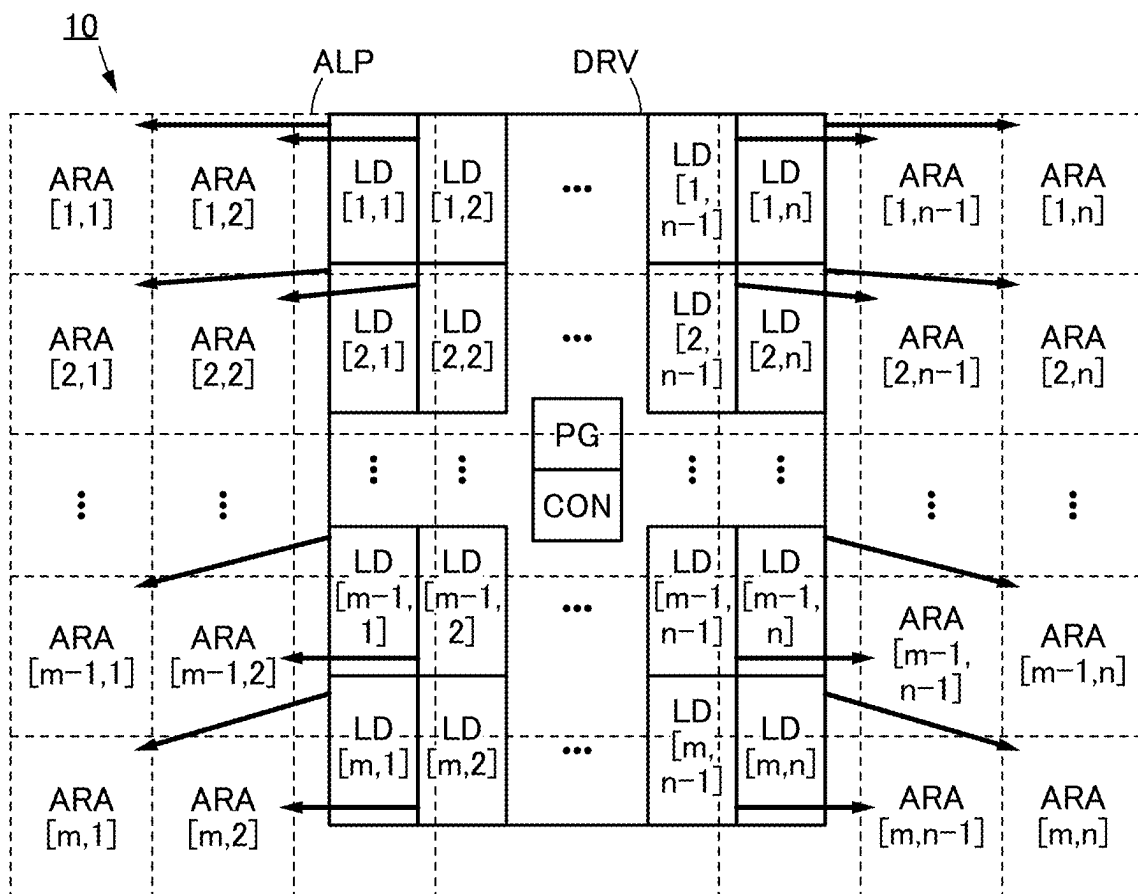


FIG. 12

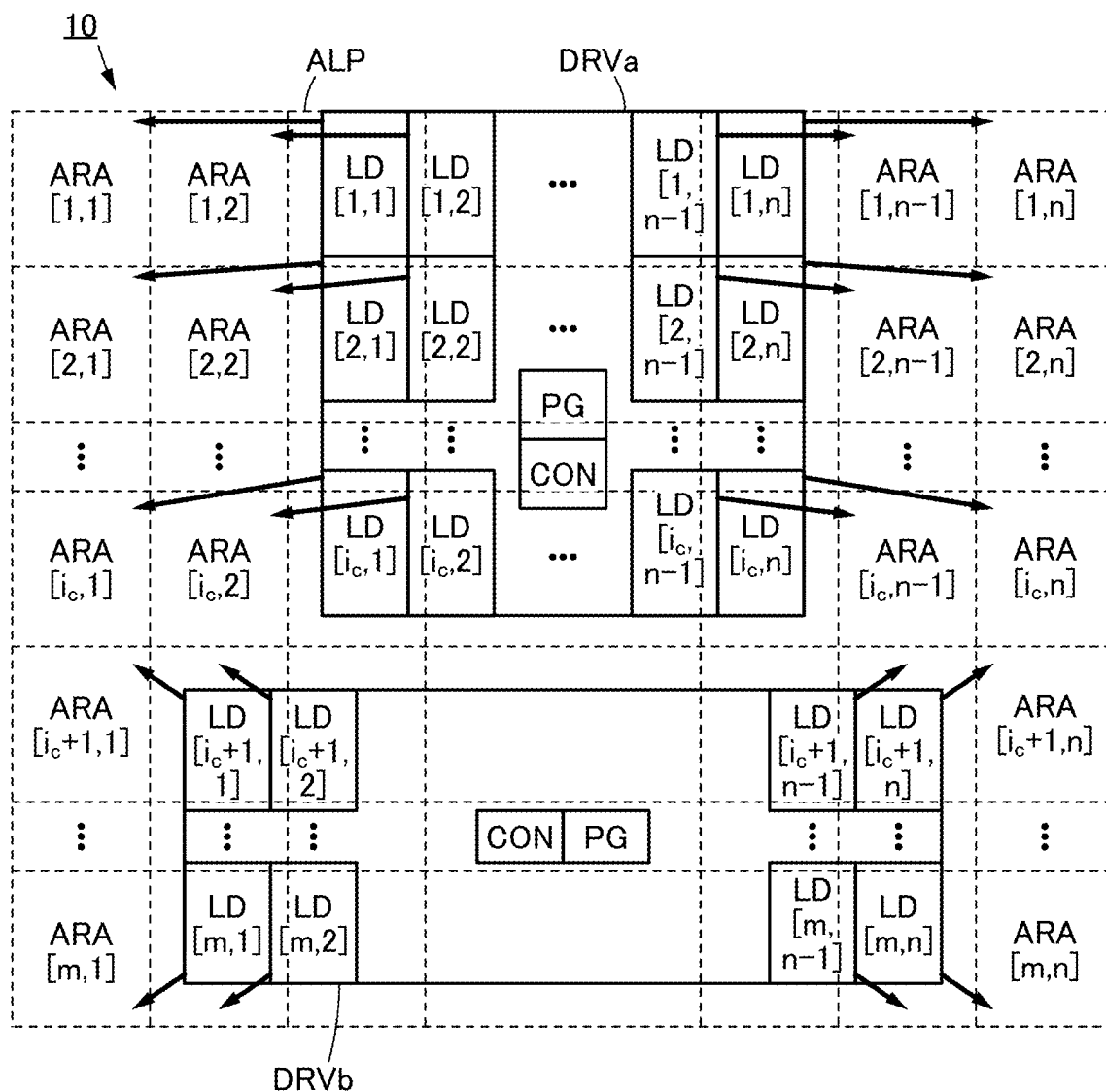


FIG. 14

11A

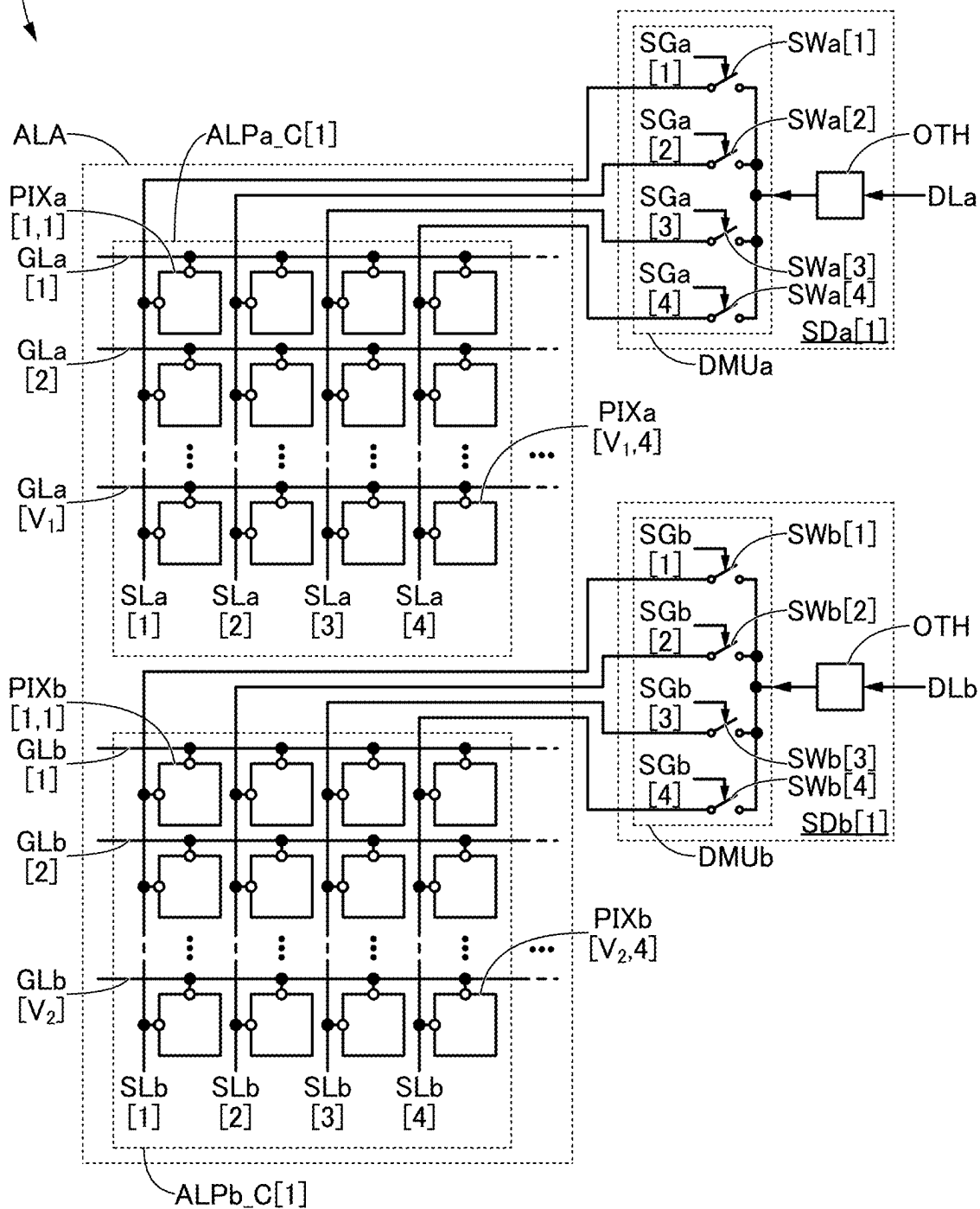


FIG. 15

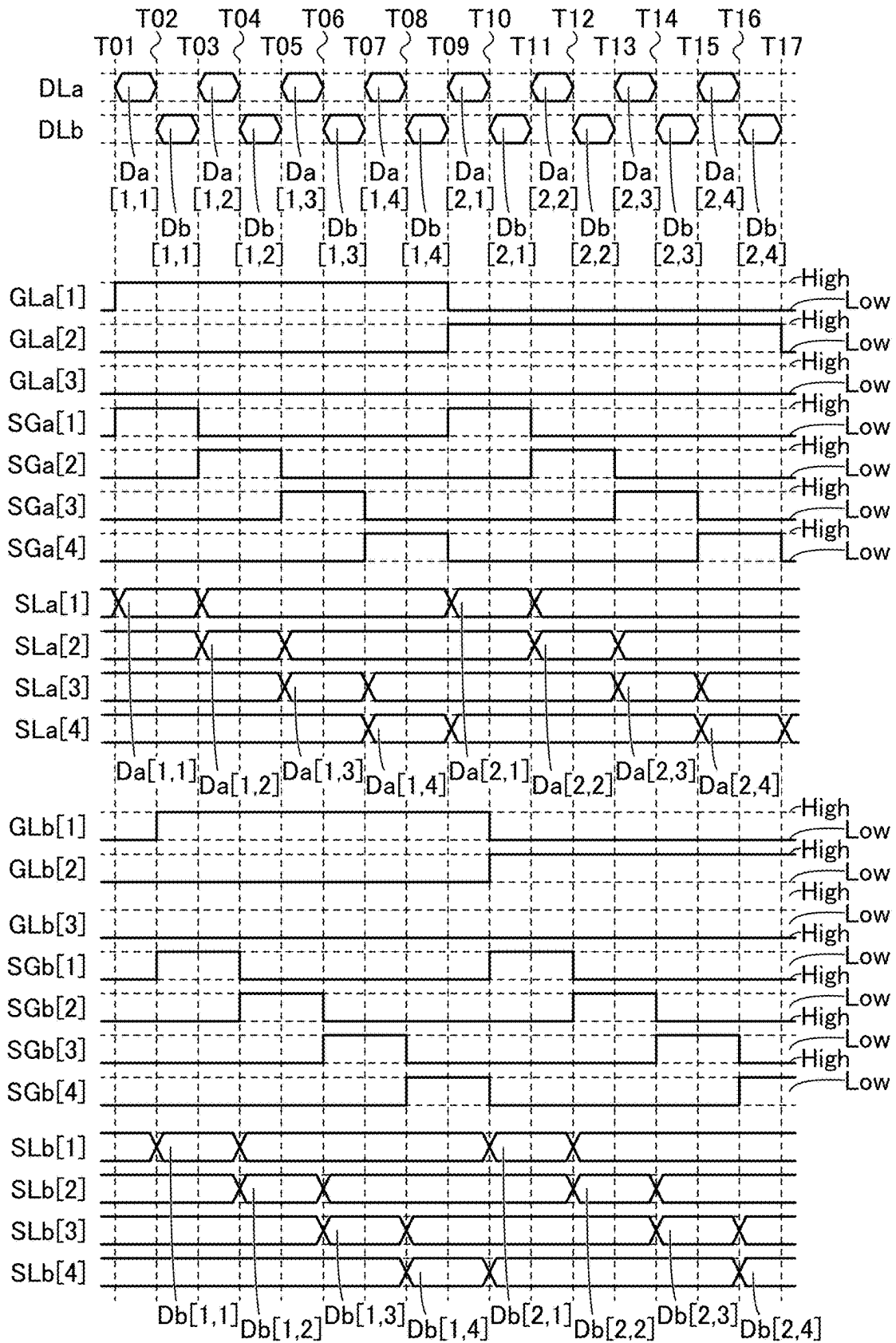


FIG. 16A

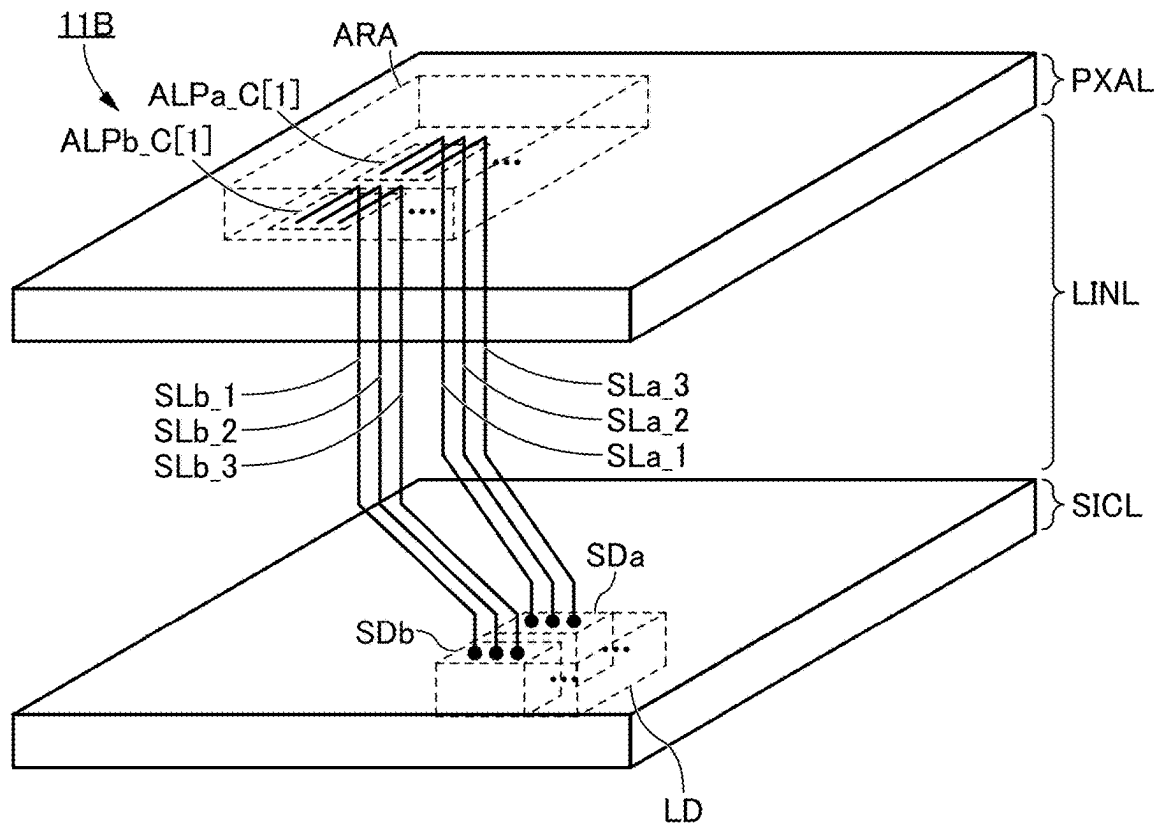


FIG. 16B

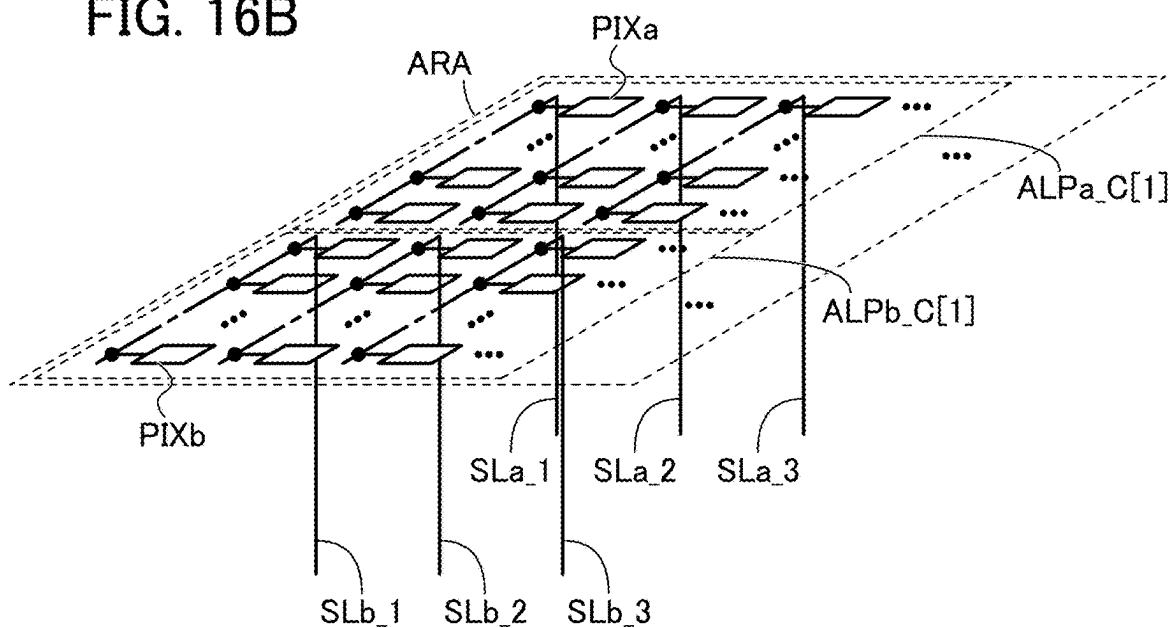


FIG. 17

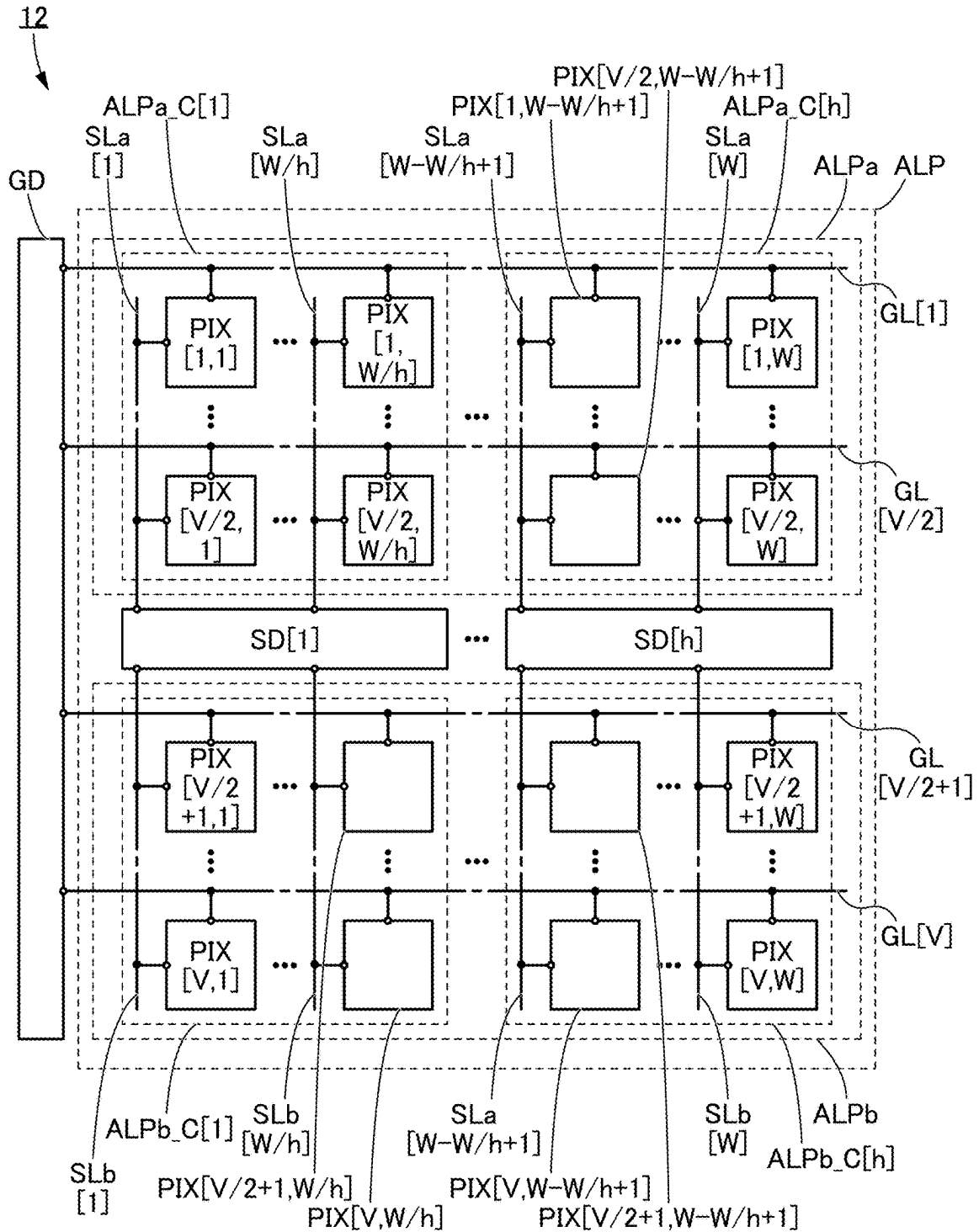


FIG. 18A

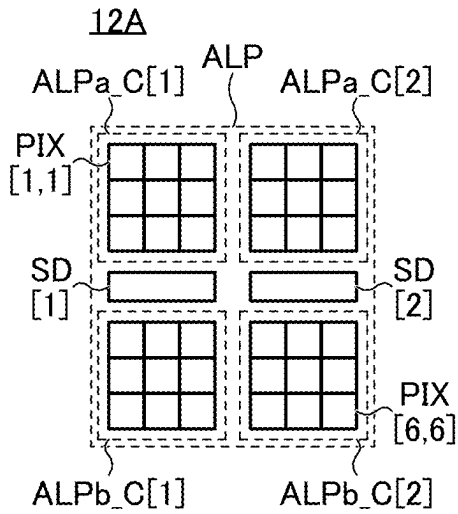


FIG. 18B

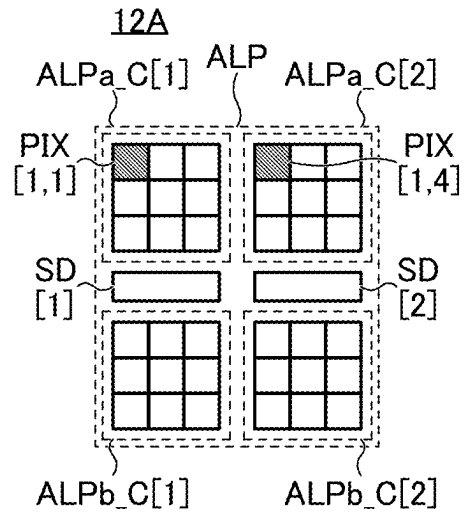


FIG. 18C

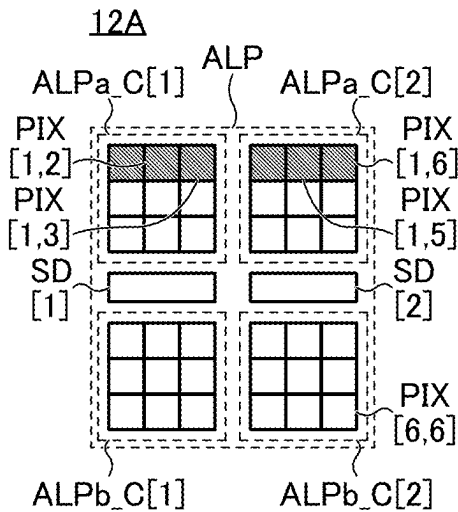


FIG. 18D

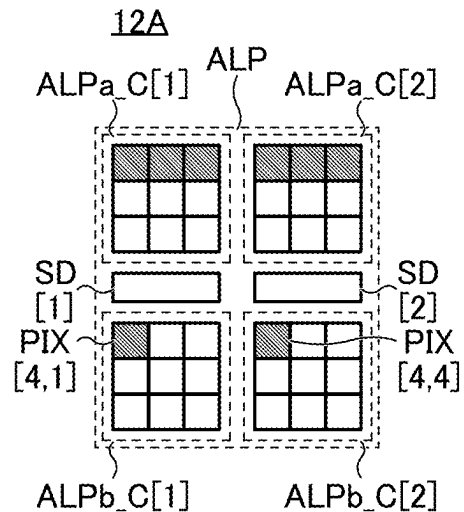


FIG. 18E

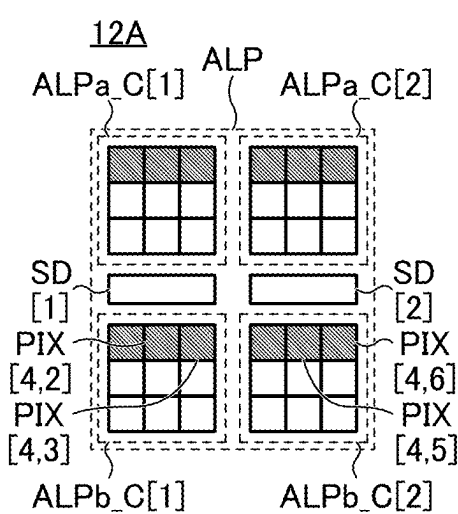


FIG. 18F

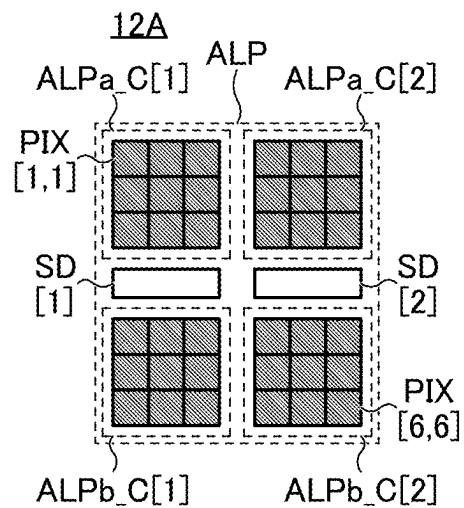


FIG. 19

13A

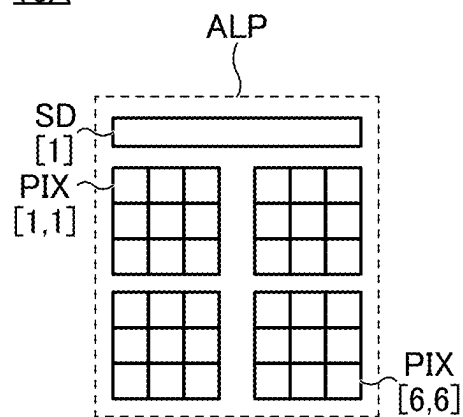


FIG. 20

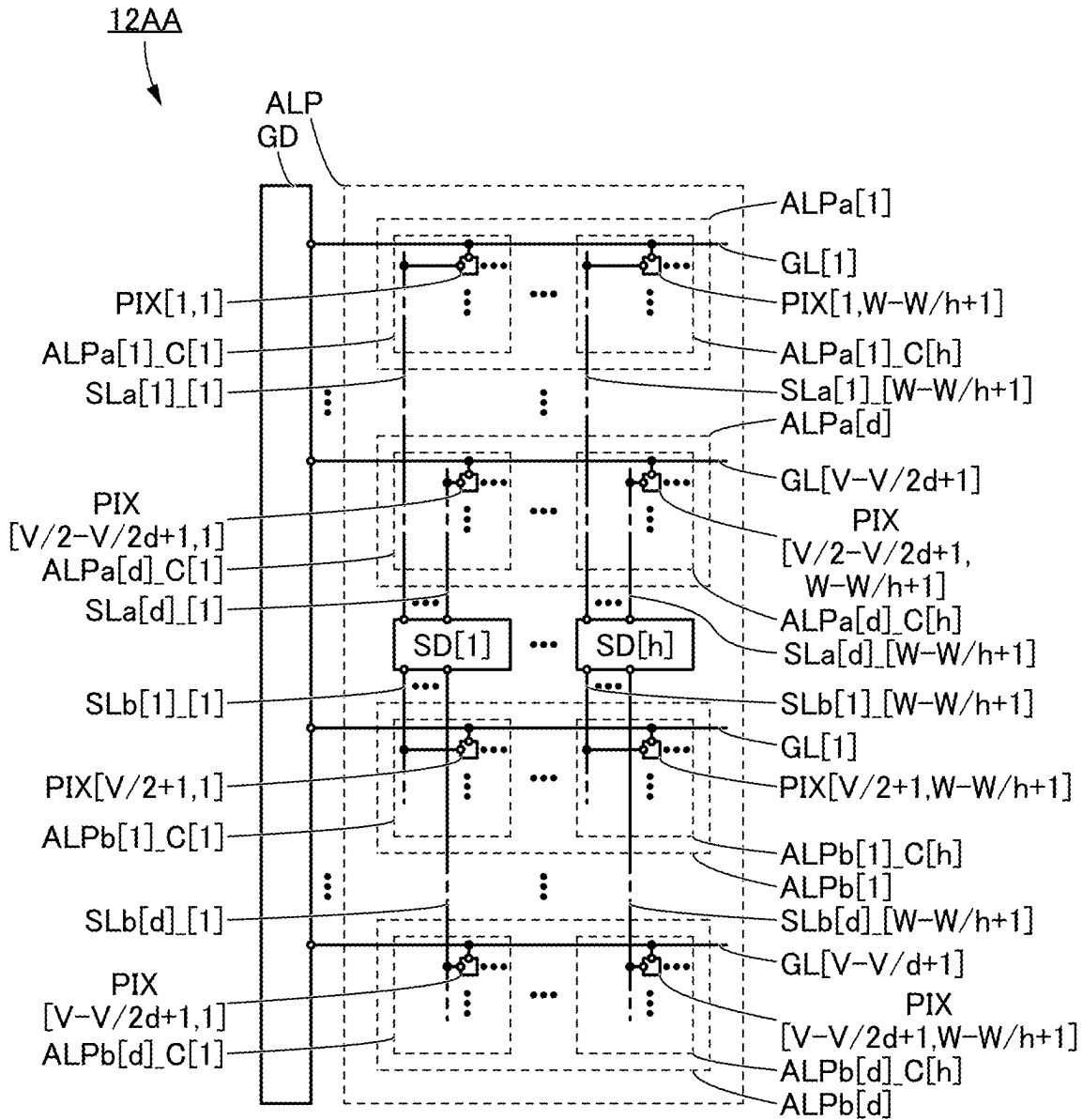


FIG. 21

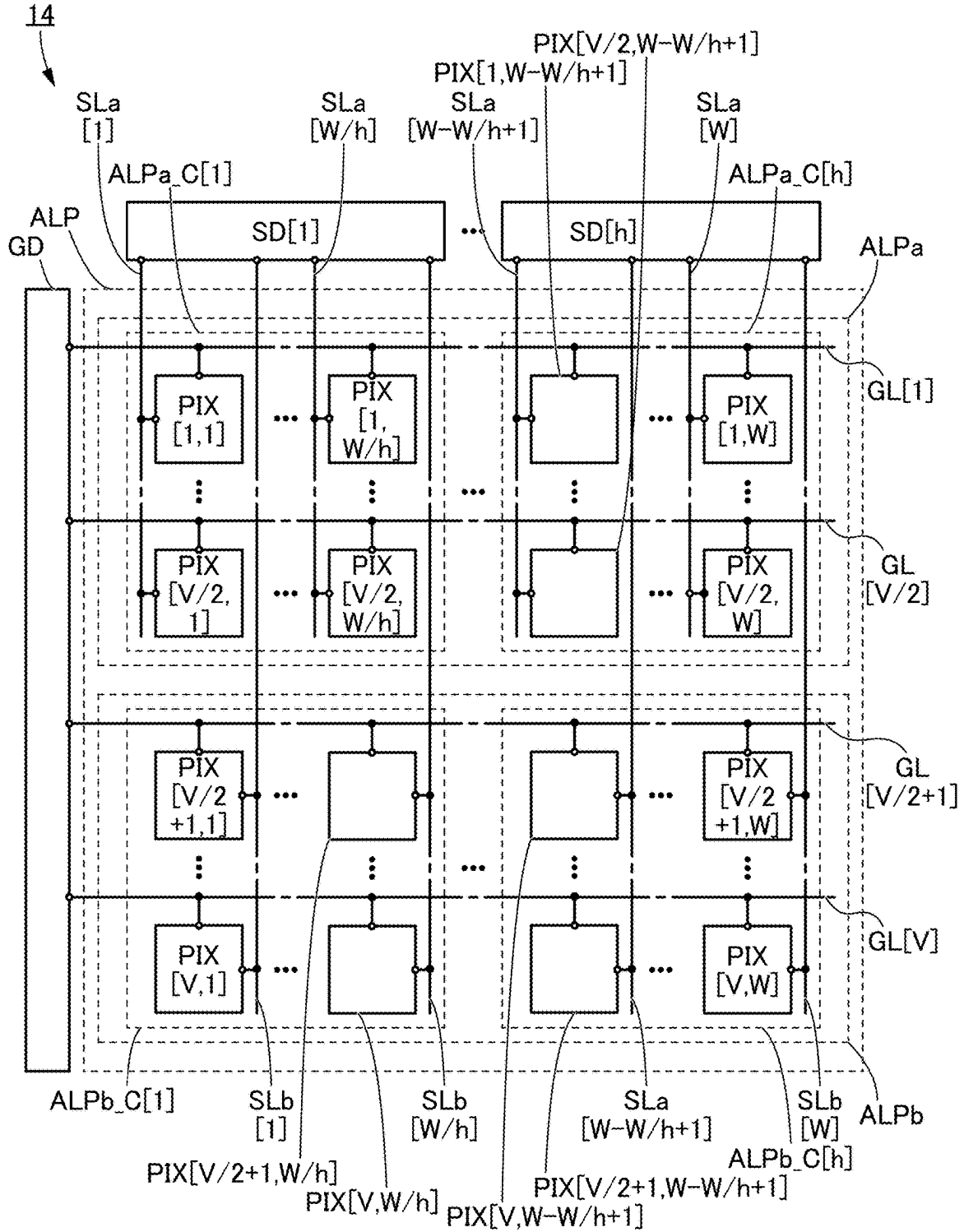


FIG. 22

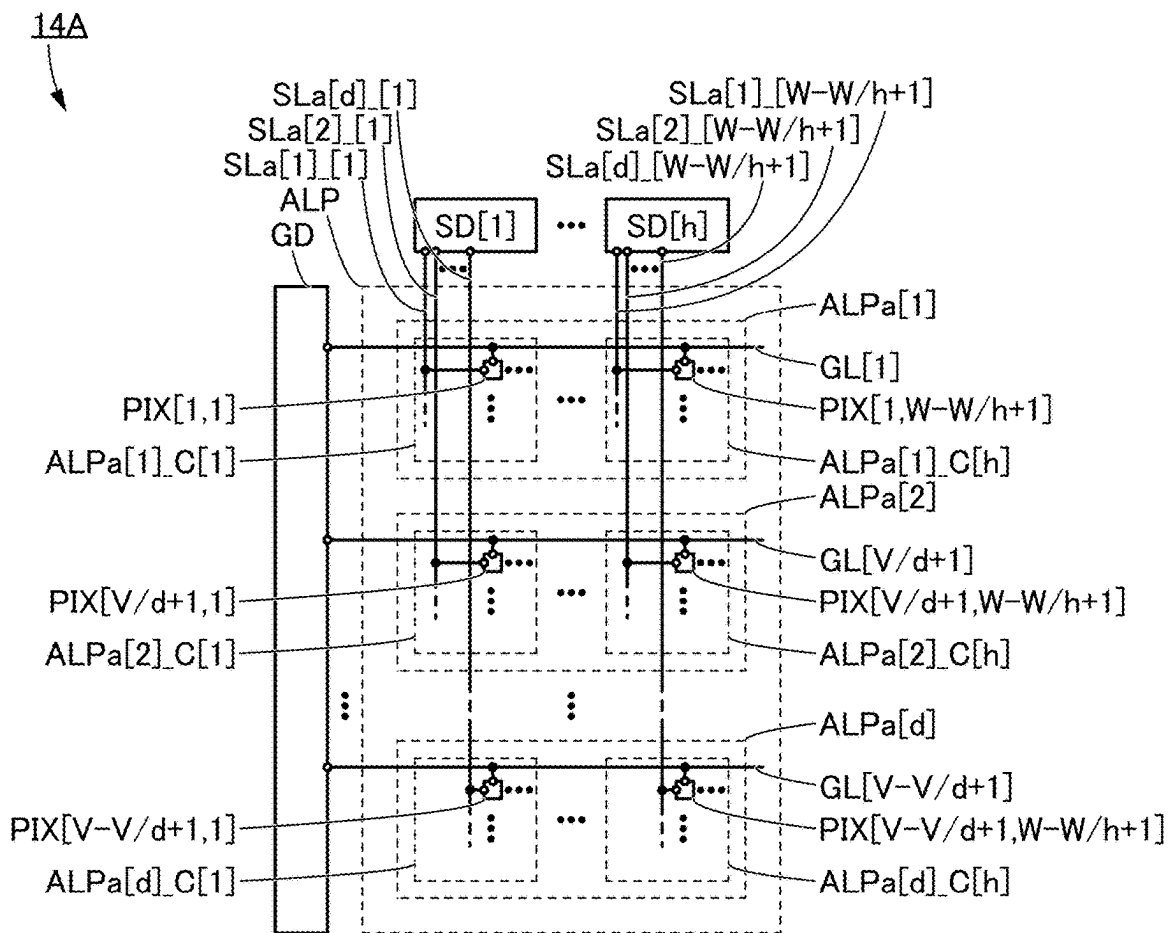


FIG. 23

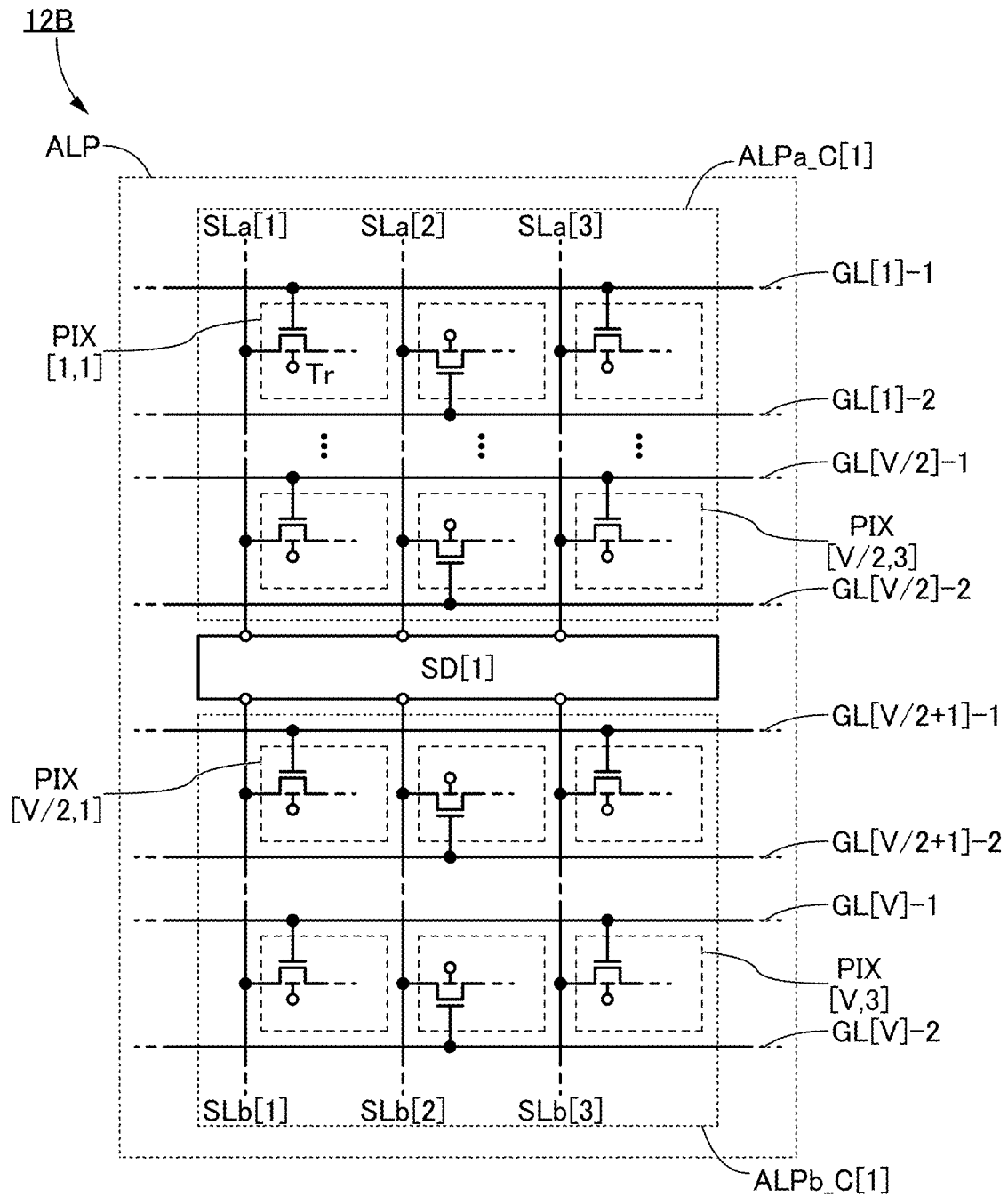


FIG. 24

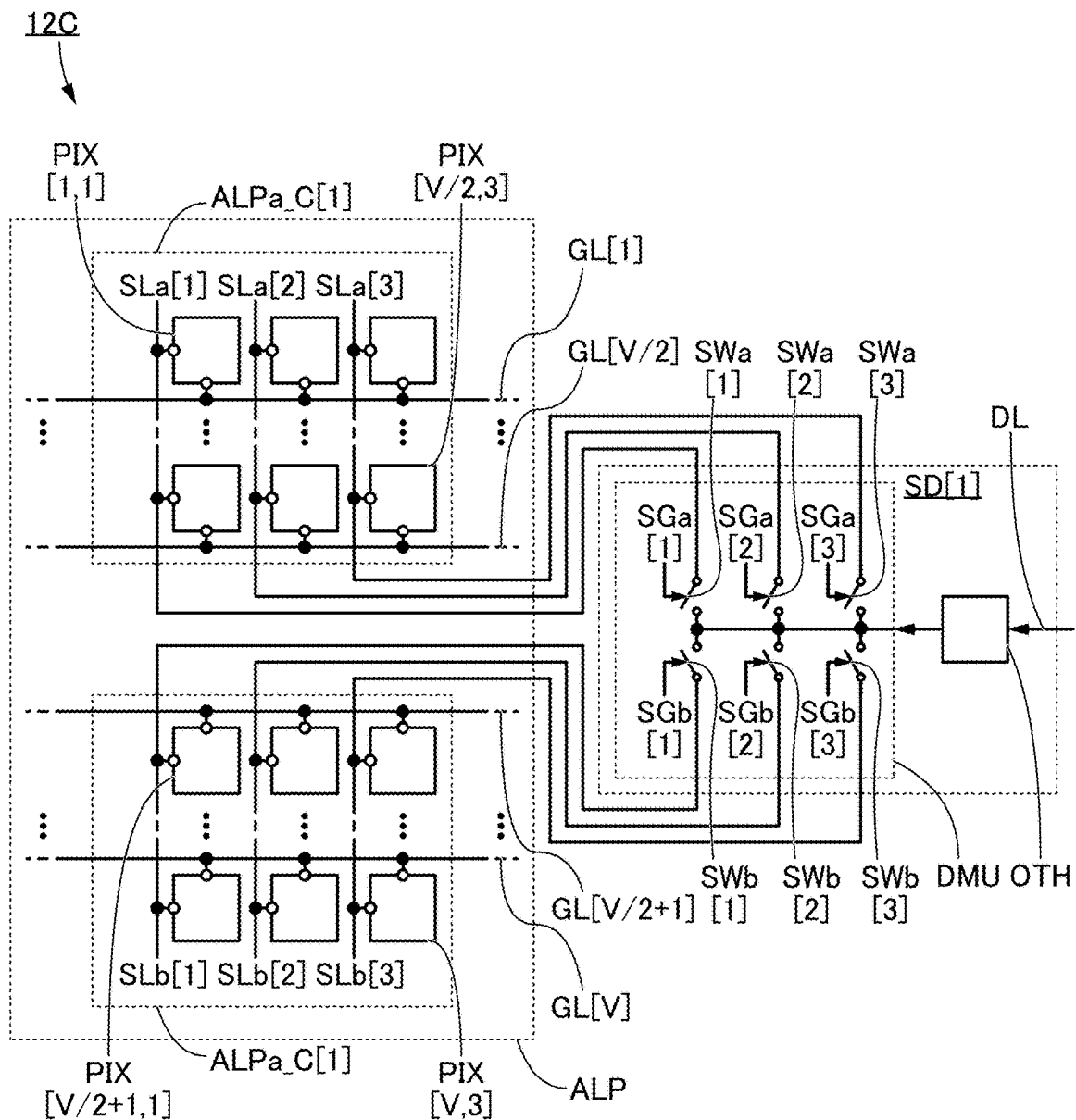


FIG. 25

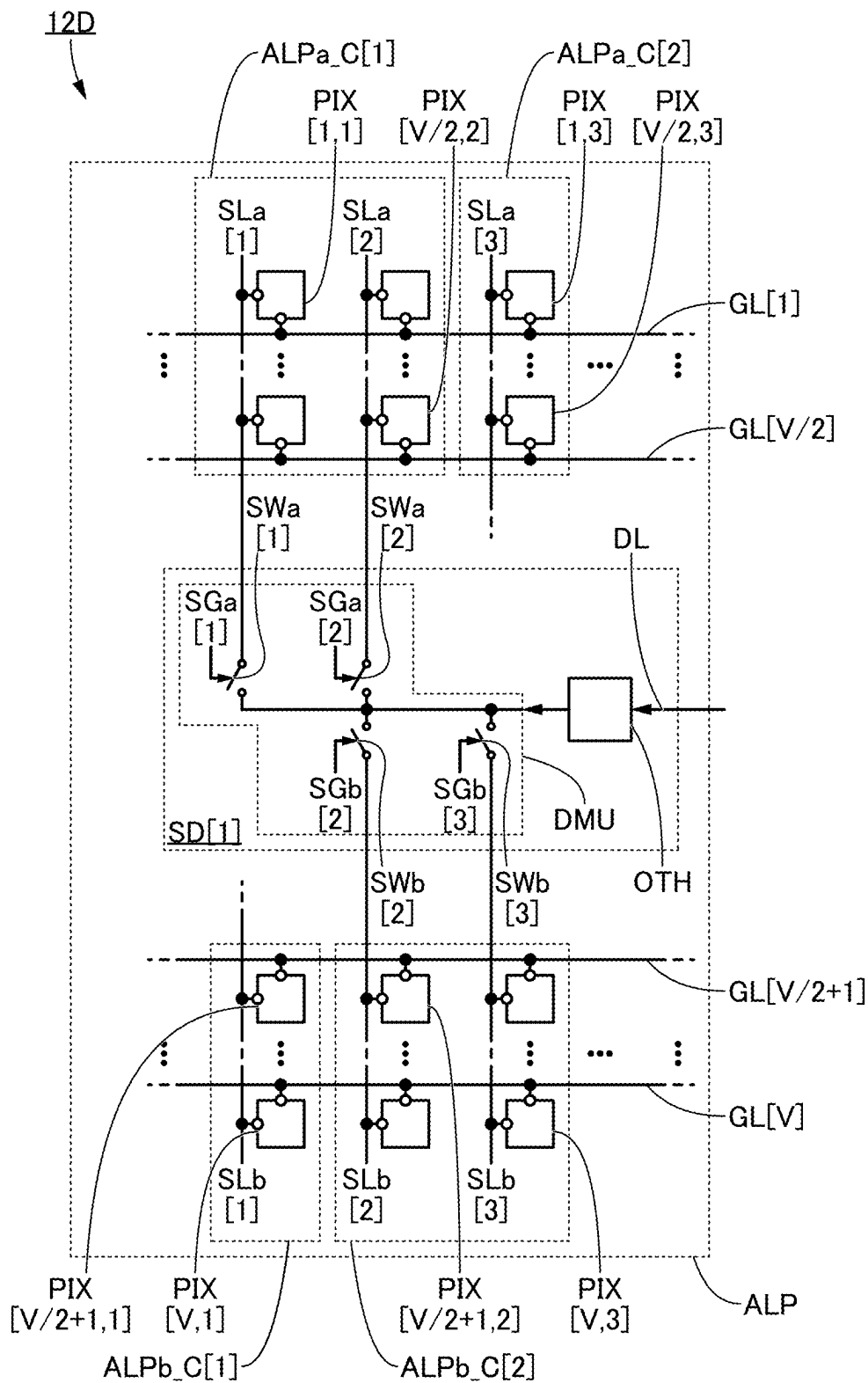


FIG. 26A

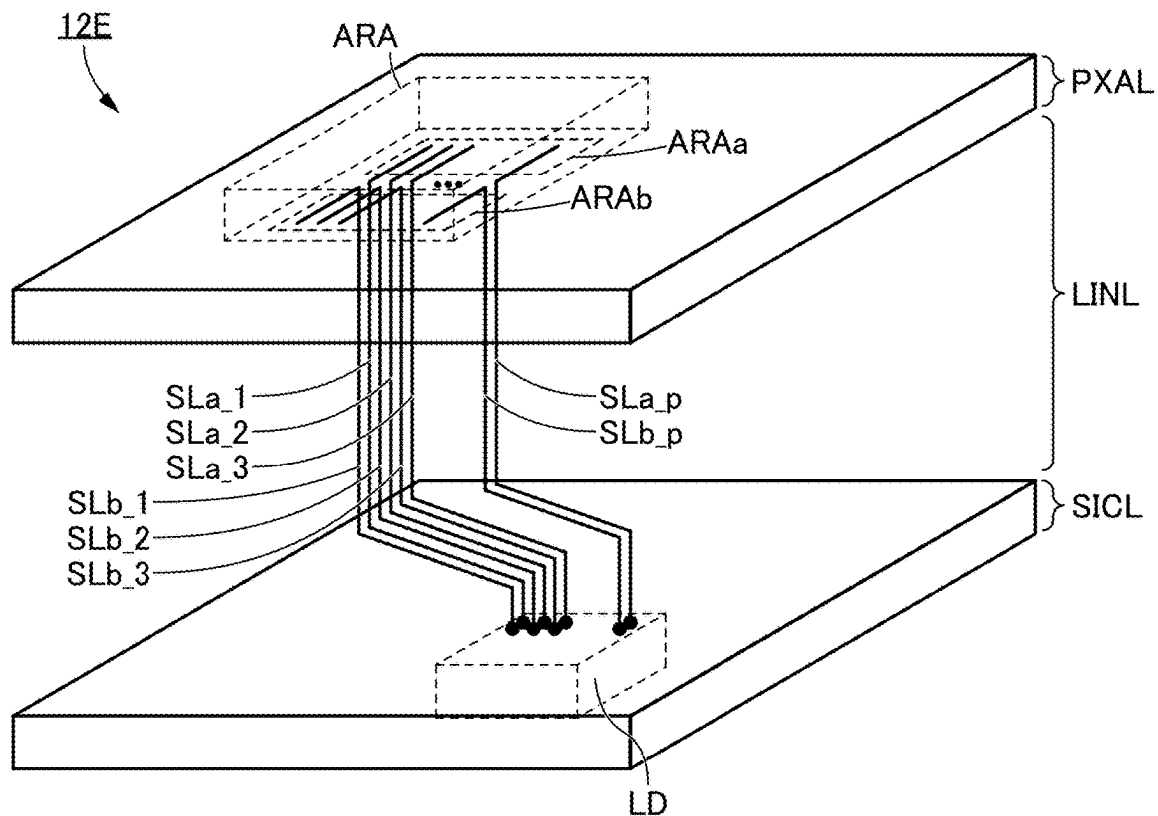


FIG. 26B

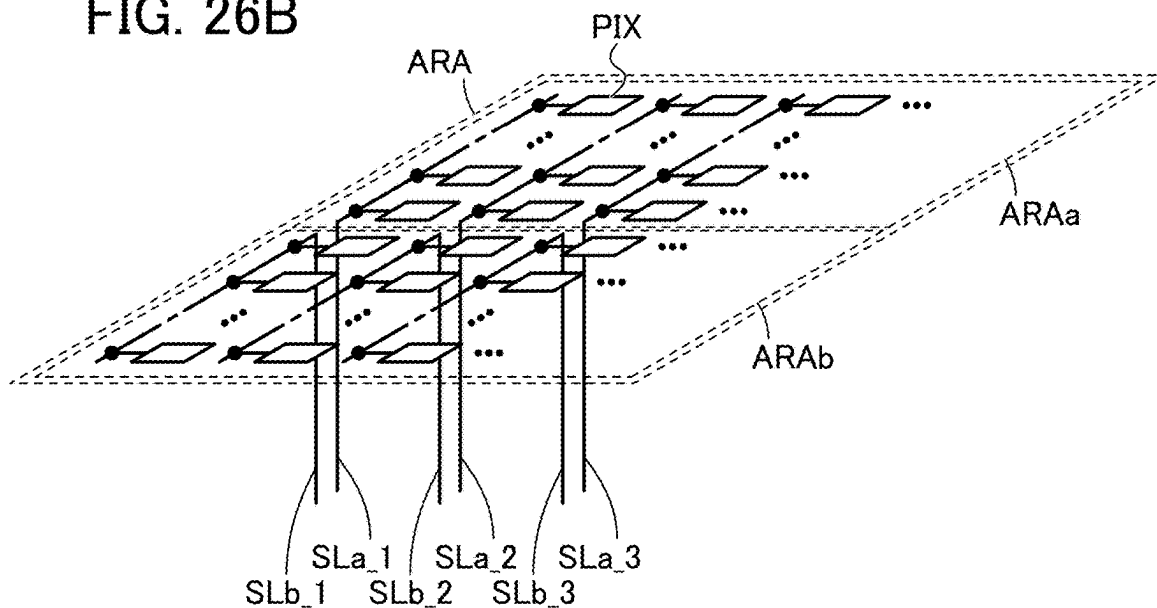


FIG. 27

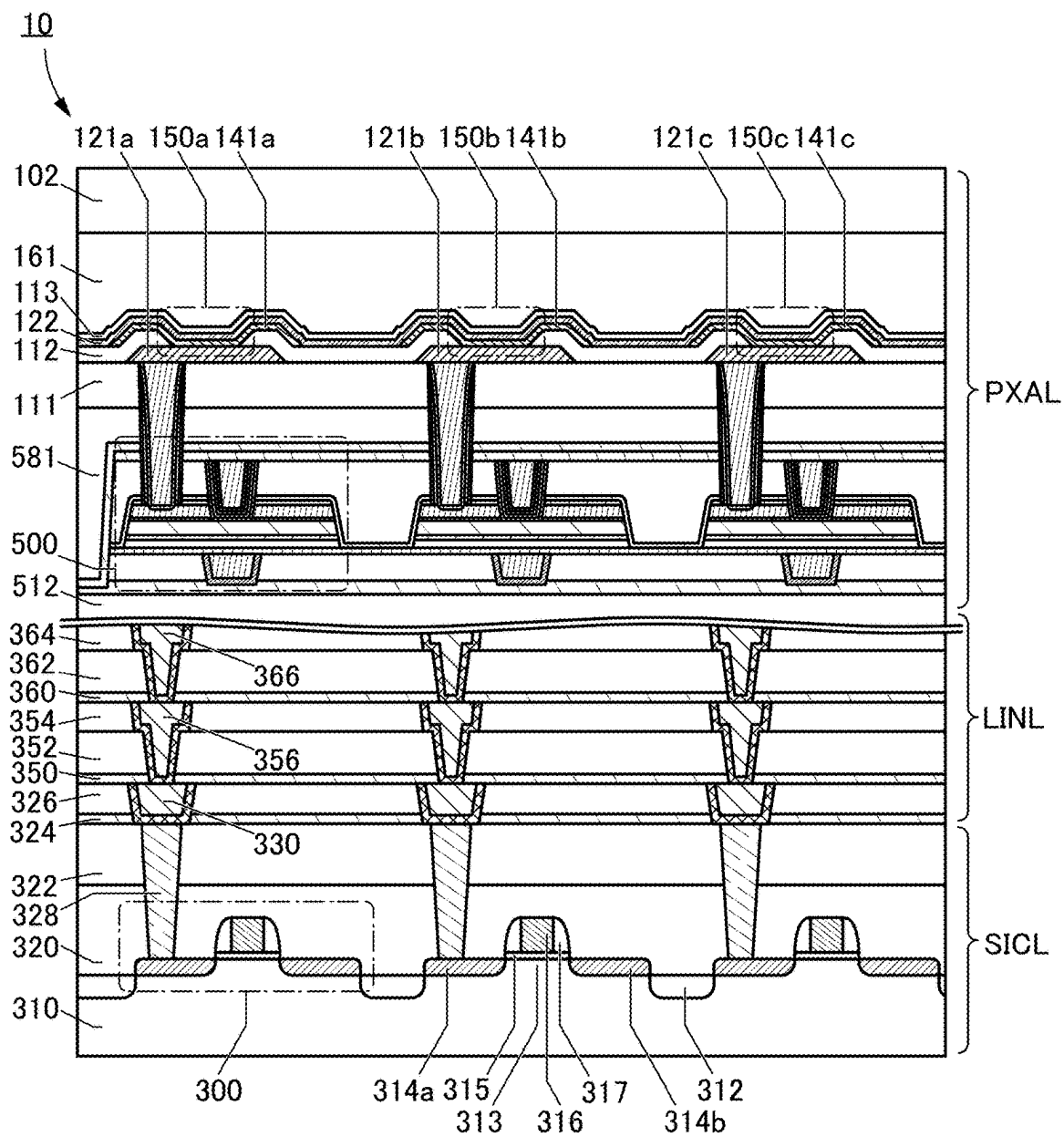


FIG. 28A

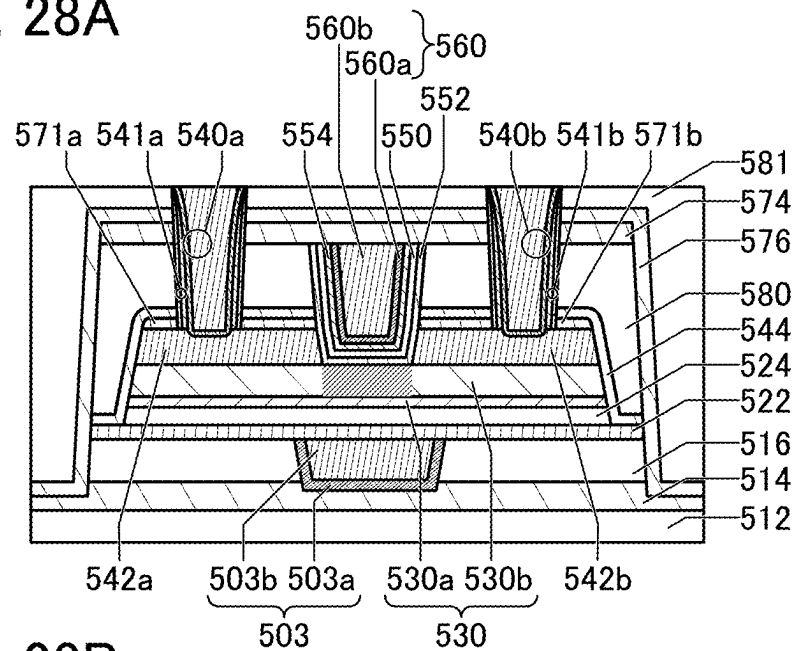
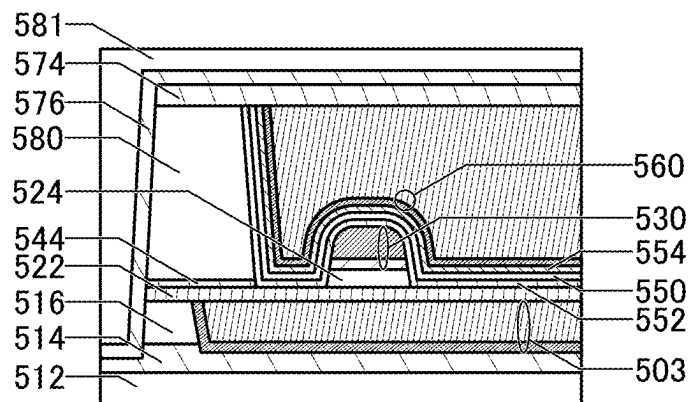


FIG. 28B



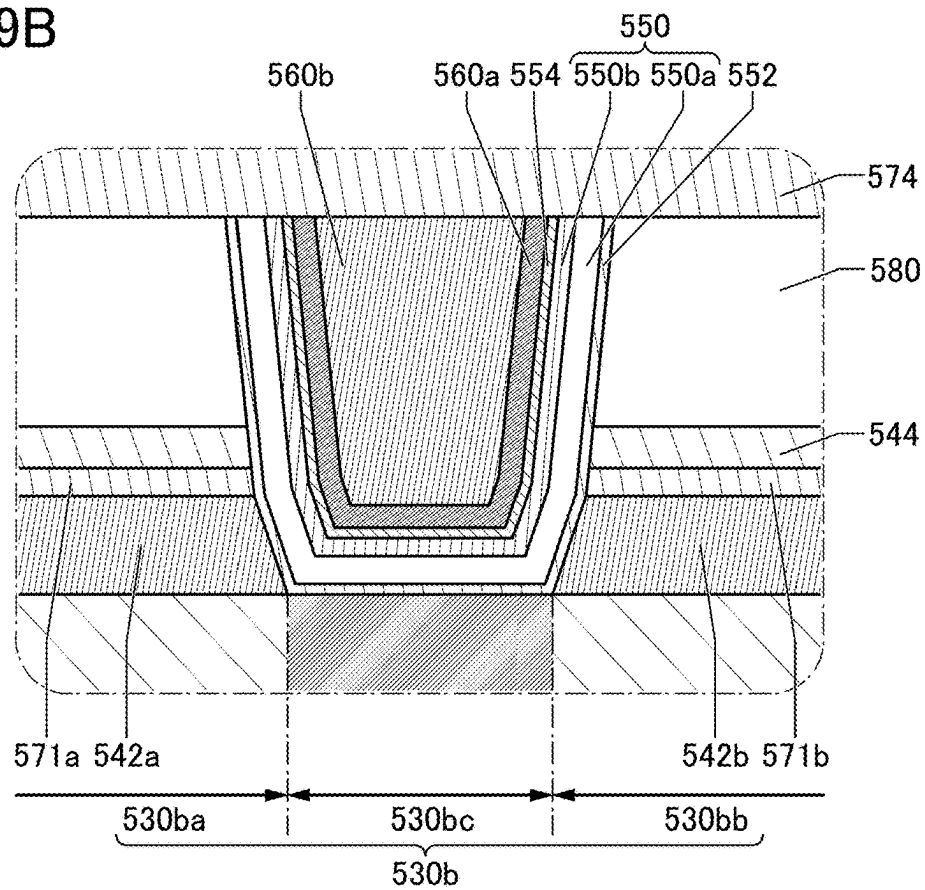


FIG. 30A

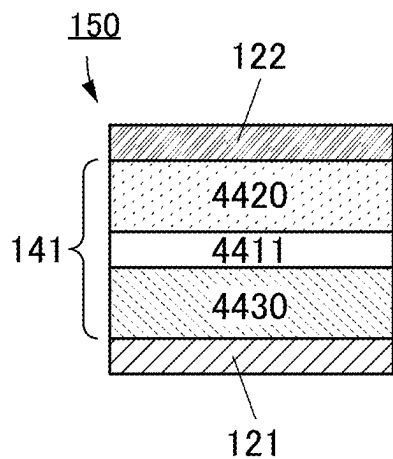


FIG. 30B

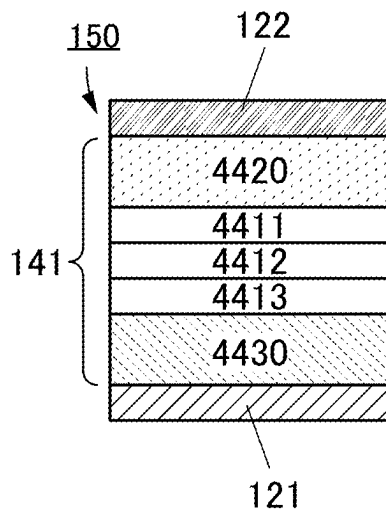


FIG. 30C

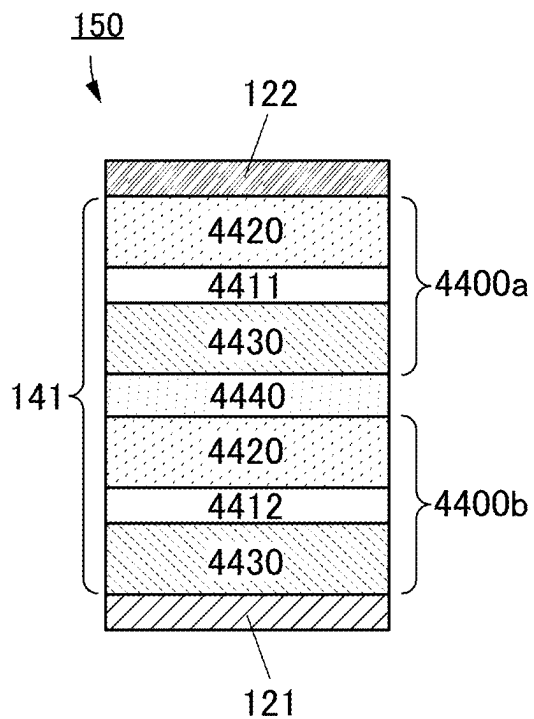


FIG. 31A

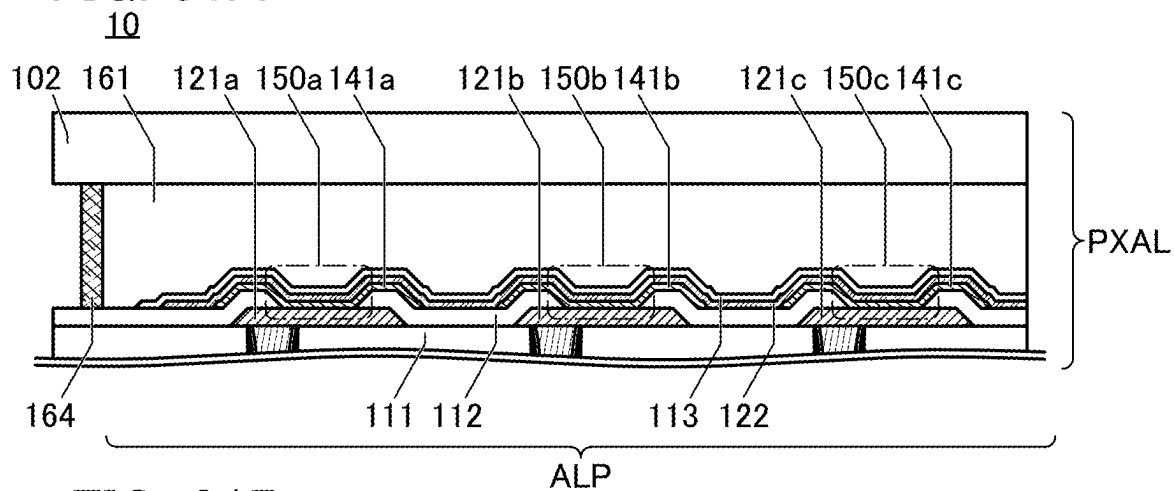


FIG. 31B

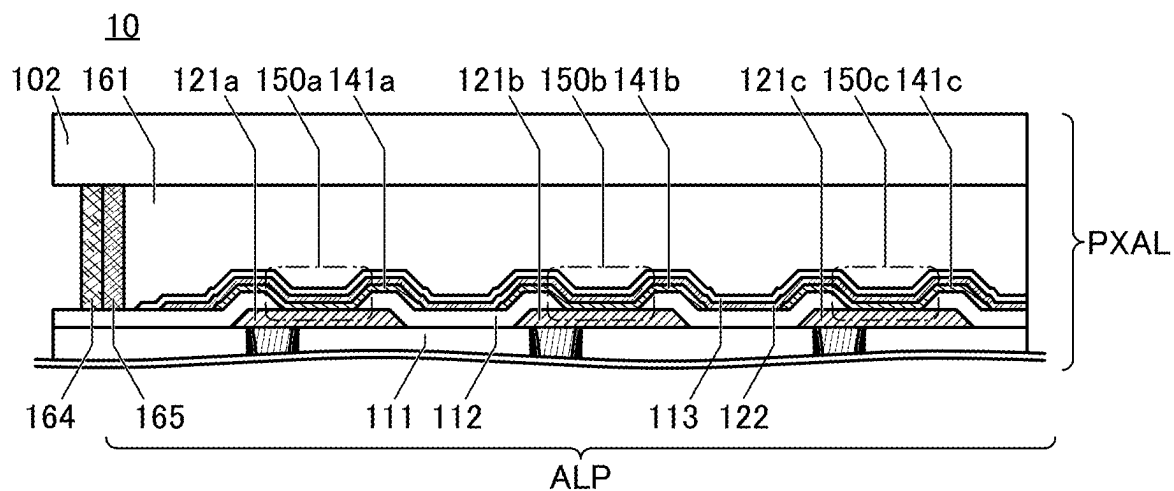


FIG. 32A

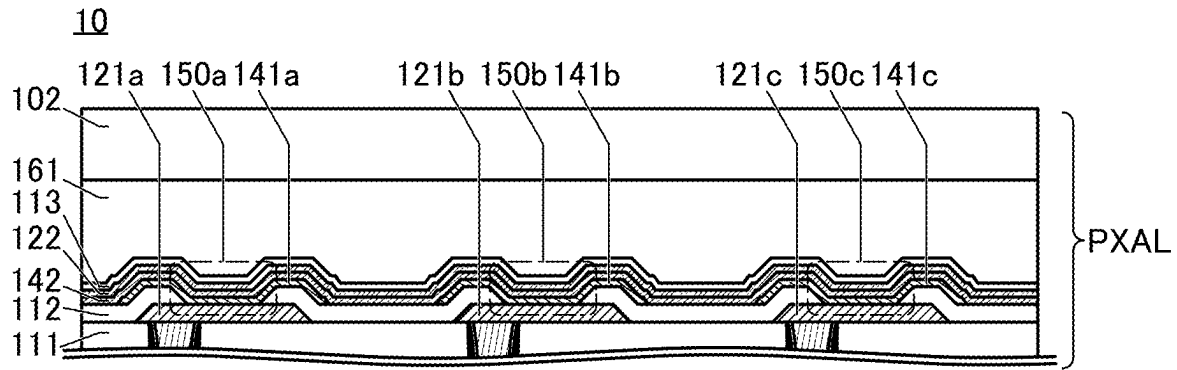


FIG. 32B

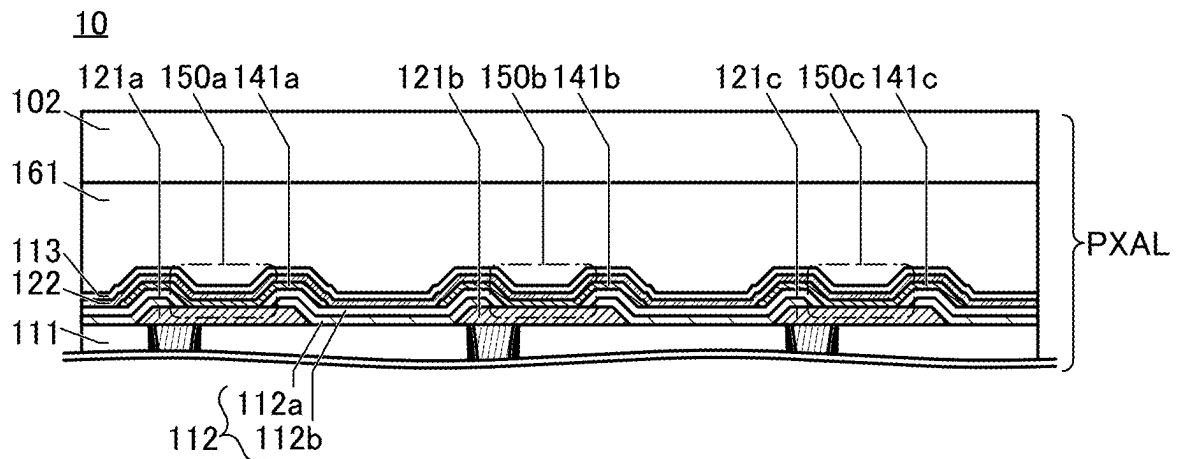


FIG. 32C

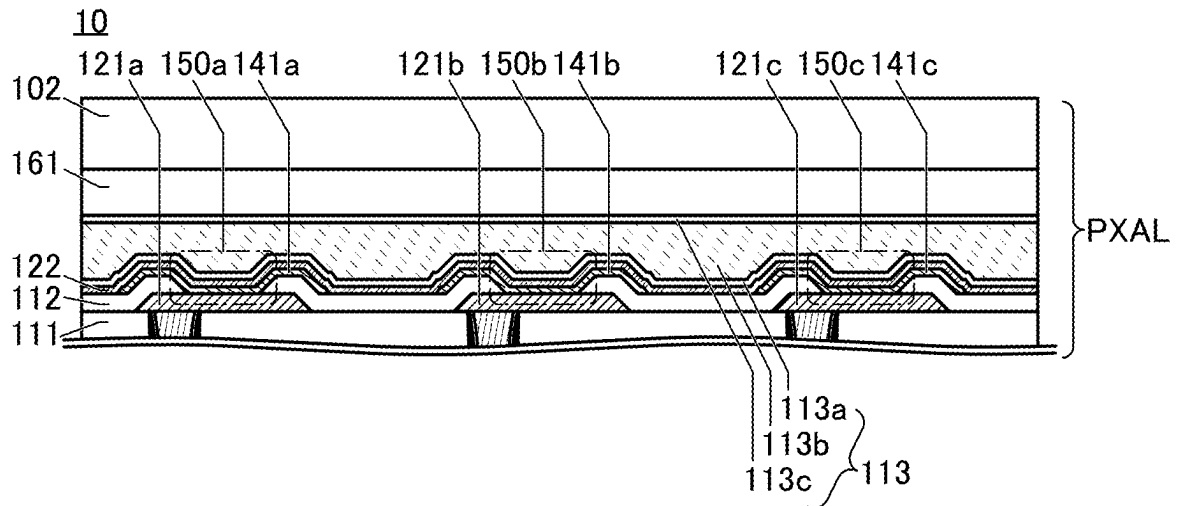


FIG. 33A

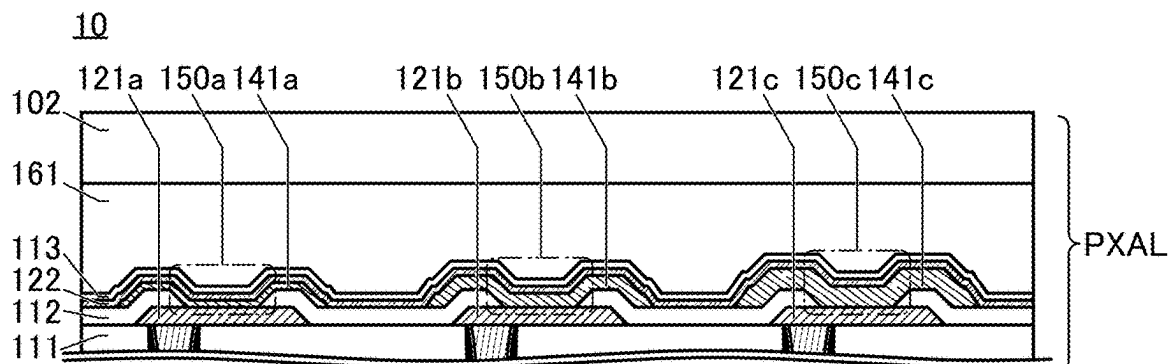


FIG. 33B

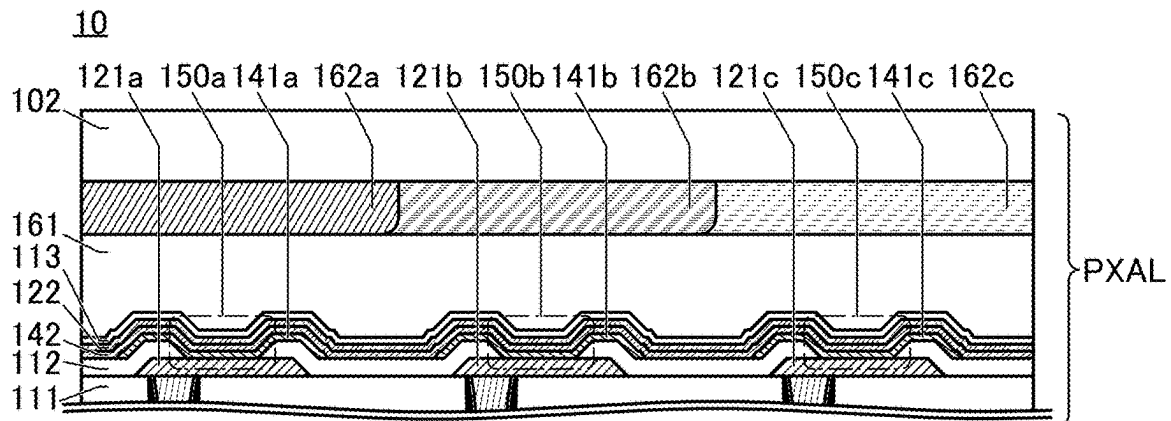


FIG. 33C

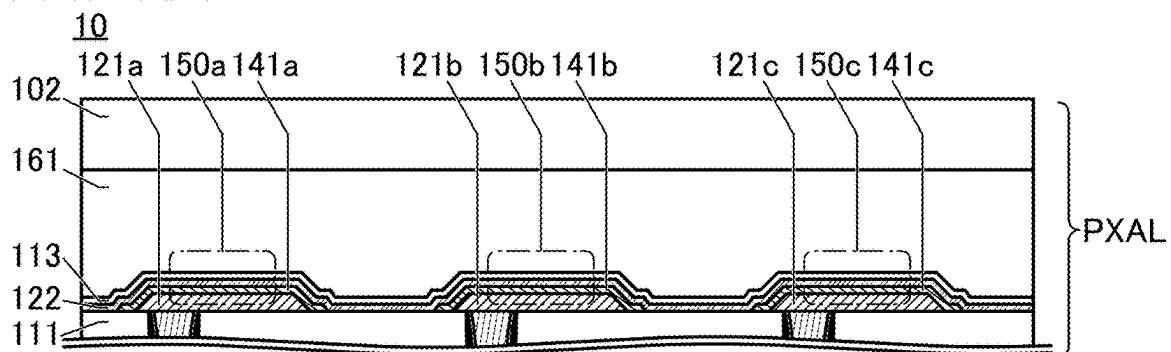


FIG. 33D

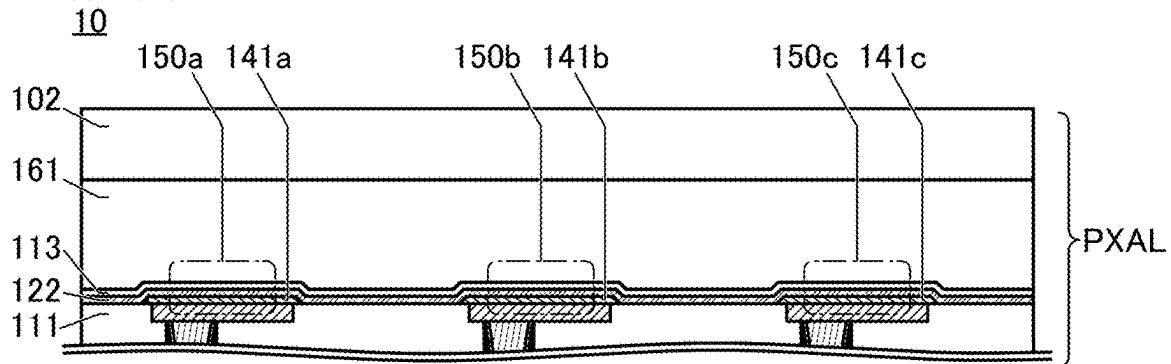


FIG. 34A

Intermediate state		
New crystalline phase		
Amorphous	Crystalline	Crystal
• completely amorphous	• CAAC • nc • CAC excluding single crystal and poly crystal	• single crystal • poly crystal

FIG. 34B

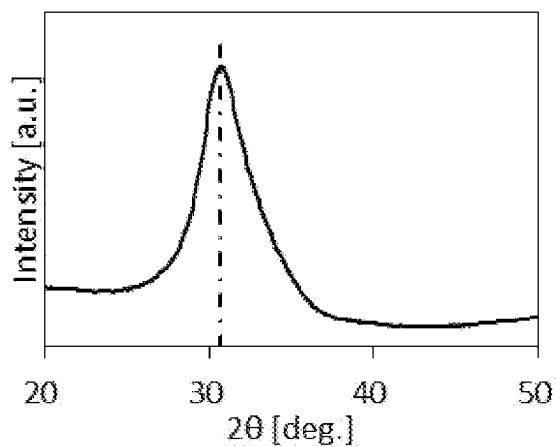


FIG. 34C

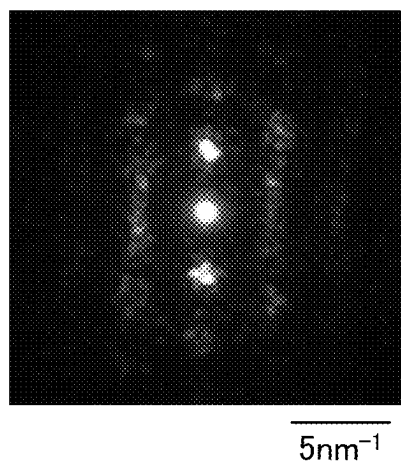


FIG. 35A

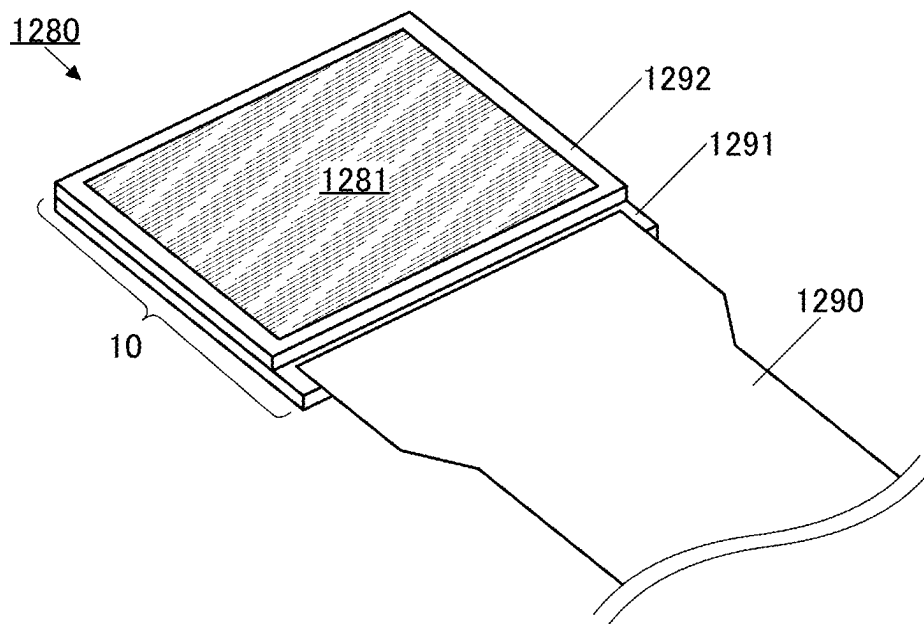


FIG. 35B

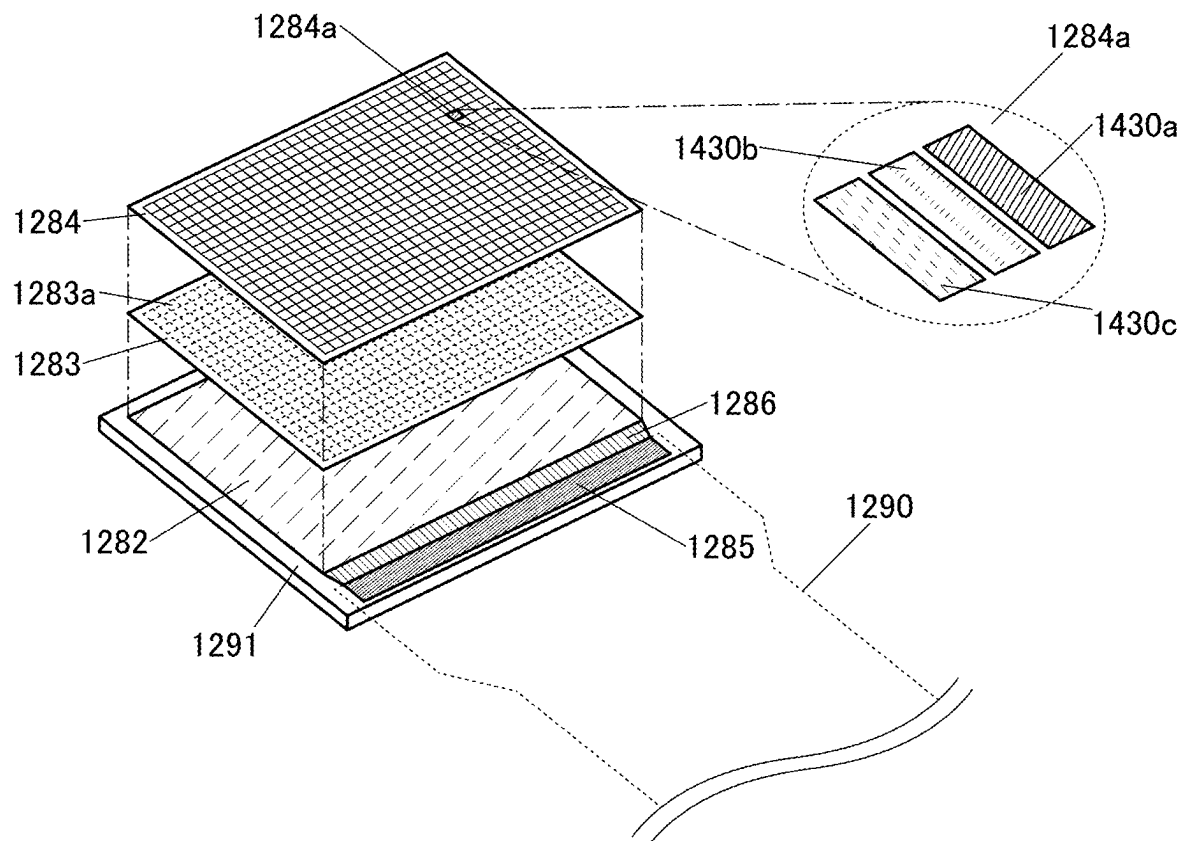


FIG. 36A

8300

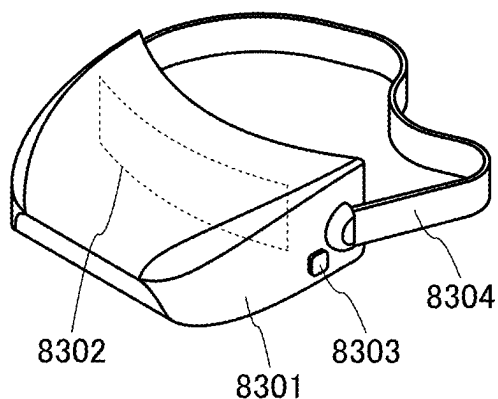


FIG. 36B

8300

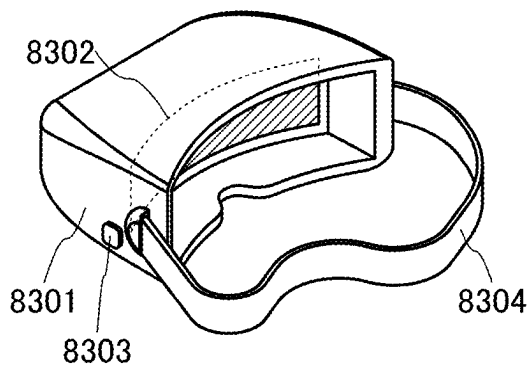


FIG. 36C

8300

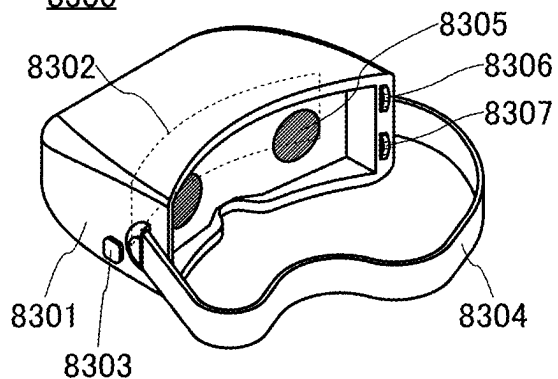


FIG. 36D

8300

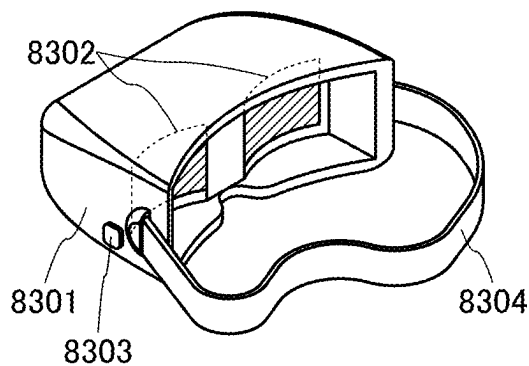


FIG. 36E

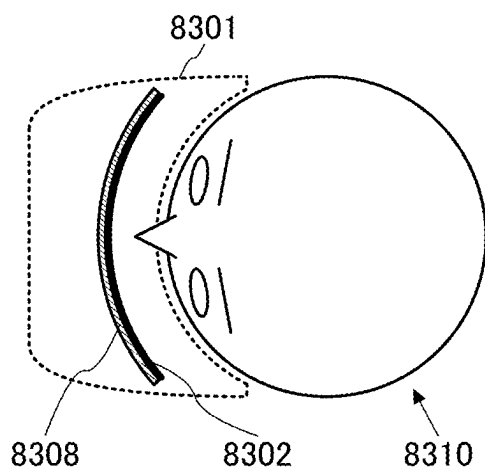


FIG. 36F

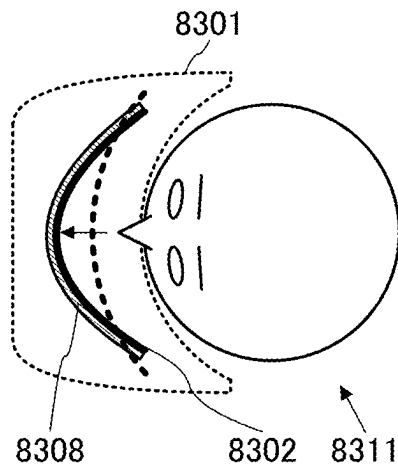


FIG. 37A

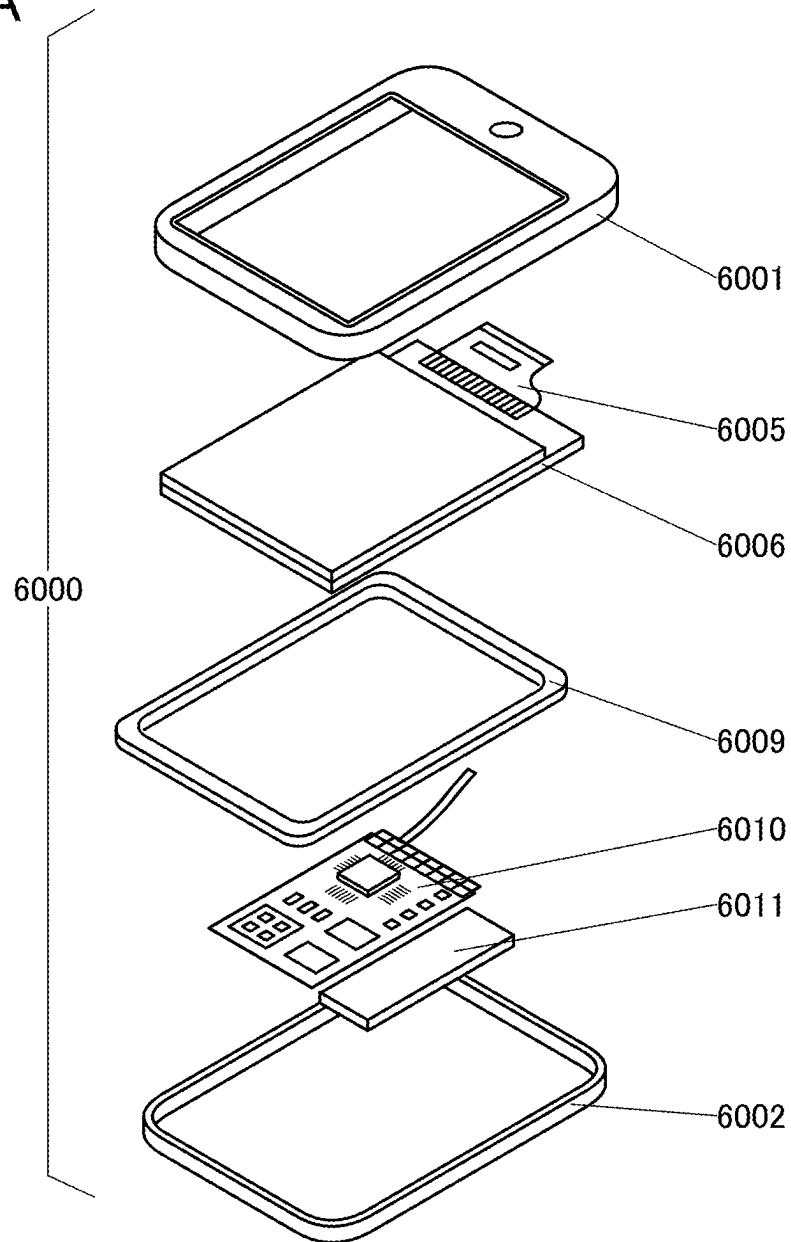


FIG. 37B

6000

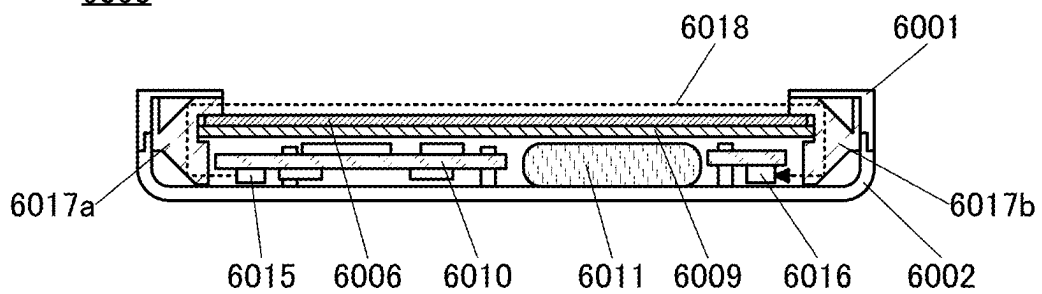


FIG. 38A

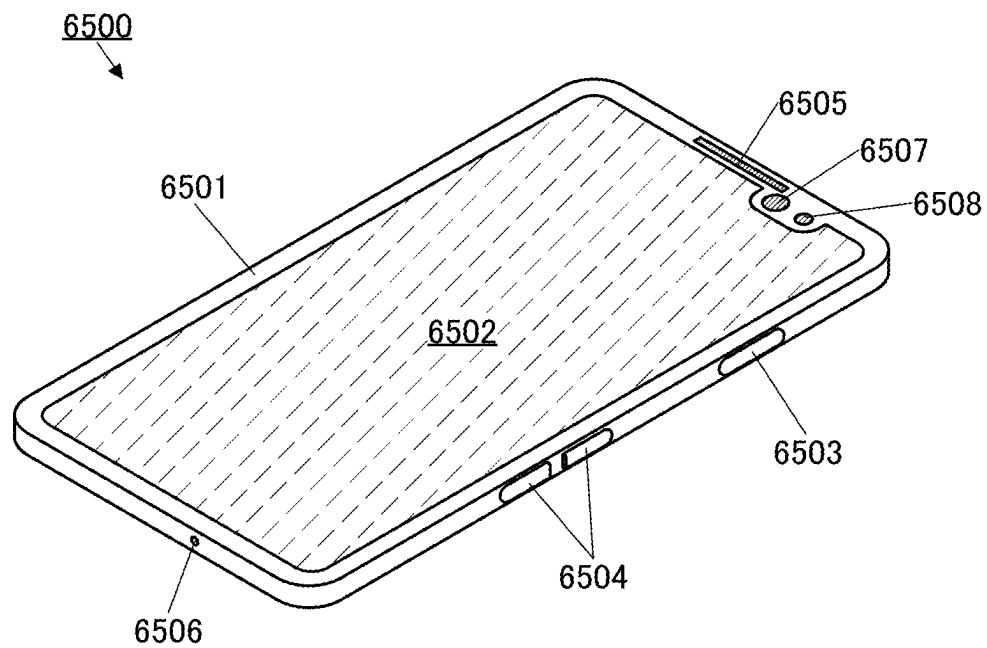


FIG. 38B

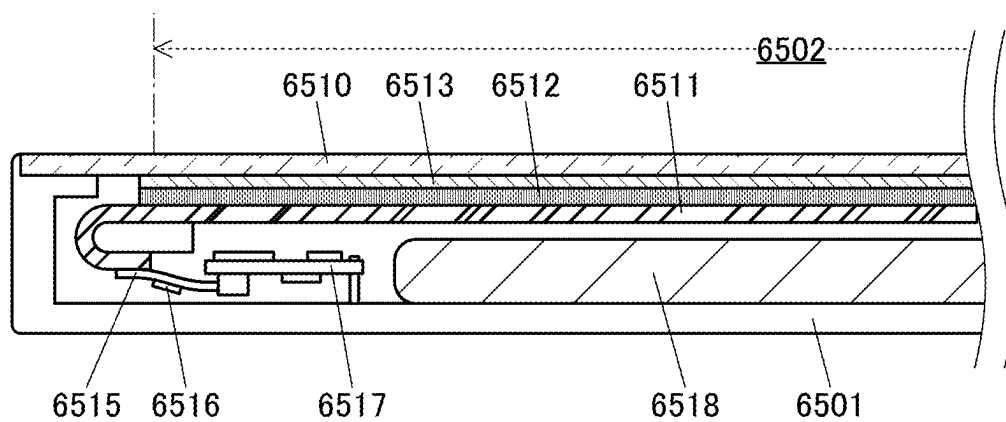


FIG. 39A

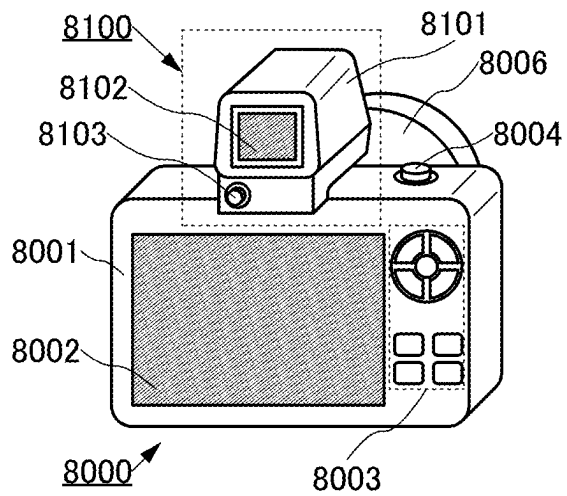


FIG. 39B

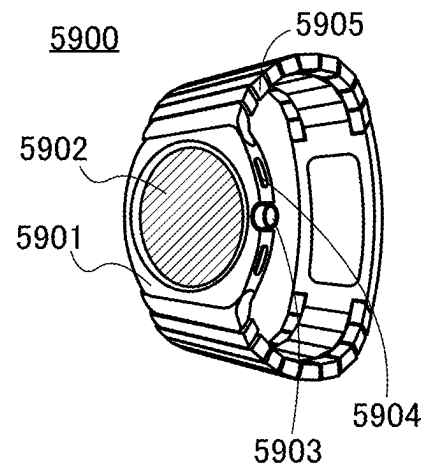


FIG. 39C

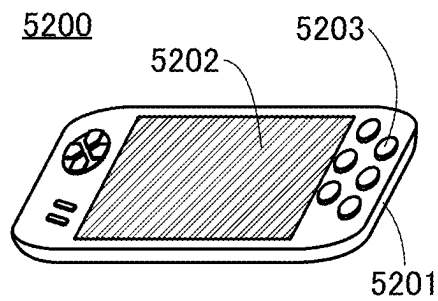


FIG. 40A

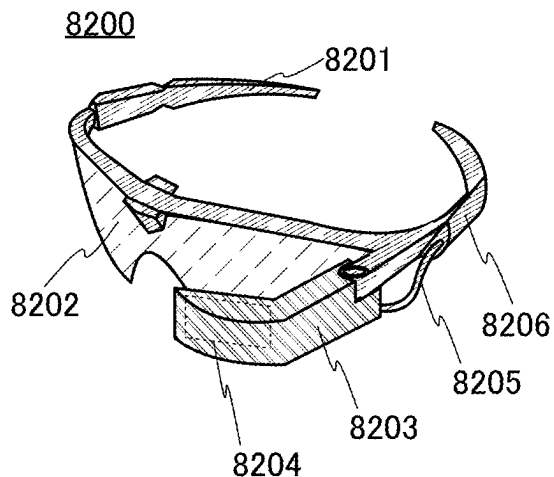


FIG. 40B

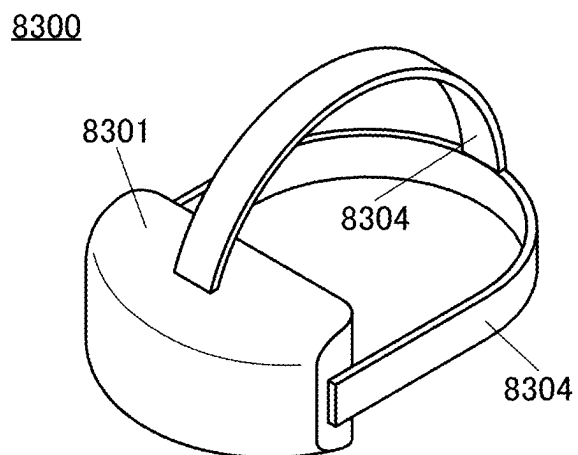


FIG. 40C

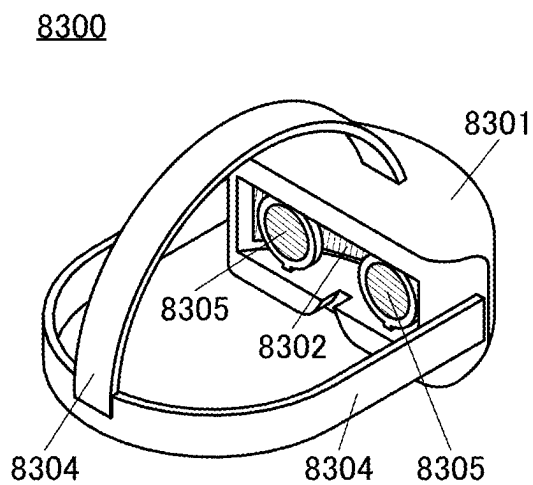
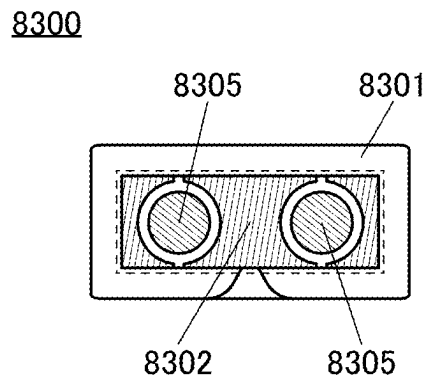


FIG. 40D



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**DISPLAY DEVICE AND ELECTRONIC
DEVICE****TECHNICAL FIELD**

One embodiment of the present invention relates to a display device and an electronic device.

Note that one embodiment of the present invention is not limited to the above technical field. The technical field of the invention disclosed in this specification and the like relates to an object, a driving method, or a manufacturing method. Alternatively, one embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter. Therefore, specific examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display apparatus, a liquid crystal display apparatus, a light-emitting apparatus, a power storage device, an imaging device, a memory device, a signal processing device, a processor, an electronic device, a system, a driving method thereof, a manufacturing method thereof, and a testing method thereof.

BACKGROUND ART

Display devices applicable for XR such as VR (virtual reality) and AR (augmented reality) have been required. For example, it is expected that such displays give a strong sense of reality and a strong sense of immersion to users. For example, an increase in the display quality, such as an increase in the resolution of the display device or an increase in the color reproducibility, allows the display device to give a strong sense of reality and a strong sense of immersion to a user.

Examples of devices applicable to such display devices include a liquid crystal display device and a light-emitting apparatus including a light-emitting device such as organic EL (Electro Luminescence) or a light-emitting diode (LED). Patent Document 1 discloses a display device with a large number of pixels and high resolution, which includes a light-emitting device containing organic EL.

REFERENCE

Patent Document

[Patent Document 1] PCT International Publication No. 2019/220278

SUMMARY OF THE INVENTION**Problems to be Solved by the Invention**

As described above, a display device having high display quality is required for XR equipment. Since a display device for XR needs to be provided in a glasses-type housing, a goggle-type housing, or the like, the size of the display device is preferably reduced, for example. Specifically, for example, in the case of VR equipment, the size (the length of a diagonal line) of the display device is preferably greater than or equal to 1 inch and less than or equal to 2 inches. For another example, in the case of AR equipment, it is preferable that the size of the display device be less than or equal to 3 inches, further preferably less than or equal to 2 inches, still further preferably less than or equal to 1.5 inches.

The display device provided in the XR equipment needs to have high definition so as to offer an enhanced sense of

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reality and an enhanced sense of immersion. In that case, designing a smaller pitch width between pixels, wirings, or the like or a smaller pixel size in a predetermined size, for example, can increase the number of pixels provided in the size. Note that when the number of pixels in the display device is large, the amount of data per one frame is increased; therefore, the speed of a driver circuit (e.g., a source driver circuit or a gate driver circuit) that drives the display device is required to be increased.

The display device provided in the XR equipment needs to have high drive frequency so as to offer an enhanced sense of reality and an enhanced sense of immersion. However, as the driving frequency becomes higher, an input time per frame is shortened, so that the amount of data that can be input to the display device in one frame is reduced in some cases.

An object of one embodiment of the present invention is to provide a display device with a reduced circuit area. Another object of one embodiment of the present invention is to provide a display device with reduced power consumption. Another object of one embodiment of the present invention is to provide a display device having high display quality. Another object of one embodiment of the present invention is to provide a display device with a high resolution and a high frame frequency. Another object of one embodiment of the present invention is to provide a novel display device. Another object of one embodiment of the present invention is to provide an electronic device including any of the display devices.

Note that the objects of one embodiment of the present invention are not limited to the objects listed above. The objects listed above do not preclude the existence of other objects. Note that the other objects are objects that are not described in this section and will be described below. The objects that are not described in this section are derived from the description of the specification, the drawings, and the like and can be extracted as appropriate from the description by those skilled in the art. Note that one embodiment of the present invention is to achieve at least one of the objects listed above and the other objects. Note that one embodiment of the present invention does not necessarily achieve all the objects listed above and the other objects.

Means for Solving the Problems

(1)

One embodiment of the present invention is a display device including a first layer, a second layer positioned above the first layer, and a third layer positioned above the second layer. The first layer includes a driver circuit and a plurality of first wirings, the second layer includes a plurality of first contact portions, and the third layer includes a pixel array and a plurality of second wirings. The pixel array includes a plurality of pixel circuits arranged in a matrix, and the driver circuit has a function of controlling the plurality of pixel circuits. The plurality of second wirings are parallel to each other and extended in the column direction of the pixel array. The plurality of pixel circuits are each electrically connected to the plurality of second wirings. The driver circuit includes a plurality of output terminals provided along a first direction. The plurality of first wirings are extended perpendicular to the first direction, and each of the plurality of output terminals is electrically connected to the plurality of first wirings. The plurality of first wirings are electrically connected to the plurality of second wirings through the plurality of first contact portions.

(2)

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In the above (1), one embodiment of the present invention may have a structure in which one of the plurality of first contact portions is positioned inside the pixel array and inside or outside the pixel circuit in a top view.

(3)

In the above (1) or (2), one embodiment of the present invention may have a structure in which the first layer includes a semiconductor substrate formed of silicon and a plurality of second contact portions. It is preferable that the driver circuit include a plurality of transistors each including silicon in a channel formation region. Furthermore, it is preferable that a plurality of low-resistance regions to be the plurality of first wirings be positioned on the top surface of the semiconductor substrate, the plurality of second contact portions be positioned between the plurality of first contact portions and the plurality of second contact portions, and the plurality of low-resistance regions include one of a source and a drain of each of the plurality of transistors.

(4)

Another embodiment of the present invention is a display device including a first layer and a third layer positioned above the first layer. The first layer includes a driver circuit region, and the third layer includes a pixel array. The pixel array includes a plurality of pixel regions, and the driver circuit region includes a plurality of local driver circuits. One of the plurality of local drivers corresponds to any one of the plurality of pixel regions. The local driver circuit has a function of driving a plurality of pixels included in the corresponding pixel region. Note that in a top view, the driver circuit region is positioned inside the pixel array, and part of the plurality of pixel regions does not overlap with the driver circuit region.

(5)

In the above (4), one embodiment of the present invention may have a structure in which each of the plurality of pixel regions includes a plurality of wirings. In the plurality of pixel regions, it is preferable that the plurality of pixels be arranged in a matrix, the plurality of wirings be positioned in the respective rows of the matrix, and one of the plurality of wirings be electrically connected to the pixel positioned in the same row. It is preferable that each of the plurality of wirings include a contact portion, and the contact portion be positioned inside the pixel or between the adjacent pixels.

(6)

In the above (4) or the above (5), one embodiment of the present invention may have a structure in which the driver circuit region includes a controller and a voltage generation circuit. It is preferable that the controller have a function of obtaining an image signal input from an outside and an address signal including a destination to which the image signal is to be transmitted and a function of selecting the plurality of local driver circuits in accordance with the address signal and transmitting the image signal to the selected local driver circuits. The voltage generation circuit preferably has a function of generating a voltage supplied to the plurality of local driver circuits or the pixel.

(7)

In any one of the above (4) to (6), one embodiment of the present invention may have a structure in which the pixel included in each of the plurality of pixel regions includes a light-emitting device with organic EL, and a first transistor, and the plurality of local driver circuits each include a second transistor. In particular, it is preferable that the first transistor include a metal oxide in a channel formation region, and the second transistor comprise silicon in a channel formation region.

(8)

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Another embodiment of the present invention is a display device including a pixel array, a driver circuit, a first wiring, and a second wiring. The pixel array includes a first region and a second region that include the same column, the first region includes a first pixel circuit, and the second region includes a second pixel circuit. The first pixel circuit and the second pixel circuit are positioned in the same column in the pixel array. The first wiring inside the first region is extended along a column direction of the pixel array, and the second wiring inside the second region is extended along the column direction of the pixel array. The driver circuit includes a demultiplexer, the first pixel circuit is electrically connected to a first output terminal of the demultiplexer through the first wiring, and the second pixel circuit is electrically connected to a second output terminal of the demultiplexer through the second wiring.

(9)

One embodiment of the present invention is a display device that has a pixel array, a driver circuit, a first wiring, and a second wiring and has a structure different from the structure of the above (8). The pixel array includes a first region and a second region that include the same column, the first region includes a first pixel circuit, and the second region includes a second pixel circuit. The first pixel circuit and the second pixel circuit are positioned in the same column in the pixel array. The first wiring passes through the inside of the first region and is extended along the column direction of the pixel array, and the second wiring passes through the inside of each of the first region and the second region and is extended along the column direction of the pixel array. The driver circuit includes a demultiplexer. The first pixel circuit is electrically connected to the first wiring, and a first output terminal of the demultiplexer is electrically connected to the first wiring. The second pixel circuit is electrically connected to the second wiring, and a second output terminal of the demultiplexer is electrically connected to the second wiring.

(10)

One embodiment of the present invention is a display device that includes a pixel array, a driver circuit, a first wiring, and a second wiring and has a structure different from the structures of the above (8) and the above (9). The pixel array includes a first region and a second region that include different columns, the first region includes a first pixel circuit, and the second region includes a second pixel circuit. The first pixel circuit and the second pixel circuit are positioned in the different columns in the pixel array, the first wiring inside the first region is extended along the column direction of the pixel array, and the second wiring inside the second region is extended along the column direction of the pixel array. The driver circuit includes a demultiplexer, the first pixel circuit is electrically connected to a first output terminal of the demultiplexer through the first wiring, and the second pixel circuit is electrically connected to a second output terminal of the demultiplexer through the second wiring.

(11)

In any one of the above (8) to (10), one embodiment of the present invention may have a structure including a third wiring and a fourth wiring. The first pixel circuit may include a first transistor, and the second pixel circuit may include a second transistor. It is preferable that the third wiring inside the first region be extended along the row direction of the pixel array, and the fourth wiring inside the second region be extended along the row direction of the pixel array. It is preferable that one of a source and a drain of the first transistor be electrically connected to the first

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wiring, and a gate of the first transistor be electrically connected to the third wiring. It is preferable that one of a source and a drain of the second transistor be electrically connected to the second wiring, and a gate of the second transistor be electrically connected to the fourth wiring (12)

In the above (11), one embodiment of the present invention may have a structure including a fifth wiring, a sixth wiring, a seventh wiring, and an eighth wiring. The first region may include a third pixel circuit, and the second region may include a fourth pixel circuit. The third pixel circuit may include a third transistor, and the fourth pixel circuit may include a fourth transistor. It is preferable that the third pixel circuit be positioned in a column different from that of the first pixel circuit in the pixel array, and the fourth pixel circuit be positioned in a column different from that of the second pixel circuit in the pixel array. It is preferable that the fifth wiring inside the first region be extended along the column direction of the pixel array, the sixth wiring inside the second region be extended along the column direction of the pixel array, the seventh wiring inside the first region be extended along the row direction of the pixel array, and the eighth wiring inside the second region be extended along the row direction of the pixel array. It is preferable that one of a source and a drain of the third transistor be electrically connected to the fifth wiring, and a gate of the third transistor be electrically connected to the seventh wiring. It is preferable that one of a source and a drain of the fourth transistor be electrically connected to the sixth wiring, and a gate of the fourth transistor be electrically connected to the eighth wiring. It is preferable that the fifth wiring be electrically connected to a third output terminal of the demultiplexer, and the sixth wiring be electrically connected to a fourth output terminal of the demultiplexer. (13)

In any one of the above (8) to (12), one embodiment of the present invention may have a structure including a first layer and a third layer positioned above the first layer. In particular, it is preferable that the first layer include the driver circuit, and the third layer include the pixel array. (14)

Another embodiment of the present invention is an electronic device including the display device according to any one of the above (1) to (13) and a housing.

Note that in this specification and the like, a semiconductor device refers to a device that utilizes semiconductor characteristics, and means a circuit including a semiconductor element (a transistor, a diode, a photodiode, or the like), a device including the circuit, and the like. The semiconductor device also means all devices that can function by utilizing semiconductor characteristics. For example, an integrated circuit, a chip including an integrated circuit, and an electronic component including a chip in a package are examples of the semiconductor device. Moreover, a memory device, a display apparatus, a light-emitting device, a lighting device, an electronic device, and the like themselves are semiconductor devices or include semiconductor devices in some cases.

In the case where there is description “X and Y are connected” in this specification and the like, the case where X and Y are electrically connected, the case where X and Y are functionally connected, and the case where X and Y are directly connected are regarded as being disclosed in this specification and the like. Accordingly, without being limited to a predetermined connection relation, for example, a connection relation shown in drawings or texts, a connection relation other than one shown in drawings or texts is

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regarded as being disclosed in the drawings or the texts. Each of X and Y denotes an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, or a layer).

For example, in the case where X and Y are electrically connected, one or more elements that allow electrical connection between X and Y (e.g., a switch, a transistor, a capacitor, an inductor, a resistor, a diode, a display device, a light-emitting device, and a load) can be connected between X and Y. Note that a switch has a function of being controlled to be turned on or off. That is, the switch has a function of being in a conduction state (on state) or a non-conduction state (off state) to control whether current flows or not.

For example, in the case where X and Y are functionally connected, one or more circuits that allow functional connection between X and Y (e.g., a logic circuit (an inverter, a NAND circuit, or a NOR circuit); a signal converter circuit (a digital-to-analog converter circuit, an analog-to-digital converter circuit, a gamma correction circuit, or the like); a potential level converter circuit (a power supply circuit (a step-up circuit, a step-down circuit, or the like), a level shifter circuit for changing the potential level of a signal, or the like); a voltage source; a current source; a switching circuit; an amplifier circuit (a circuit that can increase signal amplitude, the amount of current, or the like, an operational amplifier, a differential amplifier circuit, a source follower circuit, a buffer circuit, or the like); a signal generation circuit; a memory circuit; or a control circuit) can be connected between X and Y. For instance, even if another circuit is provided between X and Y, X and Y are regarded as being functionally connected when a signal output from X is transmitted to Y.

Note that an explicit description “X and Y are electrically connected” includes the case where X and Y are electrically connected (i.e., the case where X and Y are connected with another element or another circuit provided therebetween) and the case where X and Y are directly connected (i.e., the case where X and Y are connected without another element or another circuit provided therebetween).

It can be expressed as, for example, “X, Y, and a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor are electrically connected to each other, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected in this order”. Alternatively, it can be expressed as “a source (or a first terminal or the like) of a transistor is electrically connected to X; a drain (or a second terminal or the like) of the transistor is electrically connected to Y; and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected to each other in this order”. Alternatively, it can be expressed as “X is electrically connected to Y through a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are provided in this connection order”. When the connection order in a circuit structure is defined by an expression similar to the above examples, a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor can be distinguished from each other to specify the technical scope. Note that these expressions are examples and the expression is not limited to these expressions. Here, X and Y each denote an object

(e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, or a layer).

Even when independent components are electrically connected to each other in a circuit diagram, one component has functions of a plurality of components in some cases. For example, when part of a wiring also functions as an electrode, one conductive film has functions of both of the components that are a wiring and an electrode. Thus, electrical connection in this specification includes, in its category, such a case where one conductive film has functions of a plurality of components.

In this specification and the like, a “resistor” can be, for example, a circuit element having a resistance value higher than 0Ω or a wiring having a resistance value higher than 0Ω . Therefore, in this specification and the like, a “resistor” sometimes includes a wiring having a resistance value, a transistor in which current flows between its source and drain, a diode, and a coil. Thus, the term “resistor” can be sometimes replaced with the terms “resistance”, “load”, “region having a resistance value”, and the like; conversely, the terms “resistance”, “load”, and “region having a resistance value” can be sometimes replaced with the term “resistor” and the like. The resistance value can be, for example, preferably higher than or equal to $1\text{ m}\Omega$ and lower than or equal to 10Ω , further preferably higher than or equal to $5\text{ m}\Omega$ and lower than or equal to 5Ω , still further preferably higher than or equal to $10\text{ m}\Omega$ and lower than or equal to 1Ω . As another example, the resistance value may be higher than or equal to 1Ω and lower than or equal to $1\times 10^9\Omega$.

In this specification and the like, a “capacitor” can be, for example, a circuit element having an electrostatic capacitance value higher than 0 F , a region of a wiring having an electrostatic capacitance value higher than 0 F , parasitic capacitance, or gate capacitance of a transistor. The terms “capacitor”, “parasitic capacitance”, “gate capacitance”, and the like can be replaced with the term “capacitance” and the like in some cases. Conversely, the term “capacitance” can be replaced with the terms “capacitor”, “parasitic capacitance”, “gate capacitance”, and the like in some cases. The term “pair of electrodes” of “capacitor” can be replaced with “pair of conductors”, “pair of conductive regions”, “pair of regions”, and the like. Note that the electrostatic capacitance value can be higher than or equal to 0.05 fF and lower than or equal to 10 pF , for example. As another example, the electrostatic capacitance value may be higher than or equal to 1 pF and lower than or equal to $10\text{ }\mu\text{F}$.

In this specification and the like, a transistor includes three terminals called a gate, a source, and a drain. The gate is a control terminal for controlling the conduction state of the transistor. Two terminals functioning as the source and the drain are input/output terminals of the transistor. One of the two input/output terminals serves as the source and the other serves as the drain on the basis of the conductivity type (n-channel type or p-channel type) of the transistor and the levels of potentials applied to the three terminals of the transistor. Thus, the terms “source” and “drain” can be sometimes replaced with each other in this specification and the like. In this specification and the like, expressions “one of a source and a drain” (or a first electrode or a first terminal) and “the other of the source and the drain” (or a second electrode or a second terminal) are used in description of the connection relation of a transistor. Depending on the transistor structure, a transistor may include a back gate in addition to the above three terminals. In that case, in this specification and the like, one of the gate and the back gate of the transistor may be referred to as a first gate and the

other of the gate and the back gate of the transistor may be referred to as a second gate. Moreover, the terms “gate” and “back gate” can be replaced with each other in one transistor in some cases. In the case where a transistor includes three or more gates, the gates may be referred to as a first gate, a second gate, and a third gate, for example, in this specification and the like.

In this specification and the like, for example, a transistor with a multi-gate structure having two or more gate electrodes can be used as the transistor. With the multi-gate structure, channel formation regions are connected in series; accordingly, a plurality of transistors are connected in series. Thus, with the multi-gate structure, the amount of off-state current can be reduced, and the withstand voltage of the transistor can be increased (the reliability can be improved). Alternatively, with the multi-gate structure, drain-source current does not change very much even if drain-source voltage changes at the time of operation in a saturation region, so that a flat slope of voltage—current characteristics can be obtained. By utilizing the flat slope of the voltage—current characteristics, an ideal current source circuit or an active load having an extremely high resistance value can be obtained. Accordingly, a differential circuit, a current mirror circuit, and the like having excellent properties can be obtained.

The case where a single circuit element is illustrated in a circuit diagram may indicate a case where the circuit element includes a plurality of circuit elements. For example, the case where a single resistor is illustrated in a circuit diagram may indicate a case where two or more resistors are electrically connected to each other in series. As another example, the case where a single capacitor is illustrated in a circuit diagram may indicate a case where two or more capacitors are electrically connected to each other in parallel. As another example, the case where a single transistor is illustrated in a circuit diagram may indicate a case where two or more transistors are electrically connected to each other in series and their gates are electrically connected to each other. Similarly, as another example, the case where a single switch is illustrated in a circuit diagram may indicate a case where the switch includes two or more transistors which are electrically connected to each other in series or in parallel and whose gates are electrically connected to each other.

In this specification and the like, a node can be referred to as a terminal, a wiring, an electrode, a conductive layer, a conductor, an impurity region, or the like depending on the circuit structure and/or the device structure. Furthermore, a terminal, a wiring, or the like can be referred to as a node.

In this specification and the like, “voltage” and “potential” can be replaced with each other as appropriate. “Voltage” refers to a potential difference from a reference potential, and when the reference potential is a ground potential, for example, “voltage” can be replaced with “potential”. Note that the ground potential does not necessarily mean 0 V . Moreover, potentials are relative values, and a potential supplied to a wiring, a potential applied to a circuit and the like, and a potential output from a circuit and the like, for example, change with a change of the reference potential.

In this specification and the like, the terms “high-level potential” and “low-level potential” do not mean a particular potential. For example, in the case where two wirings are both described as “functioning as a wiring for supplying a high-level potential”, the levels of the high-level potentials supplied from the wirings are not necessarily equal to each other. Similarly, in the case where two wirings are both described as “functioning as a wiring for supplying a low-

level potential”, the levels of the low-level potentials supplied from the wirings are not necessarily equal to each other.

“Current” means a charge transfer (electrical conduction); for example, the description “electrical conduction of positively charged particles occurs” can be rephrased as “electrical conduction of negatively charged particles occurs in the opposite direction”. Therefore, unless otherwise specified, “current” in this specification and the like refers to a charge transfer (electrical conduction) accompanied by carrier movement. Examples of a carrier here include an electron, a hole, an anion, a cation, and a complex ion, and the type of carrier differs between current flow systems (e.g., a semiconductor, a metal, an electrolyte solution, and a vacuum). The “direction of current” in a wiring or the like refers to the direction in which a carrier with a positive charge moves, and the amount of current is expressed as a positive value. In other words, the direction in which a carrier with a negative charge moves is opposite to the direction of current, and the amount of current is expressed as a negative value. Thus, in the case where the polarity of current (or the direction of current) is not specified in this specification and the like, the description “current flows from element A to element B” can be rephrased as “current flows from element B to element A”, for example. The description “current is input to element A” can be rephrased as “current is output from element A”, for example.

Ordinal numbers such as “first”, “second”, and “third” in this specification and the like are used to avoid confusion among components. Thus, the terms do not limit the number of components. In addition, the terms do not limit the order of components. In this specification and the like, for example, a “first” component in one embodiment can be referred to as a “second” component in other embodiments or the scope of claims. As another example, a “first” component in one embodiment in this specification and the like can be omitted in other embodiments or the scope of claims.

In this specification and the like, the terms for describing positioning, such as “over” and “under”, are sometimes used for convenience to describe the positional relation between components with reference to drawings. The positional relation between components is changed as appropriate in accordance with the direction in which the components are described. Thus, the positional relation is not limited to the terms described in the specification and the like, and can be described with another term as appropriate depending on the situation. For example, the expression “an insulator positioned over (on) the top surface of a conductor” can be replaced with the expression “an insulator positioned under (on) a bottom surface of a conductor” when the direction of a drawing showing these components is rotated by 180°.

Furthermore, the term “over” or “under” does not necessarily mean that a component is placed directly over or directly under and in direct contact with another component. For example, the expression “electrode B over insulating layer A” does not necessarily mean that the electrode B is formed on and in direct contact with the insulating layer A, and does not exclude the case where another component is provided between the insulating layer A and the electrode B.

In this specification and the like, components arranged in a matrix and their positional relation are sometimes described using terms such as “row” and “column”. The positional relation between components is changed as appropriate in accordance with the direction in which the components are described. Thus, the positional relation is not limited to the terms described in the specification and the like, and can be described with another term as appropriate

depending on the situation. For example, the term “row direction” can be replaced with the term “column direction” when the direction of the diagram is rotated by 90°.

In this specification and the like, the terms “film”, “layer”, and the like can be interchanged with each other depending on the situation. For example, the term “conductive layer” can be changed into the term “conductive film” in some cases. As another example, the term “insulating film” can be changed into the term “insulating layer” in some cases. Alternatively, the term “film”, “layer”, or the like is not used and can be interchanged with another term depending on the case or the situation. For example, the term “conductive layer” or “conductive film” can be changed into the term “conductor” in some cases. As another example, the term “insulating layer” or “insulating film” can be changed into the term “insulator” in some cases.

In this specification and the like, the term “electrode”, “wiring”, “terminal”, or the like does not limit the function of a component. For example, an “electrode” is used as part of a “wiring” in some cases, and vice versa. Furthermore, the term “electrode” or “wiring” can also mean, for example, the case where a plurality of “electrodes” or “wirings” are formed in an integrated manner. For example, a “terminal” is used as part of a “wiring” or an “electrode” in some cases, and vice versa. Furthermore, the term “terminal” also includes the case where a plurality of “electrodes”, “wirings”, “terminals”, or the like are formed in an integrated manner, for example. Therefore, for example, an “electrode” can be part of a “wiring” or a “terminal”, and a “terminal” can be part of a “wiring” or an “electrode”. Moreover, the terms “electrode”, “wiring”, “terminal”, and the like are sometimes replaced with the term “region” or the like depending on the case.

In this specification and the like, the terms “wiring”, “signal line”, “power supply line”, and the like can be interchanged with each other depending on the case or the situation. For example, the term “wiring” can be changed into the term “signal line” in some cases. As another example, the term “wiring” can be changed into the term “power supply line” or the like in some cases. Conversely, the term “signal line”, “power supply line”, or the like can be changed into the term “wiring” in some cases. The term “power supply line” or the like can be changed into the term “signal line” or the like in some cases. Conversely, the term “signal line” or the like can be changed into the term “power supply line” or the like in some cases. The term “potential” that is applied to a wiring can be changed into the term “signal” or the like depending on the case or the situation. Conversely, the term “signal” or the like can be changed into the term “potential” in some cases.

In this specification and the like, an impurity in a semiconductor refers to, for example, an element other than a main component of a semiconductor layer. For example, an element with a concentration of lower than 0.1 atomic % is an impurity. When an impurity is contained, for example, the density of defect states in a semiconductor is increased, carrier mobility is decreased, or crystallinity is decreased in some cases. In the case where the semiconductor is an oxide semiconductor, examples of an impurity that changes characteristics of the semiconductor include Group 1 elements, Group 2 elements, Group 13 elements, Group 14 elements, Group 15 elements, and transition metals other than the main components; specific examples are hydrogen (contained also in water), lithium, sodium, silicon, boron, phosphorus, carbon, and nitrogen. Specifically, in the case where the semiconductor is a silicon layer, examples of an impurity that changes characteristics of the semiconductor include Group

1 elements, Group 2 elements, Group 13 elements, and Group 15 elements (except oxygen and hydrogen).

In this specification and the like, a switch has a function of being in a conduction state (on state) or a non-conduction state (off state) to determine whether current flows or not. Alternatively, a switch has a function of selecting and changing a current path. Thus, a switch may have two or more terminals through which current flows, in addition to a control terminal. For example, an electrical switch or a mechanical switch can be used. That is, a switch can be any element capable of controlling current, and is not limited to a particular element.

Examples of an electrical switch include a transistor (e.g., a bipolar transistor and a MOS transistor), a diode (e.g., a PN diode, a PIN diode, a Schottky diode, a MIM (Metal Insulator Metal) diode, a MIS (Metal Insulator Semiconductor) diode, and a diode-connected transistor), and a logic circuit in which such elements are combined. Note that in the case of using a transistor as a switch, a “conduction state” of the transistor refers to a state where a source electrode and a drain electrode of the transistor can be regarded as being electrically short-circuited or a state where current can be made to flow between the source electrode and the drain electrode. Furthermore, a “non-conduction state” of the transistor refers to a state where the source electrode and the drain electrode of the transistor can be regarded as being electrically disconnected. Note that in the case where a transistor operates just as a switch, there is no particular limitation on the polarity (conductivity type) of the transistor.

An example of a mechanical switch is a switch formed using a MEMS (micro electro mechanical systems) technology. Such a switch includes an electrode that can be moved mechanically, and operates by controlling conduction and non-conduction with movement of the electrode.

In this specification and the like, a device formed using a metal mask or a fine metal mask (FMM) may be referred to as a device having a metal mask (MM) structure. In this specification and the like, a device formed without using a metal mask or an FMM may be referred to as a device having a metal maskless (MML) structure.

In this specification and the like, a structure in which light-emitting layers in light-emitting devices of different colors (here, blue (B), green (G), and red (R)) are separately formed or separately patterned may be referred to as an SBS (Side By Side) structure. In this specification and the like, a light-emitting device capable of emitting white light may be referred to as a white-light-emitting device. Note that a combination of white-light-emitting devices with coloring layers (e.g., color filters) enables a full-color display device.

Light-emitting devices can be classified roughly into a single structure and a tandem structure. A device with a single structure includes one light-emitting unit between a pair of electrodes, and the light-emitting unit preferably includes one or more light-emitting layers. When white light emission is obtained using two light-emitting layers, the two light-emitting layers are selected such that emission colors of the light-emitting layers are complementary colors. For example, when emission colors of a first light-emitting layer and a second light-emitting layer are complementary colors, the light-emitting device can be configured to emit white light as a whole. When white light emission is obtained using three or more light-emitting layers, the light-emitting device is configured to emit white light as a whole by combining emission colors of the three or more light-emitting layers.

A device with a tandem structure includes two or more light-emitting units between a pair of electrodes, and each light-emitting unit preferably includes one or more light-emitting layers. To obtain white light emission, the structure is made so that light from light-emitting layers of the plurality of light-emitting units can be combined to be white light. Note that a structure for obtaining white light emission is similar to that in the case of a single structure. In the device with a tandem structure, an intermediate layer such as a charge-generation layer is suitably provided between the plurality of light-emitting units.

When the above white-light-emitting device (having a single structure or a tandem structure) and the above light-emitting device having an SBS structure are compared to each other, the light-emitting device having an SBS structure can have lower power consumption than the white-light-emitting device. The light-emitting device having an SBS structure is suitable for the case where the power consumption is required to be low. Meanwhile, the white-light-emitting device is suitable in terms of lower manufacturing cost or higher manufacturing yield because the manufacturing process of the white-light-emitting device is simpler than that of the light-emitting device having an SBS structure.

In this specification, “parallel” indicates a state where two straight lines are placed at an angle greater than or equal to -10° and less than or equal to 10° . Thus, the case where the angle is greater than or equal to -5° and less than or equal to 5° is also included. In addition, “approximately parallel” or “substantially parallel” indicates a state where two straight lines are placed at an angle greater than or equal to -30° and less than or equal to 30° . Moreover, “perpendicular” indicates a state where two straight lines are placed at an angle greater than or equal to 80° and less than or equal to 100° . Thus, the case where the angle is greater than or equal to 85° and less than or equal to 95° is also included. Furthermore, “approximately perpendicular” or “substantially perpendicular” indicates a state where two straight lines are placed at an angle greater than or equal to 60° and less than or equal to 120° .

Effect of the Invention

According to one embodiment of the present invention, a display device with a reduced circuit area can be provided. According to one embodiment of the present invention, a display device with reduced power consumption can be provided. According to one embodiment of the present invention, a display device having high display quality can be provided. According to one embodiment of the present invention, a display device with a high resolution and a high frame frequency can be provided. According to one embodiment of the present invention, a novel display device can be provided. According to one embodiment of the present invention, an electronic device including any of the display devices can be provided.

Note that the effects of one embodiment of the present invention are not limited to the effects listed above. The effects listed above do not preclude the existence of other effects. Note that the other effects are effects that are not described in this section and will be described below. The effects that are not described in this section are derived from the description of the specification, the drawings, and the like and can be extracted as appropriate from the description by those skilled in the art. Note that one embodiment of the present invention has at least one of the effects listed above and the other effects. Accordingly, depending on the case,

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one embodiment of the present invention does not have the effects listed above in some cases.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic cross-sectional view illustrating a structure example of a display device.

FIG. 2A is a schematic top view illustrating a structure example of a circuit layer included in a display device, and FIG. 2B is a schematic top view illustrating a structure example of the display device.

FIG. 3A is a perspective view schematically illustrating a structure example of a display device, and FIG. 3B is a schematic cross-sectional view illustrating a structure example of the display device.

FIG. 4A is a perspective view schematically illustrating a structure example of a display device, and FIG. 4B is a schematic cross-sectional view illustrating a structure example of the display device.

FIG. 5A is a perspective view schematically illustrating a structure example of a display device, and FIG. 5B is a schematic cross-sectional view illustrating a structure example of the display device.

FIG. 6A to FIG. 6C are diagrams illustrating electrical connection between pixels and wirings included in a display device.

FIG. 7 is a schematic top view illustrating a structure example of a display device.

FIG. 8A and FIG. 8B are schematic top views illustrating structure examples of a display device.

FIG. 9A and FIG. 9B are schematic top views illustrating structure examples of a display device.

FIG. 10A and FIG. 10B are schematic top views illustrating structure examples of a display device.

FIG. 11 is a schematic top view illustrating a structure example of a display device.

FIG. 12 is a schematic top view illustrating a structure example of a display device.

FIG. 13 is a block diagram illustrating a structure example of a display device.

FIG. 14 is a block diagram illustrating a structure example of a display device.

FIG. 15 is a timing chart showing an operation example of a display device.

FIG. 16A is a perspective view schematically illustrating a structure example of a display device, and FIG. 16B is a perspective view illustrating a structure example of a circuit included in the display device.

FIG. 17 is a block diagram illustrating a structure example of a display device.

FIG. 18A to FIG. 18F are block diagrams illustrating an operation example of a display device.

FIG. 19 is a block diagram illustrating a structure example of a display device.

FIG. 20 is a block diagram illustrating a structure example of a display device.

FIG. 21 is a block diagram illustrating a structure example of a display device.

FIG. 22 is a block diagram illustrating a structure example of a display device.

FIG. 23 is a block diagram illustrating a structure example of a display device.

FIG. 24 is a block diagram illustrating a structure example of a circuit included in a display device.

FIG. 25 is a block diagram illustrating a structure example of a display device.

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FIG. 26A is a perspective view schematically illustrating a structure example of a display device, and FIG. 26B is a perspective view illustrating a structure example of a circuit included in the display device.

FIG. 27 is a schematic cross-sectional view illustrating a structure example of a display device.

FIG. 28A and FIG. 28B are schematic cross-sectional views illustrating structure examples of transistors.

FIG. 29A and FIG. 29B are schematic cross-sectional views each illustrating a structure example of a transistor.

FIG. 30A to FIG. 30C are schematic diagrams illustrating structure examples of a light-emitting device.

FIG. 31A and FIG. 31B are schematic cross-sectional views illustrating structure examples of a display device.

FIG. 32A to FIG. 32C are schematic cross-sectional views illustrating structure examples of a display device.

FIG. 33A to FIG. 33D are schematic cross-sectional views illustrating structure examples of a display device.

FIG. 34A is a diagram showing classification of crystal structures of IGZO, FIG. 34B is a diagram showing an XRD spectrum of crystalline IGZO, and FIG. 34C is a diagram showing a nanobeam electron diffraction pattern of the crystalline IGZO.

FIG. 35A and FIG. 35B are diagrams illustrating a structure example of a display module.

FIG. 36A to FIG. 36F are diagrams illustrating structure examples of an electronic device.

FIG. 37A and FIG. 37B are diagrams illustrating a structure example of a display module.

FIG. 38A and FIG. 38B are diagrams illustrating a structure example of an electronic device.

FIG. 39A to FIG. 39C are diagrams illustrating structure examples of electronic devices.

FIG. 40A to FIG. 40D are diagrams illustrating structure examples of electronic devices.

MODE FOR CARRYING OUT THE INVENTION

In this specification and the like, a metal oxide is an oxide of metal in a broad sense. Metal oxides are classified into an oxide insulator, an oxide conductor (including a transparent oxide conductor), an oxide semiconductor (also simply referred to as an OS), and the like. For example, in the case where a metal oxide is included in a channel formation region of a transistor, the metal oxide is referred to as an oxide semiconductor in some cases. That is, when a metal oxide can form a channel formation region of a transistor that has at least one of an amplifying function, a rectifying function, and a switching function, the metal oxide can be referred to as a metal oxide semiconductor. In the case where an OS transistor is mentioned, the OS transistor can also be referred to as a transistor including a metal oxide or an oxide semiconductor.

In this specification and the like, a metal oxide containing nitrogen is also collectively referred to as a metal oxide in some cases. A metal oxide containing nitrogen may be called a metal oxynitride.

In this specification and the like, one embodiment of the present invention can be constituted by appropriately combining a structure described in an embodiment with any of the structures described in the other embodiments. In addition, in the case where a plurality of structure examples are described in one embodiment, the structure examples can be combined as appropriate.

Note that a content (or part of the content) described in one embodiment can be applied to, combined with, or replaced with at least one of another content (or part of the

content) in the embodiment and a content (or part of the content) described in one or a plurality of different embodiments.

Note that in each embodiment, a content described in the embodiment is a content described using a variety of diagrams or a content described with text disclosed in the specification.

Note that by combining a diagram (or part thereof) described in one embodiment with at least one of another part of the diagram, a different diagram (or part thereof) described in the embodiment, and a diagram (or part thereof) described in one or a plurality of different embodiments, much more diagrams can be formed.

Embodiments described in this specification are described with reference to the drawings. Note that the embodiments can be implemented in many different modes, and it will be readily appreciated by those skilled in the art that modes and details can be changed in various ways without departing from the spirit and scope thereof. Therefore, the present invention should not be interpreted as being limited to the description in the embodiments. Note that in the structures of the invention in the embodiments, the same portions or portions having similar functions are denoted by the same reference numerals in different drawings, and repeated description thereof is omitted in some cases. In perspective views and the like, some components might not be illustrated for clarity of the drawings.

In this specification and the like, when a plurality of components are denoted with the same reference numerals, and in particular need to be distinguished from each other, an identification sign such as “_1”, “[n]”, or “[m,n]” is sometimes added to the reference numerals. Components denoted with identification signs such as “_1”, “[n]”, and “[m,n]” in the drawings and the like are sometimes denoted without such identification signs in this specification and the like when the components do not need to be distinguished from each other.

In the drawings in this specification, the size, the layer thickness, or the region is exaggerated for clarity in some cases. Therefore, they are not limited to the illustrated scale. The drawings are schematic views showing ideal examples, and embodiments of the present invention are not limited to shapes, values, or the like shown in the drawings. For example, variations in signal, voltage, or current due to noise, variations in signal, voltage, or current due to difference in timing, or the like can be included.

Embodiment 1

In this embodiment, a display device of one embodiment of the present invention will be described.

FIG. 1 is a schematic cross-sectional view of the display device of one embodiment of the present invention. A display device **10** includes a pixel layer PXAL, a wiring layer LINL, and a circuit layer SICL.

The wiring layer LINL is provided over the circuit layer SICL, and the pixel layer PXAL is provided over the wiring layer LINL. Note that the pixel layer PXAL overlaps with a region including a driver circuit region DRV and a region LIA which are described later.

The circuit layer SICL includes a substrate BS, the driver circuit region DRV, and the region LIA.

As the substrate BS, a semiconductor substrate (e.g., a single crystal substrate) formed of silicon or germanium can be used, for example. Besides the semiconductor substrate, for example, an SOI (Silicon On Insulator) substrate, a glass substrate, a quartz substrate, a plastic substrate, a sapphire

glass substrate, a metal substrate, a stainless steel substrate, a substrate including stainless steel foil, a tungsten substrate, a substrate including tungsten foil, a flexible substrate, a bonding film, paper containing a fibrous material, a base material film, or the like can be used as the substrate BS. Examples of the glass substrate include barium borosilicate glass, aluminoborosilicate glass, and soda lime glass. As examples of the flexible substrate, the bonding film, the base material film, and the like, the following can be given. Examples include plastics typified by polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polyether sulfone (PES), and polytetrafluoroethylene (PTFE). Another example is a synthetic resin such as an acrylic resin. Other examples include polypropylene, polyester, polyvinyl fluoride, and polyvinyl chloride. Other examples include polyamide, polyimide, aramid, an epoxy resin, an inorganic vapor deposition film, and paper. Note that in the case where the manufacturing process of the display device **10** includes heat treatment, a highly heat-resistant material is preferably selected for the substrate BS.

In the description of this embodiment, the substrate BS is a semiconductor substrate containing silicon or the like as a material. Therefore, a transistor included in the driver circuit region DRV can be a transistor including silicon in a channel formation region (hereinafter referred to as a Si transistor).

The driver circuit region DRV and the region LIA are provided over the substrate BS.

The driver circuit region DRV includes, for example, a driver circuit for driving a pixel included in the pixel layer PXAL to be described later. Note that a specific structure example of the driver circuit region DRV will be described later.

A wiring is provided in the region LIA, as an example. The wiring included in the region LIA may be electrically connected to a wiring included in the wiring layer LINL. At this time, the display device **10** may have a structure in which a circuit included in the driver circuit region DRV and a circuit included in the pixel layer PXAL are electrically connected to each other through the wiring included in the region LIA and the wiring included in the wiring layer LINL. The display device **10** may have a structure in which the circuit included in the driver circuit region DRV is electrically connected to the wiring or a circuit included in the region LIA through the wiring included in the wiring layer LINL.

The region LIA may include a GPU (graphics processing unit), as an example. In the case where the display device **10** includes a touch panel, the region LIA may include a sensor controller that controls a touch sensor included in the touch panel. In the case where a light-emitting device using an organic EL material (sometimes the device is referred to as an organic EL element) is used as a display element of the display device **10**, an EL correction circuit that corrects variations in luminance of light emission of a plurality of organic EL elements may be included. In the case where a liquid crystal element is used as the display element of the display device **10**, a gamma correction circuit may be included.

The wiring layer LINL is provided over the circuit layer SICL.

For example, a wiring is provided in the wiring layer LINL. The wiring included in the wiring layer LINL functions as, for example, a wiring that electrically connects a driver circuit included in the driver circuit region DRV provided below the wiring layer LINL and a circuit included in the pixel layer PXAL provided above the wiring layer LINL.

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The pixel layer PXAL includes, for example, a plurality of pixels. The plurality of pixels may be arranged in a matrix in the pixel layer PXAL.

Each of the plurality of pixels can express one color or a plurality of colors. In particular, the plurality of colors can be, for example, three colors of red (R), green (G), and blue (B). Alternatively, the plurality of colors may be at least one color selected from, for example, red (R), green (G), blue (B), cyan, magenta, yellow, and white. Note that in the case where each of pixels expressing different colors is called a subpixel and white is expressed by a plurality of subpixels expressing different colors, the plurality of subpixels are collectively called a pixel in some cases. In this specification and the like, a subpixel is referred to as a pixel for convenience.

FIG. 2A is an example of a top view of the display device 10, illustrating only the circuit layer SICL. The display device 10 illustrated in FIG. 2A has a structure in which the driver circuit region DRV is surrounded by the region LIA, as an example.

In FIG. 2A, the driver circuit region DRV includes, for example, a plurality of local driver circuits LD, a controller CON, and a voltage generation circuit PG.

The controller CON has a function of processing an input signal from the outside of the display device 10, for example. Examples of the input signal include an image signal and an address signal including the destination of the image signal. In accordance with the address signal, the controller CON selects the local driver circuit LD included in the driver circuit region DRV and transmits an image signal to the selected local driver circuit LD.

Since the plurality of local driver circuits LD are provided in the driver circuit region DRV, the controller CON may be configured to transmit image signals to a plurality of local driver circuits LD at the same time.

The voltage generation circuit PG functions as a circuit that generates a power supply voltage for driving circuits (e.g., a source driver circuit and a gate driver circuit described later) included in the driver circuit region DRV, for example. Furthermore, the voltage generation circuit PG may have a function of generating a voltage to be supplied to pixels included in the pixel layer PXAL described later.

Each of the plurality of local driver circuits has a function of driving pixels included in the pixel layer PXAL, as an example. That is, for example, each of the plurality of local driver circuits can include a source driver circuit and a gate driver circuit. Since the plurality of local driver circuits are provided, a region of the pixel layer PXAL where pixels are to be driven can be determined for each of the local driver circuits.

For example, consider a case where a pixel array ALP included in the pixel layer PXAL is divided into regions of m rows and n columns (m is an integer of 1 or more and n is an integer of 1 or more) in the display device 10. In this case, the number of local driver circuits included in the driver circuit region DRV is $m \times n$. Note that as an example, FIG. 2A selectively illustrates a local driver circuit LD[1,1], a local driver circuit LD[1,2], a local driver circuit LD[2,1], a local driver circuit LD[2,2], a local driver circuit LD[m-1,1], a local driver circuit LD[m-1,2], a local driver circuit LD[m,1], a local driver circuit LD[m,2], a local driver circuit LD[1,n-1], a local driver circuit LD[1,n], a local driver circuit LD[2,n-1], a local driver circuit LD[2,n], a local driver circuit LD[m-1,n-1], a local driver circuit LD[m-1,n], a local driver circuit LD[m,n-1], and a local driver circuit LD[m,n].

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FIG. 2B illustrates pixel regions obtained by dividing the pixel array ALP included in the pixel layer PXAL into regions of m rows and n columns. FIG. 2B is a top view of the display device 10, only illustrating the driver circuit region DRV and the pixel array ALP. In particular, in FIG. 2B, the driver circuit region DRV is denoted by a solid line and the pixel array ALP is denoted by a dashed line. As illustrated in FIG. 2B, the driver circuit region DRV is positioned to overlap with the inside of the pixel array ALP in the top view. In addition, for example, the pixel array ALP is divided into a pixel region ARA[1,1] to a pixel region ARA[m,n] in FIG. 2B. Note that as an example, FIG. 2B selectively shows the reference numerals of the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[m-1,1], the pixel region ARA[m,1], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[m-1,n], and the pixel region ARA[m,n].

For example, in the case where the pixel array ALP is divided into 32 regions, m and n are 4 and 8, respectively, in FIG. 2A and FIG. 2B. In the case where the display device 10 has a resolution of 8K4K, the number of pixels is 7680×4320 . In the case where the colors of subpixels of the display device 10 are three colors, red (R), green (G), and blue (B), the total number of subpixels is $7680 \times 4320 \times 3$. Here, in the case where a pixel array of the display device 10 with a resolution of 8K4K is divided into 32 regions, the number of pixels per region is 960×1080 , and when the colors of the subpixels of the display device 10 are three colors, red (R), green (G), and blue (B), the number of subpixels per region is $960 \times 1080 \times 3$.

As illustrated in FIG. 2B, for example, the local driver circuit LD[1,1] drives pixels included in the pixel region ARA[1,1], and the local driver circuit LD[2,1] drives pixels included in the pixel region ARA[2,1]. The local driver circuit LD[m-1,1] drives pixels included in the pixel region ARA[m-1,1], and the local driver circuit LD[m,1] drives pixels included in the pixel region ARA[m,1]. The local driver circuit LD[1,n] drives pixels included in the pixel region ARA[1,n], and the local driver circuit LD[2,n] drives pixels included in the pixel region ARA[2,n]. The local driver circuit LD[m-1,n] drives pixels included in the pixel region ARA[m-1,n], and the local driver circuit LD[m,n] drives pixels included in the pixel region ARA[m,n]. That is, although not illustrated in FIG. 2B, a local driver circuit LD[i,j] positioned in the i-th row and j-th column (i is an integer greater than or equal to 1 and less than or equal to m and j is an integer greater than or equal to 1 and less than or equal to n) drives a pixel region ARA[i,j]. Note that, as an example, the relationship between the pixel region ARA and the local driver circuit LD that drives the pixels included in the pixel region ARA is indicated by a thick arrow in FIG. 2B.

As illustrated in FIG. 2B, in the case where the driver circuit region DRV is positioned to overlap with the inside of the pixel array ALP in the top view, wirings that electrically connect the pixels of the pixel region ARA and the local driver circuit LD (e.g., a source wiring, a gate wiring, and a constant voltage line) are provided as illustrated in FIG. 3A, for example. In other words, the display device of the present invention has a structure in which the wirings that electrically connect the local driver circuit LD and the pixel region ARA corresponding to the local driver circuit LD are extended in the wiring layer LINL.

Note that, as an example, a wiring group GLS[1,1] functions as a plurality of gate wirings that electrically connect the plurality of pixels included in the pixel region ARA[1,1] and a gate driver circuit included in the local

driver circuit LD[1,1] in FIG. 3A. As an example, a wiring group SLS[1,1] functions as a plurality of source wirings that electrically connect the plurality of pixels included in the pixel region ARA[1,1] and a source driver circuit included in the local driver circuit LD[1,1]. As an example, a wiring group GLS[2,1] functions as a plurality of gate wirings that electrically connect the plurality of pixels included in the pixel region ARA[2,1] and a gate driver circuit included in the local driver circuit LD[2,1]. As an example, a wiring group SLS[1,2] functions as a plurality of source wirings that electrically connect the plurality of pixels included in the pixel region ARA[1,2] and a source driver circuit included in the local driver circuit LD[1,2].

Consider a case where each of the pixel region ARA[1,1] to the pixel region ARA[m,n] includes a plurality of pixels arranged in a matrix of s rows and t columns (s is an integer of 1 or more and t is an integer of 1 or more) in FIG. 3A, as an example. In this case, for example, each of the wiring group GLS[1,1] and the wiring group GLS[2,1] has s gate wirings, and each of the wiring group SLS[1,1] and the wiring group SLS[1,2] has t source wirings. Note that in FIG. 3A, a wiring GL[1,1]₁, a wiring GL[1,1]₂, and a wiring GL[1,1]_s are selectively illustrated as wirings included in the wiring group GLS[1,1]; a wiring GL[2,1]₁, a wiring GL[2,1]₂, and a wiring GL[2,1]_s are selectively illustrated as wirings included in the wiring group GLS[2,1]; a wiring SL[1,1]₁, a wiring SL[1,1]₂, and a wiring SL[1,1]_t are selectively illustrated as wirings included in the wiring group SLS[1,1]; and a wiring SL[1,2]₁, a wiring SL[1,2]₂, and a wiring SL[1,2]_t are selectively illustrated as wirings included in the wiring group SLS[1,2].

Although not illustrated, wirings other than the gate wirings and the source wirings may be provided in the wiring layer LINL. For example, a wiring for supplying a constant voltage from the voltage generation circuit PG included in the circuit layer SICL to a pixel included in the pixel array ALP may be provided in the wiring layer LINL.

Note that the wiring layer LINL may include a plurality of layers. Specifically, for example, the wiring layer LINL may have a structure in which different wirings overlap with one another as illustrated in FIG. 3B.

As an example, FIG. 3B illustrates a cross-sectional view in which the circuit layer SICL, the wiring layer LINL, and the pixel layer PXAL are stacked. Note that only the pixel region ARA[1,1], the pixel region ARA[2,2], and the pixel region ARA[3,3] are illustrated in the pixel layer PXAL of FIG. 3B, and they are illustrated by a block diagram. Furthermore, in FIG. 3B, the local driver circuit LD[1,1] and the local driver circuit LD[2,2] are illustrated in the circuit layer SICL, and each of the local driver circuit LD[1,1] and the local driver circuit LD[2,2] includes a transistor 300. The transistor 300 in FIG. 3 includes a low-resistance region 314a and a semiconductor region 313, which are described in detail in Embodiment 4.

In FIG. 3B, one of a source and a drain of the transistor 300 of the local driver circuit LD[1,1] is electrically connected to the pixel region ARA[1,1] through the wiring GL[1,1]₁ (the wiring SL[1,1]₁). One of a source and a drain of the transistor 300 of the local driver circuit LD[2,2] is electrically connected to the pixel region ARA[2,2] through the wiring GL[2,2]₁ (the wiring SL[2,2]₁). Furthermore, in the structure illustrated in FIG. 3B, the pixel region ARA[3,3] and the wiring GL[3,3]₁ (the wiring SL[3,3]₁) are electrically connected to each other.

Note that in the structure illustrated in FIG. 3B, the wiring GL[1,1]₁ (the wiring SL[1,1]₁) includes a contact portion VA and a conductor ER. Furthermore, a conductor CD is

formed over the low-resistance region 314a of the transistor 300. Therefore, the pixel region ARA[1,1] and the local driver circuit LD[1,1] are electrically connected to each other with the contact portion VA, the conductor ER, and the conductor CD. Note that the conductor CD electrically connects the wiring GL[1,1]₁ (the wiring SL[1,1]₁) and the low-resistance region 314a of the transistor 300 and thus can be called a contact portion.

For example, at the time of designing, the wiring electrically connecting the local driver circuit LD and the pixel region ARA corresponding to the local driver circuit LD sometimes overlaps with another wiring in the same position depending on their positional relationship. In this case, the wirings overlap with each other in the same position are provided in different layers as illustrated in FIG. 3B, whereby the local driver circuit LD and the pixel region ARA corresponding to the local driver circuit LD can be electrically connected to each other without a physical contact between the different wirings. For example, a wiring electrically connecting the pixel region ARA[1,1] and the local driver circuit LD[1,1] and a wiring electrically connecting the pixel region ARA[2,2] and the local driver circuit LD[2,2] are routed so as not to be in a physical contact with each other in FIG. 3B. Furthermore, for example, a wiring electrically connecting the pixel region ARA[2,2] and the local driver circuit LD[2,2] and a wiring electrically connected to the pixel region ARA[3,3] are routed so as not to be in a physical contact with each other in FIG. 3B.

In order to reduce signal delay and/or inhibit an increase in power consumption due to parasitic resistance and the like, it is preferable that the wirings for transmitting signals from the local driver circuit LD to the pixels of the pixel region ARA (e.g., the gate wiring and the source wiring) be short. For this reason, it is preferable that the wirings electrically connecting the local driver circuit LD and the pixel region ARA in the display device 10 be designed to be short. For such a design, as an example, a contact of the wirings between each of the pixel regions ARA and the wiring layer LINL is designed to be in an optimal position.

Although the driver circuit region DRV of the circuit layer SICL and the pixel region ARA of the pixel layer PXAL are electrically connected to each other with the wirings of the wiring layer LINL in the structure illustrated in FIG. 3A and FIG. 3B, the display device of one embodiment of the present invention may have a structure in which the driver circuit region DRV of the circuit layer SICL and the pixel region ARA of the pixel layer PXAL are electrically connected to each other with wirings of the region LIA as well as the wirings of the wiring layer LINL.

For example, as illustrated in FIG. 4A, the wiring GL[1,1]₁ (the wiring SL[1,1]₁) that is included in the display device 10 and electrically connected between the local driver circuit LD[1,1] and the pixel region ARA[1,1] may be electrically connected to the pixel region ARA[1,1] from one of the source and the drain of the transistor 300 through the wirings of the wiring layer LINL, the wirings included in the region LIA (wirings indicated by thick dotted lines), and the wirings of wiring layer LINL in the order. FIG. 4B is a cross-sectional view of the display device in this case. In the structure of the display device 10 in FIG. 4B, as an example, the path of the wiring GL[1,1]₁ (the wiring SL[1,1]₁) electrically connected between the local driver circuit LD[1,1] of the driver circuit region DRV and the pixel region ARA[1,1] goes through a low-resistance region 314c provided over the substrate BS in the region LIA of the circuit layer SICL. Note that in the case where the substrate

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BS is a semiconductor substance formed of silicon, the low-resistance region 314c can be formed by being doped with an element imparting conductivity.

For example, as illustrated in FIG. 5A, the wiring GL[1, 1]_1 (the wiring SL[1,1]_1) that is included in the display device 10 and electrically connected between the local driver circuit LD[1,1] and the pixel region ARA[1,1] may be electrically connected to the pixel region ARA[1,1] from one of the source and the drain of the transistor 300 through the wirings included in the region LIA (wirings indicated by thick dotted lines) and the wirings of wiring layer LINL in the order. FIG. 5B is a cross-sectional view of the display device 10 in this case. The display device 10 of FIG. 5B includes, for example, the low-resistance region 314c in which one of the source and the drain of the transistor 300 extends to the inside of the region LIA. In other words, the low-resistance region 314c includes the one of the source and the drain of the transistor 300. In the display device 10, the local driver circuit LD[1,1] of the driver circuit region DRV and the pixel region ARA[1,1] are electrically connected to each other with the low-resistance region 314c, the wirings included in the wiring layer LINL, and the like.

<<Contact Between Pixel Region ARA and Wiring Layer LIN>>

Here, contact positions of the wirings between the pixel region ARA and the wiring layer LINL are described. FIG. 6A is a schematic view illustrating the pixel region ARA and a plurality of pixels PIX included in the pixel region ARA, as an example. Note that the plurality of pixels PIX are arranged in a matrix in the pixel region ARA, as an example. Furthermore, in the structure in FIG. 6A, as an example, each of the pixels PIX includes a transistor Tr, and the other circuit elements are not illustrated. In addition, as an example, the wiring group SLS (the wiring SL_1, the wiring SL_2, and the wiring SL_3) is provided to be extended in the X direction in the pixel region ARA in FIG. 6A. Note that although the three wirings included in the wiring group SLS are illustrated in FIG. 6A, the number of wirings may be one, two, or four or more. Furthermore, the wiring group GLS and the like are not illustrated. In this specification and the like, the X direction is sometimes rephrased as a row direction, and the Y direction is sometimes rephrased as a column direction.

In FIG. 6A, contact portions of the wirings between the pixel region ARA and the wiring layer LINL are positioned in an edge portion of the pixel region ARA, for example. Note that in FIG. 6A, the contact portions of the wirings between the pixel region ARA and the wiring layer LINL are referred to as contact portions CNT. In this case, the local driver circuit LD is preferably positioned in the positive X direction with respect to the pixel region ARA. In contrast, when the local driver circuit LD is positioned in the negative X direction with respect to the pixel region ARA, the length of the wiring between the pixel region ARA and the local driver circuit LD corresponding to the pixel region ARA is long, which can increase a tendency of occurrence of signal delay and/or power consumption due to parasitic resistance or the like in some cases, for example.

The contact positions of the wirings between the pixel region ARA and the wiring layer LINL may be inside the pixels PIX as illustrated in FIG. 6B, for example. In this case, the local driver circuit LD is preferably positioned in the positive or negative Y direction with respect to the pixel region ARA. Note that although the contact portions CNT in one wiring group SLS are provided in the inside of the pixels PIX of different columns in FIG. 6B, the contact portions

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CNT in one wiring group SLS may be provided in the inside of the pixels PIX of the same column.

Although the contact portions CNT provided in the inside of the pixels PIX are illustrated in FIG. 6B as an example, the contact portions CNT may be provided in the outside of the pixels PIX (between adjacent pixels PIX) as illustrated in FIG. 6C.

Note that in this specification and the like, as an example, the inside of the pixels PIX can be a region that overlaps with light-emitting regions of light-emitting devices (a light-emitting device 150a to a light-emitting device 150c described later) included in the pixels PIX, and the outside of the pixels PIX can be the outside of the region. Furthermore, for example, the inside of the pixels PIX can be a region that overlaps with EL layers (an EL layer 141a to an EL layer 141c described later) included in the pixels PIX, and the outside of the pixels PIX can be the outside of the region. Moreover, for example, the inside of the pixels PIX can be a region that overlaps with opening portions of an insulator reaching lower electrodes (opening portions of an insulator 112 that reach a conductor 121a to a conductor 121c described later) included in the pixels PIX, and the outside of the pixels PIX can be the outside of the region. In addition, for example, the inside of the pixels PIX can be a region that overlaps with lower electrodes (the conductor 121a to the conductor 121c described later) included in the pixels PIX, and the outside of the pixels PIX can be the outside of the region.

In this specification and the like, the boundary between the inside and outside of the pixel PIX is sometimes included in the inside of the pixel PIX. Depending on circumstances, the boundary between the inside and outside of the pixel PIX is sometimes included in the outside of the pixel PIX.

Depending on the positional relationship between the pixel region ARA and the local driver circuit LD, the positions of the plurality of contact portions CNT may be a combination of the cases illustrated in FIG. 6A to FIG. 6C. That is, one of the plurality of contact portions CNT may be positioned in the edge portion of the pixel region ARA, another thereof may be positioned in the inside of the pixel PIX, and the other thereof may be positioned in the outside of the pixel PIX.

<<Positional Relationship Between Pixel Region ARA and Local Driver Circuit LD>>

Here, the routing of the wirings that electrically connect the pixel region ARA and the local driver circuit LD in accordance with the positional relationship between the pixel region ARA and the local driver circuit LD is described.

FIG. 7 is a top view illustrating the pixel region ARA included in the pixel layer PXAL and the local driver circuit LD that is included in the circuit layer SICL and corresponds to the pixel region ARA in the display device 10.

In the pixel region ARA in FIG. 7, the pixel circuits PIX are arranged in a matrix of v rows and w columns (v is an integer of 1 or more and w is an integer of 1 or more), and a wiring SLu_1 to a wiring SLu_w are extended as source wirings in the column direction in the pixel region ARA. In a lower layer including the wiring layer LINL of FIG. 7, a wiring SLd_1 to a wiring SLd_w for electrical connection with a driver circuit SD are extended along one direction. That is, the wiring SLd_1 to the wiring SLd_w are extended substantially parallel or parallel to one another. Note that the wiring SLd_1 to the wiring SLd_w are each indicated by a dashed double-dotted line in FIG. 7.

In FIG. 7, the driver circuit SD includes a plurality of output terminals SDT. Here, the plurality of output terminals

SDT are electrically connected to the wiring SLd_1 to the wiring SLd_w. Furthermore, in FIG. 7, the arrangement direction of the plurality of output terminals SDT (the orientation of the driver circuit SD) is substantially perpendicular or perpendicular to each of the wiring SLd_1 to the wiring SLd_w. That is, the plurality of output terminals SDT are arranged along one direction, for example.

The plurality of output terminals SDT are described here. For example, in the case where the driver circuit SD includes a demultiplexer, and output terminals of the demultiplexer are directly and electrically connected to the wiring SLd_1 to the wiring SLd_w, the output terminals SDT can serve as terminals of switches included in the demultiplexer. In the case where the driver circuit SD includes a plurality of amplifiers and output terminals of the plurality of amplifiers are directly and electrically connected to the wiring SLd_1 to the wiring SLd_w, the plurality of output terminals SDT can serve as output terminals of the plurality of amplifiers. In the case where the output terminals of the amplifiers are electrically connected to sources, drains, or gates of transistors included in the amplifier, the plurality of output terminals SDT can serve as the sources, drains, or gates of the transistors. In the case where the driver circuit SD includes a plurality of digital-analog converter circuits (hereinafter referred to as DACs) and output terminals of the plurality of DACs are directly and electrically connected to the wiring SLd_1 to the wiring SLd_w, the plurality of output terminals SDT can serve as the output terminals of the plurality of DACs. In the case where the output terminals of the DACs are electrically connected to sources, drains, or gates of transistors included in the DACs, the plurality of output terminals SDT can serve as the sources, drains, or gates of the transistors.

The wiring SLd_1 to the wiring SLd_w are electrically connected to the wiring SLu_1 to the wiring SLu_w, respectively, through the contact portions CNT. Note that in some cases, the wiring SLd_1 to the wiring SLu_1 are collectively referred to as the wiring SL_1, the SLd_2 and the wiring SLu_2 are collectively referred to as the wiring SL_2, and the wiring SLd_w and the wiring SLu_w are collectively referred to as the wiring SL_w.

Note that in FIG. 7, the pixel region ARA and the driver circuit SD included in the local driver circuit LD are positioned so that the wiring SLu_1 to the wiring SLu_w are substantially perpendicular, preferably perpendicular, to the wiring SLd_1 to the wiring SLd_w. In other words, in the display device 10 in the top view of FIG. 7, the driver circuit SD is positioned substantially parallel or parallel to the column direction of the pixel region ARA of the display device 10. Therefore, the arrangement direction of the plurality of output terminals SDT (the orientation of the driver circuit SD) is substantially parallel or parallel to the columns of the pixel region ARA.

When the driver circuit SD included in the local driver circuit LD corresponding to the pixel region ARA is positioned in the column direction of the pixel region ARA in the top view, the lengths of the source wirings that electrically connect the pixel region ARA and the driver circuit SD can be as short as possible by routing the wiring SLd_1 to the wiring SLd_w as illustrated in FIG. 7. Shortening the source wirings can reduce parasitic resistance and parasitic capacitance involved in the source wirings, whereby time constant can be small. Thus, a time for transmitting signals including image data from the driver circuit SD to the pixel circuits PIX included in the pixel region ARA can be reduced.

Therefore, the display device including the pixel region ARA and the local driver circuit LD can display an image with a high frame frequency.

Note that the positional relationship between the pixel region ARA and the local driver circuit LD in the display device of one embodiment of the present invention is not limited to that in FIG. 7. For example, as illustrated in FIG. 8A, in the case where the local driver circuit LD corresponding to the pixel region ARA is not positioned along the column direction of the pixel region ARA in the top view, specifically, in the case where the local driver circuit LD is not electrically connected to the wiring SLd_1 to the wiring SLd_w when the wiring SLd_1 to the wiring SLd_w are routed in the direction substantially perpendicular, preferably perpendicular, to the wiring SLu_1 to the wiring SLu_w, the orientation of the driver circuit SD included in the local driver circuit LD (the arrangement direction of the plurality of output terminals SDT) may be changed.

The orientation of the driver circuit SD (the arrangement direction of the plurality of output terminals SDT) is changed in the inside of the local driver circuit LD, and the wiring SLd_1 to the wiring SLd_w are routed in the direction substantially perpendicular, preferably perpendicular, to the arrangement direction of the plurality of output terminals SDT, whereby the wiring SLu_1 to the wiring SLu_w of the pixel region ARA can be electrically connected to the wiring SLd_1 to the wiring SLd_w through the contact portions CNT. In other words, in the case where the positional relationship between the pixel region ARA and the local driver circuit LD is as illustrated FIG. 8A, the wiring SLu_1 to the wiring SLu_w need not be necessarily substantially perpendicular or perpendicular to the wiring SLd_1 to the wiring SLd_w in the top view.

When the driver circuit SD included in the local driver circuit LD corresponding to the pixel region ARA is not positioned along the column direction of the pixel region ARA in the top view, the orientation of the driver circuit SD (the arrangement direction of the plurality of output terminals SDT) is changed to be optimal and the wiring SLd_1 to the wiring SLd_w are routed to be in the direction substantially perpendicular, preferably perpendicular, to the arrangement direction of the plurality of output terminals SDT as illustrated in FIG. 8A, whereby the lengths of the source wirings electrically connecting the pixel region ARA and the driver circuit SD can be as short as possible.

The positions of the contact portions CNT, the arrangement order of the wiring SLd_1 to the wiring SLd_w, and the like may be changed depending on circumstances. For example, the structure illustrated in FIG. 8A may be changed to the structure illustrated in FIG. 8B. Specifically, the positions of the contact portions CNT for which the wirings SLu_1 to SLu_w are extended are changed and the arrangement order of the wiring SLd_1 to the wiring SLd_w is changed in the structure illustrated in FIG. 8B, as compared with the structure illustrated in FIG. 8A. Changing the structure illustrated in FIG. 8A to that illustrated in FIG. 8B makes it possible to minimize the difference between the longest wiring and the shortest wiring of the wiring SLd_1 to the wiring SLd_w. In other words, each length of the wiring SLd_1 to the wiring SLd_w can be made closer to the average length of the wiring SLd_1 to the wiring SLd_w. Accordingly, the differences in length among the wiring SLd_1 to the wiring SLd_w can be small, so that variations of times for transmitting image data to the wiring SL_1 to the wiring SL_w can be reduced.

Cross-sectional views of the structures illustrated in FIG. 7 to FIG. 8B can be, for example, FIG. 3B, FIG. 4B, or FIG.

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5B. In the case of FIG. 3B, the contact portion VA corresponds to, for example, the contact portion CNT in FIG. 7 to FIG. 8B, and the conductor ER corresponds to, for example, any one of the wiring SLd_1 to the wiring SLd_w in FIG. 7 to FIG. 8B. The conductor CD, the low-resistance region 314a of the transistor 300, and a region where the conductor CD is in contact with the low-resistance region 314a of the transistor 300 correspond to, for example, one of the plurality of output terminals SDT in FIG. 7 to FIG. 8B. In the case of FIG. 5B, the contact portion VA and the conductor ER correspond to, for example, the contact portion CNT in FIG. 7 to FIG. 8B. Furthermore, the conductor CD included in the circuit layer SICL can be part of the contact portion CNT in FIG. 7 to FIG. 8B. The low-resistance region 314c of the transistor 300 corresponds to, for example, any one of the wiring SLd_1 to the wiring SLd_w in FIG. 7 to FIG. 8B. The boundary between the low-resistance region 314a and the semiconductor region 313, which is described later, of the transistor 300 corresponds to one of the plurality of output terminals SDT in FIG. 7 to FIG. 8B, for example.

Although the driver circuit SD, i.e., the source driver circuit, is taken as an example in the above description with reference to FIG. 7 to FIG. 8B, for example, when the gate driver circuit is regarded in the same manner in place of the source driver circuit, it is possible to reduce a time difference among selection signals transmitted from the gate driver circuit to the wirings of the pixel region ARA.

With the display device 10 configured as illustrated in FIG. 1, FIG. 2A, and FIG. 2B, the pixel array ALP of the display device 10 is divided into the pixel region ARA[1,1] to the pixel region ARA[m,n] so that the pixel regions ARA can be driven in parallel with one another by the local driver circuit LD[1,1] to the local driver circuit LD[m,n]. In order to rewritten part of an image on the display portion of the display device 10, only necessary local driver circuits LD are driven so as to drive the pixels that are included in the pixel regions ARA displaying the part of the image. That is, the pixels included in the pixel region ARA[1,1] to the pixel region ARA[m,n] of the display portion of the display device 10 can be driven independently of one another. In this case, only the pixels included in the necessary pixel regions ARA are driven and the local driver circuits corresponding to the pixel regions ARA which need not be driven are in a resting state, so that the power consumption can be reduced. Since the pixel array ALP of the display device 10 can be divided into the pixel region ARA[1,1] to the pixel region ARA[m,n] and the pixel regions ARA can be driven independently of and in parallel with one another by the local driver circuit LD[1,1] to the local driver circuit LD[m,n], a time needed for rewriting the image displayed on the display portion of the display device 10 (e.g., a time per frame) can be reduced. Since the pixel array ALP is divided and the divided pixel regions are driven by their respective local driver circuits LD, a driving load becomes small in a divided unit (one pixel region ARA), which facilitates an increase in the operation speed and a reduction in the power. In addition, with the pixel array ALP divided, the divided units (the pixel regions ARA) can be driven at the same timing, whereby a write time of an image in one frame can be long as compared with the case where the pixel array ALP is not divided. For example, when the pixel array ALP is divided so that the plurality of gate wirings extended in the pixel array ALP are divided into four, a write time of an image can be desirably 1/4 times or near 1/4 times that of the case where the pixel array

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ALP is not divided) can be assigned to a write time. Therefore, the write time of an image can be long in one frame.

With the local driver circuits LD positioned in the center part of the circuit layer SICL or the vicinity thereof as illustrated in FIG. 1, FIG. 2A, and FIG. 2B, when an image is displayed on the display portion of the display device 10, a difference in time needed for inputting the data signal of the image to the pixels among different pixel regions ARA can be small. As for not only the data signal but also a pixel selection signal or the like supplied to the pixel region ARA, a difference in time needed for inputting the signal among different pixel regions ARA can be small in the same manner. In other words, delay of a signal transmitted from the driver circuit region to each pixel region ARA can be reduced.

Note that the structure of the display device of one embodiment of the present invention is not limited to that of the display device 10 described above. The display device of one embodiment of the present invention may have a structure changed from that of the above-described display device 10 according to circumstances.

For example, the display device 10 described above has one driver circuit region DRV included in the circuit layer SICL; however, two or more driver circuit regions DRV may be included in the circuit layer SICL. The display device 10 of FIG. 9A shows a schematic top view illustrating a structure example in which two or more driver circuit regions DRV are included in the circuit layer SICL of the display device 10.

The display device 10 of FIG. 9A has a structure in which the driver circuit region DRV is provided in part of a region overlapping with the pixel regions ARA stored in the area of two rows and n columns of the pixel array ALP. Specifically, in the display device 10 of FIG. 9A, a driver circuit region DRV[1] is provided in part of a region overlapping with the area from the pixel region ARA[1,1] to the pixel region ARA[2,n], and a driver circuit region DRV[n/2] is provided in part of a region overlapping with the area from the pixel region ARA[m-1,1] to the pixel region ARA[m,n]. That is, n/2 driver circuit regions DRV (the driver circuit region DRV[1] to the driver circuit region [n/2]) are provided in the circuit layer SICL. Note that n, the number of columns shown in FIG. 9A, is an even number.

Note that the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[m-1,1], the pixel region ARA[m,1], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[m-1,n], and the pixel region ARA[m,n] are selectively illustrated in FIG. 9A as the pixel regions ARA. Furthermore, the driver circuit region DRV[1] and the driver circuit region DRV[n/2] are selectively illustrated in FIG. 9A as the driver circuit regions DRV. Moreover, the local driver circuit LD[1,1], the local driver circuit LD[2,1], the local driver circuit LD[1,2], the local driver circuit LD[2,2], the local driver circuit LD[m-1,1], the local driver circuit LD[m,1], the local driver circuit LD[m-1,2], the local driver circuit LD[m,2], the local driver circuit LD[1,n-1], the local driver circuit LD[2,n-1], the local driver circuit LD[1,n], the local driver circuit LD[2,n], the local driver circuit LD[m-1,n-1], the local driver circuit LD[m,n-1], the local driver circuit LD[m-1,n], and the local driver circuit LD[m,n] are selectively illustrated in FIG. 9A as the local driver circuits LD.

Although the controller CON and the voltage generation circuit PG are illustrated in the center part of each of the driver circuit region DRV[1] to the driver circuit region DRV[n/2] in the display device 10 of FIG. 9A, the positions

and shapes of the controllers CON and the voltage generation circuits PG provided in the driver circuit region DRV[1] to the driver circuit region DRV[n/2] are not particularly limited.

Note that although the pixels included in each of the plurality of pixel regions ARA stored in the area of two rows and n columns in the pixel array ALP are driven by the local driver circuits LD included in one driver circuit region DRV in the display device 10 of FIG. 9A, the area of the pixel regions in the pixel array ALP which corresponds to one driver circuit region DRV may be an area of one to m rows and n columns or an area of m rows and one to n columns. Specifically, the area of the pixel regions in the pixel array ALP which corresponds to one driver circuit region DRV may be an area of three rows and n columns or the area of m rows and two columns, for example.

Although n representing the number of columns shown in the display device 10 of FIG. 9A is an even number, n may be an odd number. In this case, for example, the display device 10 of FIG. 9A may have a structure in which the whole pixel array ALP is divided into a plurality of areas of two rows and n columns and an area of one row and n columns, and a plurality of driver circuit regions DRV are provided so that one driver circuit regions DRV corresponds to one of the areas.

Although the pixels included in each of the plurality of pixel regions ARA stored in the area of two rows and n columns in the pixel array ALP are driven by the local driver circuits LD included in one driver circuit region DRV in the display device 10 of FIG. 9A, the area of the pixel regions in the pixel array ALP which corresponds to one driver circuit region DRV may be an area of one to m rows and one to n columns. Specifically, for example, one area may be set to an area of two rows and is columns so that the pixel array ALP is divided into p×q areas, and the pixels of the plurality of pixel regions ARA included in one of the areas may be driven by one driver circuit region DRV, as illustrated in FIG. 9B. Note that here, p, q, and is are each an integer of 1 or more and satisfy $n=i_a \times p$ and $m=2 \times q$. In addition, in this case, p×q driver circuit regions DRV (the driver circuit region DRV[1] to the driver circuit region [p×q]) are provided in the circuit layer SICL.

Note that the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[m-1,1], the pixel region ARA[m,1], the pixel region ARA[1, i_a], the pixel region ARA[2, i_a], the pixel region ARA[m-1, i_a], the pixel region ARA[m, i_a], the pixel region ARA[1, $n-i_a+1$], the pixel region ARA[2, $n-i_a+1$], the pixel region ARA[m-1, $n-i_a+1$], the pixel region ARA[m, $n-i_a+1$], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[m-1,n], and the pixel region ARA[m,n] are selectively illustrated in FIG. 9B as the pixel regions ARA. Furthermore, the driver circuit region DRV[1], the driver circuit region DRV[p], the driver circuit region DRV[p×q-p+1], and the driver circuit region DRV[p×q] are selectively illustrated in FIG. 9B as the driver circuit regions DRV. Moreover, the local driver circuit LD[1,1], the local driver circuit LD[2,1], the local driver circuit LD[m-1,1], the local driver circuit LD[m,1], the local driver circuit LD[1, i_a], the local driver circuit LD[2, i_a], the local driver circuit LD[m-1, i_a], the local driver circuit LD[m, i_a], the local driver circuit LD[1, $n-i_a+1$], the local driver circuit LD[2, $n-i_a+1$], the local driver circuit LD[m-1, $n-i_a+1$], the local driver circuit LD[m, $n-i_a+1$], the local driver circuit LD[1,n], the local driver circuit LD[2,n], the local driver circuit LD[m-1,n], and the local driver circuit LD[m,n] are selectively illustrated in FIG. 9B as the local driver circuits LD. Although the controller CON and the

voltage generation circuit PG are not illustrated in each of the driver circuit region DRV[1] to the driver circuit region DRV[p×q] in the display device 10 of FIG. 9B, each of the driver circuit region DRV[1] to the driver circuit region DRV[p×q] may include the controller CON and the voltage generation circuit PG. The positions and shapes of the controllers CON and the voltage generation circuits PG provided in the driver circuit region DRV[1] to the driver circuit region DRV[p×q] are not particularly limited.

Although the driver circuit regions DRV do not overlap with the edge portion of a region including all of the pixel region ARA[1,1] to the pixel region ARA[m,n] of the pixel array ALP in the display device 10 of FIG. 2B, FIG. 9A, FIG. 9B, and the like, part of the driver circuit regions DRV may overlap with part of the edge portion of the pixel array ALP. Specifically, for example, the display device 10 may be configured so that one driver circuit region DRV crosses the pixel array ALP in the column direction of the pixel array ALP in the top view as illustrated in FIG. 10A. Thus, part of the edge portion of the region including all of the pixel region ARA[1,1] to the pixel region ARA[m,n] of the pixel array ALP overlaps with the driver circuit region DRV (including the case where part of the edge portion of the pixel array ALP overlaps with part of the edge portion of the driver circuit region DRV as illustrated in FIG. 10A).

In this case, the local driver circuit LD[1,1] to the local driver circuit LD[m,n] are provided in a matrix of m rows and n columns in the driver circuit region DRV like in the driver circuit region DRV illustrated in FIG. 2B.

Note that the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[m-1,1], the pixel region ARA[m,1], the pixel region ARA[1,2], the pixel region ARA[2,2], the pixel region ARA[m-1,2], the pixel region ARA[m,2], the pixel region ARA[1,n-1], the pixel region ARA[2, n-1], the pixel region ARA[m-1, n-1], the pixel region ARA [m,n-1], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[m-1,n], and the pixel region ARA[m,n] are selectively illustrated in FIG. 10A as the pixel regions ARA. Furthermore, the local driver circuit LD[1,1], the local driver circuit LD[2,1], the local driver circuit LD[m-1,1], the local driver circuit LD[m,1], the local driver circuit LD[1,2], the local driver circuit LD[2,2], the local driver circuit LD[m-1,2], the local driver circuit LD[m,2], the local driver circuit LD[1,n-1], the local driver circuit LD[2,n-1], the local driver circuit LD[m-1, n-1], the local driver circuit LD[m,n-1], the local driver circuit LD[1,n], the local driver circuit LD[2,n], the local driver circuit LD[m-1,n], and the local driver circuit LD[m, n] are selectively illustrated in FIG. 10A as the local driver circuits LD.

Although the controller CON and the voltage generation circuit PG are illustrated in the center part of the driver circuit region DRV in the display device 10 of FIG. 10A, the positions and shapes of the controller CON and the voltage generation circuit PG provided in the driver circuit region DRV are not particularly limited.

Although one driver circuit region DRV crosses the pixel array ALP in the display device of FIG. 10A, the display device 10 may be configured so that a plurality of driver circuit regions DRV cross the pixel array ALP. Specifically, as illustrated in FIG. 10B, one area is set to an area of m rows and i_b columns so that the pixel array ALP is divided into r areas, and the pixels of the plurality of pixel regions ARA included in one of the areas may be driven by one driver circuit region DRV, for example. Note that here, r and i_b are each an integer of 1 or more and satisfy $n=i_b \times r$. In addition, in this case, r driver circuit regions DRV (the driver

circuit region DRV[1] to the driver circuit region [r]) are provided in the circuit layer SICL.

Note that the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[m-1,1], the pixel region ARA[m,1], the pixel region ARA[1,i_b], the pixel region ARA[2,i_b], the pixel region ARA[m-1,i_b], the pixel region ARA[m,i_b], the pixel region ARA[1,n-i_b+1], the pixel region ARA[2,n-i_b+1], the pixel region ARA[m-1,n-i_b+1], the pixel region ARA[m,n-i_b+1], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[m-1,n], and the pixel region ARA[m,n] are selectively illustrated in FIG. 10B as the pixel regions ARA. Furthermore, the driver circuit region DRV[1] and the driver circuit region DRV[r] are selectively illustrated in FIG. 10B as the driver circuit regions DRV. Moreover, the local driver circuit LD[1,1], the local driver circuit LD[2,1], the local driver circuit LD[m-1,1], the local driver circuit LD[m,1], a local driver circuit LD[1,i_b], the local driver circuit LD[2,i_b], the local driver circuit LD[m-1,i_b], the local driver circuit LD[m,i_b], the local driver circuit LD[1,n-i_b+1], the local driver circuit LD[2,n-i_b+1], the local driver circuit LD[m-1,n-i_b+1], the local driver circuit LD[m,n-i_b+1], the local driver circuit LD[1,n], the local driver circuit LD[2,n], the local driver circuit LD[m-1,n], and the local driver circuit LD[m,n] are selectively illustrated in FIG. 10B as the local driver circuits LD.

Although the controller CON and the voltage generation circuit PG are not illustrated in each of the driver circuit region DRV[1] to the driver circuit region DRV[r] in the display device of FIG. 10B, each of the driver circuit region DRV[1] to the driver circuit region DRV[r] may include the controller CON and the voltage generation circuit PG. The positions and shapes of the controllers CON and the voltage generation circuits PG provided in the driver circuit region DRV[1] to the driver circuit region DRV[r] are not particularly limited.

Although the pixels included in each of the plurality of pixel regions ARA stored in the area of m rows and i_b columns in the pixel array ALP are driven by the local driver circuits LD included in one driver circuit region DRV in the display device 10 of FIG. 10B, the areas of the pixel regions in the pixel array ALP which correspond to the plurality of driver circuit regions DRV may be different from one other. For example, in the case where n, the number of the columns of the pixel array, is an odd number, the display device 10 may have a structure in which the whole pixel array ALP is divided into a plurality of areas of m rows and two columns and an area of m rows and one column, and a plurality of driver circuit regions DRV are provided so that one the driver circuit region DRV corresponds to one of the areas.

Although both edge portions of the pixel array ALP that face each other overlap with part of the edge portions of the driver circuit region DRV in the display device 10 of FIG. 10A, one of the edge portions of the pixel array ALP that face each other may overlap with part of the edge portion of the driver circuit region DRV. Specifically, for example, one of the edge portions of the pixel array ALP that face each other may overlap with part of the edge portion of the driver circuit region DRV as illustrated in FIG. 11.

For the display device of one embodiment of the present invention, the above-described structure examples of the display device 10 may be combined as appropriate. As an example, the structure example of the display device 10 of FIG. 2B and the structure example of the display device 10 of FIG. 11 may be combined as shown in the display device 10 of FIG. 12. In the display device 10 of FIG. 12, the circuit layer SICL includes a driver circuit region DRVa that

overlaps with one of the edge portions of the pixel array ALP that face each other and a driver circuit region DRVb that overlaps with part of the inside region of the pixel array ALP in the top view. Specifically, in the display device 10 of FIG. 12, the driver circuit region DRVa drives pixels of a region of the pixel array ALP that includes the pixel region ARA[1,1] to the pixel region ARA[i_c,n], and the driver circuit region DRVb drives pixels of a region of the pixel array ALP that includes the pixel region ARA[i_c+1,1] to the pixel region ARA[m,n]. Note that i_c can be an integer more than or equal to 1 and less than or equal to m-1.

Note that the pixel region ARA[1,1], the pixel region ARA[2,1], the pixel region ARA[i_c,1], the pixel region ARA[i_c+1,1], the pixel region ARA[m,1], the pixel region ARA[1,2], the pixel region ARA[2,2], the pixel region ARA[i_c,2], the pixel region ARA[1,n-1], the pixel region ARA[2,n-1], the pixel region ARA[i_c,n-1], the pixel region ARA[1,n], the pixel region ARA[2,n], the pixel region ARA[i_c,n], the pixel region ARA[i_c+1,n], and the pixel region ARA[m,n] are selectively illustrated in FIG. 12 as the pixel regions ARA. In addition, the local driver circuit LD[1,1], the local driver circuit LD[2,1], the local driver circuit LD[i_c,1], the local driver circuit LD[1,2], the local driver circuit LD[2,2], the local driver circuit LD[i_c,2], the local driver circuit LD[1,n-1], the local driver circuit LD[2,n-1], the local driver circuit LD[i_c,n-1], the local driver circuit LD[1,n], the local driver circuit LD[2,n], the local driver circuit LD[i_c,n], the local driver circuit LD[i_c+1,1], the local driver circuit LD[i_c+1,2], the local driver circuit LD[m,1], the local driver circuit LD[m,2], the local driver circuit LD[i_c+1,n-1], the local driver circuit LD[i_c+1,n], the local driver circuit LD[m-1,n], and the local driver circuit LD[m,n] are selectively illustrated in FIG. 10B as the local driver circuits LD.

Each of the driver circuit region DRVa and the driver circuit region DRVb may include the controller CON and the voltage generation circuit PG in the display device 10 of FIG. 12 like in the display device 10 of FIG. 2B. The positions and shapes of the controllers CON and the voltage generation circuits PG provided in the driver circuit region DRVa and the driver circuit region DRVb are not particularly limited.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 2

In this embodiment, a structure example of a display device of one embodiment of the present invention and an example of an operation method of the display device will be described.

Structure Example 1 of Display Device

FIG. 13 illustrates a structure example of the display device of one embodiment of the present invention. A display device 11 illustrated in FIG. 13 includes, for example, a driver circuit GD, a driver circuit SDa[1] to a driver circuit SDa[h] (here h is an integer of 1 or more) and a driver circuit SDb[1] to a driver circuit SDb[h] in addition to the pixel array ALP.

For example, the driver circuit GD has a function of a gate driver circuit that selects a pixel circuit included in the pixel array ALP, which serves as a destination to which image data is to be written.

The driver circuit SDa[1] to the driver circuit SDa[h] and the driver circuit SDb[1] to the driver circuit SDb[h] each have a function of a source driver circuit that transmits image data to the pixel circuits included in the pixel array ALP, for example. Note that the driver circuit SDa[1] to the driver circuit SDa[h] and the driver circuit SDb[1] to the driver circuit SDb[h] may each further include a demultiplexer (DeMUX) for selecting a source wiring to which image data is transmitted.

The pixel array ALP includes, for example, a plurality of pixel circuits, and the plurality of pixel circuits are arranged in a matrix of V_1+V_2 rows and W columns (V_1 is an integer of 1 or more, V_2 is an integer of 1 or more, and W is a multiple of h and more than or equal to 1). Specifically, a plurality of pixel circuits PIXa are provided in the first row to the V_1 -th row of the pixel array ALP, and a plurality of pixel circuits PIXb are provided in the V_1+1 -th row to the V_1+V_2 -th row of the pixel array ALP. Note that in FIG. 13, the plurality of pixel circuits positioned in the first row to the V_1 -th row of the pixel array ALP are referred to as a pixel circuit PIXa[1,1] to a pixel circuit PIXa[V_1 ,W] and the plurality of pixel circuits positioned in the V_1+1 -th row to the V_1+V_2 -th row of the pixel array ALP are referred to as a pixel circuit PIXb[1,1] to a pixel circuit PIXb[V_2 ,W] for convenience. The plurality of pixel circuits PIXa and the plurality of pixel circuits PIXb can have the same structures.

Note that the pixel circuit PIXa[1,1], the pixel circuit PIXa[1,W/h], the pixel circuit PIXa[V_1 ,1], the pixel circuit PIXa[V_1 ,W/h], the pixel circuit PIXa[V_1 ,W-W/h+1], the pixel circuit PIXa[1,W], the pixel circuit PIXa[V_1 ,W-W/h+1], the pixel circuit PIXa[V_1 ,W], the pixel circuit PIXb[1,1], the pixel circuit PIXb[1,W/h], the pixel circuit PIXb[V_2 ,1], the pixel circuit PIXb[V_2 ,WM], the pixel circuit PIXb[1,W-W/h+1], the pixel circuit PIXb[1,W], the pixel circuit PIXb[V_2 ,W-W/h+1], and the pixel circuit PIXb[V_2 ,W] are selectively illustrated in the pixel array ALP of FIG. 13.

Consider a structure in which the pixel array ALP is divided and an image signal is written to each of the divided regions. FIG. 13 illustrates a structure example of the display device in which the pixel array ALP is divided into $2 \times h$ regions, i.e., a region ALPa_C[1] to a region ALPa_C[h] and a region ALPb_C[1] to a region ALPb_C[h]. Note that the region ALPa_C[1] to the region ALPa_C[h] correspond to a structure in which a matrix where the pixel circuit PIXa[1,1] to the pixel circuit PIXa[V_1 ,W] are arranged is divided every W/h columns, and the region ALPb_C[1] to the region ALPb_C[h] correspond to a structure in which a matrix where the pixel circuit PIXb[1,1] to the pixel circuit PIXb[V_2 ,W] are arranged is divided every W/h columns. Thus, the region ALPa_C[1] to the region ALPa_C[h] each have a structure in which the plurality of pixel circuits PIXa are arranged a matrix of V_1 rows and W/h columns, and the region ALPb_C[1] to the region ALPb_C[h] each have a structure in which the plurality of pixel circuits PIXb are arranged in a matrix of V_2 rows and W/h columns.

A wiring SLa[1] to a wiring SLa[W/h] are extended in their respective columns in the region ALPa_C[1], and each of the wiring SLa[1] to the wiring SLa[W/h] is electrically connected to the plurality of pixel circuits PIXa provided in the corresponding column in the region ALPa_C[1]. Similarly, a wiring SLb[1] to a wiring SLb[W/h] are extended in their respective columns in the region ALPb_C[1], and each of the wiring SLb[1] to the wiring SLb[W/h] is electrically connected to the plurality of pixel circuits PIXb provided in the corresponding column in the region ALPb_C[1]. Therefore, for example, both the wiring SLa[1] and the wiring SLb[1] are extended in the same column in the pixel array

ALP; for another example, both the wiring SLa[W/h] and the wiring SLb[W/h] are extended in the same column in the pixel array ALP.

Note that the wiring SLa[1] to the wiring SLa[W/h] are each electrically connected to the driver circuit SDa[1], and the wiring SLb[1] to the wiring SLb[W/h] are each electrically connected to the driver circuit SDb[1].

A wiring SLa[W-W/h+1] to a wiring SLa[W] are extended in their respective columns in the region ALPa_C[h], and each of the wiring SLa[W-W/h+1] to the wiring SLa[W] is electrically connected to the plurality of pixel circuits PIXa provided in the corresponding column in the region ALPa_C[h]. Similarly, a wiring SLb[W-W/h+1] to a wiring SLb[W] are extended in their respective columns in the region ALPb_C[h], and each of the wiring SLb[W-W/h+1] to the wiring SLb[W] is electrically connected to the plurality of pixel circuits PIXb provided in the corresponding column in the region ALPb_C[h]. Therefore, for example, both the wiring SLa[W-W/h+1] and the wiring SLb[W-W/h+1] are extended in the same column in the pixel array ALP; for another example, both the wiring SLa[W] and the wiring SLb[W] are extended in the same column in the pixel array ALP.

Note that the wiring SLa[W-W/h+1] to the wiring SLa[W] are each electrically connected to the driver circuit SDa[h], and the wiring SLb[W-W/h+1] to the wiring SLb[W] are each electrically connected to the driver circuit SDb[h].

Note that the same applies to each of the region ALPa_C[2] to the region ALPa_C[h-1] and the region ALPb_C[2] to the region ALPb_C[h-1] that are not illustrated in FIG. 13, in which the wirings are extended in their respective columns. One of the wirings in the region ALPa_C[2] to the region ALPa_C[h-1] is electrically connected to any one of the driver circuit SDa[2] to the driver circuit SDa[h-1] depending on the divided region; similarly, one of the wirings in the region ALPb_C[2] to the region ALPb_C[h-1] is electrically connected to any one of the driver circuit SDb[2] to the driver circuit SDb[h-1] depending on the divided region.

Driving Method Example of Display Device

Here, an example of a method for driving the display device 11 illustrated in FIG. 13 is described.

A display device 11A illustrated in FIG. 14 is shown as a block diagram illustrating part of the pixel array ALP of the display device 11 in FIG. 13, in which the region ALPa_C[1], the region ALPb_C[1], the driver circuit SDa[1], and the driver circuit SDb[1] are selectively illustrated. Note that, for example, W/h is 4 in the display device 11A illustrated in FIG. 14. In other words, the pixel circuits PIXa are arranged in a matrix of V_1 rows and four columns in the region ALPa_C[1], and the pixel circuits PIXb are arranged in a matrix of V_2 rows and four columns in the region ALPb_C[1] in the display device 11A illustrated in FIG. 14.

Each of the driver circuit SDa[1] and the driver circuit SDb[1] provided in the display device 11A of FIG. 14 has a function of selecting any one of four source wirings (corresponding to the wiring SLa[1] to the wiring SLa[W/h] or the wiring SLb[1] to the wiring SLb[W/h] in the display device 11 of FIG. 13) and transmitting an image signal to the source wiring.

FIG. 14 illustrates a specific structure example of the driver circuit SDa[1] and the driver circuit SDb[1]. The driver circuit SDa[1] includes, for example, a demultiplexer

DMUa and a circuit OTH, and the driver circuit SDb[1] includes, for example, a demultiplexer DMUb and a circuit OTH.

The demultiplexer DMUa includes a switch SWa[1] to a switch SWa[4], for example. The demultiplexer DMUb includes a switch SWb[1] to a switch SWb[4], for example.

A first terminal of the switch SWa[1] is electrically connected to the wiring SLa[1], a second terminal of the switch SWa[1] is electrically connected to an output terminal of the circuit OTH, and a control terminal of the switch SWa[1] is electrically connected to a wiring SGa[1]. A first terminal of the switch SWa[2] is electrically connected to the wiring SLa[2], a second terminal of the switch SWa[2] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWa[2] is electrically connected to a wiring SGa[2]. A first terminal of the switch SWa[3] is electrically connected to the wiring SLa[3], a second terminal of the switch SWa[3] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWa[3] is electrically connected to a wiring SGa[3]. A first terminal of the switch SWa[4] is electrically connected to the wiring SLa[4], a second terminal of the switch SWa[4] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWa[4] is electrically connected to a wiring SGa[4].

An input terminal of the circuit OTH included in the driver circuit SDa is electrically connected to a wiring DLa.

A first terminal of the switch SWb[1] is electrically connected to the wiring SLb[1], a second terminal of the switch SWb[1] is electrically connected to an output terminal of the circuit OTH, and a control terminal of the switch SWb[1] is electrically connected to a wiring SGb[1].

A first terminal of the switch SWb[2] is electrically connected to the wiring SLb[2], a second terminal of the switch SWb[2] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[2] is electrically connected to a wiring SGb[2]. A first terminal of the switch SWb[3] is electrically connected to the wiring SLb[3], a second terminal of the switch SWb[3] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[3] is electrically connected to a wiring SGb[3]. A first terminal of the switch SWb[4] is electrically connected to the wiring SLb[4], a second terminal of the switch SWb[4] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[4] is electrically connected to a wiring SGb[4].

An input terminal of the circuit OTH included in the driver circuit SDb is electrically connected to a wiring DLb.

For example, an analog switch or an electrical switch such as a transistor can be used as each of the switch SWa[1] to the switch SWa[4] and the switch SWb[1] to the switch SWb[4]. Alternatively, mechanical switches may be used as the switch SWa[1] to the switch SWa[4] and the switch SWb[1] to the switch SWb[4]. In the case where transistors are included in the switch SWa[1] to the switch SWa[4] and the switch SWb[1] to the switch SWb[4], for example, OS transistors, transistors including Si in their channel formation regions, or the like can be used as the transistors.

Note that in this embodiment, each of the switch SWa[1] to the switch SWa[4] and the switch SWb[1] to the switch SWb[4] is turned on when a high-level potential is input to the control terminal, and is turned off when a low-level potential is input to the control terminal.

Each of the wiring SGa[1] to the wiring SGa[4] and the wiring SGb[1] to the wiring SGb[4] functions as a wiring

that supplies a voltage for switching between the conduction state and non-conduction state of the switch whose control terminal is electrically connected to the wiring. Thus, the voltage can be, for example, a high-level potential or a low-level potential.

Accordingly, in the demultiplexer DMUa, for example, a pulse voltage with a high-level potential is applied to any one of the wiring SGa[1] to the wiring SGa[4], whereby the switch whose control terminal is supplied with the pulse voltage can be turned on only during a period when the pulse voltage is applied. Furthermore, in the demultiplexer DMUa, for example, a pulse voltage with a high-level potential is sequentially supplied to the wiring SGa[1] to the wiring SGa[4], whereby the switch SWa[1] to the switch SWa[4] can be turned on sequentially. Thus, the demultiplexer DMU can sequentially select the wiring SLa[1] to the wiring SLa[4] as destinations to which image data is to be transmitted.

The circuit OTH included in each of the demultiplexer DMUa and the demultiplexer DMUb can be, for example, an amplifier (which may be a circuit or the like including a pass transistor), a digital-analog conversion circuit (DAC), and a circuit including a latch circuit.

For example, the circuit OTH can have a function of temporarily storing a potential representing image data (which may have a digital value) transmitted from the wiring DLa (or the wiring DLb) with the latch circuit, a function of converting the image data into an analog voltage with the digital-analog conversion circuit when the image data has a digital value, and a function of amplifying the analog voltage with the amplifier and supplying it to the demultiplexer DMU.

Therefore, the circuit OTH included in the driver circuit SDa converts image data transmitted from the wiring DLa into an appropriate potential and outputs it to the output terminal of the circuit OTH. The potential output from the output terminal of the circuit OTH is supplied to the wirings (the wiring SLa[1] to the wiring SLa[4]) selected by the demultiplexer DMUa. Similarly, the circuit OTH included in the driver circuit SDb converts image data transmitted from the wiring DLb into an appropriate potential and outputs it to the output terminal of the circuit OTH. The potential output from the output terminal of the circuit OTH is supplied to the wirings (the wiring SLb [1] to the wiring SLb[4]) selected by the demultiplexer DMUb.

As illustrated in FIG. 14, when a demultiplexer is provided in each of the driver circuit SDa[1] and the driver circuit SDb[1], the number of circuits OTH can be reduced in each of the driver circuit SDa[1] and the driver circuit SDb[1]. That is, the area of each of the driver circuit SDa[1] and the driver circuit SDb [1] can be reduced.

Note that the above circuit structure of the demultiplexers DMU and the circuits OTH is an example; the circuit structure thereof may be changed as appropriate depending on the structure of the display device 11A.

As a modification example of the display device 11A, the number of source wirings (the wirings SLa and the wirings SLb) that can be selected by each of the driver circuit SDa[1] and the driver circuit SDb[1] may be changed depending on the operation speed of the circuit OTH. The operation speed of the circuit OTH is determined by the driving frequency of a transistor included in the circuit OTH, for example. With the use of a transistor with a high driving frequency in the circuit OTH, the operation speed of the circuit OTH can be increased, so that processing and output of image data can be performed at higher speed in the circuit OTH. Although each of the demultiplexer DMUa and the demultiplexer

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DMUb has four output terminals in FIG. 14, the number of output terminals of each of the demultiplexer DMUa and the demultiplexer DMUb may be changed depending on the operation speed of the circuit OTH. For example, in the case where the operation speed of the circuit OTH is high, the processing and output of image data are performed by the circuit OTH at high speed, so that a write operation of image data per column of the region ALPa_C[1] (the region ALPb_C[1]) can be shortened. Shortening the write operation of image data per column can reduce a time needed for writing image data to the pixel circuits in one column of the region ALPa_C[1] (the region ALPb_C[1]), and the write operation of image data in the column can be increased by the reduced time. In other words, shortening the write operation of image data per column allows the number of output terminals of the demultiplexer DMUa (the demultiplexer DMUb) to increase to be five or more. Furthermore, for example, in the case where the operation speed of the circuit OTH is low, the number of output terminals of each of the demultiplexer DMUa and the demultiplexer DMUb may be reduced to be two or three, for example. In other words, the number of source wirings (the wirings SLa and the wirings SLb) that can be selected by each of the driver circuit SDa[1] and the driver circuit SDb[1] can be changed depending on the operation speed of the circuit OTH.

FIG. 15 is a timing chart showing an operation example of the display device 11A illustrated in FIG. 14. Specifically, the timing chart of FIG. 15 shows changes in the potentials of a wiring GLa[1] to a wiring GLa[3], the wiring SGa[1] to the wiring SGa[4], a wiring GLb[1] to a wiring GLb[3], and the wiring SGb[1] to the wiring SGb[4], in a period from Time T01 to Time T17 and times in the vicinity thereof (note that High denotes a high-level potential and Low denotes a low-level potential in FIG. 15). In addition, the timing chart of FIG. 15 also shows image data transmitted to the wiring DLa, the wiring DLb, the wiring SLa[1] to the wiring SLa[4], and the wiring SLb[1] to the wiring SLb[4] in the period from Time T01 to Time T17 and the times in the vicinity thereof.

[Time T01 to Time T17 (the First Period)]

In the period from Time T01 to Time T17, for example, image data to be written to the pixel circuits PIXa included in the pixel array ALP of the display device 11A is sequentially transmitted from the wiring DLa to the driver circuit SDa. Similarly, image data to be written to the pixel circuits PIXb included in the pixel array ALP of the display device 11A is transmitted from the wiring DLb to the driver circuit SDb.

Specifically, in the period between Time T01 and Time T17, image data transmitted to the region ALPa_C[1] of the pixel array ALP of the display device 11A and image data transmitted to the region ALPb_C[1] of the pixel array ALP of the display device 11A are alternately supplied to the wiring DLa and the wiring DLb, respectively, for each period.

For example, when pieces of image data transmitted to the region ALPa_C[1] of the pixel array ALP are each referred to as data Da, the data Da is supplied to the wiring DLa during a period from Time T01 to Time T02, a period from Time T03 to Time T04, a period from Time T05 to Time T06, a period from Time T07 to Time T08, a period from Time T09 to Time T10, a period from Time T11 to Time T12, a period from Time T13 to T14, and a period from Time T15 to Time T16 in the timing chart of FIG. 15. For another example, when pieces of image data transmitted to the region ALPb_C[1] of the pixel array ALP are each referred to as data db, the data db is supplied to the wiring DLb in

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a period from Time T02 to Time T03, a period from Time T04 to Time T05, a period from Time T06 to Time T07, a period from Time T08 to Time T09, a period from Time T10 to Time T11, a period from Time T12 to Time T13, a period from Time T14 to Time T15, and a period from Time T16 to Time T17 in the timing chart of FIG. 15.

In this case, the input of the image data to the wiring DLa and the wiring DLb is preferably performed with a selector with one input and two outputs. Specifically, for example, in a structure where the wiring DLa is electrically connected to one output terminal of the selector with one input and two outputs and the wiring DLb is electrically connected to the other output terminal of the selector, when the data Da and the data db are alternately input as image data to the input terminal of the selector, the output of the selector is switched depending on the input data.

In the timing chart of FIG. 15, data Da[1,1] is input to the wiring DLa in the period from Time T01 to Time T02, data Da[1,2] is input thereto in the period from Time T03 to Time T04, data Da[1,3] is input thereto in the period from Time T05 to Time T06, data Da[1,4] is input thereto in the period from Time T07 to Time T08, data Da[2,1] is input thereto in the period from Time T09 to Time T10, data Da[2,2] is input thereto in the period from Time T11 to Time T12, data Da[2,3] is input thereto in the period from Time T13 to Time T14, and data Da[2,4] is input thereto in the period from Time T15 to Time T16. Furthermore, in the timing chart of FIG. 15, data db[1,1] is input to the wiring DLb in the period from Time T02 to Time T03, data db[1,2] is input thereto in the period from Time T04 to Time T05, data db[1,3] is input thereto in the period from Time T06 to Time T07, data db[1,4] is input thereto in the period from Time T08 to Time T09, data db[2,1] is input thereto in the period from Time T10 to Time T11, data db[2,2] is input thereto in the period from Time T12 to Time T13, data db[2,3] is input thereto in the period from Time T14 to Time T15, and data db[2,4] is input thereto in the period from Time T16 to Time T17.

Note that [x,y] added to the data Da denotes the address of the pixel circuit PIXa included in the region ALPa_C[1] of the pixel array ALP, which is a destination to which the data Da is to be written. Similarly, [x,y] added to the data db denotes the address of the pixel circuit PIXb included in the region ALPb_C[1] of the pixel array ALP, which is a destination to which the data db is to be written.

In the driver circuit SDa, the data Da supplied to the wiring DLa is input to the circuit OTH. The circuit OTH, for example, latches the input data Da and outputs the data Da to the output terminal of the circuit OTH. Note that the period of the latch can be a period from a time when the data Da is input to the wiring DLa to a time when the data Da to be written to a different pixel circuit is input to the wiring DLa, for example. Similarly, in the driver circuit SDb, the data db supplied to the wiring DLb is input to the circuit OTH. The circuit OTH, for example, latches the input data db and outputs the data db to the output terminal of the circuit OTH. Note that the period of the latch can be a period from a time when the data db is input to the wiring DLb to a time when the data db to be written to a different pixel circuit is input to the wiring DLb, for example.

Note that the circuit OTH included in each of the driver circuit SDa and the driver circuit SDb may have a function of performing processing such as digital-analog conversion on the input data Da.

[Time T01 to Time T09 (the Second Period)]

In a period from Time T01 to Time T09, for example, a high-level potential is input to the wiring GLa[1]. Accordingly, the high-level potential from the wiring GLa[1] is

supplied to the pixel circuit PIXa[1,1] to the pixel circuit PIXa[1,4] positioned in the first row of the region ALPa_C [1] of the display device 11A. Thus, electrical continuity is established between the pixel circuit PIXa[1,1] and the wiring SLa[1], between the pixel circuit PIXa[1,2] and the wiring SLa[2], between the pixel circuit PIXa[1,3] and the wiring SLa[3], and between the pixel circuit PIXa[1,4] and the wiring SLa[4].

Furthermore, in the period from Time T01 to Time T09, for example, a low-level potential is input to each of the wiring GLa[2] to a wiring GLa[V₁]. Accordingly, the low-level potential from the wiring GLa[2] to the wiring GLa[V₁] is supplied to the pixel circuit PIXa[2,1] to the pixel circuit PIXa[V₁,4] positioned in the second to V₁-th rows of the region ALPa_C[1] of the display device 11A. Thus, the electrical continuity between the pixel circuit PIXa[2,1] to the pixel circuit PIXa[V₁,1] and the wiring SLa[1] positioned in the first column is broken, the electrical continuity between the pixel circuit PIXa[2,2] to the pixel circuit PIXa[V₁,2] and the wiring SLa[2] positioned in the second column is broken, the electrical continuity between the pixel circuit PIXa[2,3] to the pixel circuit PIXa[V₁,3] and the wiring SLa[3] positioned in the third column is broken, and the electrical continuity between the pixel circuit PIXa[2,4] to the pixel circuit PIXa[V₁,4] and the wiring SLa[4] positioned in the fourth column is broken.

In other words, when a high-level potential is supplied to the wiring GLa[1] and a low-level potential is supplied to the wiring GLa[2] to the wiring GLa[V₁], the pixel circuit PIXa[1,1] to the pixel circuit PIXa[1,4] that serve as destinations to which image data is to be written are selected as a result.

[Time T01 to Time T03 (the Third Period)]

In a period from Time T01 to Time T03, for example, a high-level potential is input to the wiring SGa[1]. Accordingly, the high-level potential is supplied to the control terminal of the switch SWa[1] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[1] is turned on. Thus, electrical continuity is established between the wiring SLa[1] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T01 to Time T03, for example, a low-level potential is input to the wiring SGa[2] to the wiring SGa[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWa[2] to the switch SWa[4] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[2] to the switch SWa[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLa[2] to the wiring SLa[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T01 to Time T03, for example, the data Da[1,1] is input from the wiring DLa to the circuit OTH as image data. The data Da[1,1] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data Da[1,1] output from the output terminal of the circuit OTH is transmitted to the wiring SLa[1] through the switch SWa[1] that is in an on state. At this time, the pixel circuit PIXa[1,1] in the first column of the region ALPa_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLa[1]); therefore, the data Da[1,1] is written to the pixel circuit PIXa[1,1].

[Time T03 to Time T05 (the Fourth Period)]

In a period from Time T03 to Time T05, for example, a high-level potential is input to the wiring SGa[2]. Accord-

ingly, the high-level potential is supplied to the control terminal of the switch SWa[2] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[2] is turned on. Thus, electrical continuity is established between the wiring SLa[2] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T03 to Time T05, for example, a low-level potential is input to the wiring SGa[1], the wiring SGa[3], and the wiring SGa[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWa[1], the switch SWa[3], and the switch SWa[4] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[1], the switch SWa[3], and the switch SWa[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLa[1], the wiring SLa[3], and the wiring SLa[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T03 to Time T05, for example, the data Da[1,2] is input from the wiring DLa to the circuit OTH as image data. The data Da[1,2] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data Da[1,2] output from the output terminal of the circuit OTH is transmitted to the wiring SLa[2] through the switch SWa[2] that is in an on state. At this time, the pixel circuit PIXa[1,2] in the second column of the region ALPa_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLa[1]); therefore, the data Da[1,2] is written to the pixel circuit PIXa[1,2].

[Time T05 to Time T07 (the Fifth Period)]

In a period from Time T05 to Time T07, for example, a high-level potential is input to the wiring SGa[3]. Accordingly, the high-level potential is supplied to the control terminal of the switch SWa[3] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[3] is turned on. Thus, electrical continuity is established between the wiring SLa[3] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T05 to Time T07, for example, a low-level potential is input to the wiring SGa[1], the wiring SGa[2], and the wiring SGa[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWa[1], the switch SWa[2], and the switch SWa[4] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[1], the switch SWa[2], and the switch SWa[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLa[1], the wiring SLa[2], and the wiring SLa[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T05 to Time T07, for example, the data Da[1,3] is input from the wiring DLa to the circuit OTH as image data. The data Da[1,3] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data Da[1,3] output from the output terminal of the circuit OTH is transmitted to the wiring SLa[3] through the switch SWa[3] that is in an on state. At this time, the pixel circuit PIXa[1,3] in the third column of the region ALPa_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLa[1]); therefore, the data Da[1,3] is written to the pixel circuit PIXa[1,3].

[Time T07 to Time T09 (the Sixth Period)]

In a period from Time T07 to Time T09, for example, a high-level potential is input to the wiring SGa[4]. Accordingly, the high-level potential is supplied to the control

terminal of the switch SWa[4] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[4] is turned on. Thus, electrical continuity is established between the wiring SLa[4] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T07 to Time T09, for example, a low-level potential is input to the wiring SGa[1] to the wiring SGa[3]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWa[1] to the switch SWa[3] of the demultiplexer DMUa of the display device 11A, whereby the switch SWa[1] to the switch SWa[3] are each turned off. Therefore, the electrical continuity between each of the wiring SLa[1] to the wiring SLa[3] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T07 to Time T09, for example, the data Da[1,4] is input from the wiring DLa to the circuit OTH as image data. The data Da[1,4] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data Da[1,4] output from the output terminal of the circuit OTH is transmitted to the wiring SLa[4] through the switch SWa[4] that is in an on state. At this time, the pixel circuit PIXa[1,4] in the fourth column of the region ALPa_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLa[1]); therefore, the data Da[1,4] is written to the pixel circuit PIXa[1,4].

[Time T02 to Time T10 (the Seventh Period)]

In a period from Time T02 to Time T10, for example, a high-level potential is input to the wiring GLb[1]. Accordingly, the high-level potential from the wiring GLb[1] is supplied to the pixel circuit PIXb[1,1] to the pixel circuit PIXb[1,4] arranged in the first row of the region ALPb_C[1] of the display device 11A. Thus, electrical continuity is established between the pixel circuit PIXb[1,1] and the wiring SLb[1], between the pixel circuit PIXb[1,2] and the wiring SLb[2], between the pixel circuit PIXb[1,3] and the wiring SLb[3], and between the pixel circuit PIXb[1,4] and the wiring SLb[4].

Furthermore, in the period from Time T02 to Time T10, for example, a low-level potential is input to each of the wiring GLb[2] to a wiring GLb[V_i]. Accordingly, the low-level potential from the wiring GLb[2] to the wiring GLb[V_i] is supplied to the pixel circuit PIXb[2,1] to the pixel circuit PIXb[V_i,4] positioned in the first to n-th rows of the region ALPb_C[1] of the display device 11A. Thus, the electrical continuity between the pixel circuit PIXb[2,1] to the pixel circuit PIXb[V_i,1] and the wiring SLb[1] positioned in the first column is broken, the electrical continuity between the pixel circuit PIXb[2,2] to the pixel circuit PIXb[V_i,2] and the wiring SLb[2] positioned in the second column is broken, the electrical continuity between the pixel circuit PIXb[2,3] to the pixel circuit PIXb[V_i,3] and the wiring SLb[3] positioned in the third column is broken, and the electrical continuity between the pixel circuit PIXb[2,4] to the pixel circuit PIXb[V_i,4] and the wiring SLb[4] positioned in the fourth column is broken.

In other words, when a high-level potential is supplied to the wiring GLb[1] and a low-level potential is supplied to the wiring GLb[2] to the wiring GLb[V_i] the pixel circuit PIXb[1,1] to the pixel circuit PIXb[1,4] that serve as destinations to which image data is to be written are selected as a result.

[Time T02 to Time T04 (the Eighth Period)]

In a period from Time T02 to Time T04, for example, a high-level potential is input to the wiring SGb[1]. Accord-

ingly, the high-level potential is supplied to the control terminal of the switch SWb[1] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[1] is turned on. Thus, electrical continuity is established between the wiring SLb[1] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T02 to Time T04, for example, a low-level potential is input to the wiring SGb[2] to the wiring SGb[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWb[2] to the switch SWb[4] of the demultiplexer DMU of the display device 11A, whereby the switch SWb[2] to the switch SWb[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLb[2] to the wiring SLb[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T02 to Time T04, for example, the data db[1,1] is input from the wiring DLb to the circuit OTH as image data. The data db[1,1] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data db[1,1] output from the output terminal of the circuit OTH is transmitted to the wiring SLb[1] through the switch SWb[1] that is in an on state. At this time, the pixel circuit PIXb[1,1] in the first column of the region ALPb_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLb[1]); therefore, the data db[1,1] is written to the pixel circuit PIXb[1,1].

[Time T04 to Time T06 (the Ninth Period)]

In a period from Time T04 to Time T06, for example, a high-level potential is input to the wiring SGb[2]. Accordingly, the high-level potential is supplied to the control terminal of the switch SWb[2] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[2] is turned on. Thus, electrical continuity is established between the wiring SLb[2] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T04 to Time T06, for example, a low-level potential is input to the wiring SGb[1], the wiring SGb[3], and the wiring SGb[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWb[1], the switch SWb[3], and the switch SWb[4] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[1], the switch SWb[3], and the switch SWb[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLb[1], the wiring SLb[3], and the wiring SLb[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T04 to Time T06, for example, the data db[1,2] is input from the wiring DLb to the circuit OTH as image data. The data db[1,2] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data db[1,2] output from the output terminal of the circuit OTH is transmitted to the wiring SLb[2] through the switch SWb[2] that is in an on state. At this time, the pixel circuit PIXb[1,2] in the second column of the region ALPb_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLb[1]); therefore, the data db[1,2] is written to the pixel circuit PIXb[1,2].

[Time T06 to Time T08 (the 10th Period)]

In a period from Time T06 to Time T08, for example, a high-level potential is input to the wiring SGb[3]. Accordingly, the high-level potential is supplied to the control terminal of the switch SWb[3] of the demultiplexer DMUb

of the display device 11A, whereby the switch SWb[3] is turned on. Thus, electrical continuity is established between the wiring SLb[3] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T06 to Time T08, for example, a low-level potential is input to the wiring SGb[1], the wiring SGb[2], and the wiring SGb[4]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWb[1], the switch SWb[2], and the switch SWb[4] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[1], the switch SWb[2], and the switch SWb[4] are each turned off. Therefore, the electrical continuity between each of the wiring SLb[1], the wiring SLb[2], and the wiring SLb[4] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T06 to Time T08, for example, the data db[1,3] is input from the wiring DLb to the circuit OTH as image data. The data db[1,3] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data db [1,3] output from the output terminal of the circuit OTH is transmitted to the wiring SLb[3] through the switch SWb[3] that is in an on state. At this time, the pixel circuit PIXb[1,3] in the third column of the region ALPb_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLb[1]); therefore, the data db[1,3] is written to the pixel circuit PIXb[1,3].

[Time T08 to Time T10 (the 11th Period)]

In a period from Time T08 to Time T10, for example, a high-level potential is input to the wiring SGb[4]. Accordingly, the high-level potential is supplied to the control terminal of the switch SWb[4] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[4] is turned on. Thus, electrical continuity is established between the wiring SLb[4] and the output terminal of the circuit OTH.

Furthermore, in the period from Time T08 to Time T10, for example, a low-level potential is input to the wiring SGb[1] to the wiring SGb[3]. Accordingly, the low-level potential is supplied to each of the control terminals of the switch SWb[1] to the switch SWb[3] of the demultiplexer DMUb of the display device 11A, whereby the switch SWb[1] to the switch SWb[3] are each turned off. Thus, the electrical continuity between each of the wiring SLb[1] to the wiring SLb[3] and the output terminal of the circuit OTH is broken.

Moreover, in the period from Time T08 to Time T10, for example, the data db[1,4] is input from the wiring DLb to the circuit OTH as image data. The data db[1,4] is subjected to processing such as digital-analog conversion and amplification in the circuit OTH and output to the output terminal of the circuit OTH. The data db[1,4] output from the output terminal of the circuit OTH is transmitted to the wiring SLb[4] through the switch SWb[4] that is in an on state. At this time, the pixel circuit PIXb[1,4] in the fourth column of the region ALPb_C is selected as a destination to which data is to be written (because the high-level potential is input to the wiring GLb[1]); therefore, the data db [1,4] is written to the pixel circuit PIXb [1,4].

[Time T09 to Time T17 (the 12th Period)]

In a period from Time T09 to Time T17, for example, a high-level potential is input to the wiring GLa[2]. Accordingly, the high-level potential from the wiring GLa[2] is supplied to the pixel circuit PIXa[2,1] to the pixel circuit PIXa[2,4] positioned in the second row of the region ALPa_C[2] of the display device 11A. Therefore, electrical continuity is established between the pixel circuit PIXa[2,1]

and the wiring SLa[2], between the pixel circuit PIXa[2,2] and the wiring SLa[2], between the pixel circuit PIXa[2,3] and the wiring SLa[3], and between the pixel circuit PIXa[2,4] and the wiring SLa[4].

Furthermore, in the period from Time T09 to Time T17, for example, a low-level potential is input to each of the wiring GLa[1] and the wiring GLa[3] to the wiring GLa[V₁]. Accordingly, the low-level potential from the wiring GLa[1] and the wiring GLa[3] to the wiring GLa[V₁] is supplied to the pixel circuits PIXa[1,1] to PIXa[1,4] and the pixel circuits PIXa[3,1] to PIXa[V₁,4] positioned in the first row and the third to V₁-th rows of the region ALPa_C[1] of the display device 11A. Thus, the electrical continuity between the pixel circuit PIXa[1,1] and the pixel circuit PIXa[3,1] to the pixel circuit PIXa[V₁,1] and the wiring SLa[1] positioned in the first column is broken, the electrical continuity between the pixel circuit PIXa[1,2] and the pixel circuit PIXa[3,2] to the pixel circuit PIXa[V₁,2] and the wiring SLa[2] positioned in the second column is broken, the electrical continuity between the pixel circuit PIXa[1,3] and the pixel circuit PIXa[3,3] to the pixel circuit PIXa[V₁,3] and the wiring SLa[3] positioned in the third column is broken, and the electrical continuity between the pixel circuit PIXa[1,4] and the pixel circuit PIXa[3,4] to the pixel circuit PIXa[V₁,4] and the wiring SLa[4] positioned in the fourth column is broken.

In other words, when a high-level potential is supplied to the wiring GLa[2] and a low-level potential is supplied to the wiring GLa[1] and the wiring GLa[3] to the wiring GLa[V₁], the pixel circuit PIXa[2,1] to the pixel circuit PIXa[2,4] that serve as destinations to which image data is to be written are selected as a result.

In addition, in the period from Time T09 to Time T17, as in the operation performed in the third period to the sixth period, the wiring SLa[1] to the wiring SLa[4] are sequentially selected by the demultiplexer DMUa and the data Da[2,1] to the data Da[2,4] are transmitted to the wiring SLa[1] to the wiring SLa[4] in the respective periods. Thus, the data Da[2,1] to the data Da[2,4] are written to the pixel circuit PIXa[2,1] to the pixel circuit PIXa[2,4], respectively.

[Time T10 to Time T17 and Later (the 13th Period)]

In a period from Time T10 to Time T17 and later, for example, a high-level potential is input to the wiring GLb[2]. Accordingly, the high-level potential from the wiring GLb[2] is supplied to the pixel circuit PIXb[2,1] to the pixel circuit PIXb[2,4] positioned in the second row of the region ALPb_C[2] of the display device 11A. Therefore, electrical continuity is established between the pixel circuit PIXb[2,1] and the wiring SLb[2], between the pixel circuit PIXb[2,2] and the wiring SLb[2], between the pixel circuit PIXb[2,3] and the wiring SLb[3], and between the pixel circuit PIXb[2,4] and the wiring SLb[4].

Furthermore, in the period from Time T09 to Time T17, for example, a low-level potential is input to each of the wiring GLb[1] and the wiring GLb[3] to the wiring GLb[V₂]. Accordingly, the low-level potential from the wiring GLb[1] and the wiring GLb[3] to the wiring GLb[V₂] is supplied to the pixel circuits PIXb[1,1] to PIXb[1,4] and the pixel circuit PIXb[3,1] to the pixel circuit PIXb[V₂,4] positioned in the first row and the third to V₂-th rows of the region ALPb_C[1] of the display device 11A. Thus, the electrical continuity between the pixel circuit PIXb[1,1] and the pixel circuit PIXb[3,1] to the pixel circuit PIXb[V₂,1] and the wiring SLb[1] positioned in the first column is broken, the electrical continuity between the pixel circuit PIXb[1,2] and the pixel circuit PIXb[3,2] to the pixel circuit PIXb[V₂,2] and the wiring SLb[2] positioned in the second

column is broken, the electrical continuity between the pixel circuit PIXb[1,3] and the pixel circuit PIXb[3,3] to the pixel circuit PIXb[V₂,3] and the wiring SLb[3] positioned in the third column is broken, and the electrical continuity between the pixel circuit PIXb[1,4] and the pixel circuit PIXb[3,4] to the pixel circuit PIXb[V₂,4] and the wiring SLb[4] positioned in the fourth column is broken.

In other words, when a high-level potential is supplied to the wiring GLb[2] and a low-level potential is supplied to the wiring GLb[1] and the wiring GLb[3] to the wiring GLb[V₂], the pixel circuit PIXb[2,1] to the pixel circuit PIXb[2,4] that serve as destinations to which image data is to be written are selected as a result.

In addition, in the period from Time T10 to Time T17 and later, as in the operation performed in the 8th period to the 11th period, the wiring SLb[1] to the wiring SLb[4] are sequentially selected by the demultiplexer DMUb and the data db[2,1] to the data db[2,4] are transmitted to the wiring SLb[1] to the wiring SLb[4] in the respective periods. Thus, the data db[2,1] to the data db[2,4] are written to the pixel circuit PIXb[2,1] to the pixel circuit PIXb[2,4], respectively.

As for the third and following rows in the region ALPa_C[1] of the pixel array ALP of the display device 11A, the third and following rows of the region ALPa_C[1] are selected by the driver circuit GD, and as in the above-described operation, the wiring SLa[1] to the wiring SLa[4] are sequentially selected by the demultiplexer DMUa and pieces of data that correspond to the columns are transmitted to the wiring SLa[1] to the wiring SLa[4]. Similarly, also as for the third and following rows in the region ALPb_C[1] of the pixel array ALP of the display device 11A, the third and following rows of the region ALPb_C[1] are selected by the driver circuit GD, and as in the above-described operation, the wiring SLb[1] to the wiring SLb[4] are sequentially selected by the demultiplexer DMUb and pieces of image data that correspond to the columns are transmitted to the wiring SLb[1] to the wiring SLb[4].

The writing of data to the pixel circuits included in the region ALPa_C[1] and the region ALPb_C[1] of the pixel array ALP is described in the above-described operation example of the display device 11A; image data can also be written to the pixel circuits included in each of the region ALPa_C[2] to the region ALPa_C[h] and the region ALPb_C[2] to the region ALPb_C[h] in the same manner. The driver circuit SDa[1] to the driver circuit SDa[h] and the driver circuit SDb[1] to the driver circuit SDb[h], which transmit image data to the region ALPa_C[1] to the region ALPa_C[h] and the region ALPb_C[1] to the region ALPb_C[h], respectively, can be driven independently of and in parallel with one another; therefore, a time needed for rewriting an image displayed on the display portion of the display device 11 (e.g., a time per frame) can be reduced.

It is possible to combine the display device 11 of FIG. 13 with the display device 10 described in Embodiment 1. As an example of the combination structure, the pixel region ARA of the display device 10 in Embodiment 1 can be used as the pixel array ALP of the display device 11 in FIG. 13 and each of the plurality of driver circuits SDa and the plurality of driver circuits SDb in FIG. 13 can be included in the local driver circuit LD of the display device 10 in Embodiment 1.

A display device 11B illustrated in FIG. 16A has a structure of the display device 10 in Embodiment 1, in which the pixel region ARA is divided into the region ALPa_C[1] to the region ALPa_C[h] and the region ALPb_C[1] to the region ALPb_C[h]. Note that FIG. 16A selectively illustrates the region ALPa_C[1] and the region ALPb_C[1]. Source

wirings electrically connected to the pixel circuits PIXa included in the region ALPa_C[1] are referred to as a wiring SLa_1 to a wiring SLa_3, and source wirings electrically connected to the pixel circuits PIXb included in the region ALPb_C[1] are referred to as a wiring SLb_1 to the wiring SLb_3. Note that FIG. 16A selectively illustrates the wiring SLa_1, the wiring SLa_2, the wiring SLa_3, the wiring SLb_1, the wiring SLb_2 and the wiring SLb_3.

As described above, the driver circuit SDa[1] and the driver circuit SDb[1] illustrated in FIG. 13 are each included in the local driver circuit LD of the display device 11B of FIG. 16A. At least one of the others, i.e., at least one of the driver circuit SDa[2] to the driver circuit SDa[h] and the driver circuit SDb[2] to the driver circuit SDb[h], may be included in the local driver circuit LD including the driver circuit SDa[1] and the driver circuit SDb[1] or may be included in another local driver circuit LD different from the local driver circuit LD including the driver circuit SDa[1] and the driver circuit SDb[1].

FIG. 16B illustrates a specific example of the pixel region ARA included in the display device 11B of FIG. 16A.

The wiring SLa_1 to the wiring SLa_3 are extended in their respective columns in the region ALPa_C[1] of FIG. 16B, and the wiring SLb_1 to the wiring SLb_3 are extended in their respective columns in the region ALPb_C[1] of FIG. 16B. Note that FIG. 16B selectively illustrates the wiring SLa_1, the wiring SLa_2, the wiring SLa_3, the wiring SLb_1, the wiring SLb_2, and the wiring SLb_3. In the pixel region ARA of FIG. 16B, the wiring SLa_1 and the wiring SLb_1 are extended in the same first column, the wiring SLa_2 and the wiring SLb_2 are extended in the same second column, and the wiring SLa_3 and the wiring SLb_3 are extended in the same third column.

A plurality of pixel circuits PIXa are arranged in a matrix in the region ALPa_C[1] of the pixel region ARA of FIG. 16B, and a plurality of pixel circuits PIXb are arranged in a matrix in the region ALPb_C[1] of the pixel region ARA of FIG. 16B. Note that the number of pixel circuits PIXa and the number of pixel circuits PIXb that are included in the region ALPa_C[1] and the region ALPb_C[1], respectively, are preferably equal to each other. Furthermore, the plurality of pixel circuits PIXa are electrically connected to one another in the respective columns with the wiring SLa_1 to the wiring SLa_3 in the region ALPa_C[1], and the plurality of pixel circuits PIXb are electrically connected to one another in the respective columns with the wiring SLb_1 to the wiring SLb_3 in the region ALPb_C[1].

As illustrated in FIG. 16A, combining the display device 11 of FIG. 13 with the display device 10 described in Embodiment 1 enables a reduction in the write time of image data also in the display device with high resolution.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 3

In this embodiment, an example of routing of wirings in a pixel array in the display device of one embodiment of the present invention, and an example of a demultiplexer that achieves the example of the routing of wirings are described.

Structure Example of Display Device

A wiring that transmits image data (rephrased as an image signal in some cases) to a pixel array of the display device is extended in each of the columns of the pixel array, for

example. A wiring that transmits a selection signal to the pixel array of the display device is extended in each of the rows of the pixel array, for example. In this case, a wiring transmitting image data and a wiring transmitting a selection signal are determined, whereby the address of a pixel circuit to which the image data is to be written is determined.

In the case where the resolution of the display device is high, the number of pixel circuits to which image data is input is large, so that a time needed for writing in the display device is increased in some cases.

Thus, consider a structure in which the pixel array is divided into two and an image signal is written to each of the divided regions. FIG. 17A illustrates a structure example of a display device in which the pixel array ALP is divided into two regions, a region ALPa and a region ALPb. Note that in FIG. 17A, the region ALPa and the region ALPb are divided so as to include the same columns.

A display device 12 illustrated in FIG. 17 includes, for example, the driver circuit GD and a driver circuit SD[1] to a driver circuit SD[h] (here h is an integer of 1 or more) in addition to the pixel array ALP. Note that although the driver circuit SD[1] to the driver circuit SD[h] are included in the pixel array ALP in FIG. 17, one embodiment of the present invention is not limited thereto; the driver circuit SD[1] to the driver circuit SD[h] may be positioned outside the pixel array ALP as illustrated in FIG. 24 described later.

For the driver circuit GD, the description of the driver circuit GD included in the display device 11 in Embodiment 2 is referred to.

For example, the driver circuit SD[1] to the driver circuit SD[h] each have a function of a source driver circuit that transmits image data to pixel circuits included in the pixel array ALP, like the driver circuit SDa[1] to the driver circuit SDa[h] and the driver circuit SDb [1] to the driver circuit SDb[h] described in Embodiment 2. Note that the driver circuit SDa[1] to the driver circuit SDa[h] and the driver circuit SDb [1] to the driver circuit SDb[h] may each further include a demultiplexer for selecting a source wiring to which image data is transmitted.

The pixel array ALP includes, for example, a plurality of pixel circuits PIX, and the plurality of pixel circuits PIX are arranged in a matrix of V rows and W columns (V is a multiple of 2 and more than or equal to 1, W is a multiple of h and more than or equal to 1, and h is an integer of 1 or more) in the pixel array ALP. The pixel array ALP of FIG. 17 has a structure in which the pixel circuit PIX[1,1] to the pixel circuit PIX[V,W] are arranged. Note that the pixel circuit PIX[1,1], the pixel circuit PIX[V/2,1], the pixel circuit PIX[V/2+1,1], the pixel circuit PIX[V,1], the pixel circuit PIX[1,W/h], the pixel circuit PIX[V/2,W/h], the pixel circuit PIX[V/2+1,W/h], the pixel circuit PIX[V,W/h], the pixel circuit PIX[1,W-W/h+1], the pixel circuit PIX[V/2,W-W/h+1], the pixel circuit PIX[V/2+1,W-W/h+1], the pixel circuit PIX[V,W-W/h+1], the pixel circuit PIX[1,W], the pixel circuit PIX[V/2,W], the pixel circuit PIX[V/2+1,W], and the pixel circuit PIX[V,W] are selectively illustrated in the pixel array ALP of FIG. 17.

The region ALPa includes the pixel circuit PIX[1,1] to the pixel circuit PIX[V/2,W] of the pixel circuit PIX[1,1] to the pixel circuit PIX[V,W]. Meanwhile, the region ALPb includes the pixel circuit PIX[V/2+1,1] to the pixel circuit PIX[V,W] of the pixel circuit PIX[1,1] to the pixel circuit PIX[V,W].

In the region ALPa, the wiring GL[1] to the wiring GL[V/2] are extended in the respective rows. Each of the wiring GL[1] to the wiring GL[V/2] is electrically connected to the driver circuit GD. Each of the wiring GL[1] to the

wiring GL[V/2] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding row. Similarly, the wiring GL[V/2+1] to the wiring GL[V] are extended in the respective rows in the region ALPb. Each of the wiring GL[V/2+1] to the wiring GL[V] is electrically connected to the driver circuit GD. Each of the wiring GL[V/2+1] to the wiring GL[V] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding row.

Consider regions obtained by dividing the columns of the pixel array ALP into h in the display device 12. Specifically, for example, the region ALPa includes the region ALPa_C [1] to the region ALPa_C[h] as the h regions. Note that each of the region ALPa_C[1] to the region ALPa_C[h] includes the pixel circuits PIX of W/h columns. Similarly, the region ALPb includes the region ALPb_C[1] to the region ALPb_C[h] as the h regions, and each of the region ALPb_C[1] to the region ALPb_C[h] includes the pixel circuits PIX of W/h columns.

The wiring SLa[1] to the wiring SLa[W/h] are extended in their respective columns in the region ALPa_C[1], and each of the wiring SLa[1] to the wiring SLa[W/h] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding column in the region ALPa_C[1]. Similarly, the wiring SLb[1] to the wiring SLb[W/h] are extended in their respective columns in the region ALPb_C[1], and each of the wiring SLb[1] to the wiring SLb[W/h] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding column in the region ALPb_C[1]. The wiring SLa[1] to the wiring SLa[W/h] and the wiring SLb [1] to the wiring SLb[W/h] are each electrically connected to the driver circuit SD[1].

The wiring SLa[W-W/h+1] to the wiring SLa[W] are extended in their respective columns in the region ALPa_C [h], and each of the wiring SLa[W-W/h+1] to the wiring SLa[W] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding column in the region ALPa_C[h]. Similarly, the wiring SLb[W-W/h+1] to the wiring SLb[W] are extended in their respective columns in the region ALPb_C[h], and each of the wiring SLb[W-W/h+1] to the wiring SLb[W] is electrically connected to the plurality of pixel circuits PIX positioned in the corresponding column in the region ALPb_C[h]. Note that the wiring SL[W-W/h+1] to the wiring SL[W] and the wiring SLb [W-W/h+1] to the wiring SLb[W] are each electrically connected to the driver circuit SD[h].

Note that the same applies to each of the region ALPa_C [2] to the region ALPa_C[h-1] and the region ALPb_C[2] to the region ALPb_C[h-1] not illustrated in FIG. 17, in which the wirings are extended in their respective columns. One of the wirings is electrically connected to any one of the driver circuit SD[2] to the driver circuit SD[h-1] depending on the divided region.

Here, an example of a method for driving the display device 12 illustrated in FIG. 17 is described.

A display device 12A illustrated in FIG. 18A is shown as a block diagram illustrating part of the pixel array ALP of the display device 12 in FIG. 17, in which the region ALPa_C [1], the region ALPb_C[1], the driver circuit SD[1], and the driver circuit SD[2] are selectively illustrated. Note that, V, W, and W/h are 6, 6, and 3, respectively, in the display device 12A illustrated in FIG. 18A, as an example. In other words, the pixel circuits PIX are arranged in a matrix of six rows and six columns in the display device 12A illustrated in FIG. 18A.

Note that the driver circuit SD[1] provided in the display device 12A of FIG. 18A has a function of selecting any one

of six source wirings (corresponding to the wiring SLa[1] to the wiring SLa[3] and the wiring SLb[1] to the wiring SLb[3] in the case where the display device 12 of FIG. 17 is referred to) and transmitting an image signal to the source wiring. Similarly, the driver circuit SD[2] provided in the display device 12A of FIG. 18A also has a function of selecting any one of six source wiring (corresponding to the wirings SLa[4] to the wiring SLa[6] and the wiring SLb[4] to the wiring SLb[6] in the case where the display device 12 of FIG. 17 is referred to) and transmitting an image signal to the source wiring.

Consider a case where image data is written to one pixel in the display device 12A. For example, first, the driver circuit GD transmits a selection signal to the wiring GL[1] in order to select the pixel circuits PIX positioned in the first row. The driver circuit SD[1] selects the wiring SLa[1] and transmits image data to the wiring SLa[1]. Accordingly, the image data is written to the pixel circuit PIX[1,1] (see FIG. 18B). Note that in FIG. 18B to FIG. 18F, the pixel circuits PIX to which the image data is written are denoted by hatching.

Furthermore, since the driver circuit SD[2] can operate independently of the driver circuit SD[1], the driver circuit SD[2] can write image data to the pixel circuit PIX included in the region ALPa_C[2] like the driver circuit SD[1] at the same time as when the image data is written to the pixel circuit PIX[1,1]. For example, in FIG. 18B, the driver circuit SD[2] selects the wiring SLa[4] and transmits image data to the wiring SLa[4]. Thus, the image data is written to the pixel circuit PIX[1,4].

FIG. 18C illustrates an example where, after the image data is written to the pixel circuit PIX[1,1] and the pixel circuit PIX[1,4] in the display device 12A, the driver circuit SD[1] selects the wiring SLa[2] and writes image data to the pixel circuit PIX[1,2], and the driver circuit SD[2] selects the wiring SLa[5] and writes image data to the pixel circuit PIX[1,5]. In addition, in the example illustrated in FIG. 18C, after the image data is written to each of the pixel circuit PIX[1,2] and the pixel circuit PIX[1,5] in the display device 12A, the driver circuit SD[1] selects the wiring SLa[3] and writes image data to the pixel circuit PIX[1,3], and the driver circuit SD[2] selects the wiring SLa[6] and writes image data to the pixel circuit PIX[1,6].

In FIG. 18D, after the image data is written to the pixel circuit PIX[1,3] and the pixel circuit PIX[1,6] in the display device 12A, the driver circuit GD conducts an operation of transmitting a selection signal to the wiring GL[4] in order to select the pixel circuits PIX positioned in the fourth row. After that, the driver circuit SD[1] selects the wiring SLb[1] and transmits image data to the wiring SLb[1], and the driver circuit SD[2] selects the wiring SLb[4] and transmits image data to the wiring SLb[4]. Accordingly, the image data can be written to each of the pixel circuit PIX[4,1] and the pixel circuit PIX[4,4].

FIG. 18E illustrates an example where, after the image data is written to the pixel circuit PIX[4,1] and the pixel circuit PIX[4,4] in the display device 12A, the driver circuit SD[1] selects the wiring SLa[2] and writes image data to the pixel circuit PIX[4,2], and the driver circuit SD[2] selects the wiring SLa[5] and writes image data to the pixel circuit PIX[4,5]. In addition, in the example illustrated in FIG. 18E, after the image data is written to each of the pixel circuit PIX[4,2] and the pixel circuit PIX[4,5] in the display device 12A, the driver circuit SD[1] selects the wiring SLa[3] and writes image data to the pixel circuit PIX[4,3], and the driver circuit SD[2] selects the wiring SLa[6] and writes image data to the pixel circuit PIX[4,6].

As described above, in the display device 12A, the pixel circuits PIX of the first row are selected by the driver circuit GD, image data is transmitted to the wiring SLa[1] to the wiring SLa[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLa[4] to the wiring SLa[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the first row. Then, the pixel circuits PIX of the fourth row are selected by the driver circuit GD, image data is transmitted to the wiring SLb[1] to the wiring SLb[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLb[4] to the wiring SLb[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the fourth row.

Similarly, in the display device 12A, the pixel circuits PIX of the second row are selected by the driver circuit GD, image data is transmitted to the wiring SLa[1] to the wiring SLa[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLa[4] to the wiring SLa[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the second row. Then, the pixel circuits PIX of the fifth row are selected by the driver circuit GD, image data is transmitted to the wiring SLb[1] to the wiring SLb[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLb[4] to the wiring SLb[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the fifth row.

Furthermore, in the display device 12A, the pixel circuits PIX of the third row are selected by the driver circuit GD, image data is transmitted to the wiring SLa[1] to the wiring SLa[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLa[4] to the wiring SLa[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the third row. Then, the pixel circuits PIX of the sixth row are selected by the driver circuit GD, image data is transmitted to the wiring SLb[1] to the wiring SLb[3] sequentially by the driver circuit SD[1], and image data is transmitted to the wiring SLb[4] to the wiring SLb[6] sequentially by the driver circuit SD[2], whereby the image data can be written to each of the pixel circuits PIX of the sixth row.

In the above manner, the image data can be written to each of the pixel circuit PIX[1,1] to the pixel circuit PIX[6,6] in the display device 12A as illustrated in FIG. 18F.

Here, a display device 13A is illustrated in FIG. 19 as a structure example applied to a conventional display device. The driver circuit SD[1] and a plurality of pixel circuits PIX are shown in the display device 13A of FIG. 19. Note that the pixel circuits PIX are arranged in a matrix of six rows and six columns in the display device 13A illustrated in FIG. 19 as in the display device 12A of FIG. 18A.

The driver circuit SD[1] provided in the display device 13A of FIG. 19 has a function of selecting any one of six source wirings and transmitting an image signal to the source wiring.

The display device 12A of FIG. 18A is different from the display device 13A of FIG. 19, which is an example of a conventional display device, in the length of the source wiring extended in one column. Specifically, for example, the display device 12A of FIG. 18A corresponds to a structure in which the plurality of source wirings of the display device 13A of FIG. 19 are divided into a region of the first row to the third row and a region of the fourth row to the sixth row. That is, the source wiring connected to one of the output terminals of the driver circuit SD[1] (or the

driver circuit SD[2]) in the display device 12A of FIG. 18A is approximately $\frac{1}{2}$ times the source wiring connected to one of the output terminals of the driver circuit SD[1] in the display device 13A of FIG. 19; therefore, the parasitic resistance and parasitic capacitance of each source wiring in the display device 12A of FIG. 18A can be smaller than the parasitic resistance and parasitic capacitance of each source wiring in the display device 13A of FIG. 19. Therefore, the time constant of the source wiring of the display device 12A of FIG. 18A can be smaller than the time constant of the source wiring of the display device 13A of FIG. 19; thus, the operation of writing image data to the display device 12A can be performed at a higher speed.

For this reason, as illustrated in the display device 12 of FIG. 17, as an example, the pixel array ALP is divided into two, and the pixel circuits PIX included in part of each of the two divided regions are driven by the driver circuit SD, whereby writing of image data to the pixel circuits PIX included in the display device 12 can be performed at a high speed to reduce the writing time.

Furthermore, the area of the circuit layer SICL increases as the area of the display portion (i.e., the pixel array ALP) increases in the display device 12 of FIG. 17, which facilitates the arrangement of the driver circuit SD [1] to the driver circuit SD[h]. Therefore, it can be said that the structure of the display device 12 of FIG. 17 is preferred when the pixel array ALP has a large area.

Note that the structure of the display device of one embodiment of the present invention is not limited to the above-described structure example of the display device 12 described above. The display device of one embodiment of the present invention may have a structure changed from that of the display device 12 according to circumstances.

For example, the display device 12 of FIG. 17 has a structure in which one driver circuit SD transmits image data to the pixel circuits of each of the region ALPa_C and the region ALPb_C included in the pixel array ALP; however, one driver circuit SD may transmit image data to the pixel circuits of three or more regions.

Specifically, the display device 12 of FIG. 17 can be changed into the structure of a display device 12AA illustrated in FIG. 20. The display device 12AA includes the pixel array ALP including a plurality of pixel circuits PIX arranged in a matrix of V rows and W columns (here V is a multiple of $2 \times d$ and more than or equal to 1, W is a multiple of h and more than or equal to 1, d is an integer of 2 or more, and h is an integer of 1 or more) as in the display device of FIG. 17. Furthermore, the display device 12AA has a structure in which the pixel array ALP is divided every plurality of rows; for example, the pixel array ALP is divided into $2 \times d$ regions. In FIG. 20, a region ALPa[1] to a region ALPa[d] and a region ALPb [1] to a region ALPb[d] are illustrated as the divided $2 \times d$ regions. Furthermore, in the structure of the display device 12AA, each of the region ALPa[1] to the region ALPa[d] and the region ALPb[1] to the region ALPb[d] is divided every plurality of columns; for example, each of the region ALPa[1] to the region ALPa[d] and the region ALPb[1] to the region ALPb[d] is divided into h regions. FIG. 20 illustrates an example where the region ALPa[1] includes a region ALPa[1]_C[1] to a region ALPa[1]_C[h], the region ALPa[d] includes a region ALPa[d]_C[1] to a region ALPa[d]_C[h], the region ALPb [1] includes a region ALPb[1]_C[1] to a region ALPb[1]_C[h], and the region ALPb[d] includes a region ALPb[d]_C[1] to a region ALPb[d]_C[h].

In the display device 12AA of FIG. 20, the pixel circuit PIX[1,1] included in the region ALPa[1]_C[1], the pixel

circuit PIX[1,W-W/h+1] included in the region ALPa[1]_C[h], the pixel circuit PIX[V/2-V/2d+1,1] included in the region ALPa[d]_C[1], and the pixel circuit PIX[V/2-V/2d+1,W-W/h+1] included in the region ALPa[d]_C[h] are selectively illustrated. Furthermore, in the display device 12AA of FIG. 20, the pixel circuit PIX[V/2+1,1] included in the region ALPb[1]_C[1], the pixel circuit PIX[V/2+1,W-W/h+1] included in the region ALPb[1]_C[h], the pixel circuit PIX[V-V/2d+1,1] included in the region ALPb[d]_C[1], and the pixel circuit PIX[V-V/2d+1,W-W/h+1] included in the region ALPb[d]_C[h] are selectively illustrated.

Moreover, in the display device 12AA of FIG. 20, the wiring GL[1] to the wiring GL[V] are extended in the row direction in the pixel array ALP. Note that in FIG. 20, the wiring GL[1], the wiring GL[V/2-V/2d+1], the wiring GL[V/2+1], and the wiring GL[V-V/2d+1] of the wiring GL[1] to the wiring GL[V] are selectively illustrated.

In the display device 12AA of FIG. 20, $2 \times d$ source wirings are extended in each column in the pixel array ALP. Specifically, for example, a wiring SLa[1]_[1] to a wiring SLa[d]_[1] and a wiring SLb[1]_[1] to a wiring SLb[d]_[1] are extended in the first column of the pixel array ALP, a wiring SLa[1]_[W-W/h+1] to a wiring SLa[d]_[W-W/h+1] and a wiring SLb[1]_[W-W/h+1] to a wiring SLb[d]_[W-W/h+1] are extended in the W-W/h+1-th column of the pixel array ALP. Note that the wiring SLa[1]_[1], the wiring SLa[d]_[1], the wiring SLb[1]_[1], the wiring SLb[d]_[1], the wiring SLa[1]_[W-W/h+1], the wiring SLa[d]_[W-W/h+1], the wiring SLb[1]_[W-W/h+1], and the wiring SLb[d]_[W-W/h+1] are selectively illustrated in FIG. 20.

Note that [x]_[y] added to the wiring SLa (the wiring SLb) represents the address of the column of the pixel array ALP and the kind of the region ALPa (the region ALPb) including the pixel circuits to which the wiring SLa (the wiring SLb) is electrically connected. Specifically, for example, the wiring SLa[2]_[d-1] denotes that it is a wiring of the second column of the pixel array ALP and electrically connected to the pixel circuits PIX included in the region ALPa[d-1]. For another example, the wiring SLb[W]_[2] denotes that it is a wiring of the W-th column of the pixel array ALP and electrically connected to the pixel circuits PIX included in the region ALPb [2].

In the display device 12AA of FIG. 20, the plurality of pixel circuits PIX positioned in the first row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[1]. The plurality of pixel circuits PIX positioned in the V/d+1-th row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[V/d+1]. The plurality of pixel circuits PIX positioned in the V-V/d+1-th row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[V-V d+1].

In the display device 12AA of FIG. 20, the pixel circuits PIX that are positioned in the first column of the pixel array ALP and included in the region ALPa[1]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLa[1]_[1]. The pixel circuits PIX that are positioned in the first column of the pixel array ALP and included in the region ALPa[d]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLa[d]_[1]. The pixel circuits PIX that are positioned in the first column of the pixel array ALP and included in the region ALPb[1]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLb [1]_[1]. The pixel circuits PIX that are positioned in the first column of the pixel array ALP and

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included in the region ALPb[d]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLb[d]_1[1].

With the structure of the display device 12AA illustrated in FIG. 20, one driver circuit SD can transmit image data to the pixel circuits included in each of the plurality of regions. Specifically, for example, in the display device 12AA of FIG. 20, the driver circuit SD[1] can transmit image data to the pixel circuits included in each of the region ALPa[1]_C[1] to the region ALPa[d]_C[1] and the region ALPb[1]_C[1] to the region ALPb[d]_C[1].

The display device 12 has a structure in which the wirings SLa and the wirings SLb are routed so as to be electrically connected to the driver circuit SD[1] to the driver circuit SD[h] that are between the region ALPa and the region ALPb, for example; however, the display device 12 may have a structure in which each of the wirings SLa and the wirings SLb is routed in a direction of one side of the pixel array ALP, so that the wirings SLa and the wiring SLb are electrically connected to the driver circuit SD[1] to the driver circuit SD[h].

For example, a display device 14 illustrated in FIG. 21 has the structure of the display device 12 of FIG. 17, in which each of the wiring SLa[1] to the wiring SLa[W] and the wiring SLb[1] to the wiring SLb[W] is routed in a direction of one side of the pixel array ALP, i.e., in a direction toward the top portion of the drawing, so that the wiring SLa[1] to the wiring SLa[W] and the wiring SLb[1] to the wiring SLb[W] are electrically connected to the driver circuit SD[1] to the driver circuit SD[h]. Specifically, the wiring SLa[1] to the wiring SLa[W] are extended along the column direction of the pixel array ALP through the inside of the region ALPa, and the wiring SLb[1] to the wiring SLb[W] are extended along the column direction of the pixel array ALP through the inside of the region ALPa and the region ALPb.

Since the display device 14 of FIG. 21, for example, has a structure in which the wiring SLa[1] to the wiring SLa[W] and the wiring SLb[1] to the wiring SLb[W] are routed not between the region ALPa and the region ALPb but to the outside of the pixel array ALP, routing for electrical connection between the driver circuit SD[1] to the driver circuit SD[h] and the wiring SLa[1] to the wiring SLa[W] and the wiring SLb[1] to the wiring SLb[W] is not needed in the inside of the pixel array ALP of the display device 14. Therefore, the layout of the pixel array ALP is facilitated in some cases.

Note that as in the display device 12A illustrated in FIG. 18, the region ALPa_C[1] and the region ALPb_C[1] are electrically connected to the driver circuit SD[1] and the region ALPa_C[h] and the region ALPb_C[h] are electrically connected to the driver circuit SD[h] in the display device 14 of FIG. 21; therefore, the display device 14 of FIG. 21 can perform writing of image data by the same operation method as the display device 12A illustrated in FIG. 18.

For example, the display device 14 of FIG. 21 has a structure in which one driver circuit SD transmits image data to the pixel circuits of each of the region ALPa_C and the region ALPb_C included in the pixel array ALP; however, one driver circuit SD may transmit image data to the pixel circuits of three or more regions.

Specifically, the display device 14 of FIG. 21 can be changed into the structure of a display device 14A illustrated in FIG. 22. The display device 14A includes the pixel array ALP including a plurality of pixel circuits PIX arranged in a matrix of V rows and W columns (here V is a multiple of 2 and more than or equal to 1 and W is a multiple of h and

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more than or equal to 1) as in the display device of FIG. 17. Furthermore, the display device 14A has a structure in which the pixel array ALP is divided every plurality of rows; for example, the pixel array ALP is divided into d regions. In FIG. 22, the region ALPa[1] to the region ALPa[d] are illustrated as the divided d regions. Furthermore, in the structure of the display device 14A, each of the region ALPa[1] to the region ALPa[d] is divided every plurality of columns; for example, each of the region ALPa[1] to the region ALPa[d] is divided into h regions. FIG. 22 illustrates an example where the region ALPa[1] includes the region ALPa[1]_C[1] to the region ALPa[1]_C[h], the region ALPa[2] includes the region ALPa[2]_C[1] to the region ALPa[2]_C[h], and the region ALPa[d] includes the region ALPa[d]_C[1] to the region ALPa[d]_C[h].

In the display device 14A of FIG. 20, the pixel circuit PIX[1,1] included in the region ALPa[1]_C[1], the pixel circuit PIX[1,W-W/h+1] included in the region ALPa[1]_C[h], the pixel circuit PIX[V/d+1,1] included in the region ALPa[2]_C[1], the pixel circuit PIX[V/d+1, W-W/h+1] included in the region ALPa[2]_C[h], the pixel circuit PIX[V-V/d+1,1] included in the region ALPa[d]_C[1], and the pixel circuit PIX[V-V/d+1,W-W/h+1] included in the region ALPa[d]_C[h] are selectively illustrated.

Moreover, in the display device 14A of FIG. 22, the wiring GL[1] to the wiring GL[V] are extended in the row direction in the pixel array ALP. Note that the wiring GL[1], the wiring GL[V/d+1], and the wiring GL[V-V/d+1] of the wiring GL[1] to the wiring GL[V] are selectively illustrated in FIG. 22.

In the display device 14A of FIG. 22, d source wirings are extended in each column in the pixel array ALP. Specifically, for example, the wiring SLa[1]_1[1] to the wiring SLa[d]_1[1] are extended in the first column of the pixel array ALP, and the wiring SLa[1]_W-W/h+1 to the wiring SLa[d]_W-W/h+1 are extended in the W-W/h+1-th column of the pixel array ALP. Note that the wiring SLa[1]_1[1], the wiring SLa[2]_1[1], the wiring SLa[d]_1[1], the wiring SLa[1]_W-W/h+1, the wiring SLa[2]_W-W/h+1, and the wiring SLa[d]_W-W/h+1 are selectively illustrated in FIG. 22.

Note that [x]_y added to the wiring SLa represents the address of the column of the pixel array ALP and the kind of the region ALPa including the pixel circuits to which the wiring SLa (the wiring SLb) is electrically connected. Specifically, for example, the wiring SLa[2]_d-1 denotes that it is a wiring of the second column of the pixel array ALP and electrically connected to the pixel circuits PIX included in the region ALPa[d-1].

In the display device 14A of FIG. 22, the plurality of pixel circuits PIX positioned in the first row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[1]. The plurality of pixel circuits PIX positioned in the V/d+1-th row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[V/d+1]. The plurality of pixel circuits PIX positioned in the V-V/d+1-th row of the pixel array ALP are electrically connected to the driver circuit GD through the wiring GL[V-V/d+1].

In the display device 14A of FIG. 22, the pixel circuits PIX that are positioned in the first column of the pixel array ALP and included in the region ALPa[1]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLa[1]_1[1]. The pixel circuits PIX that are positioned in the first column of the pixel array ALP and included in the region ALPa[2]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLa[2]_1[1]. The pixel circuits PIX that are positioned in the first column of the

pixel array ALP and included in the region ALPa[d]_C[1] are electrically connected to the driver circuit SD[1] through the wiring SLa[d]_C[1].

With the structure of the display device 14A illustrated in FIG. 22, one driver circuit SD can transmit image data to the pixel circuits included in each of three or more regions.

Furthermore, for example, the display device 12 has a structure in which the wiring GL[1] to the wiring GL[V] are each extended in the corresponding one row as a gate wiring; however, the display device 12 may have a structure in which two gate wirings are extended in one row.

A display device 12B illustrated in FIG. 23 shows a structure example in which two gate wirings are extended in one row in the display device 12 of FIG. 17. Specifically, the display device 12B illustrated in FIG. 23 shows a structure example selectively illustrating each of the region ALPa_C[1] and the region ALPb_C[1] of the pixel array ALP and the driver circuit SD[1] in the display device 12 of FIG. 17, in which each of a wiring GL[1]-1 to a wiring GL[V]-1 and a wiring GL[1]-2 to a wiring GL[V]-2 functions as a gate wiring of the display device 12B illustrated in FIG. 23. Note that W/h is 3 in the display device 12B of FIG. 23.

In the pixel array ALP of the display device 12B of FIG. 23, the region ALPa_C[1] includes, for example, the pixel circuit PIX[1,1] to the pixel circuit PIX[V/2,3], and the region ALPb_C[1] includes, for example, the pixel circuit PIX[V/2+1,1] to the pixel circuit PIX[V,3]. Since W/h is 3, the wiring SLa[1] to the wiring SLa[3] are extended in the respective columns in the region ALPa_C[1], and the wiring SLb[1] to the wiring SLb[3] are extended in the respective columns in the region ALPb_C[1].

The display device 12B of FIG. 23 has a structure in which the wiring GL[1]-1 and the wiring GL[1]-2 are extended in the first row as a pair of gate wirings, the wiring GL[V/2]-1 and the wiring GL[V/2]-2 are extended in the V/2-th row as a pair of gate wirings, the wiring GL[V/2+1]-1 and the wiring GL[V/2+1]-2 are extended in the V/2+1-th row as a pair of gate wirings, and the wiring GL[V]-1 and the wiring GL[V]-2 are extended in the V-th row as a pair of gate wirings.

In the display device 12B of FIG. 23, each of the plurality of pixel circuits PIX included in the pixel array ALP includes the transistor Tr as a write transistor for image data. A first terminal of the transistor Tr included in each of the plurality of pixel circuits PIX is electrically connected to a source wiring of the column of the corresponding pixel circuit PIX (any one of the wiring SLa[1] to the wiring SLa[3] and the wiring SLb[1] to the wiring SLb[3]), and a second terminal of the transistor Tr is electrically connected to a circuit element or the like of the pixel circuit PIX.

A gate of the transistor Tr included in the pixel circuit PIX positioned in an odd-numbered column is electrically connected to one of the gate wirings of the row of the pixel circuit PIX (any one of the wiring GL[1]-1 to the wiring GL[V]-1). A gate of the transistor Tr included in the pixel circuit PIX positioned in an even-numbered column is electrically connected to the other of the gate wirings of the row of the pixel circuit PIX (any one of the wiring GL[1]-2 to the wiring GL[V]-2).

With the structure of the display device 12B in which two gate wirings are extended in one row as illustrated in FIG. 23, whereby, in a certain row, switching between the on state and the off state of the transistor Tr included in the pixel circuit PIX in an odd-numbered column and switching between the on state and the off state of the transistor Tr included in the pixel circuit PIX in the even-numbered column can be performed independently of each other. For

example, the display device 12B can be operated such that the transistor Tr included in the pixel circuit in one of an odd-numbered column and an even-numbered column is turned off and image data is written to the pixel circuit PIX in the other of the odd-numbered column and the even-numbered column. In other words, the structure of the display device 12B makes it possible to freely reduce the amount of data of an image to be displayed on the pixel array ALP, and thus the frame frequency of the display device 12B can be increased. Furthermore, for example, the transistor Tr of the pixel circuit PIX in an even-numbered column can be turned on in the middle of writing image data to the pixel circuit PIX in an odd-numbered column. In this manner, part of a write period of an odd-numbered column and part of a write period of an even-numbered column can overlap with each other, which can reduce a write time of an image of the whole pixel array ALP of the display device 12B.

Note that in the display device 12B of FIG. 23, the gate of the transistor Tr included in the pixel circuit PIX of an odd-numbered column is electrically connected to one of the gate wirings of the row of the pixel circuit PIX (any one of the wiring GL[1]-1 to the wiring GL[V]-1), and the gate of the transistor Tr included in the pixel circuit PIX of an even-numbered column is electrically connected to the other of the gate wirings of the row of the pixel circuit PIX (any one of the wiring GL[1]-2 to the wiring GL[V]-2); however, the structure of the display device of one embodiment of the present invention is not limited thereto. For example, the display device of one embodiment of the present invention may have a structure in which, in the pixel circuits PIX of four consecutive columns included in a certain row of the pixel array ALP, the gate of each of the transistors Tr included in the pixel circuits PIX of the initial two consecutive columns is electrically connected to one of the gate wirings of the row (any one of the wiring GL[1]-1 to the wiring GL[V]-1) and the gate of each of the transistors Tr included in the pixel circuits PIX of the remaining two consecutive columns is electrically connected to the other of the gate wirings of the row (any one of the wiring GL[1]-2 to the wiring GL[V]-2). Furthermore, for example, the display device of one embodiment of the present invention may have a structure in which, in a plurality of pixel circuits PIX included in a certain row of the pixel array ALP, the gates of the transistors Tr included in the pixel circuits PIX selected from the plurality of pixel circuits PIX are electrically connected to one of the gate wirings of the row (any one of the wiring GL[1]-1 to the wiring GL[V]-1), and the gates of the transistors Tr included in the remaining pixel circuits PIX are electrically connected to the other of the gate wirings of the row (any one of the wiring GL[1]-2 to the wiring GL[V]-2).

Next, a structure example of the driver circuit SD is described. A display device 12C illustrated in FIG. 24 shows a structure example selectively illustrating the region ALPa_C[1] and the region ALPb_C[1] of the pixel array ALP and the driver circuit SD[1] in the display device 12 of FIG. 17. Note that W/h is 3 in the display device 12C illustrated in FIG. 24.

In the pixel array ALP of the display device 12C of FIG. 24, the region ALPa_C[1] includes, for example, the pixel circuit PIX[1,1] to the pixel circuit PIX[V/2,3], and the region ALPb_C[1] includes, for example, the pixel circuit PIX[V/2+1,1] to the pixel circuit PIX[V,3]. Since W/h is 3, the wiring SLa[1] to the wiring SLa[3] are extended in the respective columns in the region ALPa_C[1], and the wiring SLb[1] to the wiring SLb[3] are extended in the respective columns in the region ALPb_C[1]. Furthermore, the wiring

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GL[1] to the wiring GL[V/2] are extended in the respective rows in the region ALPa_C[1], and the wiring GL[V/2+1] to the wiring GL[V] are extended in the respective columns in the region ALPb_C[1].

The driver circuit SD[1] illustrated in FIG. 24 includes a demultiplexer DMU and the circuit OTH, for example.

For the demultiplexer DMU illustrated in FIG. 24, the descriptions of the demultiplexer DMUa and the demultiplexer DMUb in Embodiment 2 are referred to. Note that the demultiplexer DMU illustrated in FIG. 24 differs from the demultiplexer DMUa and the demultiplexer DMUb in including six output terminals.

For the circuit OTH illustrated in FIG. 24, the description of the circuit OTH in Embodiment 2 is referred to.

The demultiplexer DMU includes the switch SWa[1] to the switch SWa[3] and the switch SWb[1] to the switch SWb[3], for example.

A first terminal of the switch SWa[1] is electrically connected to the wiring SLa[1], a second terminal of the switch SWa[1] is electrically connected to an output terminal of the circuit OTH, and a control terminal of the switch SWa[1] is electrically connected to the wiring SGa[1]. A first terminal of the switch SWa[2] is electrically connected to the wiring SLa[2], a second terminal of the switch SWa[2] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWa[2] is electrically connected to the wiring SGa[2]. A first terminal of the switch SWa[3] is electrically connected to the wiring SLa[3], a second terminal of the switch SWa[3] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWa[3] is electrically connected to the wiring SGa[3]. A first terminal of the switch SWb[1] is electrically connected to the wiring SLb[1], a second terminal of the switch SWb[1] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[1] is electrically connected to the wiring SGb[1]. A first terminal of the switch SWb[2] is electrically connected to the wiring SLb[2], a second terminal of the switch SWb[2] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[2] is electrically connected to the wiring SGb[2]. A first terminal of the switch SWb[3] is electrically connected to the wiring SLb[3], a second terminal of the switch SWb[3] is electrically connected to the output terminal of the circuit OTH, and a control terminal of the switch SWb[3] is electrically connected to the wiring SGb[3].

An input terminal of the circuit OTH is electrically connected to a wiring DL.

The switches that can be used as the switch SWa[1] to the switch SWa[3] and the switch SWb[1] to the switch SWb[3] described in Embodiment 2 can be used as the switch SWa[1] to the switch SWa[3] and the switch SWb[1] to the switch SWb[3]. Therefore, for an operation example of the switch SWa[1] to the switch SWa[3] and the switch SWb[1] to the switch SWb[3] in this embodiment, the descriptions of the switch SWa[1] to the switch SWa[4] and the switch SWb[1] to the switch SWb[4] in Embodiment 2 are referred to.

In addition, like the wiring SGa[1] to the wiring SGa[4] and the wiring SGb[1] to the wiring SGb[4] described in Embodiment 2, the wiring SGa[1] to the wiring SGa[3] and the wiring SGb[1] to the wiring SGb[3] each function as a wiring that supplies a voltage for switching between the conduction state and non-conduction state of the switch whose control terminal is electrically connected to the

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wiring. Thus, the voltage can be, for example, a high-level potential or a low-level potential.

As described above, in the demultiplexer DMU, for example, a pulse voltage with a high-level potential is applied to any one of the wiring SGa[1] to the wiring SGa[3] and the wiring SGb[1] to the wiring SGb[3], whereby the switch whose control terminal is supplied with the pulse voltage can be turned on during a period when the pulse voltage is applied. Furthermore, in the demultiplexer DMU, for example, a pulse voltage with a high-level potential is sequentially supplied to the wiring SGa[1] and the wiring SGa[3] and the wiring SGb[1] to the wiring SGb[3], whereby the switch SWa[1] to the switch SWa[3] and the switch SWb[1] to the switch SWb[3] can be turned on sequentially. Thus, the demultiplexer DMU can sequentially select the wiring SLa[1] to the wiring SLa[3] and the wiring SLb[1] to the wiring SLb[3] as destinations to which image data is to be transmitted.

The structure of the display device of one embodiment of the present invention is not limited to the structure of the display device 12C. The display device of one embodiment of the present invention can have a structure of the display device 12C that is changed appropriately, for example.

Two source wirings extended in the same column (e.g., the pair of the wiring SLa[1] and the wiring SLb[1] or the pair of the wiring SLa[2] and the wiring SLb[2]) in the pixel array ALP are electrically connected to two output terminals of the demultiplexer DMU included in the driver circuit SD in the display device 12C; however, the display device 12C may include a structure in which two output terminals of the demultiplexer DMU are not electrically connected to two source wirings extended to the same column. In other words, the display device 12C may include a structure in which one of the wiring SLa[x] and the wiring SLb[x] (here x is an integer of 1 or more and 3 or less) is electrically connected to a plurality of output terminals of the demultiplexer and the other of the wiring SLa[x] and the wiring SLb[x] is not electrically connected to the plurality of output terminals of the demultiplexer.

A display device 12D illustrated in FIG. 25 is a modification example of the display device 12C, i.e., a display device including a structure in which one of the wiring SLa[x] and the wiring SLb[x] is electrically connected to a plurality of output terminals of the demultiplexer and the other of the wiring SLa[x] and the wiring SLb[x] is not electrically connected to the plurality of output terminals of the demultiplexer. Specifically, the display device 12D has a structure of the display device 12C in which the switch SWb[1] and the switch SWa[3] are not provided in the demultiplexer DMU.

In the structure of the display device 12D of FIG. 25, the region ALPa_C[1] includes the pixel circuits PIX of a range from the first row and the first column to the V/2-th row and the second column in the pixel array ALP, the region ALPa_C[2] includes the pixel circuits PIX of a range from the first row and the third column to the V/2-th row and the third column, the region ALPb_C[1] includes the pixel circuits PIX of a range from the V/2+1-th row and the first column to the V-th row and the first column in the pixel array ALP, and the region ALPb_C[2] includes the pixel circuits PIX of a range from the V/2+1-th row and the third column to the V-th row and the third column. That is, the region ALPa_C[1] and the region ALPb_C[2] electrically connected to the driver circuit SD[1] include columns different from each other in the pixel array ALP.

Note that the pixel circuit PIX[1,1], the pixel circuit PIX[V/2,2], the pixel circuit PIX[1,3], the pixel circuit

PIX[V/2,3], the pixel circuit PIX[V/2+1,1], the pixel circuit PIX[V,1], the pixel circuit PIX[V/2+1,2], and the pixel circuit PIX[V,3] are selectively illustrated in the pixel array ALP of FIG. 25.

In the display device 12D of FIG. 25, the driver circuit SD[1] has a function of driving the pixel circuits PIX included in the region ALPa_C[1] and the region ALPb_C[2]. Note that the pixel circuits PIX included in the region ALPa_C[2] and the region ALPb_C[1] are preferably driven by another driver circuit SD different from the driver circuit SD[1]. In other words, it is preferable that each of the wiring SLa[3] and the wiring SLb[1] be electrically connected to the demultiplexer DMU of the driver circuit SD different from the driver circuit SD[1].

Although the wiring SLa[3] and the wiring SLb[1] are not electrically connected to the driver circuit SD[1] in the structure of the display device 12D of FIG. 25, the display device of one embodiment of the present invention, for example, may have a structure of the display device 12D in which one of the wiring SLa[2] and the wiring SLb[2] is not electrically connected to the driver circuit SD[1]. Furthermore, for example, a structure of the display device 12 of FIG. 17 in which the wiring SLa and the wiring SLb in the same column are electrically connected to different driver circuits SD may be employed.

The driver circuit SD[1] is electrically connected to the pixel circuits PIX in the region ALPa_C[1] and the region ALPb_C[1] and the driver circuit SD[h] is electrically connected to the pixel circuits PIX in the region ALPa_C[h] and the region ALPb_C[h] in the display device 12 of FIG. 17; however, one embodiment of the present invention is not limited thereto. For example, in the display device 12, the driver circuit SD[1] may be electrically connected to the pixel circuits PIX not in the region ALPa_C[1] and/or the region ALPb_C[1] but in another region. For example, the driver circuit SD[1] may be electrically connected not to the region ALPa_C[1] but to any one of the region ALPa_C[2] to the region ALPa_C[h]. The driver circuit SD[1] may be electrically connected not to the region ALPb_C[1] but to any one of the region ALPb_C[2] to the region ALPb_C[h]. In other words, the structure may be changed such that the driver circuit SD[x] (here x is an integer of 1 or more and h or less) included in the display device 12 is electrically connected to any one of the region ALPa_C[1] to the region ALPa_C[h] and any one of the region ALPb_C[1] to the region ALPb_C[h].

It is possible to combine the display device 12 of FIG. 17 with the display device 10 described in Embodiment 1. As an example of the combination structure, the pixel region ARA of the display device 10 of Embodiment 1 can be used as the pixel array ALP of the display device 12 of FIG. 17 and the plurality of driver circuits SD of FIG. 17 can be included in the local driver circuit LD of the display device 10 in Embodiment 1.

A display device 12E illustrated in FIG. 26A has a structure of the display device 10 in Embodiment 1, in which the pixel region ARA is divided into a region ARAa and a region ARAb. Source wirings electrically connected to the pixel circuits PIX included in the region ARAa are referred to as the wiring SLa_1 to the wiring SLa_p (here p is an integer of 1 or more), and source wirings electrically connected to the pixel circuits PIX included in the region ARAb are referred to as the wiring SLb_1 to the wiring SLb_p. Note that FIG. 26A selectively illustrates the wiring SLa_1, the wiring SLa_2, the wiring SLa_3, the wiring SLa_p, the wiring SLb_1, the wiring SLb_2, the wiring SLb_3, and the wiring SLb_p.

As described above, the plurality of driver circuits SD illustrated in FIG. 17 are included in the local driver circuit LD of the display device 12E of FIG. 26.

FIG. 26B illustrates a specific example of the pixel region ARA included in the display device 12E of FIG. 26A.

The wiring SLa_1 to the wiring SLa_p are extended in the respective columns in the region ARAa of FIG. 26B, and the wiring SLb_1 to the wiring SLb_p are extended in the respective columns in the region ARAb of FIG. 26B. Note that FIG. 26B selectively illustrates the wiring SLa_1, the wiring SLa_2, the wiring SLa_3, the wiring SLb_1, the wiring SLb_2, and the wiring SLb_3. In the pixel region ARA of FIG. 26B, the wiring SLa_1 and the wiring SLb_1 are extended in the same first column, the wiring SLa_2 and the wiring SLb_2 are extended in the same second column, and the wiring SLa_3 and the wiring SLb_3 are extended in the same third column.

The plurality of pixel circuits PIX are arranged in a matrix in the pixel region ARA of FIG. 26B. Note that the number of pixel circuits PIX included in the region ARAa and the number of pixel circuits PIX included in the region ARAb are preferably equal to each other. Furthermore, the plurality of pixel circuits PIX are electrically connected to one another in the respective columns with the wiring SLa_1 to the wiring SLa_p in the region ARAa, and the plurality of pixel circuits PIX are electrically connected to one another in the respective columns with the wiring SLb_1 to the wiring SLb_p in the region ARAb.

As illustrated in FIG. 26, combining the display device 12 of FIG. 17 with the display device 10 described in Embodiment 1 enables a reduction in the write time of image data also in the display device with high resolution.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 4

In this embodiment, a structure example of a display device of one embodiment of the present invention is described.

Structure Example of Display Device

FIG. 27 is a cross-sectional view illustrating an example of a display device of one embodiment of the present invention. The display device 10 illustrated in FIG. 27 includes a pixel circuit and a driver circuit over a substrate 310, for example.

The substrate 310 corresponds to the substrate BS described in the above embodiment, for example. Therefore, a material usable for the substrate BS can be used for the substrate 310.

In the description of this embodiment, the substrate 310 is a semiconductor substrate containing silicon or the like as a material.

The display device 10 includes the transistor 300, a transistor 500, and the light-emitting device 150a to the light-emitting device 150c over the substrate 310.

The transistor 300 is provided on the substrate 310 and includes an element isolation layer 312, a conductor 316, an insulator 315, an insulator 317, the semiconductor region 313 that is part of the substrate 310, and the low-resistance region 314a and a low-resistance regions 314b that function as source and drain regions. Accordingly, the transistor 300 is a transistor (Si transistor) whose channel formation region contains silicon. Although FIG. 27 illustrates a structure in

which one of the source region and the drain region of the transistor **300** is electrically connected to a conductor **330**, a conductor **356**, and a conductor **366**, which are described later, through a conductor **328** described later, the electrical connection in the semiconductor device of one embodiment of the present invention is not limited thereto. In the semiconductor device of one embodiment of the present invention, for example, the other of the source and the drain of the transistor **300** may be electrically connected to the conductor **330**, the conductor **356**, and the conductor **366** through the conductor **328**, or alternatively, a gate of the transistor **300** may be electrically connected to the conductor **330**, the conductor **356**, and the conductor **366** through the conductor **328**.

The transistor **300** can be a fin type when, for example, the top surface of the semiconductor region **313** and the side surface thereof in the channel width direction are covered with the conductor **316** with the insulator **315** as a gate insulating film therebetween. The effective channel width is increased in the fin-type transistor **300**, whereby the on-state characteristics of the transistor **300** can be improved. In addition, contribution of the electric field of the gate electrode can be increased, so that the off-state characteristics of the transistor **300** can be improved.

Note that the transistor **300** may be either a p-channel transistor or an n-channel transistor. Alternatively, a plurality of transistors **300** may be provided and both the p-channel transistor and the n-channel transistor may be used.

A region of the semiconductor region **313** where a channel is formed, a region in the vicinity thereof, the low-resistance region **314a** and the low-resistance region **314b** functioning as a source region and a drain region, and the like preferably contain a semiconductor such as a silicon-based semiconductor, and preferably contain single crystal silicon. Alternatively, the regions may be formed using a material containing Ge (germanium), SiGe (silicon germanium), GaAs (gallium arsenide), GaAlAs (gallium aluminum arsenide), GaN (gallium nitride), or the like. A structure using silicon whose effective mass is controlled by applying stress to the crystal lattice and changing the lattice spacing may be employed. Alternatively, the transistor **300** may be a HEMT (High Electron Mobility Transistor) with GaAs and GaAlAs, or the like.

For the conductor **316** functioning as a gate electrode, a semiconductor material such as silicon containing an element that imparts n-type conductivity, such as arsenic or phosphorus, or an element that imparts p-type conductivity, such as boron, or a conductive material such as a metal material, an alloy material, or a metal oxide material can be used.

Note that since the work function of a conductor depends on the material of the conductor, the threshold voltage of the transistor can be adjusted by selecting the material of the conductor. Specifically, it is preferable to use a material such as titanium nitride or tantalum nitride for the conductor. Moreover, in order to ensure both conductivity and embeddability, it is preferable to use stacked layers of metal materials such as tungsten and aluminum for the conductor, and it is particularly preferable to use tungsten in terms of heat resistance.

The element isolation layer **312** is provided to separate a plurality of transistors on the substrate **310** from each other. The element isolation layer can be formed by, for example, a LOCOS (Local Oxidation of Silicon) method, an STI (Shallow Trench Isolation) method, a mesa isolation method, or the like.

Note that the transistor **300** illustrated in FIG. **27** is an example and the structure is not limited thereto; an appropriate transistor is used in accordance with a circuit configuration, a driving method, or the like. For example, the transistor **300** may have a planar structure instead of a fin-type structure.

Over the transistor **300** illustrated in FIG. **27**, an insulator **320**, an insulator **322**, an insulator **324**, and an insulator **326** are sequentially stacked from the substrate **310** side.

For the insulator **320**, the insulator **322**, the insulator **324**, and the insulator **326**, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, or aluminum nitride can be used, for example.

Note that in this specification, silicon oxynitride refers to a material that has a higher oxygen content than a nitrogen content, and silicon nitride oxide refers to a material that has a higher nitrogen content than an oxygen content. Moreover, in this specification, aluminum oxynitride refers to a material that has a higher oxygen content than a nitrogen content, and aluminum nitride oxide refers to a material that has a higher nitrogen content than an oxygen content.

The insulator **322** may have a function of a planarization film for eliminating a level difference caused by the transistor **300** or the like covered with the insulator **320** and the insulator **322**. For example, the top surface of the insulator **322** may be planarized by planarization treatment using a chemical mechanical polishing (CMP) method or the like to improve planarity.

For the insulator **324**, it is preferable to use a barrier insulating film preventing diffusion of water, hydrogen, impurities and the like from the substrate **310**, the transistor **300**, or the like to a region above the insulator **324** (e.g., the region where the transistor **500**, the light-emitting device **150a** to the light-emitting device **150c**, and the like are provided). Accordingly, for the insulator **324**, it is preferable to use an insulating material that has a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, and a water molecule (through which the above-described impurities are unlikely to pass). Furthermore, depending on the situation, for the insulator **324**, it is preferable to use an insulating material having a function of inhibiting diffusion of impurities such as a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule (N_2O , NO , NO_2 , or the like), and a copper atom (through which the above-described oxygen is unlikely to pass). Alternatively, it is preferable that the insulator **324** have a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like).

For the film having a barrier property against hydrogen, silicon nitride formed by a CVD (Chemical Vapor Deposition) method can be used, for example. Here, diffusion of hydrogen into a semiconductor element including an oxide semiconductor, such as the transistor **500**, degrades the characteristics of the semiconductor element in some cases. Therefore, a film that inhibits hydrogen diffusion is preferably used between the transistor **500** and the transistor **300**. The film that inhibits hydrogen diffusion is specifically a film from which a small amount of hydrogen is released.

The amount of released hydrogen can be analyzed by thermal desorption spectroscopy (TDS), for example. The amount of hydrogen released from the insulator **324** that is converted into hydrogen atoms per unit area of the insulator **324** is less than or equal to 10×10^{15} atoms/cm², preferably less than or equal to 5×10^{15} atoms/cm² in TDS analysis in a film-surface temperature range of 50° C. to 500° C., for example.

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Note that the permittivity of the insulator 326 is preferably lower than that of the insulator 324. For example, the relative permittivity of the insulator 326 is preferably lower than 4, further preferably lower than 3. The relative permittivity of the insulator 326 is, for example, preferably 0.7 times or less, further preferably 0.6 times or less the relative permittivity of the insulator 324. When a material with a low permittivity is used for the interlayer film, the parasitic capacitance generated between wirings can be reduced.

In addition, the conductor 328, the conductor 330, and the like that are connected to the light-emitting devices and the like provided above the insulator 326 are embedded in the insulator 320, the insulator 322, the insulator 324, and the insulator 326. Note that the conductor 328, the conductor 330, and the like function as plugs or wirings. A plurality of conductors functioning as plugs or wirings are collectively denoted by the same reference numeral in some cases. Moreover, in this specification and the like, a wiring and a plug connected to the wiring may be a single component. That is, part of a conductor functions as a wiring in some cases and part of a conductor functions as a plug in other cases.

As a material of each of plugs and wirings (e.g., the conductor 328 and the conductor 330), a single layer or a stacked layer of a conductive material such as a metal material, an alloy material, a metal nitride material, or a metal oxide material can be used. It is preferable to use a high-melting-point material that has both heat resistance and conductivity, such as tungsten or molybdenum, and it is preferable to use tungsten. Alternatively, a low-resistance conductive material such as aluminum or copper is preferably used. The use of a low-resistance conductive material can reduce wiring resistance.

A wiring layer may be provided over the insulator 326 and the conductor 330. For example, in FIG. 27, an insulator 350, an insulator 352, and an insulator 354 are provided to be stacked in this order above the insulator 326 and the conductor 330. Furthermore, the conductor 356 is formed in the insulator 350, the insulator 352, and the insulator 354. The conductor 356 has a function of a plug or a wiring that is connected to the transistor 300. Note that the conductor 356 can be provided using a material similar to those for the conductor 328 and the conductor 330.

For example, like the insulator 324, the insulator 350 is preferably formed using an insulator having a barrier property against impurities such as hydrogen and water. The insulator 352 and the insulator 354 are preferably formed using an insulator having a comparatively low relative permittivity to reduce parasitic capacitance generated between wirings, like the insulator 326. The insulator 352 and the insulator 354 have functions of an interlayer insulating film and a planarization film. The conductor 356 preferably contains a conductor having a barrier property against impurities such as hydrogen and water. In particular, the conductor having a barrier property against hydrogen is formed in an opening portion included in the insulator 350 having a barrier property against hydrogen. With this structure, the transistor 300 and the transistor 500 can be separated by the barrier layer, so that diffusion of hydrogen from the transistor 300 into the transistor 500 can be inhibited.

For the conductor having a barrier property against hydrogen, tantalum nitride or the like is preferably used, for example. The use of a stack including tantalum nitride and tungsten that has high conductivity can inhibit diffusion of hydrogen from the transistor 300 while the conductivity of a wiring is kept. In that case, a tantalum nitride layer having

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a barrier property against hydrogen is preferably in contact with the insulator 350 having a barrier property against hydrogen.

An insulator 360, an insulator 362, and an insulator 364 are stacked in this order over the insulator 354 and the conductor 356.

Like the insulator 324 or the like, the insulator 360 is preferably formed using an insulator having a barrier property against impurities such as water and hydrogen. Thus, the insulator 360 can be formed using any of the materials usable for the insulator 324 or the like, for example.

The insulator 362 and the insulator 364 have functions of an interlayer insulating film and a planarization film. Like the insulator 324, the insulator 362 and the insulator 364 are preferably formed using an insulator having a barrier property against impurities such as water and hydrogen. Thus, the insulator 362 and/or the insulator 364 can be formed using any of the materials usable for the insulator 324.

An opening portion is provided in regions of the insulator 360, the insulator 362, and the insulator 364 that overlap with part of the conductor 356, and a conductor 366 is provided to fill the opening portion. The conductor 366 is also formed over the insulator 362. The conductor 366 has a function of a plug or a wiring connected to the transistor 300, for example. Note that the conductor 366 can be provided using a material similar to those for the conductor 328 and the conductor 330.

An insulator 512 is provided above the insulator 364 and the conductor 366. A substance having a barrier property against oxygen or hydrogen is preferably used for the insulator 512. The insulator 512 is formed using, for example, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, or aluminum nitride.

For the film having a barrier property against hydrogen, silicon nitride deposited by a CVD method can be used, for example. Here, diffusion of hydrogen into a semiconductor element including an oxide semiconductor, such as the transistor 500, degrades the characteristics of the semiconductor element in some cases. Therefore, a film that inhibits hydrogen diffusion is preferably used between the transistor 500 and the transistor 300. The film that inhibits hydrogen diffusion is specifically a film from which a small amount of hydrogen is released.

A material similar to that for the insulator 320 can be used for the insulator 512, for example. When a material with a comparatively low permittivity is used for these insulators, parasitic capacitance generated between wirings can be reduced. A silicon oxide film, a silicon oxynitride film, or the like can be used as the insulator 512, for example.

The transistor 500 that is an OS transistor is provided over the insulator 512, for example.

Details of the transistor 500 are described here. FIG. 28A and FIG. 28B illustrate a structure example of the transistor 500 that is an OS transistor. FIG. 28A is a cross-sectional view of the OS transistor in the channel length direction, and FIG. 28B is a cross-sectional view of the OS transistor in the channel width direction.

As illustrated in FIG. 28A and FIG. 28B, an insulator 514 and an insulator 516 are formed over the insulator 512.

As the insulator 514, it is preferable to use a film having a barrier property that prevents diffusion of hydrogen or impurities from the substrate 310, a region where the circuit element or the like below the insulator 512 is provided, or the like into a region where the transistor 500 is provided. Thus, silicon nitride deposited by a CVD method can be used for the insulator 514, for example.

For the insulator **516**, a material similar to that for the insulator **512** can be used, for example.

As illustrated in FIG. **28A** and FIG. **28B**, the transistor **500** includes the insulator **516** over the insulator **514**, a conductor **503** (a conductor **503a** and a conductor **503b**) provided to be embedded in the insulator **514** or the insulator **516**, an insulator **522** over the insulator **516** and the conductor **503**, an insulator **524** over the insulator **522**, an oxide **530a** over the insulator **524**, an oxide **530b** over the oxide **530a**, a conductor **542a** over the oxide **530b**, an insulator **571a** over the conductor **542a**, a conductor **542b** over the oxide **530b**, an insulator **571b** over the conductor **542b**, an insulator **552** over the oxide **530b**, an insulator **550** over the insulator **552**, an insulator **554** over the insulator **550**, a conductor **560** (a conductor **560a** and a conductor **560b**) that is over the insulator **554** and overlaps with part of the oxide **530b**, and an insulator **544** provided over the insulator **522**, the insulator **524**, the oxide **530a**, the oxide **530b**, the conductor **542a**, the conductor **542b**, the insulator **571a**, and the insulator **571b**. Here, as illustrated in FIG. **28A** and FIG. **28B**, the insulator **552** is in contact with the top surface of the insulator **522**, the side surface of the insulator **524**, the side surface of the oxide **530a**, the side surface and the top surface of the oxide **530b**, the side surface of the conductor **542**, the side surface of the insulator **571**, the side surface of the insulator **544**, the side surface of an insulator **580**, and the bottom surface of the insulator **550**. The top surface of the conductor **560** is placed to be substantially level with the upper portion of the insulator **554**, the upper portion of the insulator **550**, the upper portion of the insulator **552**, and the top surface of the insulator **580**. An insulator **574** is in contact with part of at least one of the top surface of the conductor **560**, the upper portion of the insulator **552**, the upper portion of the insulator **550**, the upper portion of the insulator **554**, and the top surface of the insulator **580**.

An opening reaching the oxide **530b** is provided in the insulator **580** and the insulator **544**. The insulator **552**, the insulator **550**, the insulator **554**, and the conductor **560** are provided in the opening. The conductor **560**, the insulator **552**, the insulator **550**, and the insulator **554** are provided between the conductor **542a** and the conductor **542b** and between the insulator **571a** and the insulator **571b** in the channel length direction of the transistor **500**. The insulator **554** includes a region in contact with the side surface of the conductor **560** and a region in contact with the bottom surface of the conductor **560**.

The oxide **530** preferably includes the oxide **530a** provided over the insulator **524** and the oxide **530b** provided over the oxide **530a**. Including the oxide **530a** under the oxide **530b** makes it possible to inhibit diffusion of impurities into the oxide **530b** from components formed below the oxide **530a**.

Although a structure in which two layers, the oxide **530a** and the oxide **530b**, are stacked as the oxide **530** in the transistor **500** is described, the present invention is not limited thereto. For example, the transistor **500** can include a single-layer structure of the oxide **530b** or a stacked-layer structure of three or more layers. Alternatively, the oxide **530a** and the oxide **530b** can each have a stacked-layer structure.

The conductor **560** functions as a first gate (also simply referred to as a gate) electrode, and the conductor **503** functions as a second gate (also referred to as a back gate) electrode. The insulator **552**, the insulator **550**, and the insulator **554** function as a first gate insulator, and the insulator **522** and the insulator **524** function as a second gate insulator. Note that the gate insulator is also referred to as a

gate insulating layer or a gate insulating film in some cases. The conductor **542a** functions as one of a source and a drain, and the conductor **542b** functions as the other of the source and the drain. At least part of a region of the oxide **530** that overlaps with the conductor **560** functions as a channel formation region.

FIG. **29A** is an enlarged view of the vicinity of the channel formation region in FIG. **28A**. Supply of oxygen to the oxide **530b** forms the channel formation region in a region between the conductor **542a** and the conductor **542b**. As illustrated in FIG. **29A**, the oxide **530b** includes a region **530bc** functioning as the channel formation region of the transistor **500** and a region **530ba** and a region **530bb** that are provided to sandwich the region **530bc** and function as a source region and a drain region. At least part of the region **530bc** overlaps with the conductor **560**. In other words, the region **530bc** is provided between the conductor **542a** and the conductor **542b**. The region **530ba** is provided to overlap with the conductor **542a**, and the region **530bb** is provided to overlap with the conductor **542b**.

The region **530bc** functioning as the channel formation region has a smaller amount of oxygen vacancies (an oxygen vacancy in a metal oxide is sometimes referred to as V_O in this specification and the like) or a lower impurity concentration than the region **530ba** and the region **530bb** to be a high-resistance region having a low carrier concentration. Thus, the region **530bc** can be regarded as being i-type (intrinsic) or substantially i-type.

A transistor using a metal oxide is likely to change its electrical characteristics when impurities or oxygen vacancies (V_O) exist in a region of the metal oxide where a channel is formed, which might degrade the reliability. In some cases, hydrogen in the vicinity of an oxygen vacancy (V_O) forms a defect that is an oxygen vacancy (V_O) into which hydrogen enters (hereinafter, sometimes referred to as VOH), which generates an electron serving as a carrier. Therefore, when the region of the oxide semiconductor where a channel is formed includes oxygen vacancies, the transistor tends to have normally-on characteristics (even when no voltage is applied to the gate electrode, the channel exists and current flows through the transistor). Thus, impurities, oxygen vacancies, and VOH are preferably reduced as much as possible in the region of the oxide semiconductor where a channel is formed.

The region **530ba** and the region **530bb** functioning as the source region and the drain region are each a low-resistance region with an increased carrier concentration because they include a large amount of oxygen vacancies (V_O) or have a high concentration of an impurity such as hydrogen, nitrogen, or a metal element. In other words, the region **530ba** and the region **530bb** are each an n-type region having a higher carrier concentration and a lower resistance than the region **530bc**.

The carrier concentration in the region **530bc** functioning as the channel formation region is preferably lower than or equal to $1 \times 10^{18} \text{ cm}^{-3}$, further preferably lower than $1 \times 10^{17} \text{ cm}^{-3}$, still further preferably lower than $1 \times 10^{16} \text{ cm}^{-3}$, yet further preferably lower than $1 \times 10^{13} \text{ cm}^{-3}$, yet still further preferably lower than $1 \times 10^{12} \text{ cm}^{-3}$. Note that the lower limit of the carrier concentration in the region **530bc** functioning as the channel formation region is not particularly limited and can be, for example, $1 \times 10^{-9} \text{ cm}^{-3}$.

Between the region **530bc** and the region **530ba** or the region **530bb**, a region having a carrier concentration that is lower than or substantially equal to the carrier concentrations in the region **530ba** and the region **530bb** and higher than or substantially equal to the carrier concentration in the

region **530bc** may be formed. That is, the region functions as a junction region between the region **530bc** and the region **530ba** or the region **530bb**. The hydrogen concentration in the junction region is lower than or substantially equal to the hydrogen concentrations in the region **530ba** and the region **530bb** and higher than or substantially equal to the hydrogen concentration in the region **530bc** in some cases. The amount of oxygen vacancies in the junction region is smaller than or substantially equal to the amounts of oxygen vacancies in the region **530ba** and the region **530bb** and larger than or substantially equal to the amount of oxygen vacancies in the region **530bc** in some cases.

Although FIG. 29A illustrates an example in which the region **530ba**, the region **530bb**, and the region **530bc** are formed in the oxide **530b**, the present invention is not limited thereto. For example, the above regions may be formed not only in the oxide **530b** but also in the oxide **530a**.

In the oxide **530**, the boundaries between the regions are difficult to detect clearly in some cases. The concentration of a metal element and an impurity element such as hydrogen or nitrogen, which is detected in each region, may be gradually changed not only between the regions but also in each region. That is, the region closer to the channel formation region preferably has a lower concentration of a metal element and an impurity element such as hydrogen or nitrogen.

In the transistor **500**, a metal oxide functioning as a semiconductor (such a metal oxide is hereinafter also referred to as an oxide semiconductor) is preferably used for the oxide **530** (the oxide **530a** and the oxide **530b**) including the channel formation region.

The metal oxide functioning as a semiconductor preferably has a band gap of 2 eV or more, further preferably 2.5 eV or more. With the use of a metal oxide having such a wide band gap, the off-state current of the transistor can be reduced.

As the oxide **530**, it is preferable to use, for example, a metal oxide such as an In-M-Zn oxide containing indium, the element M, and zinc (the element M is one or more kinds selected from aluminum, gallium, yttrium, tin, copper, vanadium, beryllium, boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like). Alternatively, an In—Ga oxide, an In—Zn oxide, or an indium oxide may be used as the oxide **530**.

Here, the atomic ratio of In to the element M in the metal oxide used as the oxide **530b** is preferably greater than the atomic ratio of In to the element M in the metal oxide used as the oxide **530a**.

The oxide **530a** is provided under the oxide **530b** in the above manner, whereby impurities and oxygen can be inhibited from diffusing into the oxide **530b** from components formed below the oxide **530a**.

When the oxide **530a** and the oxide **530b** contain a common element (as the main component) besides oxygen, the density of defect states at an interface between the oxide **530a** and the oxide **530b** can be made low. Since the density of defect states at the interface between the oxide **530a** and the oxide **530b** can be made low, the influence of interface scattering on carrier conduction is small, and a high on-state current can be obtained.

The oxide **530b** preferably has crystallinity. It is particularly preferable to use a CAAC-OS (c-axis aligned crystalline oxide semiconductor) as the oxide **530b**.

The CAAC-OS is a metal oxide having a dense structure with high crystallinity and small amounts of impurities and defects (e.g., oxygen vacancies (V_O)). In particular, after the

formation of a metal oxide, heat treatment is performed at a temperature at which the metal oxide does not become a polycrystal (e.g., higher than or equal to 400° C. and lower than or equal to 600° C.), whereby a CAAC-OS having a dense structure with higher crystallinity can be obtained. When the density of the CAAC-OS is increased in such a manner, diffusion of impurities or oxygen in the CAAC-OS can be further reduced.

On the other hand, a clear crystal grain boundary is difficult to observe in the CAAC-OS; thus, it can be said that a reduction in electron mobility due to the crystal grain boundary is less likely to occur. Thus, a metal oxide including a CAAC-OS is physically stable. Therefore, the metal oxide including a CAAC-OS is resistant to heat and has high reliability.

If impurities and oxygen vacancies exist in a region of an oxide semiconductor where a channel is formed, a transistor using the oxide semiconductor might have variable electrical characteristics and poor reliability. In some cases, hydrogen in the vicinity of an oxygen vacancy forms a defect that is the oxygen vacancy into which hydrogen enters (hereinafter, sometimes referred to as V_OH), which generates an electron serving as a carrier. Therefore, when the region of the oxide semiconductor where a channel is formed includes oxygen vacancies, the transistor tends to have normally-on characteristics (even when no voltage is applied to the gate electrode, the channel exists and current flows through the transistor). Thus, impurities, oxygen vacancies, and V_OH are preferably reduced as much as possible in the region of the oxide semiconductor where a channel is formed. In other words, it is preferable that the region of the oxide semiconductor where a channel is formed have a reduced carrier concentration and be of an i-type (intrinsic) or substantially i-type.

As a countermeasure to the above, an insulator containing oxygen that is released by heating (hereinafter, sometimes referred to as excess oxygen) is provided in the vicinity of the oxide semiconductor and heat treatment is performed, so that oxygen can be supplied from the insulator to the oxide semiconductor to reduce oxygen vacancies and V_OH . However, supply of an excess amount of oxygen to the source region or the drain region might cause a decrease in the on-state current or field-effect mobility of the transistor **500**. Furthermore, a variation of the amount of oxygen supplied to the source region or the drain region in the substrate plane leads to a variation in characteristics of the semiconductor device including the transistor.

Therefore, the region **530bc** functioning as the channel formation region in the oxide semiconductor is preferably an i-type or substantially i-type region with a reduced carrier concentration, whereas the region **530ba** and the region **530bb** functioning as the source region and the drain region are preferably n-type regions with high carrier concentrations. That is, it is preferable that oxygen vacancies and V_OH in the region **530bc** of the oxide semiconductor be reduced and the region **530ba** and the region **530bb** not be supplied with an excess amount of oxygen.

Thus, in this embodiment, microwave treatment is performed in an oxygen-containing atmosphere in a state where the conductor **542a** and the conductor **542b** are provided over the oxide **530b** so that oxygen vacancies and V_OH in the region **530bc** can be reduced. Here, the microwave treatment refers to, for example, treatment using an apparatus including a power source that generates high-density plasma with the use of a microwave.

The microwave treatment in an oxygen-containing atmosphere converts an oxygen gas into plasma using a high-

frequency wave such as a microwave or RF and activates the oxygen plasma. At this time, the region **530bc** can be irradiated with the high-frequency wave such as a microwave or RF. By the effect of the plasma, a microwave, or the like, V_OH in the region **530bc** can be cut; thus, hydrogen H can be removed from the region **530bc** and an oxygen vacancy V_O can be filled with oxygen. That is, the reaction " $V_OH \rightarrow H + V_O$ " occurs in the region **530bc**, so that the hydrogen concentration in the region **530bc** can be reduced. As a result, oxygen vacancies and V_OH in the region **530bc** can be reduced to lower the carrier concentration.

In the microwave treatment in an oxygen-containing atmosphere, the high-frequency wave such as the microwave or RF, the oxygen plasma, or the like is blocked by the conductor **542a** and the conductor **542b** and does not affect the region **530ba** nor the region **530bb**. In addition, the effect of the oxygen plasma can be reduced by the insulator **571** and the insulator **580** that are provided to cover the oxide **530b** and the conductor **542**. Hence, a reduction in V_OH and supply of an excess amount of oxygen do not occur in the region **530ba** and the region **530bb** in the microwave treatment, preventing a decrease in carrier concentration.

Microwave treatment is preferably performed in an oxygen-containing atmosphere after formation of an insulating film to be the insulator **552** or after formation of an insulating film to be the insulator **550**. By performing the microwave treatment in an oxygen-containing atmosphere through the insulator **552** or the insulator **550** in such a manner, oxygen can be efficiently supplied into the region **530bc**. In addition, the insulator **552** is provided to be in contact with the side surface of the conductor **542** and the surface of the region **530bc**, thereby preventing oxygen more than necessary from being supplied to the region **530bc** and preventing the side surface of the conductor **542** from being oxidized. Furthermore, the side surface of the conductor **542** can be inhibited from being oxidized when an insulating film to be the insulator **550** is formed.

The oxygen supplied into the region **530bc** has any of a variety of forms such as an oxygen atom, an oxygen molecule, and an oxygen radical (also referred to as an O radical, an atom or a molecule having an unpaired electron, or an ion). Note that the oxygen supplied into the region **530bc** preferably has any one or more of the above forms, and is particularly preferably an oxygen radical. Furthermore, the film quality of the insulator **552** and the insulator **550** can be improved, leading to higher reliability of the transistor **500**.

In the above manner, oxygen vacancies and V_OH can be selectively removed from the region **530bc** in the oxide semiconductor, whereby the region **530bc** can be an i-type or substantially i-type region. Furthermore, supply of an excess amount of oxygen to the region **530ba** and the region **530bb** functioning as the source region and the drain region can be inhibited and the state of the n-type region before the microwave treatment is performed can be maintained. As a result, a change in the electrical characteristics of the transistor **500** can be inhibited, and thus a variation in the electrical characteristics of the transistors **500** in the substrate plane can be reduced.

With the above structure, a semiconductor device with a small variation in transistor characteristics can be provided. A semiconductor device with favorable reliability can also be provided. A semiconductor device having favorable electrical characteristics can be provided.

As illustrated in FIG. **28B**, a curved surface may be provided between the side surface of the oxide **530b** and the top surface of the oxide **530b** in a cross-sectional view of the

transistor **500** in the channel width direction. In other words, an end portion of the side surface and an end portion of the top surface may be curved (hereinafter, also referred to as rounded).

The radius of curvature of the curved surface is preferably greater than 0 nm and less than the thickness of the oxide **530b** in a region overlapping with the conductor **542**, or less than half of the length of a region that does not have the curved surface. Specifically, the radius of curvature of the curved surface is greater than 0 nm and less than or equal to 20 nm, preferably greater than or equal to 1 nm and less than or equal to 15 nm, further preferably greater than or equal to 2 nm and less than or equal to 10 nm. Such a shape can improve the coverage of the oxide **530b** with the insulator **552**, the insulator **550**, the insulator **554**, and the conductor **560**.

The oxide **530** preferably has a stacked-layer structure of a plurality of oxide layers with different chemical compositions. Specifically, the atomic ratio of the element M to a metal element that is a main component of the metal oxide used as the oxide **530a** is preferably greater than the atomic ratio of the element M to a metal element that is a main component of the metal oxide used as the oxide **530b**. Moreover, the atomic ratio of the element M to In in the metal oxide used as the oxide **530a** is preferably greater than the atomic ratio of the element M to In in the metal oxide used as the oxide **530b**. Furthermore, the atomic ratio of In to the element M in the metal oxide used as the oxide **530b** is preferably greater than the atomic ratio of In to the element M in the metal oxide used as the oxide **530a**.

The oxide **530b** is preferably an oxide having crystallinity, such as a CAAC-OS. An oxide having crystallinity, such as a CAAC-OS, has a dense structure with small amounts of impurities and defects (e.g., oxygen vacancies) and high crystallinity. This can inhibit oxygen extraction from the oxide **530b** by the source electrode or the drain electrode. This can reduce oxygen extraction from the oxide **530b** even when heat treatment is performed; thus, the transistor **500** is stable with respect to high temperatures in a manufacturing process (what is called thermal budget).

Here, the conduction band minimum gradually changes at a junction portion of the oxide **530a** and the oxide **530b**. In other words, the conduction band minimum at the junction portion of the oxide **530a** and the oxide **530b** continuously changes or is continuously connected. To achieve this, the density of defect states in a mixed layer formed at the interface between the oxide **530a** and the oxide **530b** is preferably made low.

Specifically, when the oxide **530a** and the oxide **530b** contain a common element as a main component besides oxygen, a mixed layer with a low density of defect states can be formed. For example, in the case where the oxide **530b** is an In-M-Zn oxide, an In-M-Zn oxide, an M-Zn oxide, an oxide of the element M, an In-Zn oxide, an indium oxide, or the like may be used as the oxide **530a**.

Specifically, as the oxide **530a**, a metal oxide with a composition of In:M:Zn=1:3:4 [atomic ratio] or in the neighborhood thereof, or a composition of In:M:Zn=1:1:0.5 [atomic ratio] or in the neighborhood thereof is used. As the oxide **530b**, a metal oxide with a composition of In:M:Zn=1:1:1 [atomic ratio] or in the neighborhood thereof, or a composition of In:M:Zn=4:2:3 [atomic ratio] or in the neighborhood thereof is used. Note that a composition in the neighborhood includes the range of $\pm 30\%$ of an intended atomic ratio. Gallium is preferably used as the element M.

When the metal oxide is deposited by a sputtering method, the above atomic ratio is not limited to the atomic

ratio of the deposited metal oxide and may be the atomic ratio of a sputtering target used for depositing the metal oxide.

As illustrated in FIG. 28A or the like, the insulator 552 formed using aluminum oxide or the like is provided in contact with the top and side surfaces of the oxide 530, whereby indium contained in the oxide 530 is unevenly distributed, in some cases, at the interface between the oxide 530 and the insulator 552 and in its vicinity. Accordingly, the vicinity of the surface of the oxide 530 comes to have an atomic ratio close to that of an indium oxide or that of an In—Zn oxide. Such an increase in the atomic ratio of indium in the vicinity of the surface of the oxide 530, especially the vicinity of the surface of the oxide 530b, can increase the field-effect mobility of the transistor 500.

When the oxide 530a and the oxide 530b have the above structure, the density of defect states at the interface between the oxide 530a and the oxide 530b can be made low. Thus, the influence of interface scattering on carrier conduction is small, and the transistor 500 can have a high on-state current and excellent frequency characteristics.

At least one of the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, an insulator 576, and an insulator 581 preferably functions as a barrier insulating film, which inhibits diffusion of impurities such as water and hydrogen from the substrate side or above the transistor 500 into the transistor 500. Thus, for at least one of the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581, it is preferable to use an insulating material having a function of inhibiting diffusion of impurities such as hydrogen atoms, hydrogen molecules, water molecules, nitrogen atoms, nitrogen molecules, nitrogen oxide molecules (e.g., N_2O , NO , or NO_2), or copper atoms (an insulating material through which the impurities are less likely to pass). Alternatively, it is preferable to use an insulating material having a function of inhibiting diffusion of oxygen (e.g., at least one of oxygen atoms, oxygen molecules, and the like) (an insulating material through which the oxygen is less likely to pass).

An insulator having a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen is preferably used as the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581; for example, aluminum oxide, magnesium oxide, hafnium oxide, gallium oxide, indium-gallium-zinc oxide, silicon nitride, or silicon nitride oxide can be used. For example, silicon nitride, which has a higher hydrogen barrier property, is preferably used for the insulator 512, the insulator 544, and the insulator 576. For example, aluminum oxide or magnesium oxide, which has a function of capturing or fixing hydrogen well, is preferably used for the insulator 514, the insulator 571, the insulator 574, and the insulator 581. In this case, impurities such as water and hydrogen can be inhibited from diffusing to the transistor 500 side from the substrate side through the insulator 512 and the insulator 514. Impurities such as water and hydrogen can be inhibited from diffusing to the transistor 500 side from an interlayer insulating film and the like which are provided outside the insulator 581. Alternatively, oxygen contained in the insulator 524 and the like can be inhibited from diffusing to the substrate side through the insulator 512 and the insulator 514. Alternatively, oxygen contained in the insulator 580 and the like can be inhibited from diffusing to above the transistor 500 through the insulator 574 and the like. In this manner, it is preferable that the transistor 500 be surrounded by the insulator 512, the

insulator 514, the insulator 571, the insulator 544, the insulator 574, the insulator 576, and the insulator 581, which have a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen.

Here, an oxide having an amorphous structure is preferably used for the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581. For example, a metal oxide such as AlO_x (x is a given number greater than 0) or MgO_y (y is a given number greater than 0) is preferably used. In such a metal oxide having an amorphous structure, an oxygen atom has a dangling bond and sometimes has a property of capturing or fixing hydrogen with the dangling bond. When such a metal oxide having an amorphous structure is used as the component of the transistor 500 or provided around the transistor 500, hydrogen contained in the transistor 500 or hydrogen present around the transistor 500 can be captured or fixed. In particular, hydrogen contained in the channel formation region of the transistor 500 is preferably captured or fixed. The metal oxide having an amorphous structure is used as the component of the transistor 500 or provided around the transistor 500, whereby the transistor 500 and a semiconductor device, which have favorable characteristics and high reliability, can be fabricated.

Although each of the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581 preferably has an amorphous structure, a region having a polycrystalline structure may be partly formed. Alternatively, each of the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581 may have a multilayer structure in which a layer having an amorphous structure and a layer having a polycrystalline structure are stacked. For example, a stacked-layer structure in which a layer having a polycrystalline structure is formed over a layer having an amorphous structure may be employed.

The insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581 can be deposited by a sputtering method, for example. Since a sputtering method does not need to use a molecule containing hydrogen as a deposition gas, the hydrogen concentrations in the insulator 512, the insulator 514, the insulator 544, the insulator 571, the insulator 574, the insulator 576, and the insulator 581 can be reduced. Note that the deposition method is not limited to a sputtering method, and a chemical vapor deposition (CVD) method, a molecular beam epitaxy (MBE) method, a pulsed laser deposition (PLD) method, an atomic layer deposition (ALD) method, or the like may be used as appropriate.

The resistivities of the insulator 512, the insulator 544, and the insulator 576 are preferably low in some cases. For example, by setting the resistivities of the insulator 512, the insulator 544, and the insulator 576 to approximately $1 \times 10^{13} \Omega\text{cm}$, the insulator 512, the insulator 544, and the insulator 576 can sometimes reduce charge up of the conductor 503, the conductor 542, the conductor 560, or the like in treatment using plasma or the like in the fabrication process of a semiconductor device. The resistivities of the insulator 512, the insulator 544, and the insulator 576 are preferably higher than or equal to $1 \times 10^{10} \Omega\text{cm}$ and lower than or equal to $1 \times 10^{15} \Omega\text{cm}$.

The insulator 516, the insulator 574, the insulator 580, and the insulator 581 each preferably have a lower permittivity than the insulator 514. When a material with a low permittivity is used for the interlayer film, the parasitic capacitance

generated between wirings can be reduced. For the insulator **516**, the insulator **580**, and the insulator **581**, silicon oxide, silicon oxynitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, or the like is used as appropriate, for example.

The insulator **581** is preferably an insulator functioning as an interlayer film, a planarization film, or the like, for example.

The conductor **503** is provided to overlap with the oxide **530** and the conductor **560**. Here, the conductor **503** is preferably provided to be embedded in an opening formed in the insulator **516**. Part of the conductor **503** is embedded in the insulator **514** in some cases.

The conductor **503** includes the conductor **503a** and the conductor **503b**. The conductor **503a** is provided in contact with a bottom surface and a sidewall of the opening. The conductor **503b** is provided to be embedded in a recessed portion formed in the conductor **503a**. Here, the upper portion of the conductor **503b** is substantially level with the upper portion of the conductor **503a** and the upper portion of the insulator **516**.

Here, for the conductor **503a**, it is preferable to use a conductive material having a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule (N_2O , NO , NO_2 , or the like), and a copper atom. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like).

When the conductor **503a** is formed using a conductive material having a function of inhibiting diffusion of hydrogen, impurities such as hydrogen contained in the conductor **503b** can be prevented from diffusing into the oxide **530** through the insulator **524** and the like. When the conductor **503a** is formed using a conductive material having a function of inhibiting diffusion of oxygen, the conductivity of the conductor **503b** can be inhibited from being lowered because of oxidation. As the conductive material having a function of inhibiting diffusion of oxygen, for example, titanium, titanium nitride, tantalum, tantalum nitride, ruthenium, or ruthenium oxide is preferably used. Thus, a single layer or a stacked layer of the above conductive material is used as the conductor **503a**. For example, titanium nitride is used for the conductor **503a**.

Moreover, the conductor **503b** is preferably formed using a conductive material containing tungsten, copper, or aluminum as its main component. For example, tungsten is used for the conductor **503b**.

The conductor **503** sometimes functions as a second gate electrode. In that case, by changing a potential applied to the conductor **503** not in conjunction with but independently of a potential applied to the conductor **560**, the threshold voltage (V_{th}) of the transistor **500** can be controlled. In particular, V_{th} of the transistor **500** can be higher in the case where a negative potential is applied to the conductor **503**, and the off-state current can be reduced. Thus, drain current at the time when a potential applied to the conductor **560** is 0 V can be lower in the case where a negative potential is applied to the conductor **503** than in the case where the negative potential is not applied to the conductor **503**.

In the case where the oxide **530** is a highly purified intrinsic oxide and as many impurities as possible are eliminated from the oxide **530**, the transistor **500** can be expected to become normally-off (the threshold voltage of the transistor **500** can be expected to higher than 0 V) in

some cases with no potential application to the conductor **503** and/or the conductor **560**. In that case, it is suitable to connect the conductor **560** and the conductor **503** to each other such that the same potential is supplied.

The electric resistivity of the conductor **503** is designed in consideration of the potential applied to the conductor **503**, and the thickness of the conductor **503** is determined in accordance with the electric resistivity. The thickness of the insulator **516** is substantially equal to that of the conductor **503**. The conductor **503** and the insulator **516** are preferably as thin as possible in the allowable range of the design of the conductor **503**. When the thickness of the insulator **516** is reduced, the absolute amount of impurities such as hydrogen contained in the insulator **516** can be reduced, reducing the amount of the impurities to be diffused into the oxide **530**.

When seen from above, the conductor **503** is preferably provided to be larger than a region of the oxide **530** that does not overlap with the conductor **542a** or the conductor **542b**. As illustrated in FIG. 28B, it is particularly preferable that the conductor **503** extend to a region outside end portions of the oxide **530a** and the oxide **530b** in the channel width direction. That is, the conductor **503** and the conductor **560** preferably overlap with each other with the insulators therebetween on the outer side of the side surface of the oxide **530** in the channel width direction. With this structure, the channel formation region of the oxide **530** can be electrically surrounded by the electric field of the conductor **560** functioning as a first gate electrode and the electric field of the conductor **503** functioning as the second gate electrode. In this specification, a transistor structure in which a channel formation region is electrically surrounded by electric fields of a first gate and a second gate is referred to as a surrounded channel (S-channel) structure.

In this specification and the like, a transistor having the S-channel structure refers to a transistor having a structure in which a channel formation region is electrically surrounded by the electric fields of a pair of gate electrodes. The S-channel structure disclosed in this specification and the like is different from a fin-type structure and a planar structure. With the S-channel structure, resistance to a short-channel effect can be enhanced, that is, a transistor in which a short-channel effect is less likely to occur can be provided.

When the transistor **500** becomes normally-off and has the above-described S-channel structure, the channel formation region can be electrically surrounded. Accordingly, the transistor **500** can be regarded as having a GAA (Gate All Around) structure or an LGAA (Lateral Gate All Around) structure. When the transistor **500** has the S-channel structure, the GAA structure, or the LGAA structure, the channel formation region that is formed at the interface between the oxide **530** and the gate insulating film or in the vicinity of the interface can be formed in the entire bulk of the oxide **530**. In other words, the transistor **500** having the S-channel structure, the GAA structure, or the LGAA structure can be what is called a Bulk-Flow type, in which a carrier path is used as the entire bulk. A transistor structure with a Bulk-Flow type can improve the density of current flowing in the transistor and thus can be expected to improve the on-state current of the transistor or increase the field-effect mobility of the transistor.

Furthermore, as illustrated in FIG. 28B, the conductor **503** is extended to function as a wiring as well. However, without limitation to this structure, a structure in which a conductor functioning as a wiring is provided below the conductor **503** may be employed. In addition, the conductor

503 is not necessarily provided in each transistor. For example, the conductor **503** may be shared by a plurality of transistors.

Although the transistor **500** having a structure in which the conductor **503** is a stack of the conductor **503a** and the conductor **503b** is illustrated, the present invention is not limited thereto. For example, the conductor **503** may be provided to have a single-layer structure or a stacked-layer structure of three or more layers.

The insulator **522** and the insulator **524** function as a gate insulator.

It is preferable that the insulator **522** have a function of inhibiting diffusion of hydrogen (e.g., at least one of a hydrogen atom, a hydrogen molecule, and the like). In addition, it is preferable that the insulator **522** have a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like). For example, the insulator **522** preferably has a function of inhibiting diffusion of one or both of hydrogen and oxygen more than the insulator **524**.

As the insulator **522**, an insulator containing an oxide of one or both of aluminum and hafnium, which is an insulating material, is preferably used. For the insulator, aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), or the like is preferably used. In the case where the insulator **522** is formed using such a material, the insulator **522** functions as a layer that inhibits release of oxygen from the oxide **530** to the substrate side and diffusion of impurities such as hydrogen from the periphery of the transistor **500** into the oxide **530**. Thus, providing the insulator **522** can inhibit diffusion of impurities such as hydrogen into the transistor **500** and inhibit generation of oxygen vacancies in the oxide **530**. Moreover, the conductor **503** can be inhibited from reacting with oxygen contained in the insulator **524** or the oxide **530**.

Alternatively, aluminum oxide, bismuth oxide, germanium oxide, niobium oxide, silicon oxide, titanium oxide, tungsten oxide, yttrium oxide, or zirconium oxide may be added to the above insulator, for example. Alternatively, these insulators may be subjected to nitriding treatment. A stack of silicon oxide, silicon oxynitride, or silicon nitride over these insulators may be used for the insulator **522**.

For example, a single layer or stacked layers of an insulator containing what is called a high-k material such as aluminum oxide, hafnium oxide, tantalum oxide, or zirconium oxide may be used for the insulator **522**. As scaling down and high integration of transistors progress, a problem such as leakage current may arise because of a thinner gate insulator. When a high-k material is used for an insulator functioning as the gate insulator, a gate potential at the time when the transistor operates can be reduced while the physical thickness is maintained. Furthermore, a substance with a high permittivity such as lead zirconate titanate (PZT), strontium titanate (SrTiO₃), or (Ba,Sr)TiO₃ (BST) may be used for the insulator **522**.

Silicon oxide or silicon oxynitride, for example, can be used as appropriate for the insulator **524** that is in contact with the oxide **530**.

In a fabrication process of the transistor **500**, heat treatment is preferably performed with a surface of the oxide **530** exposed. For example, the heat treatment is performed at a temperature higher than or equal to 100° C. and lower than or equal to 600° C., preferably higher than or equal to 350° C. and lower than or equal to 550° C. Note that the heat treatment is performed in a nitrogen gas or inert gas atmosphere, or an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more. For example, the

heat treatment is preferably performed in an oxygen atmosphere. This can supply oxygen to the oxide **530** to reduce oxygen vacancies (V_O). The heat treatment may be performed under reduced pressure. Alternatively, the heat treatment may be performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more in order to compensate for released oxygen, after heat treatment in a nitrogen gas or inert gas atmosphere. Alternatively, the heat treatment may be performed in a nitrogen gas or inert gas atmosphere successively after heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more.

Note that oxygen adding treatment performed on the oxide **530** can promote a reaction in which oxygen vacancies in the oxide **530** are repaired with supplied oxygen, i.e., a reaction of "V_O+O→null". Furthermore, hydrogen remaining in the oxide **530** reacts with supplied oxygen, so that the hydrogen can be removed as H₂O (dehydration). This can inhibit recombination of hydrogen remaining in the oxide **530** with oxygen vacancies and formation of V_OH.

Note that the insulator **522** and the insulator **524** may each have a stacked-layer structure of two or more layers. In that case, without limitation to a stacked-layer structure formed of the same material, a stacked-layer structure formed of different materials may be employed. The insulator **524** may be formed into an island shape so as to overlap with the oxide **530a**. In this case, the insulator **544** is in contact with the side surface of the insulator **524** and the top surface of the insulator **522**.

The conductor **542a** and the conductor **542b** are provided in contact with the top surface of the oxide **530b**. The conductor **542a** and the conductor **542b** function as a source electrode and a drain electrode of the transistor **500**.

For the conductor **542** (the conductor **542a** and the conductor **542b**), for example, a nitride containing tantalum, a nitride containing titanium, a nitride containing molybdenum, a nitride containing tungsten, a nitride containing tantalum and aluminum, a nitride containing titanium and aluminum, or the like is preferably used. In one embodiment of the present invention, a nitride containing tantalum is particularly preferable. For another example, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, or an oxide containing lanthanum and nickel may be used. These materials are preferable because they are each a conductive material that is not easily oxidized or a material that maintains the conductivity even after absorbing oxygen.

Note that hydrogen contained in the oxide **530b** or the like diffuses into the conductor **542a** or the conductor **542b** in some cases. In particular, when a nitride containing tantalum is used for the conductor **542a** and the conductor **542b**, hydrogen contained in the oxide **530b** or the like is likely to diffuse into the conductor **542a** or the conductor **542b**, and the diffused hydrogen is bonded to nitrogen contained in the conductor **542a** or the conductor **542b** in some cases. That is, hydrogen contained in the oxide **530b** or the like is absorbed by the conductor **542a** or the conductor **542b** in some cases.

No curved surface is preferably formed between the side surface of the conductor **542** and the top surface of the conductor **542**. When no curved surface is formed in the conductor **542**, the conductor **542** can have a large cross-sectional area in the channel width direction. Accordingly, the conductivity of the conductor **542** is increased, so that the on-state current of the transistor **500** can be increased.

The insulator **571a** is provided in contact with the top surface of the conductor **542a**, and the insulator **571b** is

provided in contact with the top surface of the conductor **542b**. The insulator **571** preferably functions as at least a barrier insulating film against oxygen. Thus, the insulator **571** preferably has a function of inhibiting oxygen diffusion. For example, the insulator **571** preferably has a function of inhibiting diffusion of oxygen more than the insulator **580**. For example, a nitride containing silicon such as silicon nitride may be used for the insulator **571**. The insulator **571** preferably has a function of capturing impurities such as hydrogen. In that case, for the insulator **571**, a metal oxide having an amorphous structure, for example, an insulator such as aluminum oxide or magnesium oxide, may be used. It is particularly preferable to use aluminum oxide having an amorphous structure or amorphous aluminum oxide for the insulator **571** because hydrogen can be captured or fixed more effectively in some cases. Accordingly, the transistor **500** and a semiconductor device, which have favorable characteristics and high reliability, can be fabricated.

The insulator **544** is provided to cover the insulator **524**, the oxide **530a**, the oxide **530b**, the conductor **542**, and the insulator **571**. The insulator **544** preferably has a function of capturing and fixing hydrogen. In that case, the insulator **544** preferably includes silicon nitride, or a metal oxide having an amorphous structure, for example, an insulator such as aluminum oxide or magnesium oxide. Alternatively, for example, a stacked-layer film of aluminum oxide and silicon nitride over the aluminum oxide may be used as the insulator **544**.

When the above insulator **571** and the insulator **544** are provided, the conductor **542** can be surrounded by the insulators having a barrier property against oxygen. That is, oxygen contained in the insulator **524** and the insulator **580** can be prevented from diffusing into the conductor **542**. As a result, the conductor **542** can be inhibited from being directly oxidized by oxygen contained in the insulator **524** and the insulator **580**, so that an increase in resistivity and a reduction in on-state current can be inhibited.

The insulator **552** functions as part of the gate insulator. As the insulator **552**, a barrier insulating film against oxygen is preferably used. As the insulator **552**, an insulator that can be used as the insulator **574** described above may be used. An insulator containing an oxide of one or both of aluminum and hafnium is preferably used as the insulator **552**. As the insulator, aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), an oxide containing hafnium and silicon (hafnium silicate), or the like can be used. In this embodiment, aluminum oxide is used for the insulator **552**. In this case, the insulator **552** is an insulator containing at least oxygen and aluminum.

As illustrated in FIG. 28B, the insulator **552** is provided in contact with the top surface and the side surface of the oxide **530b**, the side surface of the oxide **530a**, the side surface of the insulator **524**, and the top surface of the insulator **522**. That is, the regions of the oxide **530a**, the oxide **530b**, and the insulator **524** that overlap with the conductor **560** are covered with the insulator **552** in the cross section in the channel width direction. With this structure, the insulator **552** having a barrier property against oxygen can prevent release of oxygen from the oxide **530a** and the oxide **530b** at the time of heat treatment or the like. This can inhibit formation of oxygen vacancies (V_O) in the oxide **530a** and the oxide **530b**. Therefore, oxygen vacancies (V_O) and V_{OH} formed in the region **530bc** can be reduced. Thus, the transistor **500** can have favorable electrical characteristics and higher reliability.

Even when an excess amount of oxygen is contained in the insulator **580**, the insulator **550**, and the like, oxygen can

be inhibited from being excessively supplied to the oxide **530a** and the oxide **530b**. Thus, the region **530ba** and the region **530bb** are prevented from being excessively oxidized by oxygen through the region **530bc**; a reduction in on-state current or field-effect mobility of the transistor **500** can be inhibited.

As illustrated in FIG. 28A, the insulator **552** is provided in contact with the side surfaces of the conductor **542**, the insulator **544**, the insulator **571**, and the insulator **580**. This can inhibit formation of an oxide film on the side surface of the conductor **542** by oxidization of the side surface. Accordingly, a reduction in on-state current or field-effect mobility of the transistor **500** can be inhibited.

Furthermore, the insulator **552** needs to be provided in an opening formed in the insulator **580** and the like, together with the insulator **554**, the insulator **550**, and the conductor **560**. The thickness of the insulator **552** is preferably small for scaling down the transistor **500**. The thickness of the insulator **552** is preferably greater than or equal to 0.1 nm, greater than or equal to 0.5 nm, or greater than or equal to 1.0 nm, and less than or equal to 1.0 nm, less than or equal to 3.0 nm, or less than or equal to 5.0 nm. Note that the above-described lower limits and upper limits can be combined with each other. In that case, at least part of the insulator **552** includes a region having the above-described thickness. The thickness of the insulator **552** is preferably smaller than that of the insulator **550**. In that case, at least part of the insulator **552** includes a region having a thickness smaller than that of the insulator **550**.

To form the insulator **552** having a small thickness as described above, an ALD method is preferably used for deposition. An ALD method is a method in which deposition is performed by introducing a first source gas (also referred to as a precursor or a metal precursor) and a second source gas (also referred to as a reactant, an oxidizer, or a nonmetallic precursor) for reaction alternately into a chamber, and repeating the introduction of these source gases. As an ALD method, a thermal ALD method, in which a precursor and a reactant react with each other only by a thermal energy, a PEALD (Plasma Enhanced ALD) method, in which a reactant excited by plasma is used, and the like can be used. The use of plasma in a PEALD method is sometimes preferable because deposition at a lower temperature is possible.

An ALD method, which enables an atomic layer to be deposited one by one using self-limiting characteristics by atoms, has advantages such as deposition of an extremely thin film, deposition on a component with a high aspect ratio, deposition of a film with a small number of defects such as pinholes, deposition with excellent coverage, and low-temperature deposition. Therefore, the insulator **552** can be formed on the side surface of the opening formed in the insulator **580** and the like to have a small thickness as described above and to have favorable coverage.

Note that some of precursors usable in an ALD method contain carbon or the like. Thus, in some cases, a film provided by an ALD method contains impurities such as carbon in a larger amount than a film provided by another deposition method. Note that impurities can be quantified by secondary ion mass spectrometry (SIMS) or X-ray photoelectron spectroscopy (XPS).

The insulator **550** functions as part of the gate insulator. The insulator **550** is preferably provided in contact with the top surface of the insulator **552**. The insulator **550** can be formed using silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon

oxide, or the like. In particular, silicon oxide and silicon oxynitride, which have thermal stability, are preferable. The insulator **550** in this case is an insulator containing at least oxygen and silicon.

As in the insulator **524**, the concentration of an impurity such as water or hydrogen in the insulator **550** is preferably reduced. The lower limit of the thickness of the insulator **550** is preferably 1 nm or 0.5 nm, and the upper limit is preferably 15 nm or 20 nm. Note that the above-described lower limits and upper limits can be combined with each other. For example, the thickness of the insulator **550** is preferably greater than or equal to 0.5 nm and less than or equal to 20 nm, preferably greater than or equal to 1 nm and less than or equal to 15 nm. In that case, at least part of the insulator **550** includes a region having the above-described thickness.

Although FIG. **28A**, FIG. **28B**, and the like illustrate a single-layer structure of the insulator **550**, the present invention is not limited to this structure, and a stacked-layer structure of two or more layers may be employed. For example, as illustrated in FIG. **29B**, the insulator **550** may have a stacked-layer structure including two layers of an insulator **550a** and an insulator **550b** over the insulator **550a**.

In the case where the insulator **550** has a stacked-layer structure of two layers as illustrated in FIG. **29B**, it is preferable that the insulator **550a** in a lower layer be formed using an insulator that is likely to transmit oxygen and the insulator **550b** in an upper layer be formed using an insulator having a function of inhibiting oxygen diffusion. With such a structure, oxygen contained in the insulator **550a** can be inhibited from diffusing into the conductor **560**. That is, a reduction in the amount of oxygen supplied to the oxide **530** can be inhibited. In addition, oxidation of the conductor **560** due to oxygen contained in the insulator **550a** can be inhibited. For example, it is preferable that the insulator **550a** be provided using any of the above-described materials that can be used for the insulator **550** and the insulator **550b** be provided using an insulator containing an oxide of one or both of aluminum and hafnium. As the insulator, aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), an oxide containing hafnium and silicon (hafnium silicate), or the like can be used. In this embodiment, hafnium oxide is used as the insulator **550b**. In this case, the insulator **550b** is an insulator containing at least oxygen and hafnium. The thickness of the insulator **550b** is preferably greater than or equal to 0.5 nm or greater than or equal to 1.0 nm, and less than or equal to 3.0 nm or less than or equal to 5.0 nm. Note that the above-described lower limits and upper limits can be combined with each other. In that case, at least part of the insulator **550b** includes a region having the above-described thickness.

In the case where silicon oxide, silicon oxynitride, or the like is used for the insulator **550a**, the insulator **550b** may be formed using an insulating material that is a high-k material having a high dielectric constant. The gate insulator having a stacked-layer structure of the insulator **550a** and the insulator **550b** can be thermally stable and can have a high dielectric constant. Thus, a gate potential that is applied during the operation of the transistor can be reduced while the physical thickness of the gate insulator is maintained. In addition, the equivalent oxide thickness (EOT) of the insulator functioning as the gate insulator can be reduced. Therefore, the withstand voltage of the insulator **550** can be increased.

The insulator **554** functions as part of a gate insulator. As the insulator **554**, a barrier insulating film against hydrogen is preferably used. This can prevent diffusion of impurities

such as hydrogen contained in the conductor **560** into the insulator **550** and the oxide **530b**. As the insulator **554**, an insulator that can be used as the insulator **576** described above may be used. For example, silicon nitride deposited by a PEALD method may be used as the insulator **554**. In this case, the insulator **554** is an insulator containing at least nitrogen and silicon.

Furthermore, the insulator **554** may have a barrier property against oxygen. Thus, diffusion of oxygen contained in the insulator **550** into the conductor **560** can be inhibited.

Furthermore, the insulator **554** needs to be provided in an opening formed in the insulator **580** and the like, together with the insulator **552**, the insulator **550**, and the conductor **560**. The thickness of the insulator **554** is preferably small for scaling down the transistor **500**. The thickness of the insulator **554** is preferably greater than or equal to 0.1 nm, greater than or equal to 0.5 nm, or greater than or equal to 1.0 nm, and less than or equal to 3.0 nm or less than or equal to 5.0 nm. Note that the above-described lower limits and upper limits can be combined with each other. In that case, at least part of the insulator **554** includes a region having the above-described thickness. The thickness of the insulator **554** is preferably smaller than that of the insulator **550**. In that case, at least part of the insulator **554** includes a region having a thickness smaller than that of the insulator **550**.

The conductor **560** functions as the first gate electrode of the transistor **500**. The conductor **560** preferably includes the conductor **560a** and the conductor **560b** provided over the conductor **560a**. For example, the conductor **560a** is preferably provided to cover the bottom surface and the side surface of the conductor **560b**. As illustrated in FIG. **28A** and FIG. **28B**, the top surface of the conductor **560** is substantially level with the upper portion of the insulator **550**. Note that although the conductor **560** has a two-layer structure of the conductor **560a** and the conductor **560b** in FIG. **28A** and FIG. **28B**, the conductor **560** can have, besides the two-layer structure, a single-layer structure or a stacked-layer structure of three or more layers.

For the conductor **560a**, a conductive material having a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule, and a copper atom is preferably used. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like).

In addition, when the conductor **560a** has a function of inhibiting diffusion of oxygen, the conductivity of the conductor **560b** can be inhibited from being lowered because of oxidation due to oxygen contained in the insulator **550**. As the conductive material having a function of inhibiting diffusion of oxygen, for example, titanium, titanium nitride, tantalum, tantalum nitride, ruthenium, or ruthenium oxide is preferably used.

Furthermore, the conductor **560** also functions as a wiring and thus is preferably a conductor having high conductivity. For example, a conductive material containing tungsten, copper, or aluminum as its main component can be used for the conductor **560b**. The conductor **560b** can have a stacked-layer structure. Specifically, for example, the conductor **560b** can have a stacked-layer structure of titanium or titanium nitride and the above conductive material.

In the transistor **500**, the conductor **560** is formed in a self-aligned manner to fill the opening formed in the insulator **580** and the like. The formation of the conductor **560** in this manner allows the conductor **560** to be placed

properly in a region between the conductor **542a** and the conductor **542b** without alignment.

As illustrated in FIG. **28B**, in the channel width direction of the transistor **500**, with reference to the bottom surface of the insulator **522**, the level of the bottom surface of the conductor **560** in a region where the conductor **560** and the oxide **530b** do not overlap with each other is preferably lower than the level of the bottom surface of the oxide **530b**. When the conductor **560** functioning as the gate electrode covers the side surface and the top surface of the channel formation region of the oxide **530b** with the insulator **550** and the like therebetween, the electric field of the conductor **560** can easily act on the entire channel formation region of the oxide **530b**. Thus, the on-state current of the transistor **500** can be increased and the frequency characteristics of the transistor **500** can be improved. The difference between the level of the bottom surface of the conductor **560** in a region where the oxide **530a** and the oxide **530b** do not overlap with the conductor **560** and the level of the bottom surface of the oxide **530b**, with reference to the bottom surface of the insulator **522**, is preferably greater than or equal to 0 nm, greater than or equal to 3 nm, or greater than or equal to 5 nm, and less than or equal to 20 nm, less than or equal to 50 nm, or less than or equal to 100 nm. Note that the above-described lower limits and upper limits can be combined with each other.

The insulator **580** is provided over the insulator **544**, and the opening is formed in a region where the insulator **550** and the conductor **560** are to be provided. In addition, the top surface of the insulator **580** may be planarized.

The insulator **580** functioning as an interlayer film preferably has a low permittivity. When a material with a low permittivity is used for an interlayer film, parasitic capacitance generated between wirings can be reduced. The insulator **580** is preferably provided using a material similar to that for the insulator **516**, for example. In particular, silicon oxide and silicon oxynitride, which have thermal stability, are preferable. Materials such as silicon oxide, silicon oxynitride, and porous silicon oxide are particularly preferable because a region containing oxygen to be released by heating can be easily formed.

The concentration of impurities such as water and hydrogen in the insulator **580** is preferably reduced. An oxide containing silicon, such as silicon oxide or silicon oxynitride, is used as appropriate for the insulator **580**, for example.

The insulator **574** preferably functions as a barrier insulating film that inhibits impurities such as water and hydrogen from diffusing into the insulator **580** from above and preferably has a function of capturing impurities such as hydrogen. The insulator **574** preferably functions as a barrier insulating film that inhibits passage of oxygen. For the insulator **574**, a metal oxide having an amorphous structure, for example, an insulator such as aluminum oxide, can be used. In this case, the insulator **574** is an insulator containing at least oxygen and aluminum. The insulator **574**, which has a function of capturing impurities such as hydrogen, is provided in contact with the insulator **580** in a region sandwiched between the insulator **512** and the insulator **581**, whereby impurities such as hydrogen contained in the insulator **580** and the like can be captured and the amount of hydrogen in the region can be constant. It is particularly preferable to use aluminum oxide having an amorphous structure for the insulator **574**, in which case hydrogen can sometimes be captured or fixed more effectively. Accord-

ingly, the transistor **500** and a semiconductor device, which have favorable characteristics and high reliability, can be fabricated.

The insulator **576** functions as a barrier insulating film that inhibits impurities such as water and hydrogen from diffusing into the insulator **580** from above. The insulator **576** is provided over the insulator **574**. The insulator **576** is preferably formed using a nitride containing silicon such as silicon nitride or silicon nitride oxide. For example, silicon nitride deposited by a sputtering method may be used for the insulator **576**. When the insulator **576** is deposited by a sputtering method, a high-density silicon nitride film can be formed. To obtain the insulator **576**, silicon nitride deposited by a PEALD method or a CVD method may be stacked over silicon nitride deposited by a sputtering method.

One of a first terminal and a second terminal of the transistor **500** is electrically connected to a conductor **540a** functioning as a plug, and the other of the first terminal and the second terminal of the transistor **500** is electrically connected to a conductor **540b**. Note that the conductor **540a**, the conductor **540b**, and the like may function as wirings for electrical connection to the light-emitting device **150** or the like provided thereabove. In the display device **10** of FIG. **27**, the conductor **540a**, the conductor **540b**, and the like may function as wirings for electrical connection to the transistor **300** and the like. Note that in this specification and the like, the conductor **540a** and the conductor **540b** are collectively referred to as the conductor **540**.

The conductor **540a** is provided in a region overlapping with the conductor **542a**, for example. Specifically, an opening portion is formed in the insulator **571**, the insulator **544**, the insulator **580**, the insulator **574**, the insulator **576**, and the insulator **581** illustrated in FIG. **28A** in the region overlapping with the conductor **542a**, and the conductor **540a** is provided inside the opening portion. The conductor **540b** is provided in a region overlapping with the conductor **542b**, for example. Specifically, an opening portion is formed in the insulator **571**, the insulator **544**, the insulator **580**, the insulator **574**, the insulator **576**, and the insulator **581** illustrated in FIG. **28A** in the region overlapping with the conductor **542b**, and the conductor **540b** is provided inside the opening portion.

As illustrated in FIG. **28A**, an insulator **541a** as an insulator having an impurity barrier property may be provided between the conductor **540a** and the side surface of the opening portion in the region overlapping with the conductor **542a**. Similarly, an insulator **541b** as an insulator having an impurity barrier property may be provided between the conductor **540b** and the side surface of the opening portion in the region overlapping with the conductor **542b**. Note that in this specification and the like, the insulator **541a** and the insulator **541b** are collectively referred to as an insulator **541**.

For the conductor **540a** and the conductor **540b**, a conductive material containing tungsten, copper, or aluminum as its main component is preferably used. The conductor **540a** and the conductor **540b** may each have a stacked-layer structure.

In the case where the conductor **540** has a stacked-layer structure, a conductive material having a function of inhibiting passage of impurities such as water and hydrogen is preferably used for a first conductor provided in the vicinity of the insulator **574**, the insulator **576**, the insulator **581**, the insulator **580**, the insulator **544**, and the insulator **571**. For example, tantalum, tantalum nitride, titanium, titanium nitride, ruthenium, ruthenium oxide, or the like is preferably used. The conductive material having a function of inhibit-

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ing passage of impurities such as water and hydrogen may be used as a single layer or stacked layers. Moreover, impurities such as water and hydrogen contained in a layer above the insulator 576 can be inhibited from entering the oxide 530 through the conductor 540a and the conductor 540b.

For the insulator 541a and the insulator 541b, a barrier insulating film that can be used for the insulator 544 or the like may be used. For the insulator 541a and the insulator 541b, for example, an insulator such as silicon nitride, aluminum oxide, or silicon nitride oxide may be used. Since the insulator 541a and the insulator 541b are provided in contact with the insulator 574, the insulator 576, and the insulator 571, impurities such as water and hydrogen contained in the insulator 580 or the like can be inhibited from entering the oxide 530 through the conductor 540a and the conductor 540b. In particular, silicon nitride is suitable because of its high blocking property against hydrogen. Furthermore, oxygen contained in the insulator 580 can be prevented from being absorbed by the conductor 540a and the conductor 540b.

When the insulator 541a and the insulator 541b each have a stacked-layer structure as illustrated in FIG. 28A, a first insulator in contact with an inner wall of the opening in the insulator 580 and the like and a second insulator inside the first insulator are preferably formed using a combination of a barrier insulating film against oxygen and a barrier insulating film against hydrogen.

For example, aluminum oxide deposited by an ALD method may be used as the first insulator and silicon nitride deposited by a PEALD method may be used as the second insulator. With this structure, oxidation of the conductor 540 can be inhibited, and hydrogen can be inhibited from entering the conductor 540.

Although the first insulator of the insulator 541 and the second conductor of the insulator 541 are stacked in the transistor 500, the present invention is not limited thereto. For example, the insulator 541 may have a single-layer structure or a stacked-layer structure of three or more layers. Although the first conductor of the conductor 540 and the second conductor of the conductor 540 are stacked in the transistor 500, the present invention is not limited thereto. For example, the conductor 540 may have a single-layer structure or a stacked-layer structure of three or more layers.

The structure of the transistor included in the semiconductor device of one embodiment of the present invention is not limited to that of the transistor 500 illustrated in FIG. 28A and FIG. 28B. The structure of the transistor included in the semiconductor device of one embodiment of the present invention may be changed in accordance with circumstances.

An insulator 111 is provided above the transistor 500.

An insulator having a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen is preferably used as the insulator 111; for example, aluminum oxide, magnesium oxide, hafnium oxide, gallium oxide, indium gallium zinc oxide, silicon nitride, or silicon nitride oxide can be used. For example, silicon nitride, which has a higher hydrogen barrier property, is preferably used for the insulator 111. For example, aluminum oxide or magnesium oxide, which has a function of capturing or fixing hydrogen well, is preferably used for the insulator 111.

The insulator 111 is preferably a film with high planarity. In this case, an organic material such as an acrylic resin or a polyimide can be used for the insulator 111, for example.

The light-emitting device 150a to the light-emitting device 150c are provided above the insulator 111.

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Here, the light-emitting device 150a to the light-emitting device 150c are described.

Over the insulator 111 are provided the conductor 121a to the conductor 121c functioning as respective pixel electrodes of the light-emitting device 150a to the light-emitting device 150c. In FIG. 27, a region where none of the conductor 121a to the conductor 121c is provided exists over part of the insulator 111. Note that in this specification and the like, the conductor 121a to the conductor 121c are collectively referred to as the conductor 121 in some cases.

The conductor 121a to the conductor 121c can be formed in such a manner that, for example, a conductive film is formed over the insulator 111 and the conductive film is subjected to a patterning step, an etching step, or the like.

The conductor 121a to the conductor 121c, respectively, function as anodes of the light-emitting device 150a, the light-emitting device 150b, and the light-emitting device 150c provided in the display device 10, for example.

Indium tin oxide (sometimes referred to as ITO) or the like can be used for the conductor 121a to the conductor 121c, for example.

Each of the conductor 121a to the conductor 121c may have a stacked-layer structure of two or more layers instead of a single layer. For example, a conductor having high visible-light reflectance can be used for the first-layer conductor and a conductor having a high light-transmitting property can be used for the uppermost-layer conductor. Examples of a conductor having high visible-light reflectance include silver, aluminum, and an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (Ag—Pd—Cu (APC) film). Examples of a conductor having a high light-transmitting property include indium tin oxide described above. The conductor 121a to the conductor 121c can be, for example, a stacked-layer film in which a pair of titanium films sandwich aluminum (a film in which Ti, Al, and Ti are stacked in this order), a stacked-layer film in which a pair of indium tin oxide films sandwich silver (a film in which ITO, Ag, and ITO are stacked in this order), or the like.

The insulator 112 is provided over the insulator 111 and the conductor 121a. In FIG. 27, a region where the insulator 112 is not provided exists over part of the conductor 121a, over part of the conductor 121b, and over part of the conductor 121c. The insulator 112 can be provided in such a manner that, for example, an insulating film to be the insulator 112 is formed over the insulator 111 and the conductor 121a to the conductor 121c, and the insulating film is subjected to patterning by a photolithography method or the like to be provided with opening portions reaching the conductor 121a to the conductor 121c in part of a region overlapping with the conductor 121a to the conductor 121c.

The insulator 112 can be an inorganic film with an insulating property, for example. As the inorganic film with an insulating property, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, aluminum nitride, or the like can be used, for example.

The insulator 112 may be an organic film including an insulating layer. Examples of the organic film that can be used for the insulator 112 include polyimide.

In addition, the insulator 112 may have a multilayer structure. Specifically, for example, the insulator 112 may have a multilayer structure in which the first layer is the above-described organic film and the second layer is the above-described inorganic film.

The EL layer 141a is provided over the insulator 112 and the conductor 121a. The EL layer 141b is provided over the insulator 112 and the conductor 121b. The EL layer 141c is

provided over the insulator **112** and the conductor **121c**. In FIG. **27**, a region where the EL layer **141a** to the EL layer **141c** are not provided exists over part of the insulator **112**.

The EL layer **141a** to the EL layer **141c** preferably include light-emitting layers emitting light of different colors from one another. For example, the EL layer **141a** can include a blue (B)-light-emitting layer, the EL layer **141b** can include a green (G)-light-emitting layer, and the EL layer **141c** can include a red (R)-light-emitting layer. The display device **10** may have such a structure (SBS structure) in which light-emitting layers for respective colors are provided over a plurality of pixel electrodes (the conductor **121a** to the conductor **121c**).

Note that the combination of colors of light emitted by the light-emitting layers included in the EL layer **141a** to the EL layer **141c** is not limited to the above, and a color such as cyan, magenta, yellow, or the like may also be used, for example. The number of colors of light emitted by the light-emitting devices **150** included in the display device **10**, which is three in the above example, may be two or four or more.

The EL layer **141a**, the EL layer **141b**, and the EL layer **141c** may each include one or more of an electron-injection layer, an electron-transport layer, a hole-injection layer, and a hole-transport layer in addition to a layer containing a light-emitting organic compound (light-emitting layer).

Specifically, the EL layer **141a**, the EL layer **141b**, and the EL layer **141c** can be formed, for example, by an evaporation method (a vacuum evaporation method or the like), a coating method (a dip coating method, a die coating method, a bar coating method, a spin coating method, a spray coating method, or the like), a printing method (an ink-jet method, a screen printing (stencil) method, an offset printing (planography) method, a flexography (relief printing) method, a gravure printing method, a micro-contact printing method, or the like), or the like.

In the case where a deposition method such as the coating method or the printing method is employed, a high-molecular compound (e.g., an oligomer, a dendrimer, or a polymer), a middle-molecular compound (a compound between a low-molecular compound and a high-molecular compound with a molecular weight of 400 to 4000), an inorganic compound (e.g., a quantum dot material), or the like can be used. Note that as the quantum dot material, a colloidal quantum dot material, an alloyed quantum dot material, a core-shell quantum dot material, a core quantum dot material, or the like can be used.

Like the light-emitting device **150** illustrated in FIG. **30A**, the light-emitting device **150a** to the light-emitting device **150c** in FIG. **27** can include a plurality of layers such as a layer **4420**, a light-emitting layer **4411**, and a layer **4430**, for example.

The layer **4420** can include, for example, a layer containing a substance with a high electron-injection property (electron-injection layer) and a layer containing a substance with a high electron-transport property (electron-transport layer). The light-emitting layer **4411** contains a light-emitting compound, for example. The layer **4430** can include, for example, a layer containing a substance with a high hole-injection property (hole-injection layer) and a layer containing a substance with a high hole-transport property (hole-transport layer).

The structure including the layer **4420**, the light-emitting layer **4411**, and the layer **4430**, which is provided between a pair of electrodes (the conductor **121** and a later-described conductor **122**), can function as a single light-emitting unit,

and the structure in FIG. **30A** is referred to as a single structure in this specification and the like.

Note that the structure in which a plurality of light-emitting layers (the light-emitting layer **4411**, a light-emitting layer **4412**, and a light-emitting layer **4413**) are provided between the layer **4420** and the layer **4430** as illustrated in FIG. **30B** is also a variation of the single structure.

A stack including a plurality of layers such as the layer **4420**, the light-emitting layer **4411**, and the layer **4430** is sometimes referred to as a light-emitting unit. A plurality of light-emitting units can be connected in series with an intermediate layer (a charge-generation layer) therebetween. Specifically, a light-emitting unit **4400a** and a light-emitting unit **4400b**, which are the plurality of light-emitting units, can be connected in series with an intermediate layer (charge-generation layer) **4440** therebetween as illustrated in FIG. **30C**. Such a structure is referred to as a tandem structure in this specification. A tandem structure may be rephrased as, for example, a stack structure in this specification and the like. The tandem structure enables a light-emitting device capable of high luminance light emission. By having a tandem structure, a light-emitting device presumably has increased emission efficiency and an extended lifetime. In the case where the light-emitting device **150** of the display device **10** in FIG. **27** has a tandem structure, the EL layer **141** can include, for example, the layer **4420**, the light-emitting layer **4411**, and the layer **4430** that are included in the light-emitting unit **4400a**, an intermediate layer **4440**, and the layer **4420**, the light-emitting layer **4412**, and the layer **4430** that are included in the light-emitting unit **4400b**.

In displaying white, the aforementioned SBS structure consumes lower power than the aforementioned single structure and tandem structure. To reduce power consumption, the SBS structure is thus preferably used. Meanwhile, the single structure and the tandem structure are preferable in that the manufacturing cost is low or the manufacturing yield is high because the manufacturing processes of the single structure and the tandem structure are simpler than that of the SBS structure.

The emission color of the light-emitting device **150** can be red, green, blue, cyan, magenta, yellow, white, or the like depending on the material that constitutes the EL layer **141**. Furthermore, the color purity can be further increased when the light-emitting device **150** has a microcavity structure.

The light-emitting element that emits white light preferably contains two or more kinds of light-emitting substances in the light-emitting layer. To obtain white light emission, two or more kinds of light-emitting substances are selected such that their emission colors are complementary.

The light-emitting layer preferably contains two or more selected from light-emitting substances that emit light of red (R), green (G), blue (B), yellow (Y), orange (O), and the like. Alternatively, the light-emitting layer preferably contains two or more light-emitting substances that emit light containing two or more of spectral components of R, G, and B.

As illustrated in FIG. **27**, there is a gap between the EL layers of two light-emitting devices of different colors. In this manner, the EL layer **141a**, the EL layer **141b**, and the EL layer **141c** are preferably provided so as not to be in contact with one another. This suitably prevents unintentional light emission (also referred to as crosstalk) from being caused by a current flowing through two adjacent EL layers. As a result, the contrast can be increased to achieve a display device with high display quality.

As the formation method of the EL layer **141a** to the EL layer **141c**, a method with photolithography can be given. For example, the EL layer **141a** to the EL layer **141c** can be formed in such a manner that an EL film to be the EL layer **141a** to the EL layer **141c** is formed over the insulator **111** and the conductor **121** and then subjected to patterning by a photolithography method. Alternatively, the EL layer **141a** to the EL layer **141c** may be formed in such a manner that the conductor **122** is formed over the EL film and both the conductor **122** and the EL film are subjected to patterning by a photolithography method. In this case, the EL layer **141a** to the EL layer **141c** have the same structure. Accordingly, in order that the display device **10** employing the above formation method can carry out color display, the light-emitting device **150a** to the light-emitting device **150c** respectively including the EL layer **141a** to the EL layer **141c** emit white light and the white light is extracted to the outside through coloring layers (color filters).

Alternatively, the EL layer **141a** to the EL layer **141c** are formed in the following manner: an EL film to be the EL layer **141a** is formed over the insulator **111** and the conductor **121** and processed by a photolithography method to form the EL layer **141a**. Then, the EL layer **141b** and the EL layer **141c** are formed in the predetermined regions by the same steps. The EL layer **141a** to the EL layer **141c** formed by this method can have different structures, enabling the display device **10** to have the SBS structure.

When the above method is employed, the gap between pixels can be short. As a result, the number of pixels included in the display portion can be large, increasing the resolution of the display device. The gap between the pixels is preferably less than or equal to 5 μm , further preferably less than or equal to 1 μm .

Other than a photolithography method, a nanoimprint method, a lift-off method, or the like may be employed in the formation of the EL layer **141a** to the EL layer **141c**. Alternatively, island-shaped thin films may be directly formed by a deposition method using a shielding mask such as a metal mask.

The conductor **122** is provided over the insulator **112**, the EL layer **141a**, the EL layer **141b**, and the EL layer **141c**. An insulator **113** is provided over the conductor **122**.

The conductor **122** functions as, for example, a common electrode for each of the light-emitting device **150a** to the light-emitting device **150c**. The conductor **122** preferably contains a conductive material having a light-transmitting property so that light emitted by the light-emitting device **150** can be extracted to above the display device **150**.

The conductor **122** is preferably a light-transmitting and light-reflective material having high conductivity (sometimes referred to as a transfective electrode). For example, an alloy of silver and magnesium, or indium tin oxide can be used as the conductor **122**.

The insulator **113** functions as, for example, a passivation film that protects the light-emitting device **150a**, the light-emitting device **150b**, and the light-emitting device **150c**. Thus, the insulator **113** is preferably formed using a material that prevents entry of water and the like. Any of the materials that can be used as the insulator **111** can be used as the insulator **113**, for example. Specifically, aluminum oxide, silicon nitride, silicon nitride oxide, or the like can be used.

A resin layer **161** is provided over the insulator **113**. A substrate **102** is provided over the resin layer **161**.

As the substrate **102**, a substrate having a light-transmitting property is preferably used, for example. Using a light-transmitting substrate as the substrate **102** enables extraction of light emitted from the light-emitting device

150a, the light-emitting device **150b**, and the light-emitting device **150c** to above the substrate **102**.

The display device **10** in FIG. **27** formed in the above-described manner can have a resolution preferably higher than or equal to 1000 ppi, further preferably higher than or equal to 3000 ppi, still further preferably higher than or equal to 5000 ppi.

Sealing Structure Example of Display Device

Next, a sealing structure for the light-emitting device **150a** to the light-emitting device **150c** applicable to the display device **10** in FIG. **27** is described.

FIG. **31A** is a cross-sectional view of an example of a sealing structure applicable to the display device **10** in FIG. **27**. Specifically, FIG. **31A** illustrates an end portion of the pixel array ALP of the display device **10** in FIG. **27** and components provided around the end portion. FIG. **31A** selectively illustrates only part of the pixel layer PXAL of the display device **10**. Specifically, FIG. **31A** illustrates the insulator **111**, a plug connected to the transistor **500**, and insulators, conductors, the light-emitting device **150a** to the light-emitting device **150c**, and the like that are positioned above the insulator **111**.

In the display device **10** in FIG. **31A**, an adhesive layer **164** is provided at or around the end portion of the pixel array ALP. Specifically, the display device **10** is fabricated such that the adhesive layer **164** is placed between the insulator **112** and the substrate **102**.

The adhesive layer **164** is preferably formed using, for example, a material inhibiting transmission of an impurity such as moisture. Using the material for the adhesive layer **164** can increase the reliability of the display device **10**.

The structure in which the adhesive layer **164** is used to bond the insulator **112** and the substrate **102** with the resin layer **161** positioned between the insulator **112** and the substrate **102** is sometimes referred to as a solid sealing structure. In the case where the resin layer **161** in the solid sealing structure has a function of bonding the insulator **112** and the substrate **102** like the adhesive layer **164**, the adhesive layer **164** is not necessarily provided.

A structure in which the adhesive layer **164** is used to bond the insulator **112** and the substrate **102** with not the resin layer **161** but an inert gas filling the space between the insulator **112** and the substrate **102** is sometimes referred to as a hollow sealing structure (not shown). Examples of an inert gas include a nitrogen gas and an argon gas.

In the sealing structure of the display device **10** illustrated in FIG. **31A**, two or more overlapping adhesive layers may be used. For example, as illustrated in FIG. **31B**, an adhesive layer **165** may be provided inward from the adhesive layer **164** (may be provided between the adhesive layer **164** and the resin layer **161**). Two or more overlapping adhesive layers can inhibit transmission of an impurity such as moisture more, further increasing the reliability of the display device **10**.

A desiccant may be mixed into the adhesive layer **165**. In this case, the desiccant adsorbs moisture contained in the resin layer **161**, insulators, conductors, EL layers, and the like that are provided inward from the adhesive layer **164** and the adhesive layer **165**, increasing the reliability of the display device **10**.

Although the solid sealing structure of the display device **10** is illustrated in FIG. **31B**, a hollow sealing structure may be employed.

Furthermore, in each of the sealing structures of the display device **10** in FIG. **31A** and FIG. **31B**, the resin layer

161 filling the space may be replaced with an inert liquid. Examples of the inert liquid include a fluorine-based inert liquid.

Modification Example of Display Device

One embodiment of the present invention is not limited to the above-described structure, and the above-described structure can be changed as appropriate in accordance with the situation. A modification example of the display device 10 in FIG. 27 is described with reference to FIG. 32A to FIG. 33D. Note that FIG. 32A to FIG. 33D selectively illustrate only part of the pixel layer PXAL of the display device 10. Specifically, each of FIG. 32A to FIG. 33D illustrates the insulator 111, a plug connected to the transistor 500, and insulators, conductors, the light-emitting device 150a to the light-emitting device 150c, and the like that are positioned above the insulator 111.

The display device 10 may have a structure in which the number of colors of light emitted by the light-emitting devices 150 is two, for example. Moreover, the display device 10 may have a structure in which the number of colors of light emitted by the light-emitting devices 150 is four or more, for example (not illustrated).

The display device 10 may have a structure in which, for example, an EL layer 142 is provided over the EL layer 141a to the EL layer 141c and the insulator 112 as illustrated in FIG. 32A. Specifically, for example, in FIG. 30A, the EL layer 142 can include the layer 4420 when the EL layer 141a to the EL layer 141c each include the layer 4430 and the light-emitting layer 4411. In this case, the layer 4420 included in the EL layer 142 functions as a common layer shared by the light-emitting device 150a to the light-emitting device 150c. In a similar manner, for another example, in FIG. 30B, the EL layer 142 can include the layer 4420 when the EL layer 141a to the EL layer 141c each include the layer 4430 and the light-emitting layer 4411, in which case the layer 4420 included in the EL layer 142 functions as a common layer shared by the light-emitting device 150a to the light-emitting devices 150c. For another example, in FIG. 30C, the EL layer 142 can include the layer 4420 of the light-emitting unit 4400b when the EL layer 141a to the EL layer 141c each include the layer 4430, the light-emitting layer 4412, and the layer 4420 that are included in the light-emitting unit 4400b, the intermediate layer 4440, and the layer 4430 and the light-emitting layer 4411 that are included in the light-emitting unit 4400a, in which case the layer 4420 of the light-emitting unit 4400a included in the EL layer 142 functions as a common layer shared by the light-emitting device 150a to the light-emitting devices 150c.

In the structure of the display device 10, for example, the insulator 112 may have a multilayer structure that includes an insulator made of an organic material as the first layer and an insulator made of an inorganic material as the second layer, as described above. As an example, FIG. 32B illustrates a cross-sectional view of part of the display device 10 in which the insulator 112 has a multilayer structure including an insulator 112a that is an insulator made of an organic material and an insulator 112b that is an insulator made of an inorganic material.

The organic material can be, for example, a polyimide or the like, and the inorganic material can be any of the materials usable for the insulator 112 of the display device 10 in FIG. 27 and the like.

In the structure of the display device 10, for example, the insulator 113 may have a stacked-layer structure including

two or more layers, instead of a single layer. The insulator 113 may have a three-layer structure that includes an insulator made of an inorganic material as the first layer, an insulator made of an organic material as the second layer, and an insulator made of an inorganic material as the third layer. FIG. 32C illustrates a cross-sectional view of part of the display device 10 in which the insulator 113 has a multilayer structure including an insulator 113a that is an insulator made of an inorganic material, an insulator 113b that is an insulator made of an organic material, and an insulator 113c that is an insulator made of an inorganic material.

In the structure of the display device 10, for example, the EL layer 141a to the EL layer 141c may be each provided with a microcavity structure. In the microcavity structure, for example, the conductor 122 as an upper electrode (common electrode) is formed using a light-transmitting and light-reflective conductive material, the conductor 121 as a lower electrode (pixel electrode) is formed using a light-reflective conductive material, and the distance between a bottom surface of the light-emitting layer and a top surface of the lower electrode, i.e., the thickness of the layer 4430 in FIG. 30A, is set to the thickness corresponding to the wavelength of the color of light emitted by the light-emitting layer included in the EL layer 141.

For example, light that is reflected back by the lower electrode (reflected light) considerably interferes with light that directly enters the upper electrode from the light-emitting layer (incident light). For this reason, the optical length between the lower electrode and the light-emitting layer is preferably adjusted to $(2n-1)\lambda/4$ (n is a natural number of 1 or larger and λ is a wavelength of light to be amplified). By adjusting the optical length, the phases of the reflected light and the incident light having the wavelength λ can be aligned with each other and the light emitted from the light-emitting layer can be further amplified. In the case where the reflected light and the incident light have a wavelength other than the wavelength λ , their phases are not aligned with each other, resulting in attenuation without resonance.

In the above-described structure, the EL layer may include a plurality of light-emitting layers or a single light-emitting layer. Furthermore, for example, the combination with the aforementioned tandem light-emitting device structure is possible; specifically, the above-described structure can be applied to a structure in which one light-emitting device includes a plurality of EL layers sandwiching a charge-generation layer and each EL layer includes one or more light-emitting layers.

With the microcavity structure, emission intensity with a specific wavelength in the front direction can be increased, whereby power consumption can be reduced. Particularly in the case of a device for XR such as VR and AR, light emitted from the light-emitting device in the front direction often enters the eyes of the user wearing the device; thus, a display device of a device for XR preferably includes a microcavity structure. Note that in the case of a display device which displays images with subpixels of four colors, red, yellow, green, and blue, the display device can have favorable characteristics because a microcavity structure suitable for wavelengths of the corresponding colors can be employed in each subpixel, in addition to the effect of an improvement in luminance owing to yellow light emission.

As an example, FIG. 33A illustrates a cross-sectional view of part of the display device including a microcavity structure. In the case where the light-emitting device 150a includes a blue (B)-light-emitting layer, the light-emitting

device **150b** includes a green (G)-light-emitting layer, and the light-emitting device **150c** includes a red (R)-light-emitting layer, it is preferable that the EL layer **141a** have the smallest thickness, the EL layer **141b** have the second largest thickness, and the EL layer **141c** have the largest thickness as illustrated in FIG. 33A. Specifically, the thicknesses of the layers **4430** included in the EL layer **141a**, the EL layer **141b**, and the EL layer **141c** are determined depending on the color of the light emitted by the corresponding light-emitting layer. In this case, the layer **4430** included in the EL layer **141a** has the smallest thickness and the layer **4430** included in the EL layer **141c** has the largest thickness.

In the structure of the display device **10**, for example, a coloring layer (color filter) or the like may be provided. As an example, FIG. 33B illustrates a structure in which a coloring layer **162a**, a coloring layer **162b**, and a coloring layer **162c** are included between the resin layer **161** and the substrate **102**. Note that the coloring layer **162a** to the coloring layer **162c** can be formed on the substrate **102**, for example. In the case where the light-emitting device **150a** includes a blue (B)-light-emitting layer, the light-emitting device **150b** includes a green (G)-light-emitting layer, and the light-emitting device **150c** includes a red (R)-light-emitting layer, the coloring layer **162a** is a blue coloring layer, the coloring layer **162b** is a green coloring layer, and the coloring layer **162c** is a red coloring layer.

The display device **10** illustrated in FIG. 33B can be fabricated in such a manner that the substrate **102** provided with the coloring layer **162a** to the coloring layer **162c** and the substrate **310** over which components up to the light-emitting device **150a** to the light-emitting device **150c** are formed are bonded to each other with the resin layer **161** therebetween. This bonding is preferably performed such that the light-emitting device **150a** and the coloring layer **162a** overlap with each other, the light-emitting device **150b** and the coloring layer **162b** overlap with each other, and the light-emitting device **150c** and the coloring layer **162c** overlap with each other. In the display device **10** provided with the coloring layer **162a** to the coloring layer **162c**, for example, the light emitted by the light-emitting device **150b** is extracted to above the substrate **102** not through the coloring layer **162a** or the coloring layer **162c** but through the coloring layer **162b**. That is, light emitted from the light-emitting device **150** in an oblique direction (a direction at an elevation angle with a top surface of the substrate **102** used as a horizontal plane) can be blocked in the display device **10**; thus, the viewing angle dependence of the display device **10** can be reduced, inhibiting the display quality of an image displayed by the display device **10** from decreasing when the image is viewed from an oblique direction.

The coloring layer **162a** to the coloring layer **162c** formed on the substrate **102** may be covered with, for example, a resin which is also referred to as an overcoat layer. Specifically, the resin layer **161**, the overcoat layer, the coloring layer **162a** to the coloring layer **162c**, and the substrate **102** may be stacked in this order in the display device **10** (not illustrated). Note that examples of the resin usable for the overcoat layer include a thermosetting material that has a light-transmitting property and is based on an acrylic or epoxy resin.

In the structure of the display device **10**, for example, a black matrix may be included in addition to the coloring layers (not illustrated). The black matrix **163** between the coloring layer **162a** and the coloring layer **162b**, between the coloring layer **162b** and the coloring layer **162c**, and between the coloring layer **162c** and the coloring layer **162a**

can block more light emitted from the light-emitting device **150** in an oblique direction (a direction at an elevation angle with the top surface of the substrate **102** used as a horizontal plane) in the display device **10**; thus, the display quality of an image displayed by the display device **10** can be more prevented from decreasing when the image is viewed from an oblique direction.

In the case where the display device includes coloring layers as illustrated in FIG. 33B or the like, the light-emitting device **150a** to the light-emitting device **150c** of the display apparatus may each be a white-light-emitting device (not shown). The structure of the light-emitting device can be a single structure or a tandem structure, for example.

The display device **10** may have a structure in which the insulator **112** is not provided over the conductor **121a** to the conductor **121c**, for example. FIG. 33C illustrates a structure example in which the insulator **112** is not provided in the display device **10** in FIG. 27 or the like. Furthermore, the display device **10** may have a structure in which the conductor **121a** to the conductor **121c** are embedded in the insulator **111**, for example. FIG. 33D illustrates a structure example of the display device in which the conductor **121a** to the conductor **121c** are embedded in the insulator **111**. Note that this structure can be fabricated in such a manner that, for example, opening portions in which the conductor **121a** to the conductor **121c** are to be embedded are formed in the insulator **111**, a conductive film to be the conductor **121a** to the conductor **121c** is formed, and then, chemical mechanical polishing (CMP) is performed until the insulator **111** is exposed.

In the above-described structure of the display device **10**, the conductor **121a** to the conductor **121c** serve as the anodes and the conductor **122** serves as the cathode; however, the display device **10** may have a structure in which the conductor **121a** to the conductor **121c** serve as cathodes and the conductor **122** serves as an anode. In other words, in the above-described manufacturing process, the stacking order of the hole-injection layer, hole-transport layer, light-emitting layer, electron-transport layer, and electron-injection layer that are included in the EL layer **141a** to the EL layer **141c** and the EL layer **142** can be reversed.

Note that the insulators, the conductors, the semiconductors, and the like disclosed in this specification and the like can be formed by a PVD (Physical Vapor Deposition) method or a CVD method. Examples of a PVD method include a sputtering method, a resistance heating evaporation method, an electron beam evaporation method, and a PLD method. The formation by a plasma CVD method or a thermal CVD method or the like can be given as a CVD method. As examples of the thermal CVD method, a MOCVD (Metal Organic Chemical Vapor Deposition) method, an ALD method, and the like can be particularly given.

A thermal CVD method, which is a deposition method not using plasma, has an advantage that no defect due to plasma damage is generated.

Deposition by a thermal CVD method may be performed in such a manner that a source gas and an oxidizer are supplied into a chamber at a time, the pressure in the chamber is set to an atmospheric pressure or a reduced pressure, and they are made to react with each other in the vicinity of the substrate or over the substrate to be deposited over the substrate.

Deposition by an ALD method may be performed in such a manner that the pressure in a chamber is set to an atmospheric pressure or a reduced pressure, source gases for reaction are sequentially introduced into the chamber, and

then the sequence of the gas introduction is repeated. For example, two or more kinds of source gases are sequentially supplied to the chamber by switching respective switching valves (also referred to as high-speed valves); in order to avoid mixing of the plurality of kinds of source gases, an inert gas (argon, nitrogen, or the like) or the like is introduced at the same time as or after introduction of a first source gas and then a second source gas is introduced. Note that in the case where the first source gas and the inert gas are introduced at a time, the inert gas serves as a carrier gas, and the inert gas may also be introduced at the same time as the introduction of the second source gas. Alternatively, the second source gas may be introduced after the first source gas is exhausted by vacuum evacuation instead of the introduction of the inert gas. The first source gas is adsorbed on the surface of the substrate to deposit a first thin layer; then the second source gas is introduced to react with the first thin layer; as a result, a second thin layer is stacked over the first thin layer, so that a thin film is formed. The sequence of the gas introduction is controlled and repeated a plurality of times until a desired thickness is obtained, so that a thin film with excellent step coverage can be formed. The thickness of the thin film can be adjusted by the number of repetition times of the sequence of the gas introduction; therefore, an ALD method makes it possible to accurately adjust film thickness and is thus suitable for manufacturing a minute FET.

A variety of films such as the metal film, the semiconductor film, and the inorganic insulating film disclosed in the above-described embodiments can be formed by a thermal CVD method such as an MOCVD method or an ALD method; for example, in the case of forming an In—Ga—Zn—O film, trimethylindium ($\text{In}(\text{CH}_3)_3$), trimethylgallium ($\text{Ga}(\text{CH}_3)_3$), and dimethylzinc ($\text{Zn}(\text{CH}_3)_2$) are used. Without limitation to the above combination, triethylgallium ($\text{Ga}(\text{C}_2\text{H}_5)_3$) can also be used instead of trimethylgallium, and diethylzinc ($\text{Zn}(\text{C}_2\text{H}_5)_2$) can also be used instead of dimethylzinc.

For example, in the case where a hafnium oxide film is formed with a deposition apparatus using an ALD method, two kinds of gases, ozone (O_3) as an oxidizer and a source gas which is obtained by vaporizing liquid containing a solvent and a hafnium precursor compound (hafnium alkoxide or hafnium amide such as tetrakis(dimethylamide)hafnium (TDMAH, $\text{Hf}[\text{N}(\text{CH}_3)_2]_4$)), are used. Examples of another material include tetrakis(ethylmethylamide)hafnium.

For example, in the case where an aluminum oxide film is formed by a deposition apparatus using an ALD method, two kinds of gases, H_2O as an oxidizer and a source gas that is obtained by vaporizing liquid containing a solvent and an aluminum precursor compound (trimethylaluminum (TMA, $\text{Al}(\text{CH}_3)_3$) or the like) are used. Examples of another material include tris(dimethylamide)aluminum, triisobutylaluminum, and aluminum tris(2,2,6,6-tetramethyl-3,5-heptanedionate).

For example, in the case where a silicon oxide film is formed by a deposition apparatus using an ALD method, hexachlorodisilane is adsorbed on a surface on which a film is to be formed, and radicals of an oxidizing gas (O_2 or dinitrogen monoxide) are supplied to react with the adsorbate.

For example, in the case where a tungsten film is deposited by a deposition apparatus using an ALD method, a WF_6 gas and a B_2H_6 gas are sequentially and repeatedly introduced to form an initial tungsten film, and then a WF_6 gas

and an H_2 gas are sequentially and repeatedly introduced to form a tungsten film. Note that an SiH_4 gas may be used instead of a B_2H_6 gas.

In the case where an In—Ga—Zn—O film is formed as an oxide semiconductor film with a deposition apparatus using an ALD method, a precursor (sometimes called a metal precursor or the like) and an oxidizer (sometimes called a reactant, a non-metal precursor, or the like) are sequentially and repetitively introduced. Specifically, for example, an $\text{In}(\text{CH}_3)_3$ gas as a precursor and an O_3 gas as an oxidizer are introduced to form an In—O layer; a $\text{Ga}(\text{CH}_3)_3$ gas as a precursor and an O_3 gas as an oxidizer are introduced to form a GaO layer; and then, a $\text{Zn}(\text{CH}_3)_2$ gas as a precursor and an O_3 gas as an oxidizer are introduced to form a ZnO layer. Note that the order of these layers is not limited to this example. A mixed oxide layer such as an In—Ga—O layer, an In—Zn—O layer, or a Ga—Zn—O layer may be formed with the use of these gases. Note that although an H_2O gas which is obtained by bubbling water with an inert gas such as Ar may be used instead of an O_3 gas, it is preferable to use an O_3 gas which does not contain H. Furthermore, instead of an $\text{In}(\text{CH}_3)_3$ gas, an $\text{In}(\text{C}_2\text{H}_5)_3$ gas may be used. Furthermore, instead of a $\text{Ga}(\text{CH}_3)_3$ gas, a $\text{Ga}(\text{C}_2\text{H}_5)_3$ gas may be used. Furthermore, a $\text{Zn}(\text{CH}_3)_2$ gas may be used.

The aspect ratio of a display portion of the display device of one embodiment of the present invention is not particularly limited. For example, the display device is compatible with a variety of screen ratios such as 1:1 (a square), 4:3, 16:9, and 16:10.

The shape of the display device of one embodiment of the present invention is not particularly limited. The display device can have any of various shapes such as a rectangular shape, a polygonal shape (e.g., octagon), a circular shape, and an elliptical shape.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 5

Described in this embodiment is a metal oxide (hereinafter, also referred to as an oxide semiconductor) that can be used in the OS transistor described in the above embodiment.

The metal oxide preferably contains at least indium or zinc. In particular, indium and zinc are preferably contained. In addition, aluminum, gallium, yttrium, tin, or the like is preferably contained. Furthermore, one or more kinds selected from boron, silicon, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, cobalt, and the like may be contained.

<Classification of Crystal Structure>

First, the classification of the crystal structures of an oxide semiconductor will be described with reference to FIG. 34A. FIG. 34A is a diagram showing the classification of crystal structures of an oxide semiconductor, typically IGZO (a metal oxide containing In, Ga, and Zn).

As shown in FIG. 34A, an oxide semiconductor is roughly classified into “Amorphous”, “Crystalline”, and “Crystal”. The term “Amorphous” includes completely amorphous. The term “Crystalline” includes CAAC (c-axis-aligned crystalline), nc (nanocrystalline), and CAC (Cloud-Aligned Composite) (excluding single crystal and poly crystal). Note that the term “Crystalline” excludes single crystal, poly crystal, and completely amorphous. The term “Crystal” includes single crystal and poly crystal.

Note that the structures in the thick frame in FIG. 34A are in an intermediate state between "Amorphous" and "Crystal", and belong to a new crystalline phase. That is, these structures are completely different from "Amorphous", which is energetically unstable, and "Crystal".

Note that a crystal structure of a film or a substrate can be evaluated with an X-ray diffraction (XRD) spectrum. FIG. 34B shows an XRD spectrum, which is obtained by GIXD (Grazing-Incidence XRD) measurement, of a CAAC-IGZO film classified into "Crystalline" (the vertical axis represents intensity in arbitrary unit (a.u.)). Note that a GIXD method is also referred to as a thin film method or a Seemann-Bohlin method. The XRD spectrum that is shown in FIG. 34B and obtained by GIXD measurement may be hereinafter simply referred to as an XRD spectrum. The CAAC-IGZO film in FIG. 34B has a composition in the neighborhood of In:Ga:Zn=4:2:3 [atomic ratio]. The CAAC-IGZO film in FIG. 34B has a thickness of 500 nm.

As shown in FIG. 34B, a clear peak indicating crystallinity is detected in the XRD spectrum of the CAAC-IGZO film. Specifically, a peak indicating c-axis alignment is detected at 2θ of around 31° in the XRD spectrum of the CAAC-IGZO film. As shown in FIG. 34B, the peak at 2θ of around 31° is asymmetric with respect to the axis of the angle at which the peak intensity (Intensity) is detected.

A crystal structure of a film or a substrate can also be evaluated with a diffraction pattern obtained by a nanobeam electron diffraction (NBED) method (such a pattern is also referred to as a nanobeam electron diffraction pattern). FIG. 34C shows a diffraction pattern of the CAAC-IGZO film. FIG. 34C shows a diffraction pattern obtained by the NBED method in which an electron beam is incident in the direction parallel to the substrate. The composition of the CAAC-IGZO film in FIG. 34C is In:Ga:Zn=4:2:3 [atomic ratio] or the neighborhood thereof. In the nanobeam electron diffraction method, electron diffraction is performed with a probe diameter of 1 nm.

As shown in FIG. 34C, a plurality of spots indicating c-axis alignment are observed in the diffraction pattern of the CAAC-IGZO film.

<<Structure of Oxide Semiconductor>>

Oxide semiconductors might be classified in a manner different from one shown in FIG. 34A when classified in terms of the crystal structure. Oxide semiconductors are classified into a single crystal oxide semiconductor and a non-single-crystal oxide semiconductor, for example. Examples of the non-single-crystal oxide semiconductor include the above-described CAAC-OS and nc-OS. Other examples of the non-single-crystal oxide semiconductor include a polycrystalline oxide semiconductor, an amorphous-like oxide semiconductor (a-like OS), and an amorphous oxide semiconductor.

Here, the above-described CAAC-OS, nc-OS, and a-like OS are described in detail.

[CAAC-OS]

The CAAC-OS is an oxide semiconductor that has a plurality of crystal regions each of which has c-axis alignment in a particular direction. Note that the particular direction refers to the thickness direction of a CAAC-OS film, the normal direction of the surface where the CAAC-OS film is formed, or the normal direction of the surface of the CAAC-OS film. The crystal region refers to a region having a periodic atomic arrangement. When an atomic arrangement is regarded as a lattice arrangement, the crystal region also refers to a region with a uniform lattice arrangement. The CAAC-OS has a region where a plurality of crystal regions are connected in the a-b plane direction, and

the region has distortion in some cases. Note that distortion refers to a portion where the direction of a lattice arrangement changes between a region with a uniform lattice arrangement and another region with a uniform lattice arrangement in a region where a plurality of crystal regions are connected. That is, the CAAC-OS is an oxide semiconductor having c-axis alignment and having no clear alignment in the a-b plane direction.

Note that each of the plurality of crystal regions is formed of one or more minute crystals (crystals each of which has a maximum diameter of less than 10 nm). In the case where the crystal region is formed of one minute crystal, the maximum diameter of the crystal region is less than 10 nm. In the case where the crystal region is formed of a large number of minute crystals, the size of the crystal region may be approximately several tens of nanometers.

In the case of an In-M-Zn oxide (the element M is one or more kinds selected from aluminum, gallium, yttrium, tin, titanium, and the like), the CAAC-OS tends to have a layered crystal structure (also referred to as a layered structure) in which a layer containing indium (In) and oxygen (hereinafter, an In layer) and a layer containing the element M, zinc (Zn), and oxygen (hereinafter, an (M,Zn) layer) are stacked. Indium and the element M can be replaced with each other. Therefore, indium may be contained in the (M,Zn) layer. In addition, the element M may be contained in the In layer. Note that Zn may be contained in the In layer. Such a layered structure is observed as a lattice image in a high-resolution TEM image, for example.

When the CAAC-OS film is subjected to structural analysis by out-of-plane XRD measurement with an XRD apparatus using $\theta/2\theta$ scanning, for example, a peak indicating c-axis alignment is detected at 2θ of 31° or around 31°. Note that the position of the peak indicating c-axis alignment (the value of 2θ) may change depending on the kind, composition, or the like of the metal element contained in the CAAC-OS.

For example, a plurality of bright spots are observed in the electron diffraction pattern of the CAAC-OS film. Note that one spot and another spot are observed point-symmetrically with a spot of the incident electron beam passing through a sample (also referred to as a direct spot) as the symmetric center.

When the crystal region is observed from the particular direction, a lattice arrangement in the crystal region is basically a hexagonal lattice arrangement; however, a unit lattice is not always a regular hexagon and is a non-regular hexagon in some cases. A pentagonal lattice arrangement, a heptagonal lattice arrangement, and the like are included in the distortion in some cases. Note that a clear crystal grain boundary (also referred to as grain boundary) cannot be observed even in the vicinity of the distortion in the CAAC-OS. That is, formation of a crystal grain boundary is inhibited by the distortion of lattice arrangement. This is probably because the CAAC-OS can tolerate distortion owing to a low density of arrangement of oxygen atoms in the a-b plane direction, an interatomic bond distance changed by substitution of a metal atom, and the like.

Note that a crystal structure in which a clear crystal grain boundary is observed is what is called polycrystal. It is highly probable that the crystal grain boundary becomes a recombination center and traps carriers and thus decreases the on-state current and field-effect mobility of a transistor, for example. Thus, the CAAC-OS in which no clear crystal grain boundary is observed is one of crystalline oxides having a crystal structure suitable for a semiconductor layer of a transistor. Note that Zn is preferably contained to form

the CAAC-OS. For example, an In—Zn oxide and an In—Ga—Zn oxide are suitable because they can inhibit generation of a crystal grain boundary as compared with an In oxide.

The CAAC-OS is an oxide semiconductor with high crystallinity in which no clear crystal grain boundary is observed. Thus, in the CAAC-OS, a reduction in electron mobility due to the crystal grain boundary is unlikely to occur. Moreover, since the crystallinity of an oxide semiconductor might be decreased by entry of impurities, formation of defects, and the like, the CAAC-OS can be regarded as an oxide semiconductor that has small amounts of impurities, defects (e.g., oxygen vacancies), and the like. Thus, an oxide semiconductor including the CAAC-OS is physically stable. Therefore, the oxide semiconductor including the CAAC-OS is resistant to heat and has high reliability. In addition, the CAAC-OS is stable with respect to high temperatures in the manufacturing process (what is called thermal budget). Accordingly, the use of the CAAC-OS for the OS transistor can extend the degree of freedom of the manufacturing process.

[nc-OS]

In the nc-OS, a microscopic region (e.g., a region with a size greater than or equal to 1 nm and less than or equal to 10 nm, in particular, a region with a size greater than or equal to 1 nm and less than or equal to 3 nm) has a periodic atomic arrangement. In other words, the nc-OS includes a minute crystal. Note that the size of the minute crystal is, for example, greater than or equal to 1 nm and less than or equal to 10 nm, particularly greater than or equal to 1 nm and less than or equal to 3 nm; thus, the minute crystal is also referred to as a nanocrystal. Furthermore, there is no regularity of crystal orientation between different nanocrystals in the nc-OS. Thus, the orientation in the whole film is not observed. Accordingly, the nc-OS cannot be distinguished from an a-like OS and an amorphous oxide semiconductor by some analysis methods. For example, when an nc-OS film is subjected to structural analysis by out-of-plane XRD measurement with an XRD apparatus using $\theta/2\theta$ scanning, a peak indicating crystallinity is not detected. Furthermore, a diffraction pattern like a halo pattern is observed when the nc-OS film is subjected to electron diffraction (also referred to as selected-area electron diffraction) using an electron beam with a probe diameter larger than the diameter of a nanocrystal (e.g., larger than or equal to 50 nm). Meanwhile, in some cases, a plurality of spots in a ring-like region with a direct spot as the center are observed in the obtained electron diffraction pattern when the nc-OS film is subjected to electron diffraction (also referred to as nanobeam electron diffraction) using an electron beam with a probe diameter nearly equal to or smaller than the diameter of a nanocrystal (e.g., 1 nm or larger and 30 nm or smaller).

[a-Like OS]

The a-like OS is an oxide semiconductor having a structure between those of the nc-OS and the amorphous oxide semiconductor. The a-like OS contains a void or a low-density region. That is, the a-like OS has low crystallinity as compared with the nc-OS and the CAAC-OS. Moreover, the a-like OS has a higher hydrogen concentration in the film than the nc-OS and the CAAC-OS.

<<Structure of Oxide Semiconductor>>

Next, the above-described CAC-OS will be described in detail. Note that the CAC-OS relates to the material composition.

[CAC-OS]

The CAC-OS refers to one composition of a material in which elements constituting a metal oxide are unevenly

distributed with a size greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 3 nm, or a similar size, for example. Note that a state in which one or more metal elements are unevenly distributed and regions including the metal element(s) are mixed with a size greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 1 nm and less than or equal to 3 nm, or a similar size in a metal oxide is hereinafter referred to as a mosaic pattern or a patch-like pattern.

In addition, the CAC-OS has a composition in which materials are separated into a first region and a second region to form a mosaic pattern, and the first regions are distributed in the film (this composition is hereinafter also referred to as a cloud-like composition). That is, the CAC-OS is a composite metal oxide having a composition in which the first regions and the second regions are mixed.

Note that the atomic ratios of In, Ga, and Zn to the metal elements contained in the CAC-OS in an In—Ga—Zn oxide are denoted with [In], [Ga], and [Zn], respectively. For example, the first region in the CAC-OS in the In—Ga—Zn oxide has [In] higher than [In] in the composition of the CAC-OS film. Moreover, the second region has [Ga] higher than [Ga] in the composition of the CAC-OS film. Alternatively, for example, the first region has [In] higher than [In] in the second region and [Ga] lower than [Ga] in the second region. Moreover, the second region has [Ga] higher than [Ga] in the first region and [In] lower than [In] in the first region.

Specifically, the first region includes indium oxide, indium zinc oxide, or the like as its main component. The second region includes gallium oxide, gallium zinc oxide, or the like as its main component. That is, the first region can be rephrased as a region containing In as its main component. The second region can be rephrased as a region containing Ga as its main component.

Note that a clear boundary between the first region and the second region cannot be observed in some cases.

For example, in EDX mapping obtained by energy dispersive X-ray spectroscopy (EDX), it is confirmed that the CAC-OS in the In—Ga—Zn oxide has a structure in which the region containing In as its main component (the first region) and the region containing Ga as its main component (the second region) are unevenly distributed and mixed.

In the case where the CAC-OS is used for a transistor, a switching function (on/off switching function) can be given to the CAC-OS owing to the complementary action of the conductivity derived from the first region and the insulating property derived from the second region. That is, the CAC-OS has a conducting function in part of the material and has an insulating function in another part of the material; as a whole, the CAC-OS has a function of a semiconductor. Separation of the conducting function and the insulating function can maximize each function. Accordingly, when the CAC-OS is used for a transistor, a high on-state current (I_{on}), high field-effect mobility (μ), and excellent switching operation can be achieved.

An oxide semiconductor has various structures with different properties. Two or more kinds among the amorphous oxide semiconductor, the polycrystalline oxide semiconductor, the a-like OS, the CAC-OS, the nc-OS, and the CAAC-OS may be included in an oxide semiconductor of one embodiment of the present invention.

<Transistor Including Oxide Semiconductor>

Next, the case where the above oxide semiconductor is used for a transistor will be described.

When the above oxide semiconductor is used for a transistor, a transistor with high field-effect mobility can be achieved. In addition, a transistor having high reliability can be achieved.

An oxide semiconductor having a low carrier concentration is preferably used in a transistor. For example, the carrier concentration of an oxide semiconductor is lower than or equal to $1 \times 10^{17} \text{ cm}^{-3}$, preferably lower than or equal to $1 \times 10^{15} \text{ cm}^{-3}$, further preferably lower than or equal to $1 \times 10^{13} \text{ cm}^{-3}$, still further preferably lower than or equal to $1 \times 10^{11} \text{ cm}^{-3}$, yet further preferably lower than $1 \times 10^{10} \text{ cm}^{-3}$, and higher than or equal to $1 \times 10^{-9} \text{ cm}^{-3}$. In order to reduce the carrier concentration of an oxide semiconductor film, the impurity concentration in the oxide semiconductor film is reduced so that the density of defect states can be reduced. In this specification and the like, a state with a low impurity concentration and a low density of defect states is referred to as a highly purified intrinsic or substantially highly purified intrinsic state. Note that an oxide semiconductor having a low carrier concentration may be referred to as a highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor.

A highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has a low density of defect states and thus also has a low density of trap states in some cases.

Charge trapped by the trap states in the oxide semiconductor takes a long time to disappear and might behave like fixed charge. Thus, a transistor whose channel formation region is formed in an oxide semiconductor with a high density of trap states has unstable electrical characteristics in some cases.

Accordingly, in order to obtain stable electrical characteristics of a transistor, reducing the impurity concentration in an oxide semiconductor is effective. In order to reduce the impurity concentration in the oxide semiconductor, it is preferable that the impurity concentration in an adjacent film be also reduced. Examples of impurities include hydrogen, nitrogen, an alkali metal, an alkaline earth metal, iron, nickel, and silicon.

<Impurity>

Here, the influence of each impurity in the oxide semiconductor will be described.

When silicon or carbon, which is one of Group 14 elements, is contained in the oxide semiconductor, defect states are formed in the oxide semiconductor. Thus, the concentration of silicon or carbon in the oxide semiconductor and the concentration of silicon or carbon in the vicinity of an interface with the oxide semiconductor (the concentration obtained by secondary ion mass spectrometry (SIMS)) are each set lower than or equal to $2 \times 10^{18} \text{ atoms/cm}^3$, preferably lower than or equal to $2 \times 10^{17} \text{ atoms/cm}^3$.

When the oxide semiconductor contains an alkali metal or an alkaline earth metal, defect states are formed and carriers are generated in some cases. Thus, a transistor using an oxide semiconductor that contains an alkali metal or an alkaline earth metal is likely to have normally-on characteristics. Thus, the concentration of an alkali metal or an alkaline earth metal in the oxide semiconductor, which is obtained by SIMS, is lower than or equal to $1 \times 10^{18} \text{ atoms/cm}^3$, preferably lower than or equal to $2 \times 10^{16} \text{ atoms/cm}^3$.

Furthermore, when the oxide semiconductor contains nitrogen, the oxide semiconductor easily becomes n-type by generation of electrons serving as carriers and an increase in carrier concentration. As a result, a transistor using an oxide semiconductor containing nitrogen as a semiconductor is likely to have normally-on characteristics. When nitrogen is

contained in the oxide semiconductor, a trap state is sometimes formed. This might make the electrical characteristics of the transistor unstable. Therefore, the concentration of nitrogen in the oxide semiconductor, which is obtained by SIMS, is set lower than $5 \times 10^{19} \text{ atoms/cm}^3$, preferably lower than or equal to $5 \times 10^{18} \text{ atoms/cm}^3$, further preferably lower than or equal to $1 \times 10^{18} \text{ atoms/cm}^3$, still further preferably lower than or equal to $5 \times 10^{17} \text{ atoms/cm}^3$.

Hydrogen contained in the oxide semiconductor reacts with oxygen bonded to a metal atom to be water, and thus forms an oxygen vacancy in some cases. Entry of hydrogen into the oxygen vacancy generates an electron serving as a carrier in some cases. Furthermore, bonding of part of hydrogen to oxygen bonded to a metal atom causes generation of an electron serving as a carrier in some cases. Thus, a transistor using an oxide semiconductor containing hydrogen is likely to have normally-on characteristics. Accordingly, hydrogen in the oxide semiconductor is preferably reduced as much as possible. Specifically, the hydrogen concentration in the oxide semiconductor, which is obtained by SIMS, is set lower than $1 \times 10^{20} \text{ atoms/cm}^3$, preferably lower than $1 \times 10^{19} \text{ atoms/cm}^3$, further preferably lower than $5 \times 10^{18} \text{ atoms/cm}^3$, still further preferably lower than $1 \times 10^{18} \text{ atoms/cm}^3$.

When an oxide semiconductor with sufficiently reduced impurities is used for the channel formation region of the transistor, stable electrical characteristics can be given.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 6

In this embodiment, a display module using the display device of one embodiment of the present invention is described.

Structure Example of Display Module

First, a display module including the display device of one embodiment of the present invention is described.

FIG. 35A is a perspective view of a display module 1280. The display module 1280 includes the display device 10 and an FPC 1290.

The display module 1280 includes a substrate 1291 and a substrate 1292. The display module 1280 includes a display portion 1281. The display portion 1281 is a region of the display module 1280 where an image is displayed and is a region where light emitted from pixels provided in a pixel portion 1284 described later can be seen.

FIG. 35B is a perspective view schematically illustrating a structure on the substrate 1291 side. Over the substrate 1291, a circuit portion 1282, a pixel circuit portion 1283 over the circuit portion 1282, and the pixel portion 1284 over the pixel circuit portion 1283 are stacked. In addition, a terminal portion 1285 for connection to the FPC 1290 is included in a portion not overlapping with the pixel portion 1284 over the substrate 1291. The terminal portion 1285 and the circuit portion 1282 are electrically connected to each other through a wiring portion 1286 formed of a plurality of wirings.

Note that the pixel portion 1284 and the pixel circuit portion 1283 correspond to the pixel layer PXAL described above, for example. The circuit portion 1282 corresponds to the circuit layer SICL described above, for example.

The pixel portion 1284 includes a plurality of pixels 1284a arranged periodically. An enlarged view of one pixel

1284a is shown on the right side in FIG. 35B. The pixel **1284a** includes a light-emitting device **1430a**, a light-emitting device **1430b**, and a light-emitting device **1430c** which are different in the emission color. Note that the light-emitting device **1430a**, the light-emitting device **1430b**, and the light-emitting device **1430c** correspond to the light-emitting device **150a**, the light-emitting device **150b**, and the light-emitting device **150c** described above. The above-described light emitting devices may be arranged in a stripe pattern as illustrated in FIG. 35B. Alternatively, a variety of kinds of patterns such as a delta pattern or a PenTile pattern can be employed.

The pixel circuit portion **1283** includes a plurality of pixel circuits **1283a** arranged periodically.

One pixel circuit **1283a** is a circuit that controls light emission from three light-emitting devices included in one pixel **1284a**. One pixel circuit **1283a** may be provided with three circuits each of which controls light emission of one light-emitting device. For example, the pixel circuit **1283a** can include at least one selection transistor, one current control transistor (driving transistor), and a capacitor for one light-emitting device. In this case, a gate signal is input to a gate of the selection transistor, and a source signal is input to one of a source and a drain of the selection transistor. With such a structure, an active-matrix display device is achieved.

The circuit portion **1282** includes a circuit for driving each of the pixel circuits **1283a** in the pixel circuit portion **1283**. For example, one or both of a gate line driver circuit and a source line driver circuit are preferably included. In addition, at least one of an arithmetic circuit, a memory circuit, a power supply circuit, and the like may be included.

The FPC **1290** serves as a wiring for supplying a video signal, a power supply potential, or the like to the circuit portion **1282** from the outside. An IC may be mounted on the FPC **1290**.

The display module **1280** can have a structure in which one or both of the pixel circuit portion **1283** and the circuit portion **1282** are stacked below the pixel portion **1284**; thus, the aperture ratio (the effective display area ratio) of the display portion **1281** can be significantly high. For example, the aperture ratio of the display portion **1281** can be higher than or equal to 40% and lower than 100%, preferably higher than or equal to 50% and lower than or equal to 95%, and further preferably higher than or equal to 60% and lower than or equal to 95%. Furthermore, the pixels **1284a** can be arranged extremely densely and thus the display portion **1281** can have greatly high resolution. For example, the pixels **1284a** are preferably arranged in the display portion **1281** with a resolution higher than or equal to 2000 ppi, preferably higher than or equal to 3000 ppi, further preferably higher than or equal to 5000 ppi, still further preferably higher than or equal to 6000 ppi, and lower than or equal to 20000 ppi or lower than or equal to 30000 ppi.

Such a display module **1280** has extremely high resolution, and thus can be suitably used for a device for VR such as a head-mounted display or a glasses-type device for AR. For example, even in the case of a structure in which the display portion of the display module **1280** is seen through a lens, pixels of the extremely-high-resolution display portion **1281** included in the display module **1280** are prevented from being seen when the display portion is enlarged by the lens, so that display providing a high sense of immersion can be performed. Without being limited thereto, the display module **1280** can be suitably used for electronic devices including a relatively small display portion. For example, the display module **1280** can be favorably used in a display

portion of an electronic device to be worn on a human body, such as a wrist-watch type electronic device.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 7

In this embodiment, an example of a head-mounted display including a display device will be described as an example of an electronic device of one embodiment of the present invention.

FIG. 36A and FIG. 36B illustrate external views of a head-mounted display **8300**.

The head-mounted display **8300** includes a housing **8301**, a display portion **8302**, an operation button **8303**, and a band-shaped fixing unit **8304**.

The operation button **8303** has a function of a power button or the like. The head-mounted display **8300** may include a button other than the operation button **8303**.

As illustrated in FIG. 36C, lenses **8305** may be provided between the display portion **8302** and the user's eyes. The user can see magnified images on the display portion **8302** through the lenses **8305**, leading to a higher realistic sensation. In that case, as illustrated in FIG. 36C, a dial **8306** for changing the position of the lenses and adjusting visibility may be included.

The display portion **8302** can use the display device of one embodiment of the present invention. The display device of one embodiment of the present invention has an extremely high resolution; thus, even when an image is magnified using the lenses **8305** as in FIG. 36C, the user does not perceive pixels, and a more realistic image can be displayed.

FIG. 36A to FIG. 36C each illustrate an example in which one display portion **8302** is provided. This structure can reduce the number of components.

The display portion **8302** can display an image for the right eye and an image for the left eye side by side on a right region and a left region, respectively. Thus, a three-dimensional image using binocular disparity can be displayed.

One image which can be seen by both eyes may be displayed on the entire display portion **8302**. A panorama image can thus be displayed from end to end of the field of view, which can provide a higher sense of reality.

Here, the head-mounted display **8300** preferably has a mechanism for changing the curvature of the display portion **8302** to an optimal value in accordance with the size of the user's head, the position of the user's eyes, or the like. For example, the user himself/herself may adjust the curvature of the display portion **8302** by operating a dial **8307** for adjusting the curvature of the display portion **8302**. Alternatively, a sensor for detecting the size of the user's head, the position of the user's eyes, or the like (e.g., a camera, a contact sensor, and a noncontact sensor) may be provided on the housing **8301**, and a mechanism for adjusting the curvature of the display portion **8302** on the basis of data detected by the sensor may be provided.

In the case where the lenses **8305** are used, a mechanism for adjusting the position and angle of the lenses **8305** in synchronization with the curvature of the display portion **8302** is preferably provided. Alternatively, the dial **8306** may have a function of adjusting the angle of the lenses.

FIG. 36E and FIG. 36F illustrate an example of including a driver portion **8308** that controls the curvature of the display portion **8302**. The driver portion **8308** is fixed to at least a part of the display portion **8302**. The driver portion

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8308 has a function of changing the shape of the display portion **8302** when the part that is fixed to the display portion **8302** changes in shape or moves.

FIG. 36E is a schematic diagram illustrating the case where a user **8310** having a relatively large head wears the housing **8301**. In that case, the driver portion **8308** adjusts the shape of the display portion **8302** so that the curvature is relatively small (the radius of curvature is large).

By contrast, FIG. 36F illustrates the case where a user **8311** having a smaller head than the user **8310** wears the housing **8301**. The user **8311** has a shorter distance between the eyes than the user **8310**. In that case, the driver portion **8308** adjusts the shape of the display portion **8302** so that the curvature of the display portion **8302** is large (the radius of curvature is small).

In FIG. 36F, the position and shape of the display portion **8302** in FIG. 36E are denoted by a dashed line.

When the head-mounted display **8300** has such a mechanism for adjusting the curvature of the display portion **8302**, an optimal display can be offered to a variety of users of all ages and genders.

When the curvature of the display portion **8302** is changed in accordance with contents displayed on the display portion **8302**, the user can have a more realistic sensation. For example, shaking can be expressed by vibrating the curvature of the display portion **8302**. In this way, it is possible to produce various effects according to the scene in contents, and provide the user with new experiences. A further realistic display can be provided when the display portion **8302** operates in conjunction with a vibration module provided in the housing **8301**.

Note that the head-mounted display **8300** may include two display portions **8302** as illustrated in FIG. 36D.

Since the two display portion **8302** are included, the user's eyes can see their respective display portions. Thus, a high-resolution image can be displayed even when a three-dimensional display using parallax or the like is performed. In addition, the display portion **8302** is curved around an arc with the user's eye as an approximate center. Thus, distances between the user's eye and display surfaces of the display portion become equal; thus, the user can see a more natural image. Even when the luminance or chromaticity of light from the display portion is changed depending on the angle at which the user see it, since the user's eye is positioned in a normal direction of the display surface of the display portion, the influence of the change can be substantially ignorable and thus a more realistic image can be displayed.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 8

In this embodiment, a display module that can be fabricated using the display device of one embodiment of the present invention is described.

In a display module **6000** illustrated in FIG. 37A, a display device **6006** to which an FPC **6005** is connected, a frame **6009**, a printed circuit board **6010**, and a battery **6011** are provided between an upper cover **6001** and a lower cover **6002**.

A display device fabricated using the display device of one embodiment of the present invention can be used as the display device **6006**, for example. With the display device **6006**, a display module with extremely low power consumption can be achieved.

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The shape and size of the upper cover **6001** and the lower cover **6002** can be changed as appropriate in accordance with the size of the display device **6006**.

The display device **6006** may have a function of a touch panel.

The frame **6009** may have a function of protecting the display device **6006**, a function of blocking electromagnetic waves generated by the operation of the printed circuit board **6010**, a function of a heat dissipation plate, or the like.

The printed circuit board **6010** includes a power supply circuit, a signal processing circuit for outputting a video signal and a clock signal, a battery control circuit, and the like.

FIG. 37B is a schematic cross-sectional view of the display module **6000** including an optical touch sensor.

The display module **6000** includes a light-emitting portion **6015** and a light-receiving portion **6016** that are provided on the printed circuit board **6010**. Furthermore, a pair of light guide portions (a light guide portion **6017a** and a light guide portion **6017b**) are provided in regions surrounded by the upper cover **6001** and the lower cover **6002**.

The display device **6006** overlaps with the printed circuit board **6010** and the battery **6011** with the frame **6009** therebetween. The display device **6006** and the frame **6009** are fixed to the light guide portion **6017a** and the light guide portion **6017b**.

Light **6018** emitted from the light-emitting portion **6015** travels over the display device **6006** through the light guide portion **6017a** and reaches the light-receiving portion **6016** through the light guide portion **6017b**. For example, blocking of the light **6018** by a sensing target such as a finger or a stylus enables detection of touch operation.

A plurality of light-emitting portions **6015** are provided along two adjacent sides of the display device **6006**, for example. A plurality of light-receiving portions **6016** are provided at the positions on the opposite side of the light-emitting portions **6015**. Accordingly, information about the position of touch operation can be obtained.

As the light-emitting portion **6015**, a light source such as an LED element can be used, for example, and it is particularly preferable to use a light source emitting infrared rays. As the light-receiving portion **6016**, a photoelectric element that receives light emitted from the light-emitting portion **6015** and converts it into an electric signal can be used. A photodiode that can receive infrared rays can be suitably used.

The light guide portion **6017a** and the light guide portion **6017b** which transmit the light **6018** allow the light-emitting portion **6015** and the light-receiving portion **6016** to be placed under the display device **6006**, inhibiting a malfunction of the touch sensor due to external light reaching the light-receiving portion **6016**. Particularly when a resin that absorbs visible light and transmits infrared rays is used, a malfunction of the touch sensor can be inhibited more effectively.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 9

In this embodiment, examples of an electronic device for which the display device of one embodiment of the present invention can be used are described.

An electronic device **6500** illustrated in FIG. 38A is a portable information terminal that can be used as a smartphone.

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The electronic device **6500** includes a housing **6501**, a display portion **6502**, a power button **6503**, buttons **6504**, a speaker **6505**, a microphone **6506**, a camera **6507**, a light source **6508**, and the like. The display portion **6502** has a touch panel function.

The display device of one embodiment of the present invention can be used in the display portion **6502**.

FIG. **38B** is a schematic cross-sectional view including an end portion of the housing **6501** on the microphone **6506** side.

A protective member **6510** having a light-transmitting property is provided on the display surface side of the housing **6501**, and a display panel **6511**, an optical member **6512**, a touch sensor panel **6513**, a printed circuit board **6517**, a battery **6518**, and the like are provided in a space surrounded by the housing **6501** and the protective member **6510**.

The display panel **6511**, the optical member **6512**, and the touch sensor panel **6513** are fixed to the protective member **6510** with an adhesive layer not illustrated.

Part of the display panel **6511** is bent in a region outside the display portion **6502**. An FPC **6515** is connected to the bent part. An IC **6516** is mounted on the FPC **6515**. The FPC **6515** is connected to a terminal provided for the printed circuit board **6517**.

A flexible display panel can be used as the display panel **6511**, for example. Thus, an extremely lightweight electronic device can be obtained. Furthermore, since the display panel **6511** is extremely thin, the battery **6518** with a high capacity can be provided without an increase in the thickness of the electronic device. Moreover, part of the display panel **6511** is bent to provide a connection portion with the FPC **6515** on the back side of the pixel portion, whereby an electronic device with a narrow bezel can be obtained.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

Embodiment 10

In this embodiment, electronic devices each including a display device fabricated using one embodiment of the present invention are described.

Electronic devices described below as examples each include the display device of one embodiment of the present invention in a display portion. Thus, the electronic devices achieve high resolution. In addition, the electronic devices can each achieve both high resolution and a large screen.

One embodiment of the present invention includes the display device and at least one of an antenna, a battery, a housing, a camera, a speaker, a microphone, a touch sensor, and an operation button.

The electronic device of one embodiment of the present invention may include a secondary battery, and it is preferable that the secondary battery be capable of being charged by contactless power transmission.

Examples of the secondary battery include a lithium ion secondary battery such as a lithium polymer battery using a gel electrolyte (lithium ion polymer battery), a nickel-hydride battery, a nickel-cadmium battery, an organic radical battery, a lead-acid battery, an air secondary battery, a nickel-zinc battery, and a silver-zinc battery.

The electronic device of one embodiment of the present invention may include an antenna. When a signal is received by the antenna, the electronic device can display a video, data, or the like on a display portion. When the electronic

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device includes the antenna and a secondary battery, the antenna may be used for contactless power transmission.

A display portion in an electronic device of one embodiment of the present invention can display a video with a resolution of, for example, full high definition, 4K2K, 8K4K, 16K8K, or higher.

Examples of the electronic devices include a digital camera, a digital video camera, a digital photo frame, a cellular phone, a portable game machine, a portable information terminal, and an audio reproducing device, in addition to electronic devices with comparatively large screens, such as a television device, a notebook personal computer, a monitor device, digital signage, a pachinko machine, and a game machine.

An electronic device using one embodiment of the present invention can be incorporated along a flat surface or a curved surface of an inside wall or an outside wall of a building such as a house or a building, an interior or an exterior of a car, or the like.

FIG. **39A** is a diagram illustrating appearance of a camera **8000** to which a finder **8100** is attached.

The camera **8000** includes a housing **8001**, a display portion **8002**, operation buttons **8003**, a shutter button **8004**, and the like. In addition, a detachable lens **8006** is attached to the camera **8000**.

Note that the lens **8006** and the housing may be integrated with each other in the camera **8000**.

The camera **8000** can take images by the press of the shutter button **8004** or touch on the display portion **8002** serving as a touch panel.

The housing **8001** includes a mount including an electrode, so that the finder **8100**, a stroboscope, or the like can be connected to the housing.

The finder **8100** includes a housing **8101**, a display portion **8102**, a button **8103**, and the like.

The housing **8101** is attached to the camera **8000** with the mount engaging with a mount of the camera **8000**. In the finder **8100**, a video or the like received from the camera **8000** can be displayed on the display portion **8102**.

The button **8103** has a function of a power button or the like.

The display device of one embodiment of the present invention can be used for the display portion **8002** of the camera **8000** and the display portion **8102** of the finder **8100**. Note that a finder may be incorporated in the camera **8000**.

FIG. **39B** is an external view of an information terminal **5900** that is an example of a wearable terminal. The information terminal **5900** includes a housing **5901**, a display portion **5902**, an operation button **5903**, an operator **5904**, a band **5905**, and the like.

The wearable terminal can display an image with high display quality on the display portion **5902** by including the display device described in the above embodiment.

FIG. **39C** is a diagram illustrating the appearance of a portable game machine **5200** which is an example of a game machine. The portable game machine **5200** includes a housing **5201**, a display portion **5202**, a button **5203**, and the like.

Videos displayed on the portable game machine **5200** can be output with a display device such as a television device, a personal computer display, a game display, and a head-mounted display.

The portable game machine **5200** can display an image with high display quality on the display portion **5202** by including the display device described in the above embodiment. In addition, the portable game machine **5200** with low power consumption can be provided. Moreover, heat generation from a circuit can be reduced owing to low power

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consumption; thus, the influence of heat generation on the circuit itself, a peripheral circuit, and a module can be reduced.

FIG. 40A is a diagram illustrating appearance of a head-mounted display 8200.

The head-mounted display 8200 includes a mounting portion 8201, a lens 8202, a main body 8203, a display portion 8204, a cable 8205, and the like. In addition, a battery 8206 is incorporated in the mounting portion 8201.

The cable 8205 supplies power from the battery 8206 to the main body 8203. The main body 8203 includes a wireless receiver or the like and can display received video information on the display portion 8204. In addition, the main body 8203 is provided with a camera, and information on the movement of the user's eyeball or eyelid can be used as an input means.

The mounting portion 8201 may be provided with a plurality of electrodes capable of sensing current flowing in response to the movement of the user's eyeball in a position in contact with the user to have a function of recognizing the user's sight line. Furthermore, the mounting portion 8201 may have a function of monitoring the user's pulse with the use of current flowing through the electrodes. Moreover, the mounting portion 8201 may include a variety of sensors such as a temperature sensor, a pressure sensor, and an acceleration sensor to have a function of displaying the user's biological information on the display portion 8204, a function of changing a video displayed on the display portion 8204 in accordance with the movement of the user's head, or the like.

The display device of one embodiment of the present invention can be used for the display portion 8204.

FIG. 40B, FIG. 40C, and FIG. 40D are diagrams illustrating appearance of a head-mounted display 8300. The head-mounted display 8300 includes a housing 8301, a display portion 8302, band-shaped fixing units 8304, and a pair of lenses 8305.

A user can see display on the display portion 8302 through the lenses 8305. Note that the display portion 8302 is preferably curved and placed because the user can feel a high realistic sensation. In addition, when another image displayed on a different region of the display portion 8302 is viewed through the lenses 8305, three-dimensional display using parallax or the like can also be performed. Note that the number of display portions 8302 provided is not limited to one; two display portions 8302 may be provided so that one display portion is provided for one eye of the user.

Note that the display device of one embodiment of the present invention can be used in the display portion 8302. The display device including the semiconductor device of one embodiment of the present invention has an extremely high resolution; thus, even when a video is magnified by the lenses 8305 as in FIG. 40D, the user does not perceive pixels, and a more realistic video can be displayed.

Note that this embodiment can be combined with any of the other embodiments described in this specification as appropriate.

REFERENCE NUMERALS

SICL: circuit layer, LINL: wiring layer, PXAL: pixel layer, BS: substrate, DRV: driver circuit region, DRVa: driver circuit region, DRVb: driver circuit region, LIA: region, LD: local driver circuit, ALP: pixel array, ARA: pixel region, ARAa: region, ARAb: region, PIX: pixel circuit, PIXa: pixel circuit, PIXb: pixel circuit, CON:

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controller, PG: voltage generation circuit, SLS: wiring group, SL_1: wiring, SL_2: wiring, SL_3: wiring, SLp: wiring, SLa: wiring, SLa_1: wiring, SLa_2: wiring, SLa_3: wiring, SLb: wiring, SLb_1: wiring, SLb_2: wiring, SLb_3: wiring, GLS: wiring group, SL: wiring, GL: wiring, GLa: wiring, GLb: wiring, DL: wiring, DLa: wiring, DLb: wiring, SGa: wiring, SGb: wiring, SLd_1: wiring, SLd_2: wiring, SLd_w: wiring, SLu_1: wiring, SLu_2: wiring, SLu_w: wiring, CNT: contact portion, Tr: transistor, VA: contact portion, CD: conductor, ER: conductor, ALPa: region, ALPb: region, ALPa_C: region, ALPb_C: region, 10: display device, 11: display device, 11A: display device, 11B: display device, 12: display device, 12A: display device, 12AA: display device, 12B: display device, 12C: display device, 12D: display device, 12E: display device, 13A: display device, 14: display device, 14A: display device, 102: substrate, 111: insulator, 112: insulator, 112a: insulator, 112b: insulator, 113: insulator, 113a: insulator, 113b: insulator, 113c: insulator, 121a: conductor, 121b: conductor, 121c: conductor, 122: conductor, 141a: EL layer, 141b: EL layer, 141c: EL layer, 142: EL layer, 150: light-emitting device, 150a: light-emitting device, 150b: light-emitting device, 150c: light-emitting device, 161: resin layer, 162a: coloring layer, 162b: coloring layer, 162c: coloring layer, 163: black matrix, 164: adhesive layer, 165: adhesive layer, 300: transistor, 310: substrate, 312: element isolation layer, 313: semiconductor region, 314a: low-resistance region, 314b: low-resistance region, 314c: low-resistance region, 315: insulator, 316: conductor, 317: insulator, 320: insulator, 322: insulator, 324: insulator, 326: insulator, 328: conductor, 330: conductor, 350: insulator, 352: insulator, 354: insulator, 356: conductor, 360: insulator, 362: insulator, 364: insulator, 366: conductor, 500: transistor, 503: conductor, 503a: conductor, 503b: conductor, 512: insulator, 514: insulator, 516: insulator, 522: insulator, 524: insulator, 530: oxide, 530a: oxide, 530b: oxide, 530ba: region, 530bb: region, 530bc: region, 540: conductor, 540a: conductor, 540b: conductor, 541: insulator, 541a: insulator, 541b: insulator, 542: conductor, 542a: conductor, 542b: conductor, 544: insulator, 550: insulator, 550a: insulator, 550b: insulator, 552: insulator, 554: insulator, 560: conductor, 560a: conductor, 560b: conductor, 571: insulator, 571a: insulator, 571b: insulator, 574: insulator, 576: insulator, 580: insulator, 581: insulator, 1280: display module, 1281: display portion, 1290: FPC, 1283: pixel circuit portion, 1283a: pixel circuit, 1284: pixel portion, 1284a: pixel, 1285: terminal portion, 1286: wiring portion, 1291: substrate, 1292: substrate, 1430a: light-emitting device, 1430b: light-emitting device, 1430c: light-emitting device, 4400a: light-emitting unit, 4400b: light-emitting unit, 4411: light-emitting layer, 4412: light-emitting layer, 4413: light-emitting layer, 4420: layer, 4430: layer, 4440: intermediate layer, 5200: portable game machine, 5201: housing, 5202: display portion, 5203: button, 5900: information terminal, 5901: housing, 5902: display portion, 5903: operation button, 5904: band, 5905: band, 6000: display module, 6001: upper cover, 6002: lower cover, 6005: FPC, 6006: display device, 6009: frame, 6010: printed circuit board, 6011: battery, 6015: light-emitting portion, 6016: light-receiving portion, 6017a: light guide portion, 6017b: light guide portion, 6018: light, 6500: electronic device, 6501: housing, 6502: display portion, 6503: power button, 6504: button, 6505:

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speaker, **6506**: microphone, **6507**: camera, **6508**: light source, **6510**: protective member, **6511**: display panel, **6512**: optical member, **6513**: touch sensor panel, **6515**: FPC, **6516**: IC, **6517**: printed circuit board, **6518**: battery, **8000**: camera, **8001**: housing, **8002**: display portion, **8003**: operation button, **8004**: shutter button, **8006**: lens, **8100**: finder, **8101**: housing, **8102**: display portion, **8103**: button, **8200**: head-mounted display, **8201**: mounting portion, **8202**: lens, **8203**: main body, **8204**: display portion, **8205**: cable, **8206**: battery, **8300**: head-mounted display, **8301**: housing, **8302**: display portion, **8303**: operation button, **8304**: fixing unit, **8305**: lens, **8306**: dial, **8307**: dial, **8308**: driver portion, **8310**: user, **8311**: user

The invention claimed is:

1. A display device comprising:

a first layer;

a second layer positioned above the first layer; and

a third layer positioned above the second layer,

wherein the first layer comprises a driver circuit and a plurality of first wirings,

wherein the driver circuit comprises a plurality of output terminals positioned along a first direction,

wherein the plurality of output terminals are electrically connected to the plurality of first wirings,

wherein the plurality of first wirings are extended perpendicular to the first direction,

wherein the third layer comprises a pixel array and a plurality of second wirings,

wherein the pixel array comprises a plurality of pixel circuits arranged in a matrix,

wherein the plurality of pixel circuits are electrically connected to the plurality of second wirings,

wherein the plurality of second wirings are parallel to each other and extended in a column direction of the pixel array,

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wherein the second layer comprises a plurality of first contact portions,

wherein the plurality of first wirings are electrically connected to the plurality of second wirings through the plurality of first contact portions, and

wherein the driver circuit is configured to control the plurality of pixel circuits.

2. The display device according to claim **1**,

wherein one of the plurality of first contact portions is positioned inside the pixel array and inside or outside the plurality of pixel circuits in a top view.

3. The display device according to claim **1**,

wherein the first layer comprises a semiconductor substrate comprising silicon and a plurality of second contact portions,

wherein the driver circuit comprises a plurality of transistors each including the silicon in a channel formation region,

wherein a plurality of low-resistance regions to be the plurality of first wirings are positioned on a top surface of the semiconductor substrate,

wherein the plurality of second contact portions are positioned between the plurality of first contact portions and the plurality of second contact portions, and

wherein the plurality of low-resistance regions comprise one of a source and a drain of each of the plurality of transistors.

4. The display device according to claim **1**,

wherein the third layer comprises a plurality of third wirings positioned along a second direction,

wherein the plurality of third wirings are parallel to each other and extended in a row direction of the pixel array, and

wherein the plurality of first wirings extend in a third direction intersecting the first direction and the second direction.

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